COHOMOLOGICAL STUDY OF COHERENT SHEAVES (EGA III)

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This chapter gives the fundamental theorems concerning the cohomology of coherent algebraic sheaves, with the exception of theorems explaining the theory of residues (duality theorems), which will be the subject of a later chapter. Amongst all those included here, there are essentially six fundamental theorems, and each one is the subject of one of the first six chapters. These results will prove to be essential tools in all that follows, even in questions which are not truly cohomological in their nature; the reader will see the first such examples starting from §4. §7 gives some more technical results, but ones which are constantly used in applications. Finally, in §§8–11, we will develop certain results, related to the duality of coherent sheaves, that are particularly important for applications, and which can be explained even before the introduction of the full general theory of residues.

The content of §§1 and 2 is due to J.-P. Serre, and the reader will observe that we have had only to follow (FAC). §8 and 9 are equally inspired by (FAC) (the changes necessitated by the different contexts, however, being less evident). Finally, as we said in the Introduction, §4 should be considered as the formalisation, in modern language, of the fundamental "invariance theorem" of Zariski's "theory of holomorphic functions".

We draw attention to the fact that the results of n°3.4 (and the preliminary propositions of (**0**, 13.4 to 13.7)) will not be used in what follows Chapter III, and can thus be skipped in a first reading.

[1] EGA IV does not depend on §§8–11, and will probably be published before these chapters. *[Trans.] These last four chapters were never published.*

§1. Cohomology of affine schemes

1.1. Review of the exterior algebra complex

(1.1.1). Let A be a ring, $\mathbf{f} = (f_i)_{1\leqslant i\leqslant r}$ a system of r elements of A. The *exterior algebra complex* $K_\bullet(\mathbf{f})$ corresponding to $\mathbf{f}$ is a chain complex (G, I, 2.2) defined in the following way: the graded A-module $K_\bullet(\mathbf{f})$ is equal to the *exterior algebra* $\wedge(A^r)$, graded in the usual way, and the boundary map is the *interior multiplication* $i_{\mathbf{f}}$ by $\mathbf{f}$ considered as an element of the dual $(A^r)^\vee$; we recall that $i_{\mathbf{f}}$ is an *antiderivation* of degree -1 of $\wedge(A^r)$, and if $(\mathbf{e}_i)_{1\leqslant i\leqslant r}$ is the canonical basis of A^r, then we have $i_{\mathbf{f}}(\mathbf{e}_i) = f_i$; the verification of the condition $i_{\mathbf{f}} \circ i_{\mathbf{f}} = 0$ is immediate.

An equivalent definition is the following: for each i, we consider a chain complex $K_\bullet(f_i)$ defined as follows: $K_0(f_i) = K_1(f_i) = A$, $K_n(f_i) = 0$ for $n \neq 0, 1$: the boundary map is defined by the condition that $d_1 : A \to A$ is *multiplication by* f_i. We then take $K_\bullet(\mathbf{f})$ to be the *tensor product* $K_\bullet(f_1) \otimes K_\bullet(f_2) \otimes \cdots \otimes K_\bullet(f_r)$ (G, I, 2.7) with its total degree; the verification of the isomorphism from this complex to the complex defined above is immediate.

(1.1.2). For every A-module M, we define the *chain complex*

(1.1.2.1) $$K_\bullet(\mathbf{f}, M) = K_\bullet(\mathbf{f}) \otimes_A M$$

and the *cochain complex* (G, I, 2.2)

(1.1.2.2) $$K^\bullet(\mathbf{f}, M) = \operatorname{Hom}_A(K_\bullet(\mathbf{f}, M)).$$

If g is a k-cochain of this latter complex, and if we set

$$g(i_1, \ldots, i_k) = g(\mathbf{e}_{i_1} \wedge \cdots \wedge \mathbf{e}_{i_k}),$$

then g identifies with an *alternating* map from $[1, r]^k$ to M, and it follows from the above definitions that we have

(1.1.2.3) $$d^k g(i_1, i_2, \ldots, i_{k+1}) = \sum_{h=1}^{k+1} (-1)^{h-1} f_{i_h} g(i_1, \ldots, \widehat{i_h}, \ldots, i_{k+1}).$$

(1.1.3). From the above complexes, we deduce as usual the *homology and cohomology A-modules* (G, I, 2.2) **III** | 83

(1.1.3.1) $$\mathrm{H}_\bullet(\mathbf{f}, M) = \mathrm{H}_\bullet(K_\bullet(\mathbf{f}, M)),$$

(1.1.3.2) $$\mathrm{H}^\bullet(\mathbf{f}, M) = \mathrm{H}^\bullet(K^\bullet(\mathbf{f}, M)).$$

We define an *A-isomorphism* $K_\bullet(\mathbf{f}, M) \simeq K^\bullet(\mathbf{f}, M)$ by sending each chain $z = \sum(\mathbf{e}_{i_1} \wedge \cdots \wedge \mathbf{e}_{i_k}) \otimes z_{i_1,\ldots,i_k}$ to the cochain g_z such that $g_z(j_1, \ldots, j_{r-k}) = \varepsilon z_{i_1,\ldots,i_k}$, where $(j_h)_{1\leqslant h\leqslant r-k}$ is the strictly increasing sequence complementary to the strictly increasing sequence $(i_h)_{1\leqslant h\leqslant k}$ in $[1, r]$ and $\varepsilon = (-1)^\nu$, where ν is the number of inversions of the permutation $i_1, \ldots, i_k, j_1, \ldots, j_{r-k}$ of $[1, r]$. We verify that $g_{dz} = d(g_z)$, which gives an isomorphism

(1.1.3.3) $$\mathrm{H}^i(\mathbf{f}, M) \simeq \mathrm{H}_{r-i}(\mathbf{f}, M) \text{ for } 0 \leqslant i \leqslant r.$$

In this chapter, we will especially consider the cohomology modules $\mathrm{H}^\bullet(\mathbf{f}, M)$.

For a given $\mathbf{f}$, it is immediate (G, I, 2.1) that $M \mapsto \mathrm{H}^\bullet(\mathbf{f}, M)$ is a *cohomological functor* (T, II, 2.1) from the category of A-modules to the category of graded A-modules, zero in degrees < 0 and $> r$. In addition, we have

(1.1.3.4) $$\mathrm{H}^0(\mathbf{f}, M) = \operatorname{Hom}_A(A/(\mathbf{f}), M),$$

denoting by $(\mathbf{f})$ the ideal of A generated by $f_1, \ldots, f_r$; this follows immediately from (1.1.2.3), and it is clear that $\mathrm{H}^0(\mathbf{f}, M)$ identifies with the submodule of M *killed by* $(\mathbf{f})$. Similarly, we have by (1.1.2.3) that

(1.1.3.5) $$\mathrm{H}^r(\mathbf{f}, M) = M / \left(\sum_{i=1}^r f_i M\right) = (A/(\mathbf{f})) \otimes_A M.$$

We will use the following known result, which we will recall a proof of to be complete:

Proposition (1.1.4). — *Let A be a ring, $\mathbf{f} = (f_i)_{1 \leqslant i \leqslant r}$ a finite family of elements of A, and M an A-module. If, for $1 \leqslant i \leqslant r$, the scaling $z \mapsto f_i \cdot z$ on $M_{i-1} = M/(f_1 M + \cdots + f_{i-1}M)$ is injective, then we have $\mathrm{H}^i(\mathbf{f}, M) = 0$ for $i \neq r$.*

It suffices to prove that $\mathrm{H}_i(\mathbf{f}, M) = 0$ for all $i > 0$ according to (1.1.3.3). We argue by induction on r, the case $r = 0$ being trivial. Set $\mathbf{f}' = (f_i)_{1 \leqslant i \leqslant r-1}$; this family satisfies the conditions in the statement, so if we set $L_\bullet = K_\bullet(\mathbf{f}', M)$, then we have $\mathrm{H}_i(L_\bullet) = 0$ for $i > 0$ by hypothesis, and $\mathrm{H}_0(L_\bullet) = M_{r-1}$ by virtue of (1.1.3.3) and (1.1.3.5). To abbreviate, set $K_\bullet = K_\bullet(f_r) = K_0 \oplus K_1$, with $K_0 = K_1 = A$, $d_1 : K_1 \to K_0$ multiplication by f_r; we have by definition (1.1.1) that $K_\bullet(\mathbf{f}, M) = K_\bullet \otimes_A L_\bullet$. We have the following lemma:

Lemma (1.1.4.1). — *Let $K_\bullet$ be a chain complex of free A-modules, zero except in dimensions 0 and 1. For every chain complex $L_\bullet$ of A-modules, we have an exact sequence*

$$0 \longrightarrow \mathrm{H}_0(K_\bullet \otimes \mathrm{H}_p(L_\bullet)) \longrightarrow \mathrm{H}_p(K_\bullet \otimes L_\bullet) \longrightarrow \mathrm{H}_1(K_\bullet \otimes \mathrm{H}_{p-1}(L_\bullet)) \longrightarrow 0$$

for every index p.

This is a particular case of an exact sequence of low-order terms of the Künneth spectral sequence (M, XVII, 5.2 (a) and G, I, 5.5.2); it can be proved directly as follows. Consider K_0 and K_1 as chain complexes (zero in dimensions $\neq 0$ and $\neq 1$ respectively); we then have an exact sequence of complexes **III | 84**

$$0 \longrightarrow K_0 \otimes L_\bullet \longrightarrow K_\bullet \otimes L_\bullet \longrightarrow K_1 \otimes L_\bullet \longrightarrow 0,$$

to which we can apply the exact sequence in homology

$$\cdots \longrightarrow \mathrm{H}_{p+1}(K_1 \otimes L_\bullet) \xrightarrow{\partial} \mathrm{H}_p(K_0 \otimes L_\bullet) \longrightarrow \mathrm{H}_p(K_\bullet \otimes L_\bullet) \longrightarrow \mathrm{H}_p(K_1 \otimes L_\bullet) \xrightarrow{\partial} \mathrm{H}_{p-1}(K_0 \otimes L_\bullet) \longrightarrow \cdots .$$

But it is evident that $\mathrm{H}_p(K_0 \otimes L_\bullet) = K_0 \otimes \mathrm{H}_p(L_\bullet)$ and $\mathrm{H}_p(K_1 \otimes L_\bullet) = K_1 \otimes \mathrm{H}_{p-1}(L_\bullet)$ for all p; in addition, we verify immediately that the operator $\partial : K_1 \otimes \mathrm{H}_p(L_\bullet) \to K_0 \otimes \mathrm{H}_p(L_\bullet)$ is none other than $d_1 \otimes 1$; the lemma thus follows from the above exact sequence and the definition of $\mathrm{H}_0(K_\bullet \otimes \mathrm{H}_p(L_\bullet))$ and $\mathrm{H}_1(K_\bullet \otimes \mathrm{H}_{p-1}(L_\bullet))$.

The lemma having been established, the end of the proof of Proposition (1.1.4) is immediate: the induction hypothesis of Lemma (1.1.4.1) gives $\mathrm{H}_p(K_\bullet \otimes L_\bullet) = 0$ for $p \geqslant 2$; in addition if we show that $\mathrm{H}_1(K_\bullet, \mathrm{H}_0(L_\bullet)) = 0$, then we also deduce from Lemma (1.1.4.1) that $\mathrm{H}_1(K_\bullet \otimes L_\bullet) = 0$; but by definition, $\mathrm{H}_1(K_\bullet, \mathrm{H}_0(L_\bullet))$ is none other than the kernel of the scaling $z \mapsto f_r \cdot z$ on M_{r-1}, and as by hypothesis this kernel is zero, this finishes the proof.

(1.1.5). Let $\mathbf{g} = (g_i)_{1 \leqslant i \leqslant r}$ be a second sequence of r elements of A, and set $\mathbf{fg} = (f_i g_i)_{1 \leqslant i \leqslant r}$. We can define a canonical homomorphism of complexes

(1.1.5.1) $$\phi_{\mathbf{g}} : K_\bullet(\mathbf{fg}) \longrightarrow K_\bullet(\mathbf{f})$$

as the canonical extension to the exterior algebra $\wedge(A^r)$ of the A-linear map $(x_1, \ldots, x_r) \mapsto (g_1 x_1, \ldots, g_r x_r)$ from A^r to itself. To see that we have a homomorphism of complexes, it suffices to note, in general, that if $u : E \to F$ is an A-linear map, and if $\mathbf{x} \in F^\vee$ and $\mathbf{y} = {}^t u(\mathbf{x}) \in E^\vee$, then we have the formula

(1.1.5.2) $$(\wedge u) \circ i_{\mathbf{y}} = i_{\mathbf{x}} \circ (\wedge u);$$

indeed, the two elements are antiderivations of $\wedge F$, and it suffices to check that they coincide on F, which follows immediately from the definitions.

When we identify $K_\bullet(\mathbf{f})$ with the tensor product of the $K_\bullet(f_i)$ (1.1.1), $\phi_{\mathbf{g}}$ is the tensor product of the ϕ_{g_i}, where ϕ_{g_i} is the identity in degree 0 and multiplication by g_i in degree 1.

(1.1.6). In particular, for every pair of integers m and n such that $0 \leqslant n \leqslant m$, we have homomorphisms of complexes

(1.1.6.1) $$\phi_{\mathbf{f}^{m-n}} : K_\bullet(\mathbf{f}^m) \longrightarrow K_\bullet(\mathbf{f}^n)$$

and as a result, homomorphisms

(1.1.6.2) $$\phi_{\mathbf{f}^{m-n}} : K^\bullet(\mathbf{f}^n, M) \longrightarrow K^\bullet(\mathbf{f}^m, M),$$

(1.1.6.3) $$\phi_{\mathbf{f}^{m-n}} : \mathrm{H}^\bullet(\mathbf{f}^n, M) \longrightarrow \mathrm{H}^\bullet(\mathbf{f}^m, M).$$

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The latter homomorphisms evidently satisfy the transitivity condition $\phi_{\mathbf{f}^{m-p}} = \phi_{\mathbf{f}^{m-n}} \circ \phi_{\mathbf{f}^{n-p}}$ for $p \leqslant n \leqslant m$; they therefore define two *inductive systems* of A-modules; we set

$$C^\bullet((\mathbf{f}), M) = \varinjlim_n K^\bullet(\mathbf{f}^n, M), \tag{1.1.6.4}$$

$$H^\bullet((\mathbf{f}), M) = H^\bullet(C^\bullet((\mathbf{f}), M)) = \varinjlim_n H^\bullet(\mathbf{f}^n, M), \tag{1.1.6.5}$$

the last equality following from the fact that passing to the inductive limit commutes with the functor $H^\bullet$ (G, I, 2.1). We will later see (1.4.3) that $H^\bullet((\mathbf{f}), M)$ does not depend on the *ideal* $(\mathbf{f})$ of A (and similarly on the $(\mathbf{f})$-pre-adic topology on A), which justifies the notations.

It is clear that $M \mapsto C^\bullet((\mathbf{f}), M)$ is an exact A-linear functor, and $M \mapsto H^\bullet((\mathbf{f}), M)$ is a cohomological functor.

(1.1.7). Set $\mathbf{f} = (f_i) \in A^r$ and $\mathbf{g} = (g_i) \in A^r$; denote by $e_{\mathbf{g}}$ the left multiplication by the vector $\mathbf{g} \in A^r$ on the exterior algebra $\wedge(A^r)$; we know that we have the *homotopy formula*

$$i_{\mathbf{f}} e_{\mathbf{g}} + e_{\mathbf{g}} i_{\mathbf{f}} = \langle \mathbf{g}, \mathbf{f} \rangle 1 \tag{1.1.7.1}$$

in the A-module A^r (1 denotes the identity automorphism of A^r); this relation also implies that *in the complex* $K_\bullet(\mathbf{f})$ we have

$$d e_{\mathbf{g}} + e_{\mathbf{g}} d = \langle \mathbf{g}, \mathbf{f} \rangle 1. \tag{1.1.7.2}$$

If the ideal $(\mathbf{f})$ is equal to A, then there exists a $\mathbf{g} \in A^r$ such that $\langle \mathbf{g}, \mathbf{f} \rangle = \sum_{i=1}^r g_i f_i = 1$. As a result (G, I, 2.4):

Proposition (1.1.8). — *Suppose that the ideal* $(\mathbf{f})$ *generated by the* f_i *is equal to* A. *Then the complex* $K_\bullet(\mathbf{f})$ *is homotopically trivial, and so are the complexes* $K_\bullet(\mathbf{f}, M)$ *and* $K^\bullet(\mathbf{f}, M)$ *for every* A*-module* M.

Corollary (1.1.9). — *If* $(\mathbf{f}) = A$, *then we have* $H^\bullet(\mathbf{f}, M) = 0$ *and* $H^\bullet((\mathbf{f}), M) = 0$ *for every* A*-module* M.

Proof. Indeed, we then have $(\mathbf{f}^n) = A$ for all n. □

Remark (1.1.10). — With the same notations as above, set $X = \operatorname{Spec}(A)$ and Y the closed subprescheme of X defined by the ideal $(\mathbf{f})$. We will prove in §9 that $H^\bullet((\mathbf{f}), M)$ is isomorphic to the cohomology $H^\bullet_Y(X, \widetilde{M})$ corresponding to the antifilter Φ of closed subsets of Y (T, 3.2). We will also show that Proposition (1.2.3) applied to X and to $\mathscr{F} = \widetilde{M}$ is a particular case of an exact sequence in cohomology

$$\cdots \longrightarrow H^p_Y(X, \mathscr{F}) \longrightarrow H^p(X, \mathscr{F}) \longrightarrow H^p(X - Y, \mathscr{F}) \longrightarrow H^{p+1}_Y(X, \mathscr{F}) \longrightarrow \cdots.$$

1.2. Čech cohomology of an open cover

Notation (1.2.1). — In this section, we denote:

(1) X a prescheme;
(2) $\mathscr{F}$ a quasi-coherent $\mathscr{O}_X$-module; **III** | 86
(3) $A = \Gamma(X, \mathscr{O}_X)$, $M = \Gamma(X, \mathscr{F})$;
(4) $\mathbf{f} = (f_i)_{1 \leqslant i \leqslant r}$ a finite system of elements of A;
(5) $U_i = X_{f_i}$, the open set (**0**, 5.5.2) of the $x \in X$ such that $f_i(x) \neq 0$;
(6) $U = \bigcup_{i=1}^r U_i$;
(7) $\mathfrak{U}$ the cover $(U_i)_{1 \leqslant i \leqslant r}$ of U.

(1.2.2). Suppose that X is either a prescheme whose underlying space is *Noetherian* or a *scheme* whose underlying space is *quasi-compact*. We then know (**I**, 9.3.3) that we have $\Gamma(U_i, \mathscr{F}) = M_{f_i}$. We set

$$U_{i_0 i_1 \cdots i_p} = \bigcap_{k=0}^p U_{i_k} = X_{f_{i_0} f_{i_1} \cdots f_{i_p}}$$

(**0**, 5.5.3); so we also have

$$\Gamma(U_{i_0 i_1 \cdots i_p}, \mathscr{F}) = M_{f_{i_0} f_{i_1} \cdots f_{i_p}}. \tag{1.2.2.1}$$

We have (**0**, 1.6.1) that $M_{f_{i_0}f_{i_1}\cdots f_{i_p}}$ identifies with the inductive limit $\varinjlim_n M^{(n)}_{i_0 i_1\cdots i_p}$, where the inductive system is formed by the $M^{(n)}_{i_0 i_1\cdots i_p} = M$, the homomorphisms $\phi_{nm} : M^{(m)}_{i_0 i_1\cdots i_p} \to M^{(n)}_{i_0 i_1\cdots i_p}$ being multiplication by $(f_{i_0}f_{i_1}\cdots f_{i_p})^{n-m}$ for $m \leqslant n$. We denote by $C^p_n(M)$ the set of *alternating* maps from $[1,r]^{p+1}$ to M (for all n); these A-modules also form an inductive system with respect to the ϕ_{nm}. If $C^p(\mathfrak{U},\mathscr{F})$ is the group of *alternating* Čech p-cochains relative to the cover $\mathfrak{U}$, with coefficients in $\mathscr{F}$ (G, II, 5.1), then it follows from the above that we can write

$$C^p(\mathfrak{U},\mathscr{F}) = \varinjlim_n C^p_n(M). \tag{1.2.2.2}$$

With the notations of (1.1.2), $C^p_n(M)$ identifies with $K^{p+1}(\mathbf{f}^n, M)$, and the map ϕ_{nm} identifies with the map $\phi_{\mathbf{f}^{n-m}}$ defined in (1.1.6). We thus have, for every $p \geqslant 0$, a canonical functorial isomorphism

$$C^p(\mathfrak{U},\mathscr{F}) \simeq C^{p+1}((\mathbf{f}), M). \tag{1.2.2.3}$$

In addition, the formula (1.1.2.3) and the definition of the cohomology of a cover (G, II, 5.1) shows that the isomorphisms (1.2.2.3) are compatible with the coboundary maps.

Proposition (1.2.3). — *If X is a prescheme whose underlying space in Noetherian or a scheme whose underlying space is quasi-compact, then there exists a canonical functorial isomorphism in $\mathscr{F}$*

$$\mathrm{H}^p(\mathfrak{U},\mathscr{F}) \simeq \mathrm{H}^{p+1}((\mathbf{f}), M) \text{ for } p \geqslant 1. \tag{1.2.3.1}$$

In addition, we have a functorial exact sequence in $\mathscr{F}$

$$0 \longrightarrow \mathrm{H}^0((\mathbf{f}), M) \longrightarrow M \longrightarrow \mathrm{H}^0(\mathfrak{U},\mathscr{F}) \longrightarrow \mathrm{H}^1((\mathbf{f}), M) \longrightarrow 0. \tag{1.2.3.2}$$

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Proof. The isomorphisms (1.2.3.1) are immediate consequences of what we saw in (1.2.2). On the other hand, we have $C^0(\mathfrak{U},\mathscr{F}) = C^1((\mathbf{f}), M)$; as a result, $\mathrm{H}^0(\mathfrak{U},\mathscr{F})$ identifies with the subgroup of 1-cocycles of $C^1((\mathbf{f}), M)$; as $M = C^0((\mathbf{f}), M)$, the exact sequence (1.2.3.2) is none other than the one given by the definition of the cohomology groups $\mathrm{H}^0((\mathbf{f}), M)$ and $\mathrm{H}^1((\mathbf{f}), M)$. □

Corollary (1.2.4). — *Suppose that the X_{f_i} are quasi-compact and that there exists $g_i \in \Gamma(U,\mathscr{F})$ such that $\sum_i g_i(f_i|U) = 1|U$. Then for every quasi-coherent $(\mathscr{O}_X|U)$-module $\mathscr{G}$, we have $\mathrm{H}^p(\mathfrak{U},\mathscr{G}) = 0$ for $p > 0$; if in addition $U = X$, then the canonical homomorphism (1.2.3.2) $M \to \mathrm{H}^0(\mathfrak{U},\mathscr{F})$ is bijective.*

Proof. As by hypothesis the $U_i = X_{f_i}$ are quasi-compact, so is U, and we can reduce to the case where $U = X$; the hypothesis then implies that $\mathrm{H}^p((\mathbf{f}), M) = 0$ for all $p \geqslant 0$ (1.1.9). The corollary then follows immediately from (1.2.3.1) and (1.2.3.2). □

We note that since $\mathrm{H}^0(\mathfrak{U},\mathscr{F}) = \mathrm{H}^0(U,\mathscr{F})$ (G, II, 5.2.2), we have again proved (**I**, 1.3.7) as a special case.

Remark (1.2.5). — Suppose that X is an *affine scheme*; then the $U_i = X_{f_i} = D(f_i)$ are affine open sets, as well as the $U_{i_0 i_1\cdots i_p}$ (but U is not necessarily affine). In this case, the functors $\Gamma(X,\mathscr{F})$ and $\Gamma(U_{i_0 i_1\cdots i_p},\mathscr{F})$ are exact in $\mathscr{F}$ (**I**, 1.3.11). If we have an exact sequence $0 \to \mathscr{F}' \to \mathscr{F} \to \mathscr{F}'' \to 0$ of quasi-coherent $\mathscr{O}_X$-modules, then the sequence of complexes

$$0 \longrightarrow C^\bullet(\mathfrak{U},\mathscr{F}') \longrightarrow C^\bullet(\mathfrak{U},\mathscr{F}) \longrightarrow C^\bullet(\mathfrak{U},\mathscr{F}'') \longrightarrow 0$$

is exact, and thus gives an exact sequence in cohomology

$$\cdots \longrightarrow \mathrm{H}^p(\mathfrak{U},\mathscr{F}) \longrightarrow \mathrm{H}^p(\mathfrak{U},\mathscr{F}) \longrightarrow \mathrm{H}^p(\mathfrak{U},\mathscr{F}'') \xrightarrow{\partial} \mathrm{H}^{p+1}(\mathfrak{U},\mathscr{F}') \longrightarrow \cdots.$$

On the other hand, if we set $M' = \Gamma(X,\mathscr{F}')$ and $M'' = \Gamma(X,\mathscr{F}'')$, then the sequence $0 \to M' \to M \to M'' \to 0$ is exact; as $C^\bullet((\mathbf{f}), M)$ is an exact functor in M, we also have the exact sequence in cohomology

$$\cdots \longrightarrow \mathrm{H}^p((\mathbf{f}), M') \longrightarrow \mathrm{H}^p((\mathbf{f}), M) \longrightarrow \mathrm{H}^p((\mathbf{f}), M'') \xrightarrow{\partial} \mathrm{H}^{p+1}((\mathbf{f}), M') \longrightarrow \cdots.$$

This being so, as the diagram

$$\begin{tikzcd} 0 \arrow[r] & C^\bullet(\mathfrak{U},\mathscr{F}') \arrow[r] \arrow[d] & C^\bullet(\mathfrak{U},\mathscr{F}) \arrow[r] \arrow[d] & C^\bullet(\mathfrak{U},\mathscr{F}'') \arrow[r] \arrow[d] & 0 \\ 0 \arrow[r] & C^\bullet((\mathbf{f}),M') \arrow[r] & C^\bullet((\mathbf{f}),M) \arrow[r] & C^\bullet((\mathbf{f}),M'') \arrow[r] & 0 \end{tikzcd}$$

is commutative, we conclude that the diagrams **III** | 88

(1.2.5.1)
$$\begin{tikzcd} \mathrm{H}^p(\mathfrak{U},\mathscr{F}'') \arrow[r, "\partial"] \arrow[d] & \mathrm{H}^{p+1}(\mathfrak{U},\mathscr{F}') \arrow[d] \\ \mathrm{H}^{p+1}((\mathbf{f}),M'') \arrow[r, "\partial"] & \mathrm{H}^{p+2}((\mathbf{f}),M') \end{tikzcd}$$

are commutative for all p (G, I, 2.1.1).

1.3. Cohomology of an affine scheme

Theorem (1.3.1). — *Let X be an affine scheme. For every quasi-coherent $\mathscr{O}_X$-module $\mathscr{F}$, we have $\mathrm{H}^p(X,\mathscr{F})=0$ for all $p>0$.*

Proof. Let $\mathfrak{U}$ be a finite cover of X by the affine open sets $X_{f_i}=D(f_i)$ $(1\leqslant i\leqslant r)$; we then know that the ideal of $A=\Gamma(X,\mathscr{O}_X)$ generated by the f_i is equal to A. We thus conclude from Corollary (1.2.4) that we have $\mathrm{H}^p(\mathfrak{U},\mathscr{F})=0$ for $p>0$. As there are finite covers of X by affine open sets which are arbitrarily fine (**I**, 1.1.10), the definition of Čech cohomology (G, II, 5.8) shows that we also have $\check{\mathrm{H}}^p(X,\mathscr{F})=0$ for $p>0$. But this also applies to every prescheme X_f for $f\in A$ (**I**, 1.3.6), hence $\check{\mathrm{H}}^p(X_f,\mathscr{F})=0$ for $p>0$. As we have $X_f\cap X_g=X_{fg}$, we deduce that we also have $\mathrm{H}^p(X,\mathscr{F})=0$ for all $p>0$, by virtue of (G, II, 5.9.2). □

Corollary (1.3.2). — *Let Y be a prescheme, $f:X\to Y$ an affine morphism* (**II**, 1.6.1). *For every quasi-coherent $\mathscr{O}_X$-module $\mathscr{F}$, we have $\mathrm{R}^q f_*(\mathscr{F})=0$ for $q>0$.*

Proof. By definition $\mathrm{R}^q f_*(\mathscr{F})$ is the $\mathscr{O}_Y$-module associated to the presheaf $U\mapsto \mathrm{H}^q(f^{-1}(U),\mathscr{F})$, where U varies over the open subsets of Y. But the affine open sets form a basis for Y, and for such an open set U, $f^{-1}(U)$ is affine (**II**, 1.3.2), hence $\mathrm{H}^q(f^{-1}(U),\mathscr{F})=0$ by Theorem (1.3.1), which proves the corollary. □

Corollary (1.3.3). — *Let Y be a prescheme, $f:X\to Y$ an affine morphism. For every quasi-coherent $\mathscr{O}_X$-module $\mathscr{F}$, the canonical homomorphism $\mathrm{H}^p(Y,f_*(\mathscr{F}))\to\mathrm{H}^p(X,\mathscr{F})$ (**0**, 12.1.3.1) is bijective for all p.*

Proof. It suffices (by (**0**, 12.1.7)) to show that the edge homomorphisms $''E_2^{p0}=\mathrm{H}^p(Y,f_*(\mathscr{F}))\to \mathrm{H}^p(X,\mathscr{F})$ of the second spectral sequence of the composite functor Γf_* are bijective. But the E_2-term of this spectral sequence is given by $''E_2^{pq}=\mathrm{H}^p(Y,\mathrm{R}^q f_*(\mathscr{F}))$ (G, II, 4.17.1), so it follows from Corollary (1.3.2) that $''E_2^{pq}=0$ for $q>0$, and the spectral sequence degenerates; hence our assertion (**0**, 11.1.6). □

Corollary (1.3.4). — *Let $f:X\to Y$ be an affine morphism, $g:Y\to Z$ a morphism. For every quasi-coherent $\mathscr{O}_X$-module $\mathscr{F}$, the canonical homomorphism $\mathrm{R}^p g_*(f_*(\mathscr{F}))\to\mathrm{R}^p(g\circ f)_*(\mathscr{F})$ (**0**, 12.2.5.1) is bijective for all p.* **III** | 89

Proof. It suffices to note that, according to Corollary (1.3.3), for every affine open subset W of Z, the canonical homomorphism $\mathrm{H}^p(g^{-1}(W),f_*(\mathscr{F}))\to\mathrm{H}^p(f^{-1}(g^{-1}(W)),\mathscr{F})$ is bijective; this proves that the homomorphism of presheaves defining the canonical homomorphism $\mathrm{R}^p g_*(f_*(\mathscr{F}))\to\mathrm{R}^p(g\circ f)_*(\mathscr{F})$ is bijective (**0**, 12.2.5). □

1.4. Application to the cohomology of arbitrary preschemes

Proposition (1.4.1). — *Let X be a scheme, $\mathfrak{U} = (U_\alpha)$ be a cover of X by affine open sets. For every quasi-coherent $\mathscr{O}_X$-module $\mathscr{F}$, the cohomology modules $\mathrm{H}^\bullet(X, \mathscr{F})$ and $\mathrm{H}^\bullet(\mathfrak{U}, \mathscr{F})$ (over $\Gamma(X, \mathscr{O}_X)$) are canonically isomorphic.*

Proof. As X is a scheme, every finite intersection V of open sets in the cover $\mathfrak{U}$ is affine (**I**, 5.5.6), so $\mathrm{H}^q(V, \mathscr{F}) = 0$ for $g \geqslant 1$ by Theorem (1.3.1). The proposition then follows from a theorem of Leray (G, II, 5.4.1). □

Remark (1.4.2). — We note that the result of Proposition (1.4.1) is still true when the finite intersections of the sets U_α are affine, even when we do not necessarily assume that X is a scheme.

Corollary (1.4.3). — *Let X be a scheme with quasi-compact underlying space, $A = \Gamma(X, \mathscr{O}_X)$, and $\mathbf{f} = (f_i)_{1 \leqslant i \leqslant r}$ a finite sequence of elements of A such that the X_{f_i} (notation of (1.2.1)) are affine. Then (with the notations of (1.2.1)), for every quasi-coherent $\mathscr{O}_X$-module $\mathscr{F}$, we have a canonical isomorphism which is functorial in $\mathscr{F}$*

$$\mathrm{H}^q(U, \mathscr{F}) \simeq \mathrm{H}^{q+1}((\mathbf{f}), M) \text{ for } q \geqslant 1, \tag{1.4.3.1}$$

and an exact sequence which is functorial in $\mathscr{F}$

$$0 \longrightarrow \mathrm{H}^0((\mathbf{f}), M) \longrightarrow M \longrightarrow \mathrm{H}^0(U, \mathscr{F}) \longrightarrow \mathrm{H}^1((\mathbf{f}), M) \longrightarrow 0. \tag{1.4.3.2}$$

Proof. This follows immediately from Propositions (1.4.1) and (1.2.3). □

(1.4.4). If X is an *affine scheme*, then it follows from Remark (1.2.5) and Proposition (1.4.1) that for all $q \geqslant 0$, the diagrams

$$\begin{tikzcd} \mathrm{H}^q(U, \mathscr{F}'') \arrow[r, "\partial"] \arrow[d] & \mathrm{H}^{q+1}(U, \mathscr{F}') \arrow[d] \\ \mathrm{H}^{q+1}((\mathbf{f}), M'') \arrow[r, "\partial"] & \mathrm{H}^{q+2}((\mathbf{f}), M') \end{tikzcd} \tag{1.4.4.1}$$

corresponding to an exact sequence $0 \to \mathscr{F}' \to \mathscr{F} \to \mathscr{F}'' \to 0$ of quasi-coherent $\mathscr{O}_X$-modules (with the notations of Remark (1.2.5)) are commutative.

Proposition (1.4.5). — *Let X be a quasi-compact scheme, $\mathscr{L}$ an invertible $\mathscr{O}_X$-module, and consider the* III | 90 *graded ring $A_\bullet = \Gamma_\bullet(\mathscr{L})$ (**0**, 5.4.6); then $\mathrm{H}^\bullet(\mathscr{F}, \mathscr{L}) = \bigoplus_{n \in \mathbf{Z}} \mathrm{H}^\bullet(X, \mathscr{F} \otimes \mathscr{L}^{\otimes n})$ is a graded $A_\bullet$-module, and for all $f \in A_n$, we have a canonical isomorphism*

$$\mathrm{H}^\bullet(X_f, \mathscr{F}) \simeq (\mathrm{H}^\bullet(\mathscr{F}, \mathscr{L}))_{(f)} \tag{1.4.5.1}$$

of $(A_\bullet)_{(f)}$-modules.

Proof. As X is a quasi-compact scheme, we can calculate the cohomology of all the $\mathscr{O}_X$-modules $\mathscr{F} \otimes \mathscr{L}^{\otimes n}$ using the same finite cover $\mathfrak{U} = (U_i)$ consisting of the affine open sets such that the restriction $\mathscr{L}|U_i$ is isomorphic to $\mathscr{O}_X|U_i$ for each i (1.4.1). It is then immediate that the $U_i \cap X_f$ are affine open sets (**I**, 1.3.6), and we can thus calculate the cohomology $\mathrm{H}^\bullet(X_f, \mathscr{F} \otimes \mathscr{L}^{\otimes n})$ using the cover $\mathfrak{U}|X_f = (U_i \cap X_f)$ (1.4.1). It is immediate that for all $f \in A_n$, multiplication by f defines a homomorphism $\mathrm{C}^\bullet(\mathfrak{U}, \mathscr{F} \otimes \mathscr{L}^m) \to \mathrm{C}^\bullet(\mathfrak{U}, \mathscr{F} \otimes \mathscr{L}^{\otimes(m+n)})$, hence a homomorphism $\mathrm{H}^\bullet(\mathfrak{U}, \mathscr{F} \otimes \mathscr{L}^{\otimes m}) \to \mathrm{H}^\bullet(\mathfrak{U}, \mathscr{F} \otimes \mathscr{L}^{\otimes(m+n)})$, which establishes the first assertion. On the other hand, for a given $f \in A_n$, it follows from (**I**, 9.3.2) that we have an isomorphism of complexes of $(A_\bullet)_{(f)}$-modules

$$\mathrm{C}^\bullet(\mathfrak{U}|X_f, \mathscr{F}) \simeq \left(\mathrm{C}^\bullet\left(\mathfrak{U}, \bigoplus_{n \in \mathbf{Z}} \mathscr{F} \otimes \mathscr{L}^{\otimes n}\right)\right)_{(f)},$$

taking into account (**I**, 1.3.9, ii). Passing to the cohomology of these complexes, we induce the isomorphism (1.4.5.1), recalling that the functor $M \mapsto M_{(f)}$ is exact on the category of graded $A_\bullet$-modules. □

Corollary (1.4.6). — *Suppose that the hypotheses of Proposition (1.4.5) are satisfied, and in addition suppose that $\mathscr{L} = \mathscr{O}_X$. If we set $A = \Gamma(X, \mathscr{O}_X)$, then for all $f \in A$, we have a canonical isomorphism $\mathrm{H}^\bullet(X_f, \mathscr{F}) \simeq (\mathrm{H}^\bullet(X, \mathscr{F}))_f$ of A_f-modules.*

Corollary (1.4.7). — *Let X be a quasi-compact scheme, f an element of $\Gamma(X, \mathscr{O}_X)$.*

(i) *Suppose that the open set X_f is affine. Then for every quasi-coherent $\mathscr{O}_X$-module $\mathscr{F}$, every $i > 0$, and every $\xi \in \mathrm{H}^i(X, \mathscr{F})$, there exists an integer $n > 0$ such that $f^n \xi = 0$.*
(ii) *Conversely, suppose that X_f is quasi-compact and that for every quasi-coherent sheaf of ideals $\mathscr{J}$ of $\mathscr{O}_X$ and every $\zeta \in \mathrm{H}^1(X, \mathscr{J})$, there exists an $n > 0$ such that $f^n \zeta = 0$. Then X_f is affine.*

Proof.

(i) If X_f is affine, then we have $\mathrm{H}^i(X_f, \mathscr{F}) = 0$ for all $i > 0$ (1.3.1), so the assertion follows directly from Corollary (1.4.6).
(ii) By virtue of Serre's criterion (**II**, 5.2.1), it suffices to prove that for every quasi-coherent sheaf of ideals $\mathscr{K}$ of $\mathscr{O}_X|X_f$, we have $\mathrm{H}^1(X_f, \mathscr{K}) = 0$. As X_f is a quasi-compact open set in a quasi-compact scheme X, there exists a quasi-coherent sheaf of ideals $\mathscr{J}$ of $\mathscr{O}_X$ such that $\mathscr{K} = \mathscr{J}|X_f$ (**I**, 9.4.2). According to Corollary (1.4.6), we have $\mathrm{H}^1(X_f, \mathscr{K}) = (\mathrm{H}^1(X, \mathscr{J}))_f$, and the hypothesis implies that the right hand side is zero, hence the assertion.

□

Remark (1.4.8). — We note that Corollary (1.4.7, i) gives a simpler proof of the relation (**II**, 4.5.13.2).

Lemma (1.4.9). — *Let X be a quasi-compact scheme, $\mathfrak{U} = (U_i)_{1 \leqslant i \leqslant n}$ a finite cover of X by affine open sets,* III | 91
and $\mathscr{F}$ a quasi-coherent $\mathscr{O}_X$-module. The complex of sheaves $\mathscr{C}^\bullet(\mathfrak{U}, \mathscr{F})$ defined by the cover $\mathfrak{U}$ (G, II, 5.2) is then a quasi-coherent $\mathscr{O}_X$-module.

Proof. It follows from the definitions (G, II, 5.2) that $\mathscr{C}^p(\mathfrak{U}, \mathscr{F})$ is the direct sum of the direct image sheaves of the $\mathscr{F}|U_{i_0 \cdots i_p}$ under the canonical injection $U_{i_0 \cdots i_p} \to X$. The hypothesis that X is a scheme implies that these injections are affine morphisms (**I**, 5.5.6), hence the $\mathscr{C}^p(\mathfrak{U}, \mathscr{F})$ are quasi-coherent (**II**, 1.2.6). □

Proposition (1.4.10). — *Let $u : X \to Y$ be a separated and quasi-compact morphism. For every quasi-coherent $\mathscr{O}_X$-module $\mathscr{F}$, the $\mathrm{R}^q u_*(\mathscr{F})$ are quasi-coherent $\mathscr{O}_Y$-modules.*

Proof. The question is local on Y, so we can suppose that Y is affine. Then X is a finite union of affine open sets U_i ($1 \leqslant i \leqslant n$); let $\mathfrak{U}$ be the cover (U_i). In addition, as Y is a scheme, it follows from (**I**, 5.5.10) that for every affine open $V \subset Y$, the canonical injection $u^{-1}(V) \to X$ is an affine morphism; we conclude (Proposition (1.4.1) and (G, II, 5.2)) that we have a canonical isomorphism

(1.4.10.1) $$\mathrm{H}^\bullet(u^{-1}(V), \mathscr{F}) \simeq \mathrm{H}^\bullet(\Gamma(V, \mathscr{K}^\bullet)),$$

where we set $\mathscr{K}^\bullet = u_*(\mathscr{C}^\bullet(\mathfrak{U}, \mathscr{F}))$. According to Lemma (1.4.1) and (**I**, 9.2.2), $\mathscr{K}^\bullet$ is a quasi-coherent $\mathscr{O}_Y$-module; moreover, it constitutes a *complex of sheaves* since so is $\mathscr{C}^\bullet(\mathfrak{U}, \mathscr{F})$. It then follows from the definition of the cohomology $\mathscr{H}^\bullet(\mathscr{K}^\bullet)$ (G, II, 4.1) that the latter consists of quasi-coherent $\mathscr{O}_Y$-modules (**I**, 4.1.1). As (for V affine in Y) the functor $\Gamma(V, \mathscr{G})$ is exact in $\mathscr{G}$ on the category of quasi-coherent $\mathscr{O}_Y$-modules, we have (G, II, 4.1)

(1.4.10.2) $$\mathrm{H}^\bullet(\Gamma(V, \mathscr{K}^\bullet)) = \Gamma(V, \mathscr{H}^\bullet(\mathscr{K}^\bullet)).$$

Finally, we note that it follows from the definition of the canonical homomorphism

$$\mathrm{H}^\bullet(\mathfrak{U}, \mathscr{F}) \longrightarrow \mathrm{H}^\bullet(X, \mathscr{F}),$$

given in (G, II, 5.2), that if $V' \subset V$ is a second affine open subset of Y, then the diagram

$$\begin{tikzcd}
\mathrm{H}^\bullet(u^{-1}(V), \mathscr{F}) \arrow[r, "\sim"] \arrow[d] & \mathrm{H}^\bullet(\Gamma(V, \mathscr{K}^\bullet)) \arrow[d] \\
\mathrm{H}^\bullet(u^{-1}(V'), \mathscr{F}) \arrow[r, "\sim"] & \mathrm{H}^\bullet(\Gamma(V', \mathscr{K}^\bullet))
\end{tikzcd}$$

is commutative. We thus conclude from the above that the isomorphisms (1.4.10.1) define an isomorphism of $\mathscr{O}_Y$-modules

(1.4.10.3) $$\mathrm{R}^\bullet u_*(\mathscr{F}) \simeq \mathscr{H}^\bullet(\mathscr{K}^\bullet),$$

and as a result, $\mathrm{R}^\bullet u_*(\mathscr{F})$ is quasi-coherent. □ III | 92

In addition, it follows from (1.4.10.3), (1.4.10.2), and (1.4.10.1) that:

Corollary (1.4.11). — *Under the hypotheses of Proposition (1.4.10), for every affine open set V of Y, the canonical homomorphism*

$$\mathrm{H}^q(u^{-1}(V), \mathscr{F}) \longrightarrow \Gamma(V, \mathrm{R}^1 u_*(\mathscr{F})) \tag{1.4.11.1}$$

is an isomorphism for all $q \geqslant 0$.

Corollary (1.4.12). — *Suppose that the hypotheses of Proposition (1.4.10) are satisfied, and in addition suppose that Y is quasi-compact. Then there exists an integer $r > 0$ such that for every quasi-coherent $\mathscr{O}_X$-module $\mathscr{F}$ and every integer $q > r$, we have $\mathrm{R}^q u_*(\mathscr{F}) = 0$. If Y is affine, then we can take for r an integer such that there exists a cover of X consisting of r affine open sets.*

Proof. As we can cover Y by a finite number of affine open sets, we can reduce to proving the second assertion, by virtue of Corollary (1.4.11). If $\mathfrak{U}$ is a cover of X by r affine open sets, then we have $\mathrm{H}^q(\mathfrak{U}, \mathscr{F}) = 0$ for $q > r$, since the cochains of $C^q(\mathfrak{U}, \mathscr{F})$ are alternating; the assertion thus follows from Proposition (1.4.1). □

Corollary (1.4.13). — *Suppose that the hypotheses of Proposition (1.4.10) are satisfied, and in addition suppose that $Y = \mathrm{Spec}(A)$ is affine. Then for every quasi-coherent $\mathscr{O}_X$-module $\mathscr{F}$ and every $f \in A$, we have*

$$\Gamma(Y_f, \mathrm{R}^q u_*(\mathscr{F})) = (\Gamma(Y, \mathrm{R}^q u_*(\mathscr{F}))_f$$

up to canonical isomorphism.

Proof. This follows from the fact that $\mathrm{R}^q u_*(\mathscr{F})$ is a quasi-coherent $\mathscr{O}_Y$-module (**I**, 1.3.7). □

Proposition (1.4.14). — *Let $f : X \to Y$ be a separated and quasi-compact morphism, $g : Y \to Z$ an affine morphism. For every quasi-coherent $\mathscr{O}_X$-module $\mathscr{F}$, the canonical homomorphism $\mathrm{R}^p(g \circ f)_*(\mathscr{F}) \to g_*(\mathrm{R}^p f_*(\mathscr{F}))$ (**0**, 12.2.5.2) is bijective for all p.*

Proof. For every affine open subset W of Z, $g^{-1}(W)$ is an affine open subset of Y. The homomorphism of presheaves defining the canonical homomorphism

$$\mathrm{R}^p(g \circ f)_*(\mathscr{F}) \longrightarrow g_*(\mathrm{R}^p f_*(\mathscr{F}))$$

(**0**, 12.2.5) is thus bijective by Corollary (1.4.11). □

Proposition (1.4.15). — *Let $u : X \to Y$ be a separated morphism of finite type, $v : Y' \to Y$ a flat morphism of preschemes* (**0**, 6.7.1)*; let $u' = u_{(Y')}$, such that we have the commutative diagram*

$$\begin{tikzcd} X \arrow[d, "u"'] & X' = X_{(Y')} \arrow[l, "v'"'] \arrow[d, "u'"] \\ Y & Y'. \arrow[l, "v"] \end{tikzcd} \tag{1.4.15.1}$$

Then for every quasi-coherent $\mathscr{O}_X$-module $\mathscr{F}$, $\mathrm{R}^q u'_*(\mathscr{F}')$ is canonically isomorphic to $\mathrm{R}^q u_*(\mathscr{F}) \otimes_{\mathscr{O}_Y} \mathscr{O}_{Y'} = v^*(\mathrm{R}^q u_*(\mathscr{F}))$ for all $q \geqslant 0$, where $\mathscr{F}' = v'^*(\mathscr{F}) = \mathscr{F} \otimes_{\mathscr{O}_Y} \mathscr{O}_{Y'}$.

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Proof. The canonical homomorphism $\rho : \mathscr{F} \to v'_*(v'^*(\mathscr{F}))$ (**0**, 4.4.3.2) defines by functoriality a homomorphism

$$\mathrm{R}^q u_*(\mathscr{F}) \longrightarrow \mathrm{R}^q u_*(v'_*(\mathscr{F}')). \tag{1.4.15.2}$$

On the other hand, we have, by setting $w = u \circ v' = v \circ u'$, the canonical homomorphisms (**0** ,12.2.5.1 and 12.2.5.2)

$$\mathrm{R}^q u_*(v'_*(\mathscr{F}')) \longrightarrow \mathrm{R}^q w_*(\mathscr{F}') \longrightarrow v_*(\mathrm{R}^q u'_*(\mathscr{F}')). \tag{1.4.15.3}$$

Composing (1.4.15.3) and (1.4.15.2), we have a homomorphism

$$\psi : \mathrm{R}^q u_*(\mathscr{F}) \longrightarrow v_*(\mathrm{R}^q u'_*(\mathscr{F}')),$$

and finally we obtain a canonical homomorphism (whose definition does not make *any assumptions* on v)

$$\psi^{\sharp} : v^{*}(\mathrm{R}^{q}u_{*}(\mathscr{F})) \longrightarrow \mathrm{R}^{q}u'_{*}(\mathscr{F}'), \tag{1.4.15.4}$$

and it is necessary to prove that it is an isomorphism when v is *flat*. It is clear that the question is local on Y and Y', and we can therefore suppose that $Y = \mathrm{Spec}(A)$ and $Y' = \mathrm{Spec}(B)$; we will also use the following lemma:

Lemma (1.4.15.5). — *Let $\phi : A \to B$ be a ring homomorphism, $Y = \mathrm{Spec}(A)$, $X = \mathrm{Spec}(B)$, $f : X \to Y$ the morphism corresponding to ϕ, and M a B-module. For the $\mathscr{O}_X$-module $\widetilde{M}$ to be f-flat* (**0**, 6.7.1)*, it is necessary and sufficient for M to be a flat A-module. In particular, for the morphism f to be flat, it is necessary and sufficient for B to be a flat A-module.*

This follows from the definition (**0**, 6.7.1) and from (**0**, 6.3.3), taking into account (**I**, 1.3.4).

This being so, it follows from (1.4.11.1) and the definitions of the homomorphisms (1.4.15.3) (cf. (**0**, 12.2.5)) that ψ then corresponds to the composite morphism

$$\mathrm{H}^q(X,\mathscr{F}) \xrightarrow{\rho_q} \mathrm{H}^q(X, v'_*(v'^*(\mathscr{F}))) \xrightarrow{\theta_q} \mathrm{H}^q(X', v'^*(v'_*(v'^*(\mathscr{F})))) \xrightarrow{\sigma_q} \mathrm{H}^q(X', v'^*(\mathscr{F})),$$

where ρ_q and σ_q are the homomorphisms in cohomology corresponding to the canonical morphisms ρ and $\sigma : v'^*(v'_*(\mathscr{G}')) \to \mathscr{G}'$, and θ_q is the ϕ-morphism (**0**, 12.1.3.1) relative to the $\mathscr{O}_X$-module $v'_*(v'^*(\mathscr{F}))$. But by the functoriality of θ_q, we have the commutative diagram

$$\begin{tikzcd}
\mathrm{H}^q(X,\mathscr{F}) \arrow[r, "\rho_q"] \arrow[d, "\theta_q"'] & \mathrm{H}^q(X, v'_*(v'^*(\mathscr{F}))) \arrow[d, "\theta_q"] \\
\mathrm{H}^q(X', v'^*(\mathscr{F})) \arrow[r, "v'^*(\rho_q)"] & \mathrm{H}^q(X', v'^*(v'_*(v'^*(\mathscr{F})))),
\end{tikzcd}$$

and as by definition (**0**, 4.4.3) $v'^*(\rho)$ is the inverse of σ, we see that the composite morphism III | 94
considered above is finally none other than θ_q; as a result, $\psi^{\sharp}$ is the associated B-homomorphism $\mathrm{H}^q(X,\mathscr{F}) \otimes_A B \to \mathrm{H}^q(X',\mathscr{F}')$. As u is of finite type, X is a finite union of affine open sets U_i ($1 \leqslant i \leqslant r$); let $\mathfrak{U}$ be the cover (U_i). As v is an affine morphism, so is v' (**II**, 1.6.2, iii), and as a result the $U'_i = v'^{-1}(U_i)$ form an affine open cover $\mathfrak{U}'$ of X'. We then know (**0**, 12.1.4.2) that the diagram

$$\begin{tikzcd}
\mathrm{H}^q(\mathfrak{U},\mathscr{F}) \arrow[r, "\theta_q"] \arrow[d] & \mathrm{H}^q(\mathfrak{U}',\mathscr{F}') \arrow[d] \\
\mathrm{H}^q(X,\mathscr{F}) \arrow[r, "\theta_q"] & \mathrm{H}^q(X',\mathscr{F}')
\end{tikzcd}$$

is commutative, and the vertical arrows are isomorphisms since X and X' are schemes (**I**, 1.4.1). As a result, it suffices to prove that the canonical ϕ-morphism $\theta_q : \mathrm{H}^q(\mathfrak{U},\mathscr{F}) \to \mathrm{H}^q(\mathfrak{U}',\mathscr{F}')$ is such that the associated B-homomorphism

$$\mathrm{H}^q(\mathfrak{U},\mathscr{F}) \otimes_A B \longrightarrow \mathrm{H}^q(\mathfrak{U}',\mathscr{F}')$$

is an isomorphism. For every sequence $\mathbf{s} = (i_k)_{0 \leqslant k \leqslant p}$ of $p+1$ indices of $[1,r]$, set $U_{\mathbf{s}} = \bigcap_{k=0}^{p} U_{i_k}$, $U'_{\mathbf{s}} = \bigcap_{k=0}^{p} U'_{i_k} = v'^{-1}(U_{\mathbf{s}})$, $M_{\mathbf{s}} = \Gamma(U_{\mathbf{s}},\mathscr{F})$, and $M'_{\mathbf{s}} = \Gamma(U'_{\mathbf{s}},\mathscr{F}')$. The canonical map $M_{\mathbf{s}} \otimes_A B \to M'_{\mathbf{s}}$ is an isomorphism (**I**, 1.6.5), hence the canonical map $C^p(\mathfrak{U},\mathscr{F}) \otimes_A B \to C^p(\mathfrak{U}',\mathscr{F}')$ is an isomorphism, by which $d \otimes 1$ identifies with the coboundary map $C^p(\mathfrak{U}',\mathscr{F}') \to C^{p+1}(\mathfrak{U}',\mathscr{F}')$. As B is a *flat* A-module, it follows from the definition of the cohomology modules that the canonical map $\mathrm{H}^q(\mathfrak{U},\mathscr{F}) \otimes_A B \to \mathrm{H}^q(\mathfrak{U}',\mathscr{F}')$ is an isomorphism (**0**, 6.1.1). This result will later be generalized in §6. □

Corollary (1.4.16). — *Let A be a ring, X an A-scheme of finite type, and B an A-algebra which is faithfully flat over A. For X to be affine, it is necessary and sufficient for $X \otimes_A B$ to be.*

Proof. The condition is evidently necessary (**I**, 3.2.2); we show that it is sufficient. As X is separated over A and the morphism $\mathrm{Spec}(B) \to \mathrm{Spec}(A)$ is flat, it follows from Proposition (1.4.1) that we have

(1.4.16.1) $$\mathrm{H}^i(X \otimes_A B, \mathscr{F} \otimes_A B) = \mathrm{H}^i(X, \mathscr{F}) \otimes_A B$$

for every $i \geqslant 0$ and every quasi-coherent $\mathscr{O}_X$-module $\mathscr{F}$. If $X \otimes_A B$ is affine, the left hand side of (1.4.16.1) is zero for $i = 1$, hence so is $\mathrm{H}^1(X, \mathscr{F})$ since B is a faithfully flat A-module. As X is a quasi-compact scheme, we finish the proof by Serre's criterion (**II**, 5.2.1). □

Proposition (1.4.17). — *Let X be a prescheme, $0 \to \mathscr{F} \xrightarrow{u} \mathscr{G} \xrightarrow{v} \mathscr{H} \to 0$ an exact sequence of $\mathscr{O}_X$-modules. If $\mathscr{F}$ and $\mathscr{H}$ are quasi-coherent, then so is $\mathscr{G}$.*

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Proof. The question is local on X, so we can suppose that $X = \mathrm{Spec}(A)$ is affine, and it then suffices to prove that $\mathscr{G}$ satisfies the conditions (d1) and (d2) of (**I**, 1.4.1) (with $V = X$). The verification of (d2) is immediate, because if $t \in \Gamma(X, \mathscr{G})$ is zero when restricted to $D(f)$, then so is its image $v(t) \in \Gamma(X, \mathscr{H})$; therefore there exists an $m > 0$ such that $f^m v(t) = v(f^m t) = 0$ (**I**, 1.4.1), and as Γ is left exact, $f^m t = u(s)$, where $s \in \Gamma(X, \mathscr{F})$; as u is injective, the restriction of s to $D(f)$ is zero, hence (**I**, 1.4.1) there exists an integer $n > 0$ such that $f^n s = 0$; we finally deduce that $f^{m+n} t = u(f^n s) = 0$.

We now check (d1); let $t' \in \Gamma(D(f), \mathscr{G})$; as $\mathscr{H}$ is quasi-coherent, there exists an integer m such that $f^m v(t') = v(f^m t')$ extends to a section $z \in \Gamma(X, \mathscr{H})$ (**I**, 1.4.1). But in virtue of Theorem (1.3.1) (or (**I**, 5.1.9.2)) applied to the quasi-coherent $\mathscr{O}_X$-module $\mathscr{F}$, the sequence $\Gamma(X, \mathscr{G}) \to \Gamma(X, \mathscr{H}) \to 0$ is exact, so there exists $t \in \Gamma(X, \mathscr{G})$ such that $z = v(t)$; we thus see that $v(f^m t' - t'') = 0$, denoting by t'' the restriction of t to $D(f)$; thus we have $f^m t' - t'' = u(s')$, where $s' \in \Gamma(D(f), \mathscr{F})$. But as $\mathscr{F}$ is quasi-coherent, there exists an integer $n > 0$ such that $f^n s'$ extends to a section $s \in \Gamma(X, \mathscr{F})$; as $f^{m+n} t' - f^n t'' = u(f^n s')$, we see that $f^{m+n} t'$ is the restriction to $D(f)$ of a section $f^n t + u(f^n s) \in \Gamma(X, \mathscr{G})$, which finishes the proof. □

§2. Cohomological study of projective morphisms

2.1. Explicit calculations of certain cohomology groups

(2.1.1). Let X be a prescheme, $\mathscr{L}$ an invertible $\mathscr{O}_X$-module; consider the graded ring (**0**, 5.4.6) **III** | 95

(2.1.1.1) $$S = \Gamma_*(X, \mathscr{L}) = \bigoplus_{n \in \mathbf{Z}} \Gamma(X, \mathscr{L}^{\otimes n}).$$

Let $(f_i)_{1 \leqslant i \leqslant r}$ be a finite family of *homogeneous* elements of S, say $f_i \in S_{d_i}$; set $U_i = X_{f_i}$, $U = \bigcup_i U_i$, and denote by $\mathfrak{U}$ the covering (U_i) of U. For every quasi-coherent $\mathscr{O}_X$-module $\mathscr{F}$, we set

(2.1.1.2) $$\mathrm{H}^\bullet(U, \mathscr{F}(*)) = \bigoplus_{n \in \mathbf{Z}} \mathrm{H}^\bullet(U, \mathscr{F} \otimes \mathscr{L}^{\otimes n})$$

(2.1.1.3) $$\mathrm{H}^\bullet(\mathfrak{U}, \mathscr{F}(*)) = \bigoplus_{n \in \mathbf{Z}} \mathrm{H}^\bullet(\mathfrak{U}, \mathscr{F} \otimes \mathscr{L}^{\otimes n}).$$

The abelian groups (2.1.1.2) and (2.1.1.3) are *bigraded*, by taking

$$(\mathrm{H}^\bullet(U, \mathscr{F}(*)))_{mn} = \mathrm{H}^m(U, \mathscr{F} \otimes \mathscr{L}^{\otimes n})$$

and an analogous definition for (2.1.1.3). For the second grading, it is clear that these groups are graded S-modules, as follows, for example, from the fact that $\mathscr{F} \mapsto \mathrm{H}^m(U, \mathscr{F})$ and $\mathscr{F} \mapsto \mathrm{H}^m(\mathfrak{U}, \mathscr{F})$ are functors.

(2.1.2). Consider now the graded S-module (**0**, 5.4.6)

(2.1.2.1) $$M = \Gamma_*(\mathscr{L}, \mathscr{F}) = \mathrm{H}^0(X, \mathscr{F}(*)) = \bigoplus_{n \in \mathbf{Z}} \Gamma(X, \mathscr{F} \otimes \mathscr{L}^{\otimes n}).$$

If X is a prescheme whose underlying space is Noetherian, or a quasi-compact scheme, it follows from (**I**, 9.3.1) that, setting as usual $U_{i_0 i_1 \cdots i_p} = \bigcap_{k=0}^p U_{i_k}$, we have, up to canonical isomorphism, **III** | 96

$$\Gamma(U_{i_0 i_1 \cdots i_p}, \mathscr{F}(*)) = \mathrm{H}^0(U_{i_0 i_1 \cdots i_p}, \mathscr{F}(*)) = M_{f_{i_0} f_{i_1} \cdots f_{i_p}}.$$

We can further, with the notation of (1.2.2), identify $M_{f_{i_0}f_{i_1}\cdots f_{i_p}}$ with $\varinjlim_n M^{(n)}_{i_0 i_1\cdots i_p}$. This identification is an isomorphism of *graded* S-modules if we define the degree of a homogeneous element $z \in \varinjlim_n M^{(n)}_{i_0 i_1\cdots i_p}$ as follows: z is the canonical image of a homogeneous element $x \in M^{(n)}_{i_0 i_1\cdots i_p} = M$ of degree m, and we take the degree of z to be the number $m - n(d_{i_0} + d_{i_1} + \cdots + d_{i_p})$. Taking into account the definition of the homomorphisms $\phi_{kh} : M^{(h)}_{i_0 i_1\cdots i_p} \to M^{(k)}_{i_0 i_1\cdots i_p}$ (1.2.2), we see immediately that this definition does not depend on the "representative" x of z that was chosen. Denoting, as in (1.2.2), by $C^p_n(M)$ the set of alternating maps from $[1,r]^{p+1}$ to M (for every n), we similarly define a graded S-module structure on $\varinjlim_n C^p_n(M)$; as in (1.2.2), we then have

$$C^p(\mathfrak{U}, \mathscr{F}(*)) = \varinjlim_n C^p_n(M) \tag{2.1.2.2}$$

with the isomorphism between the two sides *respecting degrees*. We also have, as in (1.2.2),

$$C^p(\mathfrak{U}, \mathscr{F}(*)) = C^{p+1}((\mathbf{f}), M) = \varinjlim_n K^{p+1}(\mathbf{f}^n, M) \tag{2.1.2.3}$$

with the isomorphism *preserving degrees*: the degree of an element of $\varinjlim_n K^{p+1}(\mathbf{f}^n, M)$, which is the canonical image of a cochain $g \in K^{p+1}(\mathbf{f}^n, M)$ whose values $g(i_0, \ldots, i_p)$ lie in one and the same homogeneous component M_m of M, is $m - n(d_{i_0} + \cdots + d_{i_p})$, independently of the choice of this cochain as a representative of the element in question.

Since the preceding isomorphisms are compatible with the coboundary operators, we conclude, as in (1.2.2), that:

Proposition (2.1.3). — *Let X be a prescheme whose underlying space is Noetherian, or a quasi-compact scheme. There is a canonical isomorphism, functorial in $\mathscr{F}$,*

$$\mathrm{H}^p(\mathfrak{U}, \mathscr{F}(*)) \xrightarrow{\sim} \mathrm{H}^{p+1}((\mathbf{f}), M) \qquad \textit{for every } p \geqslant 1. \tag{2.1.3.1}$$

Moreover, there is an exact sequence, functorial in $\mathscr{F}$,

$$0 \longrightarrow \mathrm{H}^0((\mathbf{f}), M) \longrightarrow M \longrightarrow \mathrm{H}^0(\mathfrak{U}, \mathscr{F}(*)) \longrightarrow \mathrm{H}^1((\mathbf{f}), M) \longrightarrow 0. \tag{2.1.3.2}$$

All the homomorphisms introduced are of degree 0 for the graded S-module structures, where S is the ring (2.1.1.1).

Corollary (2.1.4). — *If X is a quasi-compact scheme and the $U_i = X_{f_i}$ are affine, there is a canonical isomorphism, functorial in $\mathscr{F}$ and of degree 0,* **III** | 97

$$\mathrm{H}^p(U, \mathscr{F}(*)) \xrightarrow{\sim} \mathrm{H}^{p+1}((\mathbf{f}), M) \qquad \textit{for every } p \geqslant 1, \tag{2.1.4.1}$$

and an exact sequence, functorial in $\mathscr{F}$,

$$0 \longrightarrow \mathrm{H}^0((\mathbf{f}), M) \longrightarrow M \longrightarrow \mathrm{H}^0(U, \mathscr{F}(*)) \longrightarrow \mathrm{H}^1((\mathbf{f}), M) \longrightarrow 0, \tag{2.1.4.2}$$

in which all the homomorphisms are of degree 0.

Proof. It suffices to apply (1.4.1) to the result of (2.1.3). ☐

The "local" proposition analogous to (2.1.3) is the following.

Proposition (2.1.5). — *Let S be a positively graded ring, let the f_i ($1 \leqslant i \leqslant r$) be homogeneous elements of S_+ of degrees d_i, and let M be a graded S-module. Let $X = \mathrm{Proj}(S)$ be the homogeneous prime spectrum of S, and set $U_i = \mathrm{D}_+(f_i)$, $U = \bigcup_i U_i$, and $\mathrm{H}^\bullet(U, \widetilde{M}(*)) = \bigoplus_{n\in\mathbf{Z}} \mathrm{H}^\bullet(U, (M(n))^\sim)$. There are canonical isomorphisms, functorial in M and of degree 0 for the graded S-module structures,*

$$\mathrm{H}^p(U, \widetilde{M}(*)) \xrightarrow{\sim} \mathrm{H}^{p+1}((\mathbf{f}), M) \qquad \textit{for every } p \geqslant 1, \tag{2.1.5.1}$$

and an exact sequence, functorial in M,

$$0 \longrightarrow \mathrm{H}^0((\mathbf{f}), M) \longrightarrow M \longrightarrow \mathrm{H}^0(U, \widetilde{M}(*)) \longrightarrow \mathrm{H}^1((\mathbf{f}), M) \longrightarrow 0, \tag{2.1.5.2}$$

in which all the homomorphisms are of degree 0.

Proof. Indeed, by definition (**II**, 2.5.2),

$$\Gamma(U_{i_0 i_1 \cdots i_p}, (M(n))^{\sim}) = (M_{f_{i_0}\cdots f_{i_p}})_n,$$

and therefore $\Gamma(U_{i_0 i_1 \cdots i_p}, \widetilde{M}(*)) = M_{f_{i_0}\cdots f_{i_p}}$. The rest of the argument is the same as in the proof of (2.1.4), taking into account that X is a scheme. ☐

(2.1.6). *Remarks.* — (i) Under the hypotheses of (2.1.5), the functors $\Gamma(U_{i_0 i_1 \cdots i_p}, \widetilde{M}(*))$ are exact in M, by (**0**, 1.3.2). The same argument as in (1.2.5) therefore shows that, if $0 \to M' \to M \to M'' \to 0$ is an exact sequence of graded S-modules whose homomorphisms have degree 0, then for every $p \geqslant 0$ there are commutative diagrams

(2.1.6.1)
$$\begin{tikzcd} \mathrm{H}^p(U, \widetilde{M''}(*)) \arrow[r, "\partial"] \arrow[d] & \mathrm{H}^{p+1}(U, \widetilde{M'}(*)) \arrow[d] \\ \mathrm{H}^{p+1}((\mathbf{f}), M'') \arrow[r, "\partial"] & \mathrm{H}^{p+2}((\mathbf{f}), M') \end{tikzcd}$$

(ii) Proposition (2.1.5) is especially useful when S is an A-algebra generated by finitely many elements of degree 1, with A Noetherian; in that case every quasi-coherent $\mathscr{O}_X$-module is of the form $\widetilde{M}$ (**II**, 2.7.7).

(2.1.7). We now apply (2.1.5) in the case $S = A[\mathrm{T}_0, \ldots, \mathrm{T}_r]$, where A is an arbitrary ring and the T_i are indeterminates, taking $M = S$ and $f_i = \mathrm{T}_i$. We are thus essentially reduced to computing $\mathrm{H}^\bullet((\mathbf{T}), S)$, where $\mathbf{T} = (\mathrm{T}_i)_{0 \leqslant i \leqslant r}$. **III | 98**

Lemma (2.1.8). — *If $S = A[\mathrm{T}_0, \ldots, \mathrm{T}_r]$, then, with $\mathbf{T} = (\mathrm{T}_i)_{0\leqslant i \leqslant r}$,*

(2.1.8.1) $$\mathrm{H}^i(\mathbf{T}^n, S) = 0 \qquad \textit{if } i \neq r+1,$$

(2.1.8.2) $$\mathrm{H}^{r+1}(\mathbf{T}^n, S) = S/(\mathbf{T}^n).$$

The A-module $\mathrm{H}^{r+1}(\mathbf{T}^n, S)$ therefore has an A-basis formed by the classes modulo $(\mathbf{T}^n)$ of the monomials

$$\mathrm{T}_{\mathbf{p}} = \mathrm{T}_0^{p_0} \cdots \mathrm{T}_r^{p_r}, \qquad \mathbf{p} = (p_0, \ldots, p_r), \quad 0 \leqslant p_i < n \textit{ for every } i.$$

Proof. This follows immediately from (1.1.3.5) and Proposition (1.1.4), whose hypotheses are trivially satisfied. ☐

(2.1.9). Passing to the inductive limit over n, relations (2.1.8.1) give $\mathrm{H}^i((\mathbf{T}), S) = 0$ for $i \neq r+1$. For $i = r+1$, the inductive system consists of the modules $S/(\mathbf{T}^n)$, the homomorphism

$$\phi_{nm} : S/(\mathbf{T}^n) \longrightarrow S/(\mathbf{T}^m) \qquad (0 \leqslant n \leqslant m)$$

being multiplication by $(\mathrm{T}_0 \cdots \mathrm{T}_r)^{m-n}$. For $n \geqslant \sup(p_i)_{0 \leqslant i \leqslant r}$, denote by $\xi_{\mathbf{p}}^{(n)} = \xi_{p_0,\ldots,p_r}^{(n)}$ the class of $\mathrm{T}_0^{n-p_0} \cdots \mathrm{T}_r^{n-p_r}$ modulo $(\mathbf{T}^n)$. Then $\phi_{nm}(\xi_{\mathbf{p}}^{(n)}) = \xi_{\mathbf{p}}^{(m)}$, so these elements have the same canonical image $\xi_{\mathbf{p}} = \xi_{p_0,\ldots,p_r}$ in the inductive limit $\mathrm{H}^{r+1}((\mathbf{T}), S)$. By the definition of degree in (2.1.2), the degree of $\xi_{\mathbf{p}}$ is $-|\mathbf{p}| = -(p_0 + p_1 + \cdots + p_r)$. It is clear that the $\xi_{\mathbf{p}}^{(n)}$ with $0 < p_i \leqslant n$ for $0 \leqslant i \leqslant r$ form a basis of $S/(\mathbf{T}^n)$. We therefore deduce immediately from (2.1.8):

Corollary (2.1.10). — *With the notation of (2.1.8),*

(2.1.10.1) $$\mathrm{H}^i((\mathbf{T}), S) = 0 \qquad \textit{for } i \neq r+1,$$

and $\mathrm{H}^{r+1}((\mathbf{T}), S)$ is a free A-module with basis the elements $\xi_{p_0,\ldots,p_r}$ such that $p_i > 0$ for $0 \leqslant i \leqslant r$.

(2.1.11). *Remark.* — Let N be any A-module and let $M = S \otimes_A N$. The argument of (2.1.8) shows more generally that

(2.1.11.1) $$\mathrm{H}^i(\mathbf{T}^n, M) = 0 \qquad \text{if } i \neq r+1,$$

(2.1.11.2) $$\mathrm{H}^{r+1}(\mathbf{T}^n, M) = (S/(\mathbf{T}^n)) \otimes_A N.$$

Indeed, the latter formula follows directly from (1.1.3.5). Moreover, it is clear that

$$M/(\mathrm{T}_0^n M + \cdots + \mathrm{T}_{i-1}^n M) = \big(S/(\mathrm{T}_0^n S + \cdots + \mathrm{T}_{i-1}^n S)\big) \otimes_A N,$$

since the ideal $\mathrm{T}_0^n S + \cdots + \mathrm{T}_{i-1}^n S$ is a direct summand of the A-module S. This permits us to apply (1.1.4) to M, and yields (2.1.11.1).

Combining (2.1.10) and (2.1.5), we obtain:

Proposition (2.1.12). — *Let A be a ring, let r be an integer > 0, and let $X = \mathbf{P}_A^r$ (**II**, 4.1.1). Then:*

(i) $\mathrm{H}^i(X, \mathscr{O}_X(*)) = 0$ *for* $i \neq 0, r$.

(ii) *The canonical homomorphism* $\alpha : S \to \mathrm{H}^0(X, \mathscr{O}_X(*))$ (**II**, 2.6.2) *is bijective.* **III** | 99

(iii) $\mathrm{H}^r(X, \mathscr{O}_X(*))$ *is a free A-module with basis the elements $\xi_{p_0,\ldots,p_r}$, where $p_i > 0$ for $0 \leqslant i \leqslant r$; the element $\xi_{p_0,\ldots,p_r}$ has degree $-|\mathbf{p}| = -(p_0 + \cdots + p_r)$, and*

$$\mathrm{T}_i \xi_{p_0,\ldots,p_r} = \xi_{p_0,\ldots,p_{i-1},p_i-1,p_{i+1},\ldots,p_r}.$$

Proof. In the exact sequence (2.1.5.2) applied to $M = S = A[\mathrm{T}_0, \ldots, \mathrm{T}_r]$, we have $\mathrm{H}^0((\mathbf{T}), S) = 0$ and $\mathrm{H}^1((\mathbf{T}), S) = 0$ by (2.1.10.1); moreover, Proposition (2.1.5) applies with $U = X$, since X is the union of the $\mathrm{D}_+(\mathrm{T}_i)$ (**II**, 2.3.14). It remains to identify the map $S \to \mathrm{H}^0(X, \mathscr{O}_X(*))$ in (2.1.5.2) with the canonical map α. This follows from the canonical identification of $\mathrm{H}^0(U, \mathscr{O}_X(*))$ and $\mathrm{H}^0(\mathfrak{U}, \mathscr{O}_X(*))$. ☐

Corollary (2.1.13). — *The only pairs (i, n) for which $\mathrm{H}^i(X, \mathscr{O}_X(n)) \neq 0$ can occur are $i = 0$, $n \geqslant 0$, and $i = r$, $n \leqslant -(r+1)$.*

If $A \neq 0$, then $\mathrm{H}^i(X, \mathscr{O}_X(n)) \neq 0$ does indeed hold for all the pairs listed in (2.1.13). This follows from (2.1.12), since $S_n \neq 0$ for every degree $n \geqslant 0$.

In the applications made in this chapter, we will mostly use the following less precise result.

Corollary (2.1.14). — *The A-modules $\mathrm{H}^i(X, \mathscr{O}_X(n))$ are free of finite type; if $i > 0$, they vanish for $n > 0$.*

Proposition (2.1.15). — *Let Y be a prescheme, let $\mathscr{E}$ be a locally free $\mathscr{O}_Y$-module of rank $r+1$, let $X = \mathbf{P}(\mathscr{E})$ be the projective bundle defined by $\mathscr{E}$, and let $f : X \to Y$ be the structural morphism. The only values of i and n for which $\mathrm{R}^i f_*(\mathscr{O}_X(n)) \neq 0$ can occur are $i = 0$, $n \geqslant 0$, and $i = r$, $n \leqslant -(r+1)$. Moreover, the canonical homomorphism* (**II**, 3.3.2)

$$\alpha : \mathbf{S}_{\mathscr{O}_Y}(\mathscr{E}) \longrightarrow \Gamma_*(\mathscr{O}_X) = \mathrm{R}^0 f_*(\mathscr{O}_X(*)) = \bigoplus_{n \in \mathbf{Z}} f_*(\mathscr{O}_X(n))$$

is an isomorphism.

Proof. The question is local on Y, so we may assume that Y is affine with ring A and that $\mathscr{E} = \widetilde{E}$, where $E = A^{r+1}$. We are then immediately reduced to (2.1.12), taking (1.4.11) into account. ☐

(2.1.16). *Remark.* — We will later complete the results of (2.1.15) by proving the following propositions. Set

$$\boldsymbol{\omega} = f^*\big(\bigwedge^{r+1} \mathscr{E}\big)(-r-1),$$

which is an invertible $\mathscr{O}_X$-module. Then:

(i) There is a canonical isomorphism

$$\rho : \mathscr{O}_Y \xrightarrow{\sim} \mathrm{R}^r f_*(\boldsymbol{\omega}). \tag{2.1.16.1}$$

(ii) The cup-product pairing (**0**, 12.2.2)

$$\mathrm{R}^r f_*(\mathscr{O}_X(n)) \times \mathrm{R}^0 f_*(\boldsymbol{\omega}(-n)) \longrightarrow \mathrm{R}^r f_*(\boldsymbol{\omega}), \tag{2.1.16.2}$$

composed with ρ^{-1}, defines an *isomorphism* from $\mathrm{R}^r f_*(\mathscr{O}_X(n))$ onto the *dual* of the locally free $\mathscr{O}_Y$-module

$$\mathrm{R}^0 f_*(\boldsymbol{\omega}(-n)) = \big(\bigwedge^{r+1} \mathscr{E}\big) \otimes_{\mathscr{O}_Y} \big(\mathbf{S}_{\mathscr{O}_Y}(\mathscr{E})\big)_{-n}.$$

2.2. The fundamental theorem of projective morphisms III | 100

Theorem (2.2.1). — *(Serre). Let Y be a Noetherian prescheme, let $f : X \to Y$ be a proper morphism, and let $\mathscr{L}$ be an invertible $\mathscr{O}_X$-module ample relative to f. For every $\mathscr{O}_X$-module $\mathscr{F}$, set*

$$\mathscr{F}(n) = \mathscr{F} \otimes_{\mathscr{O}_X} \mathscr{L}^{\otimes n} \qquad (n \in \mathbf{Z}).$$

Then, for every coherent $\mathscr{O}_X$-module $\mathscr{F}$:

(i) *the $\mathscr{O}_Y$-modules $\mathrm{R}^q f_*(\mathscr{F})$ are coherent;*
(ii) *there exists an integer N such that, for $n \geqslant N$,*

$$\mathrm{R}^q f_*(\mathscr{F}(n)) = 0 \qquad \textit{for every } q > 0;$$

(iii) *there exists an integer N such that, for $n \geqslant N$, the canonical homomorphism*

$$f^*\big(f_*(\mathscr{F}(n))\big) \longrightarrow \mathscr{F}(n)$$

is surjective.

Proof. First note that if the theorem is true after replacing $\mathscr{L}$ by $\mathscr{L}^{\otimes d}$, where $d > 0$, then it is true in its original form. Indeed, writing

$$\mathscr{F}(n) = \big(\mathscr{F} \otimes \mathscr{L}^{\otimes r}\big) \otimes \mathscr{L}^{\otimes hd}, \qquad h \geqslant 0, \quad 0 \leqslant r < d,$$

the hypothesis gives, for each r, an integer N_r such that (ii) and (iii) hold for $\mathscr{F} \otimes \mathscr{L}^{\otimes r}$ whenever $h \geqslant N_r$. Taking N to be the largest of the integers $dN_r + r$, assertions (ii) and (iii) therefore hold for $n \geqslant N$.

We may consequently suppose that $\mathscr{L}$ is very ample relative to f (**II**, 4.6.11). There is then a dominant open immersion over Y

$$i : X \longrightarrow P, \qquad P = \mathrm{Proj}(\mathscr{S}),$$

where $\mathscr{S}$ is a quasi-coherent $\mathscr{O}_Y$-algebra graded in positive degrees, $\mathscr{S}_1$ is of finite type and generates $\mathscr{S}$, and $\mathscr{L}$ is isomorphic to $i^*(\mathscr{O}_P(1))$ (**II**, 4.4.7). Since f is proper, i is proper as well (**II**, 5.4.4), and hence is an isomorphism $X \xrightarrow{\sim} P$. We are thus reduced to the case $X = \mathrm{Proj}(\mathscr{S})$ and $\mathscr{L} = \mathscr{O}_X(1)$. The theorem is then a consequence of the following proposition. □

Proposition (2.2.2). — *Let A be a Noetherian ring and let S be an A-algebra graded in positive degrees such that the A-module S_1 has a system of $r+1$ generators and generates the algebra S. Set $X = \mathrm{Proj}(S)$. For every coherent $\mathscr{O}_X$-module $\mathscr{F}$:*

(i) *the A-modules $\mathrm{H}^q(X, \mathscr{F})$ are of finite type;*
(ii) $\mathrm{H}^q(X, \mathscr{F}) = 0$ *for $q > r$;*
(iii) *there exists an integer N such that, for $n \geqslant N$,*

$$\mathrm{H}^q(X, \mathscr{F}(n)) = 0 \qquad \textit{for every } q > 0;$$

(iv) *there exists an integer N such that, for $n \geqslant N$, $\mathscr{F}(n)$ is generated by its sections over X.*

Proof. Let us first show how (2.2.2) implies (2.2.1). In the special case $X = \mathrm{Proj}(\mathscr{S})$ to which (2.2.1) was reduced above, Y is quasi-compact. It can therefore be covered by finitely many affine open subsets U_α, with Noetherian rings, such that the restriction of $\mathscr{S}_1$ to each U_α is generated by finitely many sections over U_α. Once (2.2.2) is known, it is enough to take in parts (ii) and (iii) of (2.2.1) the largest of the corresponding integers for the U_α, taking (1.4.11) and (**II**, 3.4.7) into account.

To prove (2.2.2), observe that X identifies with a closed subprescheme of $P = \mathbf{P}^r_A$ (**II**, 3.6.2). Moreover, if $j : X \to P$ is the canonical injection, then $j_*(\mathscr{F})$ is a coherent $\mathscr{O}_P$-module and

$$j_*(\mathscr{F}(n)) = (j_*\mathscr{F})(n)$$

by (**II**, 3.4.5) and (**II**, 3.5.2). In view of (G, II, Corollary to Theorem 4.9.1), we are therefore reduced to proving (2.2.2) in the special case

$$X = \mathbf{P}^r_A, \qquad S = A[T_0, \ldots, T_r].$$

Since X is covered by the $r+1$ affine open subsets $D_+(T_i)$, assertion (ii) follows from (1.4.12). III | 101
Assertion (iv) has already been proved in (**II**, 2.7.9).

We now prove (i) and (iii) simultaneously. They hold for $\mathscr{F} = \mathscr{O}_X(m)$ by (2.1.13), and hence also when $\mathscr{F}$ is a direct sum of finitely many $\mathscr{O}_X$-modules of the form $\mathscr{O}_X(m_j)$. They are trivially true for $q > r$ by (ii). We proceed by descending induction on q.

The sheaf $\mathscr{F}$ is isomorphic to a quotient of a direct sum $\mathscr{E}$ of finitely many sheaves $\mathscr{O}_X(m_j)$ (**II**, 2.7.10). Thus there is an exact sequence

$$0 \longrightarrow \mathscr{R} \longrightarrow \mathscr{E} \longrightarrow \mathscr{F} \longrightarrow 0,$$

where $\mathscr{R}$ is coherent (**0**, 5.3.3) and $\mathscr{E}$ satisfies (i) and (iii). Since twisting is exact, for every $n \in \mathbf{Z}$ we also have an exact sequence

$$0 \longrightarrow \mathscr{R}(n) \longrightarrow \mathscr{E}(n) \longrightarrow \mathscr{F}(n) \longrightarrow 0,$$

and hence an exact cohomology sequence

$$\mathrm{H}^{q-1}(X, \mathscr{E}(n)) \longrightarrow \mathrm{H}^{q-1}(X, \mathscr{F}(n)) \longrightarrow \mathrm{H}^{q}(X, \mathscr{R}(n)).$$

Because $\mathscr{E}(n)$ is a direct sum of sheaves $\mathscr{O}_X(n + m_j)$ (**II**, 2.5.14), the module $\mathrm{H}^{q-1}(X, \mathscr{E}(n))$ is of finite type. The module $\mathrm{H}^{q}(X, \mathscr{R}(n))$ is likewise of finite type by the induction hypothesis. Since A is Noetherian, it follows that $\mathrm{H}^{q-1}(X, \mathscr{F}(n))$ is of finite type, proving (i).

By the induction hypothesis we may choose N so that $\mathrm{H}^{q}(X, \mathscr{R}(n)) = 0$ for $n \geqslant N$. We may also choose N so that $\mathrm{H}^{q-1}(X, \mathscr{E}(n)) = 0$ for $n \geqslant N$, since $\mathscr{E}$ satisfies (iii). The exact cohomology sequence then gives

$$\mathrm{H}^{q-1}(X, \mathscr{F}(n)) = 0 \qquad (n \geqslant N),$$

which completes the proof. □

Corollary (2.2.3). — *Under the hypotheses of (2.2.1), let*

$$\mathscr{F} \longrightarrow \mathscr{G} \longrightarrow \mathscr{H}$$

be an exact sequence of coherent $\mathscr{O}_X$-modules. There exists an integer N such that, for $n \geqslant N$, the sequence

$$f_*(\mathscr{F}(n)) \longrightarrow f_*(\mathscr{G}(n)) \longrightarrow f_*(\mathscr{H}(n))$$

is exact.

Proof. Let $\mathscr{F}'$, $\mathscr{G}'$, and $\mathscr{G}''$ be respectively the kernel, image, and cokernel of $\mathscr{F} \to \mathscr{G}$. Then $\mathscr{G}'$ is the kernel and $\mathscr{G}''$ the image of $\mathscr{G} \to \mathscr{H}$; let $\mathscr{H}''$ be the cokernel of this latter homomorphism. All these $\mathscr{O}_X$-modules are coherent (**0**, 5.3.4). Since twisting is exact, it is enough to prove that, for n sufficiently large, each of the sequences

$$0 \longrightarrow f_*(\mathscr{F}'(n)) \longrightarrow f_*(\mathscr{F}(n)) \longrightarrow f_*(\mathscr{G}'(n)) \longrightarrow 0,$$
$$0 \longrightarrow f_*(\mathscr{G}'(n)) \longrightarrow f_*(\mathscr{G}(n)) \longrightarrow f_*(\mathscr{G}''(n)) \longrightarrow 0,$$
$$0 \longrightarrow f_*(\mathscr{G}''(n)) \longrightarrow f_*(\mathscr{H}(n)) \longrightarrow f_*(\mathscr{H}''(n)) \longrightarrow 0$$

is exact. Consequently, we may suppose that

$$0 \longrightarrow \mathscr{F} \longrightarrow \mathscr{G} \longrightarrow \mathscr{H} \longrightarrow 0$$

is exact. The cohomology sequence is then

$$0 \longrightarrow f_*(\mathscr{F}(n)) \longrightarrow f_*(\mathscr{G}(n)) \longrightarrow f_*(\mathscr{H}(n)) \longrightarrow \mathrm{R}^1 f_*(\mathscr{F}(n)) \longrightarrow \cdots,$$

and the conclusion follows from (2.2.1), (ii). □

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Corollary (2.2.4). — *Let Y be a Noetherian prescheme, let $f : X \to Y$ be a morphism of finite type, and let $\mathscr{L}$ be an invertible $\mathscr{O}_X$-module ample relative to f. For every $\mathscr{O}_X$-module $\mathscr{F}$, set*

$$\mathscr{F}(n) = \mathscr{F} \otimes_{\mathscr{O}_X} \mathscr{L}^{\otimes n} \qquad (n \in \mathbf{Z}).$$

Let

$$\mathscr{F} \longrightarrow \mathscr{G} \longrightarrow \mathscr{H}$$

be an exact sequence of coherent $\mathscr{O}_X$-modules such that the supports of $\mathscr{F}$ and $\mathscr{H}$ are proper over Y (**II**, 5.4.10)*. Then there exists an integer N such that, for $n \geqslant N$, the sequence*

$$f_*(\mathscr{F}(n)) \longrightarrow f_*(\mathscr{G}(n)) \longrightarrow f_*(\mathscr{H}(n))$$

is exact.

Proof. The argument at the beginning of (2.2.1) shows that if the corollary is true for $\mathscr{L}^{\otimes d}$, where $d > 0$, then it is true for $\mathscr{L}$. We may therefore suppose that $\mathscr{L}$ is very ample relative to f (**II**, 4.6.11). We can then identify X with an open subset of a Y-prescheme

$$Z = \operatorname{Proj}(\mathscr{S}),$$

where $\mathscr{S}$ is a quasi-coherent $\mathscr{O}_Y$-algebra graded in positive degrees, $\mathscr{S}_1$ is of finite type and generates $\mathscr{S}$, and

$$\mathscr{L} = i^*(\mathscr{O}_Z(1)),$$

with $i : X \to Z$ the canonical immersion (**II**, 4.4.7).

Now $\operatorname{Supp}(\mathscr{G})$ is closed in X and contained in

$$\operatorname{Supp}(\mathscr{F}) \cup \operatorname{Supp}(\mathscr{H});$$

it is therefore proper over Y. The supports of $\mathscr{F}$, $\mathscr{G}$, and $\mathscr{H}$ are consequently closed in Z (**II**, 5.4.10). Thus

$$\mathscr{F}' = i_*(\mathscr{F}), \qquad \mathscr{G}' = i_*(\mathscr{G}), \qquad \mathscr{H}' = i_*(\mathscr{H})$$

are coherent $\mathscr{O}_Z$-modules, and the sequence

$$\mathscr{F}' \longrightarrow \mathscr{G}' \longrightarrow \mathscr{H}'$$

is exact. If $g : Z \to Y$ is the structural morphism, then $f = g \circ i$, and clearly

$$\mathscr{F}'(n) = i_*(\mathscr{F}(n)),$$

with analogous identities for $\mathscr{G}'$ and $\mathscr{H}'$. The conclusion follows from (2.2.3), applied to $\mathscr{F}'$, $\mathscr{G}'$, and $\mathscr{H}'$. □

(2.2.5). *Remarks.*

(i) Assertion (i) of (2.2.1) remains true if one assumes only that Y is *locally Noetherian*. Indeed, the property is plainly local on Y. Moreover, the hypotheses of (2.2.1) imply that, for every open subset $U \subset Y$, the restriction of f to $f^{-1}(U)$ is a projective morphism

$$f^{-1}(U) \longrightarrow U$$

(**II**, 5.5.5, iii), and $\mathscr{L}|_{f^{-1}(U)}$ is ample for this morphism (**II**, 4.6.4).

(ii) Assertion (iii) of (2.2.1) remains valid, as we have seen, if one assumes only that X is a quasi-compact scheme, or a prescheme whose underlying space is Noetherian, and that $f : X \to Y$ is quasi-compact (**II**, 4.6.8). It should be noted, however, that even when Y is the *spectrum of a field K* and f is *quasi-projective*, assertion (ii) of (2.2.1) need no longer hold.

For example, let

$$X' = \operatorname{Spec}(K[T_0, \ldots, T_r])$$

and let X be the union of the affine open subsets $D(T_i)$ of X', for $0 \leqslant i \leqslant r$. Since the immersion $X \to X'$ is quasi-compact, the structural morphism $f : X \to Y$ is quasi-affine (**II**, 5.1.10), and hence $\mathscr{O}_X$ is very ample relative to f (**II**, 5.1.6). But

$$\Gamma(X, \mathscr{O}_X) = \bigcap_{i=0}^{r} (K[T_0, \ldots, T_r])_{T_i} = K[T_0, \ldots, T_r]$$

by (**I**, 8.2.1.1). It follows from (1.4.3.1) and (1.1.3.5) that

$$\mathrm{H}^r(X, \mathscr{O}_X^{\otimes n}) = \mathrm{H}^r(X, \mathscr{O}_X) = A \neq 0$$

for every n.

2.3. Application to graded sheaves of algebras and modules

Theorem (2.3.1). — *Let Y be a Noetherian prescheme, let $\mathscr{S}$ be a quasi-coherent $\mathscr{O}_Y$-algebra of finite type, graded in positive degrees, let*

$$X = \operatorname{Proj}(\mathscr{S}),$$

and let $q : X \to Y$ be the structural morphism. Let $\mathscr{M}$ be a quasi-coherent graded $\mathscr{S}$-module satisfying condition **(TF)**. *There then exists an integer N such that, for $n \geqslant N$, the canonical homomorphism* (**II**, 8.14.5.1)

$$\alpha_n : \mathscr{M}_n \longrightarrow q_*\big(\mathscr{P}roj_0(\mathscr{M}(n))\big) = q_*\big((\mathscr{P}roj(\mathscr{M}))_n\big)$$

is bijective. In other words, the canonical homomorphism **III** | 103

$$\alpha : \mathscr{M} \longrightarrow \mathbf{\Gamma}_*(\mathscr{P}roj(\mathscr{M}))$$

is a **(TN)***-isomorphism.*

Proof. We may restrict to the case in which $\mathscr{M}$ is an $\mathscr{S}$-module of finite type (**II**, 3.4.2). Since Y is quasi-compact, there is an integer $d > 0$ such that $\mathscr{S}^{(d)}$ is generated by the quasi-coherent $\mathscr{O}_Y$-module $\mathscr{S}_d$, which is of finite type (**II**, 3.1.10), and hence coherent because Y is Noetherian. Now $\mathscr{M}$ is the direct sum of the $\mathscr{M}^{(d,k)}$, for $0 \leqslant k < d$, and each $\mathscr{M}^{(d,k)}$ is a quasi-coherent $\mathscr{S}^{(d)}$-module of finite type, by (**II**, 2.1.6, iii). Since the question is local on Y, it plainly suffices to prove that each canonical homomorphism

$$\alpha : \mathscr{M}^{(d,k)} \longrightarrow \mathbf{\Gamma}_*\big((\mathscr{P}roj(\mathscr{M}))^{(d,k)}\big)$$

is a **(TN)**-isomorphism. Taking (**II**, 8.14.13) into account, and in particular diagram (**II**, 8.14.13.4), we are thus reduced to proving the theorem when $\mathscr{S}$ is generated by $\mathscr{S}_1$ and $\mathscr{S}_1$ is a coherent $\mathscr{O}_Y$-module.

Since Y is Noetherian, the argument used at the beginning of the proof of (2.2.2) shows that we may reduce to the case

$$Y = \operatorname{Spec}(A), \qquad \mathscr{S} = \widetilde{S}, \qquad \mathscr{M} = \widetilde{M},$$

where A is a Noetherian ring, S_1 is an A-module of finite type, and M is a graded S-module of finite type. We shall show that it then suffices to prove the theorem for $M = S$. Indeed, in the general case there is an exact sequence

$$L' \longrightarrow L \longrightarrow M \longrightarrow 0,$$

where L and L' are direct sums of graded modules of the form $S(m)$. If the result holds for $M = S$, it also holds for $M = S(m)$, and hence for L and L'. Consider the commutative diagram

$$\begin{tikzcd}
\widetilde{L'_n} \arrow[r] \arrow[d, "\alpha_n"'] & \widetilde{L_n} \arrow[r] \arrow[d, "\alpha_n"] & \widetilde{M_n} \arrow[r] \arrow[d, "\alpha_n"] & 0 \\
q_*(\widetilde{L'}(n)) \arrow[r] & q_*(\widetilde{L}(n)) \arrow[r] & q_*(\widetilde{M}(n)) \arrow[r] & 0.
\end{tikzcd}$$

The second row is exact by (2.2.3) once n is sufficiently large. The same is true of the first row, and **III** | 104
since the two left-hand vertical arrows are isomorphisms, the third is as well.

It therefore remains to prove the theorem for $M = S$. Suppose first that

$$S = A[T_0, \ldots, T_r],$$

where the T_i are indeterminates. In this case the assertion is precisely (2.1.11, ii). In the general case, S identifies with a quotient of a ring

$$S' = A[T_0, \ldots, T_r]$$

by a graded ideal, so X is a closed subscheme of

$$X' = \mathbf{P}^r_A$$

(**II**, 2.9.2). If $j : X \to X'$ is the canonical injection, then $j_*(\widetilde{S}(n))$ is the $\mathscr{O}_{X'}$-module

$$(\mathscr{P}roj(\widetilde{S}))(n),$$

where S is regarded as a graded S'-module; this follows at once from (**II**, 2.8.7). Since $j_*(\widetilde{S}(n))$ is an $\mathscr{O}_{X'}$-module satisfying **(TF)**, the canonical homomorphism

$$\alpha_n : S_n \longrightarrow \Gamma\big(X', j_*(\widetilde{S}(n))\big)$$

is bijective for all sufficiently large n, by the polynomial case just proved. This completes the proof, since

$$\Gamma\big(X', j_*(\widetilde{S}(n))\big) = \Gamma(X, \widetilde{S}(n)).$$

□

Corollary (2.3.2). — *Under the hypotheses of (2.3.1), let*

$$\mathscr{S}_X = \bigoplus_{n\in\mathbf{Z}} \mathscr{O}_X(n),$$

*and let $\mathscr{F}$ be a quasi-coherent graded $\mathscr{S}_X$-module of finite type. Then $\mathbf{\Gamma}_*(\mathscr{F})$ satisfies condition **(TF)**.*

Proof. In the proof of (2.3.1) we saw that X, which is isomorphic to $\operatorname{Proj}(\mathscr{S}^{(d)})$ (**II**, 3.1.8), is of finite type over Y (**II**, 3.4.1). It follows from (**II**, 8.14.9) that $\mathscr{F}$ is isomorphic to a graded $\mathscr{S}_X$-module of the form $\mathscr{P}roj(\mathscr{M})$, where $\mathscr{M}$ is a quasi-coherent graded $\mathscr{S}$-module of finite type. By (2.3.1), $\mathbf{\Gamma}_*(\mathscr{F})$ is **(TN)**-isomorphic to $\mathscr{M}$ and consequently satisfies **(TF)**. □

(2.3.3). *Scholium.* Let Y be a Noetherian prescheme, let $\mathscr{S}$ be a graded $\mathscr{O}_Y$-algebra satisfying the hypotheses of (2.3.1), and let $X = \operatorname{Proj}(\mathscr{S})$. Let $\mathtt{K}_{\mathscr{S}}$ be the abelian category of quasi-coherent graded $\mathscr{S}$-modules satisfying **(TF)**, and let $\mathtt{K}'_{\mathscr{S}}$ be the subcategory of $\mathtt{K}_{\mathscr{S}}$ consisting of the $\mathscr{S}$-modules satisfying **(TN)**. Finally, let $\mathtt{K}_X$ be the category of quasi-coherent graded $\mathscr{S}_X$-modules $\mathscr{F}$ of finite type; since $\mathscr{S}_X$ is periodic (**II**, 8.14.4) and (**II**, 8.14.12), this is equivalent to requiring the $\mathscr{F}_i$ to be coherent $\mathscr{O}_X$-modules.

By (**II**, 8.14.8), (**II**, 8.14.10), and (2.3.2), the functors

$$\mathscr{M} \longmapsto \mathscr{P}roj(\mathscr{M}), \qquad \mathscr{F} \longmapsto \mathbf{\Gamma}_*(\mathscr{F})$$

define an equivalence (T, I, 1.2) between the quotient category

$$\mathtt{K}_{\mathscr{S}}/\mathtt{K}'_{\mathscr{S}}$$

(T, I, 1.11) and the category $\mathtt{K}_X$. When $\mathscr{S}$ is generated by $\mathscr{S}_1$, the category $\mathtt{K}_X$ may be replaced by the category of coherent $\mathscr{O}_X$-modules (**II**, 8.14.12).

Proposition (2.3.4). — *Let Y be a Noetherian prescheme.*

(i) Let $\mathscr{S}$ be a quasi-coherent $\mathscr{O}_Y$-algebra of finite type, graded in positive degrees. Set

$$X = \operatorname{Proj}(\mathscr{S}), \qquad \mathscr{S}_X = \mathscr{P}roj(\mathscr{S}) = \bigoplus_{n\in\mathbf{Z}} \mathscr{O}_X(n).$$

Then $\mathscr{S}_X$ is a periodic quasi-coherent graded $\mathscr{O}_X$-algebra (**II**, 8.14.12), *its homogeneous components*

$$(\mathscr{S}_X)_n = \mathscr{O}_X(n)$$

are coherent $\mathscr{O}_X$-modules, and, if $d > 0$ is a period of $\mathscr{S}_X$, then

$$(\mathscr{S}_X)_d = \mathscr{O}_X(d)$$

is an invertible Y-ample $\mathscr{O}_X$-module. Moreover, the canonical homomorphism

$$\alpha : \mathscr{S} \longrightarrow \mathbf{\Gamma}_*(\mathscr{S}_X)$$

*is a **(TN)**-isomorphism.*

(ii) Conversely, let $q : X \to Y$ be a projective morphism, and let $\mathscr{S}'$ be a graded $\mathscr{O}_X$-algebra whose homogeneous components $\mathscr{S}'_n$, for $n \in \mathbf{Z}$, are coherent $\mathscr{O}_X$-modules. Suppose that $\mathscr{S}'$ has a period $d > 0$ such that $\mathscr{S}'_d$ is an invertible $\mathscr{O}_X$-module ample for q. Then

$$\mathscr{S} = \bigoplus_{n\geqslant 0} q_*(\mathscr{S}'_n)$$

is a quasi-coherent $\mathscr{O}_Y$-algebra of finite type, graded in positive degrees, and there is a Y-isomorphism

$$r : X \xrightarrow{\sim} \operatorname{Proj}(\mathscr{S})$$

such that $r^*(\mathscr{P}roj(\mathscr{S}))$ is isomorphic to $\mathscr{S}'$ as a graded $\mathscr{O}_X$-algebra.

Proof. (i) Almost all the assertions have already been proved, the last being a special case of (2.3.2). The periodicity of $\mathscr{S}_X$ was proved in (**II**, 8.14.14), and the assertion that there is a period $d > 0$ for which $\mathscr{O}_X(d)$ is invertible and Y-ample is precisely (**II**, 4.6.18). Finally, for $0 \leqslant k < d$, $(\mathscr{S}_X)^{(d,k)}$ is an $(\mathscr{S}_X)^{(d)}$-module of finite type (**II**, 8.14.14). It follows that each $(\mathscr{S}_X)_n$ is a quasi-coherent $\mathscr{O}_X$-module of finite type (**II**, 2.1.6, ii); since $\mathscr{O}_X$ is coherent, so is each $(\mathscr{S}_X)_n$.

(ii) After replacing the period d by one of its multiples, we may suppose that

$$\mathscr{L} = \mathscr{S}'_d$$

is very ample relative to q (**II**, 4.6.11). By hypothesis,

$$\mathscr{S}'^{(d)} = \bigoplus_{n\in\mathbf{Z}} \mathscr{L}^{\otimes n},$$

and hence

$$\mathscr{S}^{(d)} = \bigoplus_{n\geqslant 0} q_*(\mathscr{L}^{\otimes n}).$$

By (**II**, 3.1.8) and (**II**, 3.2.9), there is a Y-isomorphism

$$s : X' = \operatorname{Proj}(\mathscr{S}) \xrightarrow{\sim} X'' = \operatorname{Proj}(\mathscr{S}^{(d)})$$

such that **III** | 105

$$s^*(\mathscr{O}_{X''}(n)) = \mathscr{O}_{X'}(nd)$$

for every n (**II**, 3.5.2). We shall therefore establish the existence of a Y-isomorphism $X \xrightarrow{\sim} X'$ by proving the following assertion.

(2.3.4.1). Let Y be a Noetherian prescheme, let $q : X \to Y$ be a projective morphism, and let $\mathscr{L}$ be an invertible $\mathscr{O}_X$-module very ample for q. Then

$$\mathscr{S} = \bigoplus_{n\geqslant 0} q_*(\mathscr{L}^{\otimes n})$$

is a quasi-coherent graded $\mathscr{O}_Y$-algebra of finite type such that

$$\mathscr{S}_n = \mathscr{S}_1^n$$

for all sufficiently large n, and there is a Y-isomorphism

$$r : X \xrightarrow{\sim} P = \operatorname{Proj}(\mathscr{S})$$

such that

$$\mathscr{L} = r^*(\mathscr{O}_P(1)).$$

Since q is projective, it follows from (**II**, 5.4.4) and (**II**, 4.4.7) that there is a Y-isomorphism

$$r' : X \xrightarrow{\sim} P' = \operatorname{Proj}(\mathscr{T}),$$

where $\mathscr{T}$ is a quasi-coherent $\mathscr{O}_Y$-algebra, $\mathscr{T}_1$ is of finite type and generates $\mathscr{T}$, and

$$\mathscr{L} = r'^*(\mathscr{O}_{P'}(1)).$$

If $q' : P' \to Y$ is the structural morphism, then

$$\mathscr{S} = \bigoplus_{n\geqslant 0} q'_*(\mathscr{O}_{P'}(n)).$$

By (2.3.1), the canonical homomorphism

$$\alpha_n : \mathscr{T}_n \longrightarrow \mathscr{S}_n = q'_*(\mathscr{O}_{P'}(n))$$

is bijective for all sufficiently large n. Since $\mathscr{T}_n = \mathscr{T}_1^n$, it follows *a fortiori* that $\mathscr{S}_n = \mathscr{S}_1^n$ for all sufficiently large n.

The canonical homomorphism $\alpha : \mathscr{T} \to \mathscr{S}$ of graded $\mathscr{O}_Y$-algebras is a **(TN)**-isomorphism. Consequently

$$\Phi = \operatorname{Proj}(\alpha) : \operatorname{Proj}(\mathscr{S}) \longrightarrow \operatorname{Proj}(\mathscr{T})$$

is an isomorphism (**II**, 3.6.1). Moreover, (**II**, 3.5.2) and the fact that the graded $\mathscr{T}$-modules obtained from $\mathscr{S}(n)$ by restriction of scalars are **(TN)**-isomorphic to $\mathscr{T}(n)$ give

$$\Phi_*(\mathscr{O}_P(n)) = \mathscr{O}_{P'}(n)$$

for every n (**II**, 3.4.2). It remains only to show that $\mathscr{S}$ is of finite type over $\mathscr{O}_Y$. The modules

$$\mathscr{S}_n = q'_*(\mathscr{O}_{P'}(n))$$

are coherent by (2.2.1); and if $\mathscr{S}_n = \mathscr{S}_1^n$ for $n \geqslant n_0$, then $\mathscr{S}$ is generated by the coherent module

$$\bigoplus_{i \leqslant n_0} \mathscr{S}_i.$$

This proves the assertion (**I**, 9.6.2).

We return to the proof of (2.3.4), retaining its notation. We have proved that there is a Y-isomorphism

$$r'' : X \xrightarrow{\sim} X''$$

such that

$$r''_*(\mathscr{L}^{\otimes n}) = \mathscr{O}_{X''}(n)$$

for every $n \in \mathbf{Z}$. Let $q'' : X'' \to Y$ be the structural morphism. The algebra $\mathscr{S}'$ is the direct sum of the graded $\mathscr{S}'^{(d)}$-modules $\mathscr{S}'^{(d,k)}$. Each of these is a quasi-coherent $\mathscr{S}'^{(d)}$-module of finite type, by the periodicity of $\mathscr{S}'$ and the hypothesis that the $\mathscr{S}'_n$ are $\mathscr{O}_X$-modules of finite type (**II**, 8.14.12). Set

$$\mathscr{F}^{(k)} = r''_*(\mathscr{S}'^{(d,k)}),$$

so that the $\mathscr{F}^{(k)}$ are quasi-coherent graded $\mathscr{S}_{X''}$-modules of finite type. By (**II**, 8.14.8), the canonical homomorphism

$$\beta : \mathscr{P}roj(\mathbf{\Gamma}_*(\mathscr{F}^{(k)})) \longrightarrow \mathscr{F}^{(k)}$$

is an isomorphism of $\mathscr{S}_{X''}$-modules. On the other hand,

$$q''_*((\mathscr{F}^{(k)})_n) = q_*((\mathscr{S}'^{(d,k)})_n),$$

and for $n \geqslant 0$ the last $\mathscr{O}_Y$-module is, by definition, equal to $(\mathscr{S}^{(d,k)})_n$. In other words, the canonical injection

$$\mathscr{S}^{(d,k)} \longrightarrow \mathbf{\Gamma}_*(\mathscr{F}^{(k)})$$

is a **(TN)**-isomorphism. Hence

$$\mathscr{P}roj(\mathscr{S}^{(d,k)}) = \mathscr{P}roj(\mathbf{\Gamma}_*(\mathscr{F}^{(k)})),$$

and consequently

$$r''^*(\mathscr{P}roj(\mathscr{S}^{(d,k)})) \simeq \mathscr{S}'^{(d,k)}.$$

It remains to observe that, up to canonical isomorphism,

$$\mathscr{P}roj(\mathscr{S}^{(d,k)}) = s_*((\mathscr{P}roj(\mathscr{S}))^{(d,k)})$$

(**II**, 8.14.13.1); this proves that $r^*(\mathscr{P}roj(\mathscr{S}))$ is isomorphic to $\mathscr{S}'$.

Finally, by (2.3.2), each $\mathbf{\Gamma}_*(\mathscr{F}^{(k)})$ satisfies condition **(TF)**, and hence so does each $\mathscr{S}^{(d,k)}$. Moreover, the $\mathscr{S}_n = q_*(\mathscr{S}'_n)$ are coherent by (2.2.1), because the $\mathscr{S}'_n$ are coherent. It follows at once that the $\mathscr{S}^{(d,k)}$ are $\mathscr{S}^{(d)}$-modules of finite type. Since (2.3.4.1) shows that $\mathscr{S}^{(d)}$ is an $\mathscr{O}_Y$-algebra of finite type, $\mathscr{S}$ is likewise an $\mathscr{O}_Y$-algebra of finite type. □

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Proposition (2.3.5). — *Let Y be an integral Noetherian prescheme, let X be an integral prescheme, and let*

$$f : X \longrightarrow Y$$

be a projective birational morphism. There is then a coherent fractional ideal

$$\mathscr{J} \subset \mathscr{R}(Y)$$

(**II**, 8.1.2) *such that X is Y-isomorphic to the prescheme obtained by blowing up $\mathscr{J}$* (**II**, 8.1.3). *Moreover, there is an open subset U of Y such that the restriction of f to $f^{-1}(U)$ is an isomorphism from $f^{-1}(U)$ onto U (cf.* **I**, *6.5.5), and such that $\mathscr{J}|U$ is invertible.*

Proof. There is an invertible $\mathscr{O}_X$-module $\mathscr{L}$ very ample for f (**II**, 4.4.2) and (**II**, 5.3.2). Applying (2.3.4.1), we identify X with $\operatorname{Proj}(\mathscr{S})$, where

$$\mathscr{S} = \bigoplus_{n\geqslant 0} f_*(\mathscr{L}^{\otimes n}).$$

The modules $f_*(\mathscr{L}^{\otimes n})$ are coherent and torsion-free over $\mathscr{O}_Y$ (**I**, 7.4.5). Thus $\mathscr{S}$ identifies canonically with a sub-$\mathscr{O}_Y$-module of

$$\mathscr{S} \otimes_{\mathscr{O}_Y} \mathscr{R}(Y)$$

(**I**, 7.4.1). The latter is a simple sheaf (**I**, 7.3.6), and is therefore determined by its restriction to a nonempty open subset.

Choose a nonempty open subset $U' \subset U$ on which $\mathscr{L}|f^{-1}(U')$ is isomorphic to $\mathscr{O}_X|f^{-1}(U')$, and hence to $\mathscr{O}_Y|U'$. The restrictions $f_*(\mathscr{L}^{\otimes n})|U'$ are then isomorphic to $\mathscr{O}_Y|U'$. It follows that

$$\mathscr{S} \otimes_{\mathscr{O}_Y} \mathscr{R}(Y) \simeq \mathscr{R}(Y)[T],$$

where T is an indeterminate. By (2.3.4.1), $\mathscr{S}$ is **(TN)**-isomorphic to the sub-$\mathscr{O}_Y$-algebra generated by the canonical image of $f_*(\mathscr{L})$ in this polynomial algebra. Under the displayed identification, that image is $\mathscr{J} \cdot T$, where $\mathscr{J}$ is a coherent sub-$\mathscr{O}_Y$-module of $\mathscr{R}(Y)$ (2.2.1). Its restriction to U' is isomorphic to $\mathscr{O}_Y|U'$, and, after shrinking U if necessary, $\mathscr{J}|U$ is invertible. Consequently

$$\mathscr{S} \quad \text{is } \textbf{(TN)}\text{-isomorphic to} \quad \bigoplus_{n\geqslant 0} \mathscr{J}^n,$$

which completes the proof. □

Corollary (2.3.6). — *Under the hypotheses of (2.3.5), suppose in addition that for every nonzero coherent sub-$\mathscr{O}_Y$-module*

$$\mathscr{J} \subset \mathscr{R}(Y)$$

there is an invertible $\mathscr{O}_Y$-module $\mathscr{L}$ such that

$$\Gamma\big(Y, \mathscr{L} \otimes_{\mathscr{O}_Y} \mathscr{H}om(\mathscr{J}, \mathscr{O}_Y)\big) \neq 0.$$

Then, in the statement of (2.3.5), $\mathscr{J}$ may be taken to be an ideal of $\mathscr{O}_Y$. This additional condition is always satisfied if there exists an ample $\mathscr{O}_Y$-module.

Proof. Indeed, by (**0**, 5.4.2),

$$\mathscr{L} \otimes \mathscr{H}om(\mathscr{J}, \mathscr{O}_Y) = \mathscr{H}om\big(\mathscr{L}^{-1}, \mathscr{H}om(\mathscr{J}, \mathscr{O}_Y)\big) = \mathscr{H}om(\mathscr{J} \otimes \mathscr{L}^{-1}, \mathscr{O}_Y).$$

The hypothesis therefore says that there is a nonzero homomorphism

$$u : \mathscr{J} \otimes \mathscr{L}^{-1} \longrightarrow \mathscr{O}_Y.$$

For every $y \in Y$, the module $(\mathscr{J} \otimes \mathscr{L}^{-1})_y$ identifies with a sub-$\mathscr{O}_y$-module of the field of fractions $\mathscr{R}(Y)_y$ of $\mathscr{O}_y$ (**I**, 7.1.5). Thus u_y is necessarily injective, and u identifies $\mathscr{J} \otimes \mathscr{L}^{-1}$ with an ideal $\mathscr{J}'$ of $\mathscr{O}_Y$. But

$$\operatorname{Proj}\left(\bigoplus_{n\geqslant 0} \mathscr{J}^n\right) \quad \text{and} \quad \operatorname{Proj}\left(\bigoplus_{n\geqslant 0} (\mathscr{J} \otimes \mathscr{L}^{-1})^n\right)$$

are Y-isomorphic (**II**, 3.1.8); this proves the first assertion.

For the second, note that

$$\mathscr{F} = \mathscr{H}om_{\mathscr{O}_Y}(\mathscr{J}, \mathscr{O}_Y)$$

is coherent and nonzero, since there is an open subset U of Y on which $\mathscr{J}|U$ is invertible. If $\mathscr{L}$ is an ample $\mathscr{O}_Y$-module, there is an integer n such that

$$\mathscr{F}(n) = \mathscr{F} \otimes \mathscr{L}^{\otimes n}$$

is generated by its sections over Y (**II**, 4.5.5). *A fortiori*,

$$\Gamma(Y, \mathscr{F}(n)) \neq 0,$$

which proves the assertion. □

Corollary (2.3.7). — *Let X and Y be integral schemes projective over a field k, and let*

$$f : X \longrightarrow Y$$

be a birational k-morphism. Then X is k-isomorphic to a Y-scheme obtained by blowing up a closed subscheme Y' of Y, not necessarily reduced.

Proof. Indeed, f is projective (**II**, 5.5.5, v), and since Y is projective over k, the additional condition of **III** | 107
(2.3.6) is satisfied. It therefore suffices to take the closed subscheme Y' of Y defined by the coherent ideal $\mathscr{J}$ of (2.3.6). □

(2.3.8). *Remark.* In Chapter IV, in studying the notion of divisor, we shall see that if, in the statement of (2.3.5), the local rings $\mathscr{O}_y$, for $y \in Y$, are assumed factorial—as is the case, for example, when Y is nonsingular—then X may be obtained from Y by blowing up a closed subprescheme Y' of Y whose underlying space is contained in $Y - U$.

2.4. A generalisation of the fundamental theorem

Theorem (2.4.1). — *Let Y be a Noetherian prescheme, let $\mathscr{S}$ be a quasi-coherent $\mathscr{O}_Y$-algebra of finite type, let*

$$f : X \longrightarrow Y$$

be a projective morphism, set $\mathscr{S}' = f^*(\mathscr{S})$, and let $\mathscr{M}$ be a quasi-coherent $\mathscr{S}'$-module of finite type. Then:

(i) For every $p \in \mathbf{Z}$, $\mathrm{R}^p f_*(\mathscr{M})$ is an $\mathscr{S}$-module of finite type.

(ii) Suppose in addition that $\mathscr{L}$ is an invertible $\mathscr{O}_X$-module ample for f, and, for every $n \in \mathbf{Z}$, set

$$\mathscr{M}(n) = \mathscr{M} \otimes \mathscr{L}^{\otimes n}.$$

There is an integer N such that, for $n \geqslant N$,

$$\mathrm{R}^p f_*(\mathscr{M}(n)) = 0 \tag{2.4.1.1}$$

for every $p > 0$, and the canonical homomorphism

$$f^*\big(f_*(\mathscr{M}(n))\big) \longrightarrow \mathscr{M}(n)$$

(**0**, 4.4.3) *is surjective.*

Proof. Set

$$Y' = \operatorname{Spec}(\mathscr{S}), \qquad X' = \operatorname{Spec}(\mathscr{S}'),$$

so that $X' = X \times_Y Y'$ (**II**, 1.5.5). Let

$$g : Y' \longrightarrow Y, \qquad g' : X' \longrightarrow X$$

be the structural morphisms, which are affine by definition, and let

$$f' = f_{(Y')} : X' \longrightarrow Y'.$$

We thus have a commutative diagram

$$\begin{tikzcd} X \arrow[d, "f"'] & X' \arrow[l, "g'"'] \arrow[d, "f'"] \arrow[dl, "h"] \\ Y & Y', \arrow[l, "g"] \end{tikzcd}$$

where f' is projective (**II**, 5.5.5, iii) and

$$h = f \circ g' = g \circ f'.$$

(i) Let $\widetilde{\mathscr{M}}$ be the $\mathscr{O}_{X'}$-module associated with the quasi-coherent $\mathscr{S}'$-module $\mathscr{M}$ when X' is regarded as an affine X-scheme (**II**, 1.4.3); thus

$$\mathscr{M} = g'_*(\widetilde{\mathscr{M}}).$$

Since $\mathscr{M}$ is an $\mathscr{S}'$-module of finite type, $\widetilde{\mathscr{M}}$ is an $\mathscr{O}_{X'}$-module of finite type (**II**, 1.4.5). Moreover, h is of finite type, since g and f' are of finite type (**II**, 1.3.7) and (**I**, 6.3.4, ii). Hence X' is Noetherian (**I**, 6.3.7), and $\widetilde{\mathscr{M}}$ is therefore coherent.

Since g' is affine, the canonical homomorphism

$$\mathrm{R}^p f_*(\mathscr{M}) \longrightarrow \mathrm{R}^p h_*(\widetilde{\mathscr{M}})$$

is bijective (1.3.4). It is moreover a homomorphism of $\mathscr{S}$-modules. Indeed, the canonical homomorphism

(2.4.1.2) $$g'_*(\mathscr{O}_{X'}) \otimes_{\mathscr{O}_X} g'_*(\widetilde{\mathscr{M}}) \longrightarrow g'_*(\widetilde{\mathscr{M}})$$

defines the $\mathscr{S}'$-module structure on $\mathscr{M}$, since

$$\mathscr{S}' = g'_*(\mathscr{O}_{X'}).$$

It canonically gives rise to a homomorphism

$$f_*\big(g'_*(\mathscr{O}_{X'})\big) \otimes \mathrm{R}^p f_*\big(g'_*(\widetilde{\mathscr{M}})\big) \longrightarrow \mathrm{R}^p f_*\big(g'_*(\widetilde{\mathscr{M}})\big).$$

By (**0**, 12.2.2), and because (2.4.1.2) itself is obtained, by applying (**0**, 4.2.2.1), from the homomorphism **III** | 108

$$\mathscr{O}_{X'} \otimes \widetilde{\mathscr{M}} \longrightarrow \widetilde{\mathscr{M}}$$

that defines the $\mathscr{O}_{X'}$-module structure on $\widetilde{\mathscr{M}}$, the diagram

$$\begin{tikzcd}
f_*\big(g'_*(\mathscr{O}_{X'})\big) \otimes \mathrm{R}^p f_*\big(g'_*(\widetilde{\mathscr{M}})\big) \arrow[r] \arrow[d] & \mathrm{R}^p f_*\big(g'_*(\widetilde{\mathscr{M}})\big) \arrow[d] \\
h_*(\mathscr{O}_{X'}) \otimes \mathrm{R}^p h_*(\widetilde{\mathscr{M}}) \arrow[r] & \mathrm{R}^p h_*(\widetilde{\mathscr{M}})
\end{tikzcd}$$

is commutative (**0**, 12.2.6). Composing the horizontal arrows with the homomorphism induced by the canonical homomorphism

$$\mathscr{S} \longrightarrow f_*(f^*(\mathscr{S})) = f_*(\mathscr{S}') = f_*\big(g'_*(\mathscr{O}_{X'})\big) = h_*(\mathscr{O}_{X'}),$$

we obtain the assertion.

On the other hand, since g is affine and f' is separated and quasi-compact, the canonical homomorphism

$$\mathrm{R}^p h_*(\widetilde{\mathscr{M}}) \longrightarrow g_*\big(\mathrm{R}^p f'_*(\widetilde{\mathscr{M}})\big)$$

is bijective (1.4.14). As above, it is an isomorphism of $\mathscr{S}$-modules, this time by using the commutativity in (**0**, 12.6.2). Now f' is projective and $\widetilde{\mathscr{M}}$ is coherent, so $\mathrm{R}^p f'_*(\widetilde{\mathscr{M}})$ is a coherent $\mathscr{O}_{Y'}$-module by (2.2.1). It follows that

$$g_*\big(\mathrm{R}^p f'_*(\widetilde{\mathscr{M}})\big)$$

is an $\mathscr{S}$-module of finite type (**II**, 1.4.5).

(ii) Put

$$\mathscr{L}' = g'^*(\mathscr{L}),$$

an invertible $\mathscr{O}_{X'}$-module. For every $n \in \mathbf{Z}$, up to canonical isomorphism,

$$g'_*\big(\widetilde{\mathscr{M}} \otimes \mathscr{L}'^{\otimes n}\big) = g'_*(\widetilde{\mathscr{M}}) \otimes \mathscr{L}^{\otimes n} = \mathscr{M}(n)$$

(**0**, 5.4.10). Applying the argument of *(i)* to

$$\widetilde{\mathscr{M}} \otimes \mathscr{L}'^{\otimes n}$$

shows that

$$\mathrm{R}^p f_*\big(g'_*(\widetilde{\mathscr{M}} \otimes \mathscr{L}'^{\otimes n})\big) \simeq g_*\big(\mathrm{R}^p f'_*(\widetilde{\mathscr{M}} \otimes \mathscr{L}'^{\otimes n})\big).$$

Now $\mathscr{L}'$ is ample for f' (**II**, 4.6.13, iii). It follows from (2.2.1) that there is an integer N such that

$$\mathrm{R}^p f'_*\big(\widetilde{\mathscr{M}} \otimes \mathscr{L}'^{\otimes n}\big) = 0$$

for every $p > 0$ and every $n \geqslant N$. This proves (2.4.1.1).

It follows once more from (2.2.1) that N may be chosen so that, for $n \geqslant N$, the canonical homomorphism

$$f'^*\big(f'_*(\widetilde{\mathscr{M}} \otimes \mathscr{L}'^{\otimes n})\big) \longrightarrow \widetilde{\mathscr{M}} \otimes \mathscr{L}'^{\otimes n}$$

is surjective. Since g'_* is exact (**II**, 1.4.4), the corresponding homomorphism

$$g'_*\Big(f'^*\big(f'_*(\widetilde{\mathscr{M}}\otimes\mathscr{L}'^{\otimes n})\big)\Big)\longrightarrow g'_*(\widetilde{\mathscr{M}}\otimes\mathscr{L}'^{\otimes n})=\mathscr{M}(n)$$

is surjective. But (**II**, 1.5.2) gives

$$g'_*\Big(f'^*\big(f'_*(\widetilde{\mathscr{M}}\otimes\mathscr{L}'^{\otimes n})\big)\Big)=f^*\Big(g_*\big(f'_*(\widetilde{\mathscr{M}}\otimes\mathscr{L}'^{\otimes n})\big)\Big),$$

and $g_*\circ f'_*=f_*\circ g'_*$. Consequently,

$$g'_*\Big(f'^*\big(f'_*(\widetilde{\mathscr{M}}\otimes\mathscr{L}'^{\otimes n})\big)\Big)=f^*\big(f_*(\mathscr{M}(n))\big),$$

which completes the proof. □

(2.4.2). We shall apply (2.4.1), in particular, when $\mathscr{S}$ is an $\mathscr{O}_Y$-algebra graded in positive degrees and

$$\mathscr{M}=\bigoplus_{k\in\mathbf{Z}}\mathscr{M}_k$$

is a graded $\mathscr{S}'$-module. Under the same finiteness hypotheses on $\mathscr{S}$ and $\mathscr{M}$, it follows from (2.4.1) that **III** | 109

$$\bigoplus_{k\in\mathbf{Z}}\mathrm{R}^p f_*(\mathscr{M}_k)$$

is an $\mathscr{S}$-module of finite type for every p. Under the additional hypotheses of (2.4.1, ii), there is moreover an integer N such that, for $n\geqslant N$,

$$\mathrm{R}^p f_*(\mathscr{M}_k(n))=0$$

for every $p>0$ and every $k\in\mathbf{Z}$, and the canonical homomorphism

$$f^*\big(f_*(\mathscr{M}_k(n))\big)\longrightarrow\mathscr{M}_k(n)$$

is surjective for every $k\in\mathbf{Z}$.

2.5. Euler–Poincaré characteristic and the Hilbert polynomial

(2.5.1). Let A be an Artinian ring and let X be an A-scheme projective over

$$Y=\mathrm{Spec}(A).$$

For every coherent $\mathscr{O}_X$-module $\mathscr{F}$, the A-modules

$$\mathrm{H}^i(X,\mathscr{F})\qquad(i\geqslant 0)$$

are of finite type (2.2.1), and hence have finite length, since A is Artinian. Moreover, (2.2.1) shows that $\mathrm{H}^i(X,\mathscr{F})=0$ for all but finitely many $i\geqslant 0$. Thus the integer

$$\chi_A(\mathscr{F})=\sum_{i=0}^{\infty}(-1)^i\,\mathrm{length}_A\big(\mathrm{H}^i(X,\mathscr{F})\big)\tag{2.5.1.1}$$

is defined for every coherent $\mathscr{O}_X$-module $\mathscr{F}$. When A is an Artinian local ring, $\chi_A(\mathscr{F})$ is called the *Euler–Poincaré characteristic* of $\mathscr{F}$ with respect to A. For $\mathscr{F}=\mathscr{O}_X$, the integer $\chi_A(\mathscr{O}_X)$ is called the *arithmetic genus* of X with respect to A.

Proposition (2.5.2). — *Let*

$$0\longrightarrow\mathscr{F}'\longrightarrow\mathscr{F}\longrightarrow\mathscr{F}''\longrightarrow 0$$

be an exact sequence of coherent $\mathscr{O}_X$-modules. Then

$$\chi_A(\mathscr{F})=\chi_A(\mathscr{F}')+\chi_A(\mathscr{F}'').\tag{2.5.2.1}$$

Proof. The cohomology modules of $\mathscr{F}$, $\mathscr{F}'$, and $\mathscr{F}''$ vanish in all sufficiently large degrees. There is therefore an integer $r>0$ for which the long exact cohomology sequence has the form

$$0\longrightarrow\mathrm{H}^0(X,\mathscr{F}')\longrightarrow\mathrm{H}^0(X,\mathscr{F})\longrightarrow\mathrm{H}^0(X,\mathscr{F}'')\longrightarrow\mathrm{H}^1(X,\mathscr{F}')\longrightarrow\cdots$$

$$\cdots\longrightarrow\mathrm{H}^r(X,\mathscr{F}')\longrightarrow\mathrm{H}^r(X,\mathscr{F})\longrightarrow\mathrm{H}^r(X,\mathscr{F}'')\longrightarrow 0.$$

In an exact sequence of A-modules of finite length with zero terms at both ends, the alternating sum of the lengths is zero (**0**, 11.10.1). Applying this fact gives (2.5.2.1) immediately. □

The result of (2.5.2) applies whenever X is a quasi-compact A-scheme and the A-modules $\mathrm{H}^i(X, \mathscr{F})$ are of finite type for every coherent $\mathscr{O}_X$-module $\mathscr{F}$ (1.4.12).

Theorem (2.5.3). — *Let A be an Artinian local ring, let X be a scheme projective over*

$$Y = \mathrm{Spec}(A),$$

let $\mathscr{L}$ be an invertible $\mathscr{O}_X$-module very ample relative to Y, and let $\mathscr{F}$ be a coherent $\mathscr{O}_X$-module. For every $n \in \mathbf{Z}$, put

$$\mathscr{F}(n) = \mathscr{F} \otimes_{\mathscr{O}_X} \mathscr{L}^{\otimes n}.$$

(i) There is a unique polynomial $P \in \mathbf{Q}[T]$ such that

$$\chi_A(\mathscr{F}(n)) = P(n)$$

for every $n \in \mathbf{Z}$. The polynomial P is called the Hilbert polynomial *of $\mathscr{F}$ with respect to A.*

(ii) For all sufficiently large n,

$$\chi_A(\mathscr{F}(n)) = \mathrm{length}_A \Gamma(X, \mathscr{F}(n)).$$

(iii) The coefficient of the highest-degree term of $\chi_A(\mathscr{F}(n))$ is nonnegative.

We shall moreover prove in Chapter IV, in the section devoted to dimension, that *the degree of $\chi_A(\mathscr{F}(n))$ equals the dimension of the support of $\mathscr{F}$.*

Proof. For all sufficiently large n, we have

$$\mathrm{H}^i(X, \mathscr{F}(n)) = 0$$

for every $i > 0$ (2.2.1). Consequently, **III** | 110

$$\chi_A(\mathscr{F}(n)) = \mathrm{length}_A \mathrm{H}^0(X, \mathscr{F}(n)) = \mathrm{length}_A \Gamma(X, \mathscr{F}(n))$$

for all sufficiently large n, which proves *(ii)*. It follows that $\chi_A(\mathscr{F}(n)) \geqslant 0$ for all sufficiently large n, and *(iii)* therefore follows from *(i)*. The uniqueness assertion in *(i)* is immediate, so it remains to prove the existence of P.

We first show that we may assume

$$\mathfrak{m}\mathscr{F} = 0,$$

where $\mathfrak{m}$ is the maximal ideal of A. Indeed, there is an integer $s > 0$ such that $\mathfrak{m}^s = 0$, and $\mathscr{F}(n)$ therefore has the finite filtration

$$\mathscr{F}(n) \supset \mathfrak{m}\mathscr{F}(n) \supset \cdots \supset \mathfrak{m}^{s-1}\mathscr{F}(n) \supset 0.$$

Induction using (2.5.2.1) gives

$$\chi_A(\mathscr{F}(n)) = \sum_{k=1}^{s} \chi_A\left(\mathfrak{m}^{k-1}\mathscr{F}(n)/\mathfrak{m}^k\mathscr{F}(n)\right).$$

But

$$\mathfrak{m}^{k-1}\mathscr{F}(n)/\mathfrak{m}^k\mathscr{F}(n) = \mathscr{F}'_k(n), \qquad \mathscr{F}'_k = \mathfrak{m}^{k-1}\mathscr{F}/\mathfrak{m}^k\mathscr{F},$$

which proves the reduction.

Suppose, then, that $\mathfrak{m}\mathscr{F} = 0$. Let X' be the closed subscheme of X obtained as the inverse image, under the structural morphism

$$X \longrightarrow \mathrm{Spec}(A),$$

of the unique closed point of $\mathrm{Spec}(A)$, and let $j : X' \to X$ be the canonical injection. Then

$$\mathscr{F} = j_*(\mathscr{F}')$$

for a coherent $\mathscr{O}_{X'}$-module $\mathscr{F}'$, and X' is projective over $\mathrm{Spec}(K)$, where

$$K = A/\mathfrak{m}.$$

If $\mathscr{L}' = j^*(\mathscr{L})$, then $\mathscr{L}'$ is very ample relative to $\mathrm{Spec}(K)$ (**II**, 4.4.10), and

$$\mathscr{F}(n) = j_*(\mathscr{F}'(n)), \qquad \mathscr{F}'(n) = \mathscr{F}' \otimes_{\mathscr{O}_{X'}} \mathscr{L}'^{\otimes n}$$

(**0**, 5.4.10). It follows that

$$\chi_A(\mathscr{F}(n)) = \chi_K(\mathscr{F}'(n))$$

(G, II, 4.9.1), and we are reduced to the case in which A is a field.

Now X may be regarded as a closed subscheme of

$$P = \mathbf{P}^r_A$$

for a suitable r (**II**, 5.5.4, ii). If $i : X \to P$ is the canonical injection, the same argument gives

$$\chi_A(\mathscr{F}(n)) = \chi_A\big(i_*(\mathscr{F})(n)\big).$$

We may therefore restrict to the case

$$X = \mathbf{P}^r_A = \operatorname{Proj}(S), \qquad S = A[T_0, \ldots, T_r],$$

where A is a field.

In this case $\mathscr{F} = \widetilde{M}$ for a graded S-module M of finite type (**II**, 2.7.8). By the Hilbert syzygy theorem (M, VIII, 6.5), M has a finite resolution by graded free S-modules of finite type:

$$0 \longrightarrow L_q \longrightarrow L_{q-1} \longrightarrow \cdots \longrightarrow L_1 \longrightarrow M \longrightarrow 0.$$

Since the functor $M \mapsto \widetilde{M}$ is exact (**II**, 2.5.4), we obtain an exact sequence

$$0 \longrightarrow \widetilde{L}_q \longrightarrow \widetilde{L}_{q-1} \longrightarrow \cdots \longrightarrow \widetilde{L}_1 \longrightarrow \widetilde{M} \longrightarrow 0,$$

and hence, for every $n \in \mathbf{Z}$, an exact sequence

$$0 \longrightarrow \widetilde{L}_q(n) \longrightarrow \widetilde{L}_{q-1}(n) \longrightarrow \cdots \longrightarrow \widetilde{L}_1(n) \longrightarrow \widetilde{M}(n) \longrightarrow 0.$$

Induction on q, using (2.5.2), yields

$$\chi_A(\widetilde{M}(n)) = \sum_{j=1}^{q} (-1)^{j+1} \chi_A(\widetilde{L}_j(n)).$$

To prove *(i)*, we are therefore reduced to the case in which M is graded, free, and of finite type, and hence to the case

$$M = S(h)$$

for some $h \in \mathbf{Z}$. Then

$$\widetilde{M}(n) = \widetilde{M(n)} = \widetilde{S(n+h)}$$

(**II**, 2.5.15), so the theorem follows from the lemma below. □

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Lemma (2.5.3.1). — *Let A be a field, let $r > 0$ be an integer, and let*

$$X = \mathbf{P}^r_A.$$

Then

$$\chi_A(\mathscr{O}_X(n)) = \binom{n+r}{r}$$

for every $n \in \mathbf{Z}$.

Proof. For $n \geqslant 0$,

$$\chi_A(\mathscr{O}_X(n)) = \operatorname{length}_A \mathrm{H}^0(X, \mathscr{O}_X(n)),$$

which is the number of monomials in the T_i of total degree n, namely

$$\binom{n+r}{r}$$

(2.1.12).

For $n \leqslant -r-1$, we likewise have

$$\chi_A(\mathscr{O}_X(n)) = (-1)^r \operatorname{length}_A \mathrm{H}^r(X, \mathscr{O}_X(n)).$$

Writing $n = -r-h$, the dimension of $\mathrm{H}^r(X, \mathscr{O}_X(n))$ over A is the number of sequences

$$(p_i)_{0 \leqslant i \leqslant r}$$

of positive integers satisfying

$$\sum_{i=0}^{r} p_i = r+h$$

(2.1.12). Equivalently, it is the number of sequences of nonnegative integers

$$(q_i)_{0\leqslant i\leqslant r}$$

such that

$$\sum_{i=0}^{r} q_i = h-1.$$

This number is

$$\binom{h+r-1}{r} = (-1)^r\binom{n+r}{r}.$$

Finally, if $-r \leqslant n < 0$, then

$$\binom{n+r}{r} = 0,$$

and $\mathrm{H}^i(X, \mathscr{O}_X(n)) = 0$ for every $i \geqslant 0$ (2.1.12). This proves the lemma. □

Corollary (2.5.4). — *Let A be an Artinian local ring, let S be a graded A-algebra of finite type generated by S_1, let M be a graded S-module of finite type, and let*

$$X = \mathrm{Proj}(S).$$

Then, for all sufficiently large n,

$$\chi_A(\widetilde{M}(n)) = \mathrm{length}_A(M_n).$$

Proof. For all sufficiently large n, the modules

$$M_n \quad \text{and} \quad \Gamma(X, \widetilde{M}(n))$$

are isomorphic (2.3.1). □

2.6. Application: ampleness criteria

Proposition (2.6.1). — *Let Y be a Noetherian prescheme, let $f : X \to Y$ be a proper morphism, and let $\mathscr{L}$ be an invertible $\mathscr{O}_X$-module. The following conditions are equivalent:*

a) *$\mathscr{L}$ is ample for f.*
b) *For every coherent $\mathscr{O}_X$-module $\mathscr{F}$, there exists an integer N such that, for $n \geqslant N$,*

$$\mathrm{R}^q f_*(\mathscr{F} \otimes \mathscr{L}^{\otimes n}) = 0$$

for every $q > 0$.

c) *For every coherent ideal $\mathscr{J}$ of $\mathscr{O}_X$, there exists an integer N such that, for $n \geqslant N$,*

$$\mathrm{R}^1 f_*(\mathscr{J} \otimes \mathscr{L}^{\otimes n}) = 0.$$

Proof. We have seen that *a)* implies *b)* (2.2.1), (ii). It is immediate that *b)* implies *c)*, and it remains to prove that *c)* implies *a)*. We may restrict to the case where Y is affine (**II**, 4.6.4). It is enough to show that, as h ranges through the sections of $\mathscr{L}^{\otimes n}$ over X, with $n > 0$, the affine open subsets X_h form a covering of X (**II**, 4.5.2).

For this purpose, let x be a closed point of X and let U be an affine open neighbourhood of x. We shall find an integer n and a section

$$h \in \Gamma(X, \mathscr{L}^{\otimes n})$$

such that $x \in X_h \subset U$. The open subset X_h will then be affine (**I**, 1.3.6). The union of all such X_h is an open subset of X containing every closed point, and therefore is all of X, since X is Noetherian (**I**, 6.3.7) and (**0**, 2.1.3).

Let $\mathscr{J}$, respectively $\mathscr{J}'$, be the quasi-coherent ideal of $\mathscr{O}_X$ defining the reduced closed sub-prescheme whose underlying space is $X - U$, respectively $(X - U) \cup \{x\}$ (**I**, 5.2.1). The ideals $\mathscr{J}$ and $\mathscr{J}'$ are coherent (**I**, 6.1.1), $\mathscr{J}' \subset \mathscr{J}$, and

$$\mathscr{J}'' = \mathscr{J} / \mathscr{J}'$$

is a coherent $\mathscr{O}_X$-module (**0**, 5.3.3), supported by $\{x\}$, with $\mathscr{J}''_x = k(x)$. Since $\mathscr{L}$ is locally free, the sequence

$$0 \longrightarrow \mathscr{J}' \otimes \mathscr{L}^{\otimes n} \longrightarrow \mathscr{J} \otimes \mathscr{L}^{\otimes n} \longrightarrow \mathscr{J}'' \otimes \mathscr{L}^{\otimes n} \longrightarrow 0$$

is exact for every n. By hypothesis, for all sufficiently large n,

$$\mathrm{H}^1(X, \mathscr{J}' \otimes \mathscr{L}^{\otimes n}) = 0.$$

The exact cohomology sequence therefore shows that **III** | 112

$$\Gamma(X, \mathscr{J} \otimes \mathscr{L}^{\otimes n}) \longrightarrow \Gamma(X, \mathscr{J}'' \otimes \mathscr{L}^{\otimes n})$$

is surjective. A section g of $\mathscr{J}'' \otimes \mathscr{L}^{\otimes n}$ over X such that $g(x) \neq 0$ is consequently the image of a section

$$h \in \Gamma(X, \mathscr{J} \otimes \mathscr{L}^{\otimes n}) \subset \Gamma(X, \mathscr{L}^{\otimes n});$$

the inclusion follows from (**0**, 5.4.1). By construction, $h(x) \neq 0$, while $h(z) = 0$ for every $z \notin U$. Thus $x \in X_h \subset U$, which proves the assertion. □

Proposition (2.6.2). — *Let Y be a Noetherian prescheme, let $f : X \to Y$ be a morphism of finite type, let $g : X' \to X$ be a finite surjective morphism, let $\mathscr{L}$ be an invertible $\mathscr{O}_X$-module, and set*

$$\mathscr{L}' = g^*(\mathscr{L}).$$

Suppose that there exists a subset Z of X, proper over Y (**II**, 5.4.10)*, such that, for every $x \in X - Z$, either X is normal at x, or*

$$\big(g_*(\mathscr{O}_{X'})\big)_x$$

is a free $\mathscr{O}_{X,x}$-module. Then $\mathscr{L}$ is ample for f if and only if $\mathscr{L}'$ is ample for $f \circ g$.

Proof.

(2.6.2.1). Since g is affine, the condition is necessary (**II**, 5.1.12). To prove that it is sufficient, we may suppose that Y is affine (**II**, 4.6.4). We first show that we may also restrict to the case in which X is reduced.

Let $j : X_{\mathrm{red}} \to X$ be the canonical immersion, and put

$$X_1 = X_{\mathrm{red}}, \qquad X_1' = X' \times_X X_1.$$

We then have the commutative diagram

(2.6.2.2)
$$\begin{tikzcd} X' \arrow[d, "g"'] & X_1' \arrow[l, "j'"'] \arrow[d, "g_1"] \\ X & X_1 \arrow[l, "j"] \end{tikzcd}$$

The morphism $f \circ j$ is of finite type (**I**, 6.3.4), and g_1 is finite (**II**, 6.1.5), (iii). If $\mathscr{L}'$ is ample for $f \circ g$, then $j'^*(\mathscr{L}')$ is ample for $f \circ g \circ j'$, because j' is a closed immersion (**II**, 5.1.12) and (**I**, 4.3.2). If $Z_1 = j^{-1}(Z)$, then Z_1 is proper over Y (**II**, 5.4.10). Moreover, if X is normal at a point x, then so is X_{red} at that point; and if $\big(g_*(\mathscr{O}_{X'})\big)_x$ is a free $\mathscr{O}_{X,x}$-module, then (**II**, 1.5.2) shows that

$$\big((g_1)_*(\mathscr{O}_{X_1'})\big)_x$$

is a free $\mathscr{O}_{X_1,x}$-module. Finally, since X is Noetherian (**I**, 6.3.7), the ampleness of $j^*(\mathscr{L})$ implies that of $\mathscr{L}$ (**II**, 4.5.14). As

$$j'^*(\mathscr{L}') = g_1^*(j^*(\mathscr{L})),$$

this proves the announced reduction. Henceforth we suppose that Y is affine and that X is reduced.

The hypotheses of (**II**, 6.6.11) are now satisfied. There exist a reduced Y-prescheme X_2 and a finite birational Y-morphism

$$h : X_2 \longrightarrow X$$

whose restriction to $h^{-1}(X - Z)$ is an isomorphism onto $X - Z$, and such that $h^*(\mathscr{L})$ is ample. Replacing X' by X_2, we are reduced to proving the proposition under the additional assumption that g has the properties just listed for h. We continue to write Z for a closed subprescheme of X, proper over Y (**II**, 5.4.10), whose underlying space is the given subset Z.

(2.6.2.3). Let X_1 be a closed subprescheme of X, let $j : X_1 \to X$ be the canonical immersion, and let

$$X'_1 = g^{-1}(X_1) = X' \times_X X_1,$$

with canonical immersion $j' : X'_1 \to X'$. The corresponding square is again (2.6.2.2). Put

$$\mathscr{L}_1 = j^*(\mathscr{L}), \qquad \mathscr{L}'_1 = j'^*(\mathscr{L}') = g_1^*(\mathscr{L}_1).$$

Then $\mathscr{L}'_1$ is ample for $f \circ g \circ j'$ (**II**, 5.1.12). If $Z_1 = j^{-1}(Z)$, the closed subprescheme Z_1 of X_1 is proper over Y (**II**, 5.4.2), (ii). In other words, the hypotheses of (2.6.2) hold for X_1, $\mathscr{L}_1$, g_1, and Z_1.

This permits us to prove (2.6.2) by Noetherian induction (**0**, 2.2.2) when the restriction of g to $g^{-1}(X - Z)$ is an isomorphism onto $X - Z$; as seen in (2.6.2.1), that case is sufficient. It is enough to show that if the conclusion of (2.6.2) holds for $\mathscr{L}_1$ on every closed subprescheme X_1 of X whose underlying space is not X, then it also holds for $\mathscr{L}$. **III** | 113

(2.6.2.4). Set

$$\mathscr{A} = \mathscr{O}_X, \qquad \mathscr{B} = g_*(\mathscr{O}_{X'}).$$

Since g is birational and X is reduced, $\mathscr{B}$ is a sub-$\mathscr{A}$-algebra of the sheaf $\mathscr{R}(X)$ of total quotient rings, and it is a coherent $\mathscr{A}$-module. Moreover,

$$\mathscr{B}|_{X-Z} = \mathscr{A}|_{X-Z}.$$

Let $\mathscr{K}$ be the conductor of $\mathscr{B}$ over $\mathscr{A}$, that is, the largest quasi-coherent sub-$\mathscr{A}$-module of $\mathscr{A}$ such that

$$\mathscr{B}\mathscr{K} \subset \mathscr{A}.$$

Equivalently, $\mathscr{K}$ is the annihilator of the $\mathscr{A}$-module $\mathscr{B}/\mathscr{A}$ (**0**, 5.3.7). It follows that

$$\mathscr{B}\mathscr{K} = \mathscr{K}.$$

Clearly $\mathscr{K}_x = \mathscr{A}_x$ at every point admitting a neighbourhood W_x on which $g : g^{-1}(W_x) \to W_x$ is an isomorphism. This holds in particular at every point of $X - Z$, and in a neighbourhood of every generic point of an irreducible component of X.

Let

$$Z_1 = \operatorname{Spec}(\mathscr{A}/\mathscr{K})$$

be the closed subprescheme of X defined by $\mathscr{K}$. It is proper over Y, because its underlying space is closed in Z (**II**, 5.4.10). The definition of $\mathscr{K}$ also gives

$$\mathscr{B}|_{X-Z_1} = \mathscr{A}|_{X-Z_1}.$$

We may therefore reduce to the case

$$Z = \operatorname{Spec}(\mathscr{A}/\mathscr{K}).$$

Since $X - Z_1$ is a nonempty open subset of X, we may in addition suppose that the underlying space of Z is distinct from X.

(2.6.2.5). Regard X' as $\operatorname{Spec}(\mathscr{B})$, since g is affine, and let

$$\mathscr{K}' = \widetilde{\mathscr{K}}.$$

This is a coherent ideal of $\mathscr{O}_{X'}$ satisfying $g_*(\mathscr{K}') = \mathscr{K}$ (**II**, 1.4.1). The closed subprescheme

$$Z' = g^{-1}(Z) = Z \times_X X'$$

of X' is defined by $\mathscr{K}'$ and is equal to

$$\operatorname{Spec}(\mathscr{B}/\mathscr{K})$$

(**II**, 1.4.10). The morphism $h : Z' \to Z$ is finite (**II**, 6.1.5), (iii); consequently Z' is proper over Y (**II**, 6.1.11) and (**II**, 5.4.2), (ii).

We must prove that, for every $x \in X$ and every open neighbourhood U of x, there exist an integer $n > 0$ and a section s of $\mathscr{L}^{\otimes n}$ over X such that

$$x \in X_s \subset U$$

(**II**, 4.5.2). We distinguish two cases.

1° Suppose first that $x \in X - Z$. We may clearly suppose that $U \subset X - Z$, so that $U' = g^{-1}(U)$ does not meet Z'. Since $\mathscr{L}'$ is ample, there exist $n > 0$ and a section s' of $\mathscr{L}'^{\otimes n}$ over X' such that

$$x' = g^{-1}(x) \in X'_{s'} \subset g^{-1}(U)$$

(**II**, 4.5.2). We may also suppose that $\mathscr{K}' \otimes \mathscr{L}'^{\otimes n}$ is generated by its global sections (**II**, 4.5.5). As $\mathscr{K}'_{x'} = \mathscr{O}_{X',x'}$, one of these sections, say s'', is nonzero at x'. Multiplying it by s', and thus replacing n by $2n$, we may arrange that

$$x' \in X'_{s''} \subset g^{-1}(U).$$

By (**0**, 5.4.10), there is a canonical isomorphism

$$\Gamma(X, \mathscr{K} \otimes \mathscr{L}^{\otimes n}) \xrightarrow{\sim} \Gamma(X', \mathscr{K}' \otimes \mathscr{L}'^{\otimes n}).$$

The section s of $\mathscr{K} \otimes \mathscr{L}^{\otimes n}$ corresponding to s'' plainly has the required properties.

2° Suppose now that $x \in Z$. Let $\mathscr{J}$ be the coherent ideal of $\mathscr{O}_X$ defining the reduced closed subprescheme whose underlying space is $X - U$, and consider in $\mathscr{B}$ the coherent ideals **III** | 114

$$\mathscr{J}\mathscr{B} \qquad \text{and} \qquad \mathscr{J}\mathscr{K} = \mathscr{J}(\mathscr{K}\mathscr{B}) = \mathscr{K}(\mathscr{J}\mathscr{B}).$$

We have the diagram of inclusions

(2.6.2.6)
$$\begin{tikzcd} & \mathscr{J}\mathscr{B} \arrow[r, hook] & \mathscr{B} \\ & \mathscr{J} \arrow[u, hook] \arrow[r, hook] & \mathscr{A} \arrow[u, hook] \\ \mathscr{J}\mathscr{K}\mathscr{B} = & \mathscr{J}\mathscr{K} \arrow[u, hook] \arrow[r, hook] & \mathscr{K} \arrow[u, hook] \end{tikzcd}$$

Let $\mathscr{J}' = (\mathscr{J}\mathscr{B})^{\sim}$ be the coherent ideal of $\mathscr{O}_{X'}$ such that

$$g_*(\mathscr{J}') = \mathscr{J}\mathscr{B}.$$

Then

$$\mathscr{J}'\mathscr{K}' = (\mathscr{J}\mathscr{K}\mathscr{B})^{\sim}, \qquad \mathscr{J}'/\mathscr{J}'\mathscr{K}' = (\mathscr{J}\mathscr{B}/\mathscr{J}\mathscr{K}\mathscr{B})^{\sim}$$

(**II**, 1.4.4). Since $\mathscr{J}|_V = \mathscr{J}\mathscr{K}|_V$ on every open subset V disjoint from Z, the support of $\mathscr{J}'/\mathscr{J}'\mathscr{K}'$ is contained in Z'. Because Z' is proper over Y, (2.2.4) shows that, for all sufficiently large n, the canonical map

$$\Gamma(X', \mathscr{J}' \otimes \mathscr{L}'^{\otimes n}) \longrightarrow \Gamma\big(X', (\mathscr{J}'/\mathscr{J}'\mathscr{K}') \otimes \mathscr{L}'^{\otimes n}\big)$$

is surjective. By (**0**, 5.4.10), it follows that the canonical map

$$\Gamma(X, \mathscr{J}\mathscr{B} \otimes \mathscr{L}^{\otimes n}) \longrightarrow \Gamma\big(X, (\mathscr{J}\mathscr{B}/\mathscr{J}\mathscr{K}\mathscr{B}) \otimes \mathscr{L}^{\otimes n}\big)$$

is surjective.

Let $i : Z \to X$ and $i' : Z' \to X'$ be the canonical immersions. They fit into the commutative diagram

$$\begin{tikzcd} X' \arrow[d, "g"'] & Z' \arrow[l, "i'"'] \arrow[d, "h"] \\ X & Z \arrow[l, "i"] \end{tikzcd}$$

Put

$$\mathscr{M} = i^*(\mathscr{L}), \qquad \mathscr{M}' = i'^*(\mathscr{L}').$$

Since $\mathscr{L}'$ is ample, $\mathscr{M}'$ is ample (**II**, 5.1.12), and

$$\mathscr{M}' = h^*(\mathscr{M}).$$

The Noetherian induction hypothesis, applied because $Z \neq X$, now shows that $\mathscr{M}$ is ample. Consequently, for all sufficiently large n,

$$i^*(\mathscr{J}/\mathscr{J}\mathscr{K}) \otimes \mathscr{M}^{\otimes n}$$

is generated by its sections over Z (**II**, 4.5.5). Since

$$\mathscr{J}\,/\,\mathscr{J}\mathscr{K} = i_*\big(i^*(\mathscr{J}\,/\,\mathscr{J}\mathscr{K})\big),$$

(**0**, 5.4.10) gives a section s of

$$(\mathscr{J}\,/\,\mathscr{J}\mathscr{K})\otimes\mathscr{L}^{\otimes n}$$

over X, for some sufficiently large n, such that $s(x)\neq 0$. Indeed, $\mathscr{J}_x = \mathscr{A}_x$ by the definition of $\mathscr{J}$, whereas $\mathscr{K}_x \neq \mathscr{A}_x$ by hypothesis.

The diagram (2.6.2.6) shows that s is also a section of

$$(\mathscr{J}\mathscr{B}/\mathscr{J}\mathscr{K}\mathscr{B})\otimes\mathscr{L}^{\otimes n}$$

over X. It is therefore the canonical image of a section

$$t\in\Gamma(X,\mathscr{J}\mathscr{B}\otimes\mathscr{L}^{\otimes n}).$$

By definition, the image of t modulo $\mathscr{J}\mathscr{K}\mathscr{B}\otimes\mathscr{L}^{\otimes n}$ lies in

$$\frac{\mathscr{J}\otimes\mathscr{L}^{\otimes n}}{\mathscr{J}\mathscr{K}\mathscr{B}\otimes\mathscr{L}^{\otimes n}}.$$

The diagram (2.6.2.6) therefore implies that t is a section of $\mathscr{J}\otimes\mathscr{L}^{\otimes n}$, hence in particular a section of $\mathscr{L}^{\otimes n}$. We have $t(x)\neq 0$, so $x\in X_t$. By the definition of $\mathscr{J}$, the section t vanishes on $X-U$, the support of $\mathscr{O}_X/\mathscr{J}$. Thus $X_t\subset U$, and the proof is complete.

□

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(2.6.3). *Remark.* When X is proper over Y, Proposition (2.6.2) can be proved more simply by arguing as in Chevalley's theorem (**II**, 6.7.1), with the aid of Proposition (2.6.1) and the lemma (**II**, 6.7.1.1).

§3. Finiteness theorem for proper morphisms

3.1. The dévissage lemma

Definition (3.1.1). — Let $\mathtt{C}$ be an abelian category. We say that a subset $\mathtt{C}'$ of the set of objects of $\mathtt{C}$ is *exact* if $0\in\mathtt{C}'$ and if, for every exact sequence $0\to A'\to A\to A''\to 0$ in $\mathtt{C}$ such that two of the objects A, A', A'' are in $\mathtt{C}'$, then the third is also in $\mathtt{C}'$.

Theorem (3.1.2). — *Let X be a Noetherian prescheme; we denote by $\mathtt{C}$ the abelian category of coherent $\mathscr{O}_X$-modules. Let $\mathtt{C}'$ be an exact subset of $\mathtt{C}$, X' a closed subset of the underlying space of X. Suppose that for every closed irreducible subset Y of X', with generic point y, there exists an $\mathscr{O}_X$-module $\mathscr{G}\in\mathtt{C}'$ such that $\mathscr{G}_y$ is a $k(y)$-vector space of dimension 1. Then every coherent $\mathscr{O}_X$-module with support contained in X' is in $\mathtt{C}'$ (and in particular, if $X'=X$, then we have $\mathtt{C}'=\mathtt{C}$).*

Proof. Consider the following property $\mathbf{P}(Y)$ of a closed subset Y of X': every coherent $\mathscr{O}_X$-module with support contained in Y is in $\mathtt{C}'$. By virtue of the principle of Noetherian induction (**0**, 2.2.2), we see that we can reduce to showing that *if Y is a closed subset of X' such that the property $\mathbf{P}(Y')$ is true for every closed subset Y' of Y, distinct from Y, then $\mathbf{P}(Y)$ is true.*

Therefore, let $\mathscr{F}\in\mathtt{C}$ have support contained in Y, and we show that $\mathscr{F}\in\mathtt{C}'$. Denote also by Y the closed reduced subprescheme of X having Y for its underlying space (**I**, 5.2.1); it is defined by a coherent sheaf of ideals $\mathscr{J}$ of $\mathscr{O}_X$. We know (**I**, 9.3.4) that there exists an integer $n>0$ such that $\mathscr{J}^n\mathscr{F}=0$; for $1\leqslant k\leqslant n$, we thus have an exact sequence

$$0\longrightarrow\mathscr{J}^{k-1}\mathscr{F}/\mathscr{J}^k\mathscr{F}\longrightarrow\mathscr{F}/\mathscr{J}^k\mathscr{F}\longrightarrow\mathscr{F}/\mathscr{J}^{k-1}\mathscr{F}\longrightarrow 0$$

of coherent $\mathscr{O}_X$-modules ((**I**, 5.3.6) and (**I**, 5.3.3)); as $\mathtt{C}'$ is exact, we see, by induction on k, that it suffices to show that each of the $\mathscr{F}_k=\mathscr{J}^{k-1}\mathscr{F}/\mathscr{J}^k\mathscr{F}$ is in $\mathtt{C}'$. We thus reduce to proving that $\mathscr{F}\in\mathtt{C}'$ under the additional hypothesis that $\mathscr{J}\mathscr{F}=0$; it is equivalent to say that $\mathscr{F}=j_*(j^*(\mathscr{F}))$, where j is the canonical injection $Y\to X$. Let us now consider two cases:

(a) Y is *reducible*. Let $Y = Y' \cap Y''$, where Y' and Y'' are closed subsets of Y, distinct from Y; denote also by Y' and Y'' the closed reduced subpreschemes of X having Y and Y'' for their respective underlying spaces, which are defined respectively by sheaves of ideals $\mathscr{J}'$ and $\mathscr{J}''$ of $\mathscr{O}_X$. Set $\mathscr{F}' = \mathscr{F} \otimes_{\mathscr{O}_X} (\mathscr{O}_X/\mathscr{J}')$ and $\mathscr{F}'' = \mathscr{F} \otimes_{\mathscr{O}_X} (\mathscr{O}_X/\mathscr{J}'')$. The canonical homomorphisms $\mathscr{F} \to \mathscr{F}'$ and $\mathscr{F} \to \mathscr{F}''$ thus define a homomorphism $u : \mathscr{F} \to \mathscr{F}' \oplus \mathscr{F}''$. We show that for every $z \notin Y' \cap Y''$, the homomorphism $u_z : \mathscr{F}_z \to \mathscr{F}'_z \oplus \mathscr{F}''_z$ is *bijective*. Indeed, we have $\mathscr{J}' \cap \mathscr{J}'' = \mathscr{J}$, since the question is local and the above equality follows **III** | 116
from ((**I**, 5.2.1) and (**I**, 1.1.5)); if $z \notin Y''$, then we have $\mathscr{J}'_z = \mathscr{J}_z$, hence $\mathscr{F}'_z = \mathscr{F}_z$ and $\mathscr{F}''_z = 0$, which establishes our assertion in this case; we reason similarly for $z \notin Y'$. As a result, the kernel and cokernel of u, which are in $\mathtt{C}$ (**0**, 5.3.4), have their support in $Y' \cap Y''$, and thus is in $\mathtt{C}'$ by hypothesis; for the same reason, $\mathscr{F}'$ and $\mathscr{F}''$ are in $\mathtt{C}'$, hence also $\mathscr{F}' \oplus \mathscr{F}''$, as $\mathtt{C}'$ is exact. The conclusion then follows from the consideration of the two exact sequences

$$0 \longrightarrow \operatorname{Im} u \longrightarrow \mathscr{F}' \oplus \mathscr{F}'' \longrightarrow \operatorname{Coker} u \longrightarrow 0,$$

$$0 \longrightarrow \operatorname{Ker} u \longrightarrow \mathscr{F} \longrightarrow \operatorname{Im} u \longrightarrow 0,$$

and the hypothesis that $\mathtt{C}'$ is exact.

(b) Y is irreducible, and as a result, the subprescheme Y of X is *integral*. If y is its generic point, then we have $(\mathscr{O}_Y)_y = k(y)$, and as $j^*(\mathscr{F})$ is a coherent $\mathscr{O}_Y$-module, $\mathscr{F}_y = (j^*(\mathscr{F}))_y$ is a $k(y)$-vector space of finite dimension m. By hypothesis, there is a coherent $\mathscr{O}_X$-module $\mathscr{G} \in \mathtt{C}'$ (necessarily of support Y) such that $\mathscr{G}_y$ is a $k(y)$-vector space of dimension 1. As a result, there is a $k(y)$-isomorphism $(\mathscr{G}_y)^m \simeq \mathscr{F}_y$, which is also an $\mathscr{O}_Y$-isomorphism, and as $\mathscr{G}^m$ and $\mathscr{F}$ are coherent, there exists an open neighbourhood W of y in X and an isomorphism $\mathscr{G}^m|W \simeq \mathscr{F}|W$ (**0**, 5.2.7). Let $\mathscr{H}$ be the graph of this isomorphism, which is a coherent $(\mathscr{O}_X|W)$-submodule of $(\mathscr{G}^m \oplus \mathscr{F})|W$, canonically isomorphic to $\mathscr{G}^m|W$ and to $\mathscr{F}|W$; there thus exists a coherent $\mathscr{O}_X$-submodule $\mathscr{H}_0$ of $\mathscr{G}^m \oplus \mathscr{F}$, inducing $\mathscr{H}$ on W and 0 on $X - Y$, since $\mathscr{G}^m$ and $\mathscr{F}$ have Y for their support (**I**, 9.4.7). The restrictions $v : \mathscr{H}_0 \to \mathscr{G}^m$ and $w : \mathscr{H}_0 \to \mathscr{F}$ of the canonical projections of $\mathscr{G}^m \oplus \mathscr{F}$ are then homomorphisms of coherent $\mathscr{O}_X$-modules, which, on W and on $X - Y$, reduce to isomorphisms; in other words, the kernels and cokernels of v and w have their support in the closed set $Y - (Y \cap W)$, distinct from Y. They are in $\mathtt{C}'$; on the other hand, we have $\mathscr{G}^m \in \mathtt{C}'$ since $\mathscr{G} \in \mathtt{C}'$ and since $\mathtt{C}'$ is exact. We conclude successively, by the exactness of $\mathtt{C}'$, that $\mathscr{H}_0 \in \mathtt{C}'$, then $\mathscr{F} \in \mathtt{C}'$. Q.E.D.

□

Corollary (3.1.3). — *Suppose that the exact subset $\mathtt{C}'$ of $\mathtt{C}$ has in addition the property that any coherent direct factor of a coherent $\mathscr{O}_X$-module $\mathscr{M} \in \mathtt{C}'$ is also in $\mathtt{C}'$. In this case, the conclusion of Theorem (3.1.2) is still valid when the condition "$\mathscr{G}_y$ is a $k(y)$-vector space of dimension 1" is replaced by $\mathscr{G}_y \neq 0$ (this is equivalent to* $\operatorname{Supp}(\mathscr{G}) = Y$*).*

Proof. The reasoning of Theorem (3.1.2) must be modified only in the case (b); now $\mathscr{G}_y$ is a $k(y)$-vector space of dimension $q > 0$, and as a result, we have an $\mathscr{O}_Y$-isomorphism $(\mathscr{G}_y)^m \simeq (\mathscr{F}_y)^q$; the end of the reasoning in Theorem (3.1.2) then proves that $\mathscr{F}^q \in \mathtt{C}'$, and the additional hypothesis on $\mathtt{C}'$ implies that $\mathscr{F} \in \mathtt{C}'$. □

3.2. The finiteness theorem: the case of usual schemes

Theorem (3.2.1). — *Let Y be a locally Noetherian prescheme, $f : X \to Y$ a proper morphism. For every coherent $\mathscr{O}_X$-module $\mathscr{F}$, the $\mathscr{O}_Y$-modules $\mathrm{R}^q f_*(\mathscr{F})$ are coherent for $q \geqslant 0$.*

Proof. Since the questions is local on Y, we can suppose Y Noetherian, thus X Noetherian (**I**, 6.3.7). The coherent $\mathscr{O}_X$-modules $\mathscr{F}$ for which the conclusion of Theorem (3.2.1) is true forms an *exact* subset $\mathtt{C}'$ of the category $\mathtt{C}$ of coherent $\mathscr{O}_X$-modules. Indeed, let $0 \to \mathscr{F}' \to \mathscr{F} \to \mathscr{F}'' \to 0$ is an **III** | 117
exact sequence of coherent $\mathscr{O}_X$-modules; suppose for example that $\mathscr{F}'$ and $\mathscr{F}''$ belong to $\mathtt{C}'$; we have the long exact sequence in cohomology

$$\mathrm{R}^{q-1} f_*(\mathscr{F}'') \xrightarrow{\partial} \mathrm{R}^q f_*(\mathscr{F}') \longrightarrow \mathrm{R}^q f_*(\mathscr{F}) \longrightarrow \mathrm{R}^q f_*(\mathscr{F}'') \xrightarrow{\partial} \mathrm{R}^{q+1} f_*(\mathscr{F}'),$$

in which by hypothesis the outer four terms are coherent; it is the same for the middle term $\mathrm{R}^q f_*(\mathscr{F})$ by ((**0**, 5.3.4) and (**0**, 5.3.3)). We show in the same way that when $\mathscr{F}$ and $\mathscr{F}'$ (resp. $\mathscr{F}$ and $\mathscr{F}''$)

are in $\mathtt{C}'$, then so is $\mathscr{F}''$ (resp. $\mathscr{F}'$). In addition, every coherent *direct factor* $\mathscr{F}'$ of an $\mathscr{O}_X$-module $\mathscr{F} \in \mathtt{C}'$ belongs to $\mathtt{C}'$: indeed, $\mathrm{R}^q f_*(\mathscr{F}')$ is then a direct factor of $\mathrm{R}^q f_*(\mathscr{F})$ (G, II, 4.4.4), therefore it is of finite type, and as it is quasi-coherent (1.4.10), it is coherent, as Y is Noetherian. By virtue of Corollary (3.1.3), we reduce to proving that when X is *irreducible* with generic point x, there exists *one* coherent $\mathscr{O}_X$-module $\mathscr{F}$ belonging to $\mathtt{C}'$, such that $\mathscr{F}_x \neq 0$: indeed, if this point is established, then it can be applied to any irreducible closed subprescheme Y of X, since if $j : Y \to X$ is the canonical injection, then $f \circ j$ is proper (**II**, 5.4.2), and if $\mathscr{G}$ is a coherent $\mathscr{O}_Y$-module with support Y, then $j_*(\mathscr{G})$ is a coherent $\mathscr{O}_X$-module such that $\mathrm{R}^q(f \circ j)_*(\mathscr{G}) = \mathrm{R}^q f_*(j_*(\mathscr{G}))$ (G, II, 4.9.1), therefore we can apply Corollary (3.1.3).

By virtue of Chow's lemma (**II**, 5.6.2), there exists an irreducible prescheme X' an a *projective* and surjective morphism $g : X' \to X$ such that $f \circ g : X' \to Y$ is *projective*. There exists an ample $\mathscr{O}_X$-module $\mathscr{L}$ for g (**II**, 5.3.1); we apply the fundamental theorem of projective morphisms (2.2.1) to $g : X' \to X$ and with $\mathscr{L}$: there thus exists an integer n such that $\mathscr{F} = g_*(\mathscr{O}_{X'}(n))$ is a coherent $\mathscr{O}_X$-module and $\mathrm{R}^q g_*(\mathscr{O}_{X'}(n)) = 0$ for all $q > 0$; in addition, as $g^*(g_*(\mathscr{O}_{X'}(n))) \to \mathscr{O}_{X'}(n)$ is surjective for n large enough (2.2.1), we see that we can suppose, at the generic point x of X, that we have $\mathscr{F}_x \neq 0$ (**II**, 3.4.7). On the other hand, as $f \circ g$ is projective as Y is Noetherian, the $\mathrm{R}^q(f \circ g)_*(\mathscr{O}_{X'}(n))$ are *coherent* (2.2.1). This being so, $\mathrm{R}^\bullet(f \circ g)_*(\mathscr{O}_{X'}(n))$ is the abutment of a Leray spectral sequence, whose E_2-term is given by $E_2^{pq} = \mathrm{R}^p f_*(\mathrm{R}^q g_*(\mathscr{O}_{X'}(n)))$; the above shows that this spectral sequence degenerates, and we then know (**0**, 11.1.6) that $E_2^{p0} = \mathrm{R}^p f_*(\mathscr{F})$ is isomorphic to $\mathrm{R}^p(f \circ g)_*(\mathscr{O}_{X'}(n))$, which finishes the proof. □

Corollary (3.2.2). — *Let Y be a locally Noetherian prescheme. For every proper morphism $f : X \to Y$, the direct image under f of any coherent $\mathscr{O}_X$-module is a coherent $\mathscr{O}_Y$-module.*

Corollary (3.2.3). — *Let A be a Noetherian ring, X a proper scheme over A; for every coherent $\mathscr{O}_X$-module $\mathscr{F}$, the $\mathrm{H}^p(X, \mathscr{F})$ are A-modules of finite type, and there exists an integer $r > 0$ such that for every coherent $\mathscr{O}_X$-module $\mathscr{F}$ and all $p > r$, $\mathrm{H}^p(X, \mathscr{F}) = 0$.*

Proof. The second assertion has already been proved (1.4.12); the first follows from the finiteness theorem (3.2.1), taking into account Corollary (1.4.11). □

In particular, if X is a *proper algebraic scheme* over a field k, then, for every coherent $\mathscr{O}_X$-module $\mathscr{F}$, the $\mathrm{H}^p(X, \mathscr{F})$ are *finite-dimensional* k-vector spaces.

Corollary (3.2.4). — *Let Y be a locally Noetherian prescheme, $f : X \to Y$ a morphism of finite type. For every coherent $\mathscr{O}_X$-module $\mathscr{F}$ whose support in proper over Y* (**II**, 5.4.10), *the $\mathscr{O}_Y$-modules $\mathrm{R}^q f_*(\mathscr{F})$ are coherent.*

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Proof. Since the questions is local on Y, we can suppose Y Noetherian, and it is the same for X (**I**, 6.3.7). By hypothesis, every closed subprescheme Z of X whose underlying space is $\mathrm{Supp}(\mathscr{F})$ is proper over Y, in other words, if $j : Z \to X$ is the canonical injection, then $f \circ j : Z \to Y$ is proper. We can suppose that Z is such that $\mathscr{F} = j_*(\mathscr{G})$, where $\mathscr{G} = j^*(\mathscr{F})$ is a coherent $\mathscr{O}_Z$-module (**I**, 9.3.5); as we have $\mathrm{R}^q f_*(\mathscr{F}) = \mathrm{R}^q(f \circ j)_*(\mathscr{G})$ by Corollary (1.3.4), the conclusion follows immediately from Theorem (3.2.1). □

3.3. Generalization of the finiteness theorem (usual schemes)

Proposition (3.3.1). — *Let Y be a Noetherian prescheme, $\mathscr{S}$ a quasi-coherent $\mathscr{O}_Y$-algebra of finite type, graded in positive degrees, $Y' = \mathrm{Proj}(\mathscr{S})$, and $g : Y' \to Y$ the structure morphism. Let $f : X \to Y$ be a proper morphism, $\mathscr{S}' = f^*(\mathscr{S})$, $\mathscr{M} = \bigoplus_{k\in\mathbf{Z}} \mathscr{M}_k$ a quasi-coherent graded $\mathscr{S}'$-module of finite type. Then the $\mathrm{R}^p f_*(\mathscr{M}) = \bigoplus_{k\in\mathbf{Z}} \mathrm{R}^p f_*(\mathscr{M}_k)$ are graded $\mathscr{S}$-modules of finite type for all p. Suppose in addition that the $\mathscr{S}$ are generated by $\mathscr{S}_1$; then, for every $p \in \mathbf{Z}$, there exists an integer k_p such that for all $k \geqslant k_p$ and all $r > 0$, we have*

$$\mathrm{R}^p f_*(\mathscr{M}_{k+r}) = \mathscr{S}_r \mathrm{R}^p f_*(\mathscr{M}_k). \tag{3.3.1.1}$$

Proof. The first assertion is identical to the statement of Theorem (2.4.1, i), where we have simply replaced "projective morphism" by "proper morphism". In the proof of Theorem (2.4.1, i), the hypothesis on f was only used to show (with the notation of this proof) that $\mathrm{R}^p f'_*(\widetilde{\mathscr{M}})$ is a coherent $\mathscr{O}_{Y'}$-module. With the hypothesis of Proposition (3.3.1), f' is proper (**II**, 5.4.2, iii), so we can resume without change in the proof of Theorem (2.4.1, i), thanks to the finiteness theorem (3.2.1).

As for the second assertion, it suffices to remark that there is a finite affine open cover (U_i) of Y such that the restrictions to the U_i of the two sides of (3.3.1.1) are equal for all $k \geqslant k_{p,i}$ (**II**, 2.1.6, ii); it suffices to take for k_p the largest of the $k_{p,i}$. □

Corollary (3.3.2). — *Let A be a Noetherian ring, $\mathfrak{m}$ an ideal of A, X a proper A-scheme, and $\mathscr{F}$ a coherent $\mathscr{O}_X$-module. Then, for all $p \geqslant 0$, the direct sum $\bigoplus_{k\geqslant 0} \mathrm{H}^p(X, \mathfrak{m}^k\mathscr{F})$ is a module of finite type over the ring $S = \bigoplus_{k\geqslant 0} \mathfrak{m}^k$; in particular, there exists an integer $k_p \geqslant 0$ such that for all $k \geqslant k_p$ and all $r > 0$, we have*

$$\mathrm{H}^p(X, \mathfrak{m}^{k+r}\mathscr{F}) = \mathfrak{m}^r \mathrm{H}^p(X, \mathfrak{m}^k\mathscr{F}). \tag{3.3.2.1}$$

Proof. It suffices to apply Proposition (3.3.1) with $Y = \mathrm{Spec}(A)$, $\mathscr{S} = \widetilde{S}$, $\mathscr{M}_k = \mathfrak{m}^k\mathscr{F}$, taking into account Corollary (1.4.11). □

It should be remembered that the S-module structure on $\bigoplus_{k\geqslant 0} \mathrm{H}^p(X, \mathfrak{m}^k\mathscr{F})$ is obtained by considering, for every $a \in \mathfrak{m}^r$, the map $\mathrm{H}^p(X, \mathfrak{m}^k\mathscr{F}) \to \mathrm{H}^p(X, \mathfrak{m}^{k+r}\mathscr{F})$, which comes from the passage to cohomology of the multiplication map $\mathfrak{m}^r\mathscr{F} \to \mathfrak{m}^{k+r}\mathscr{F}$ defined by a (2.4.1).

3.4. Finiteness theorem: the case of formal schemes The results of this section (except the defini- **III** | 119
tion (3.4.1)) will not be used in the rest of this chapter.

(3.4.1). Let $\mathfrak{X}$ and $\mathfrak{S}$ be two locally Noetherian formal preschemes (**I**, 10.4.2), $f : \mathfrak{X} \to \mathfrak{S}$ a morphism of formal preschemes. We say that f is a *proper* morphism if it satisfies the following conditions:

1st. *f is a morphism of finite type* (**I**, 10.13.3).

2nd. *If $\mathscr{K}$ is a sheaf of ideals of definition for $\mathfrak{S}$ and if we set $\mathscr{J} = f^*(\mathscr{K})\mathscr{O}_{\mathfrak{X}}$, $X_0 = (\mathfrak{X}, \mathscr{O}_{\mathfrak{X}}/\mathscr{J})$, $S_0 = (\mathfrak{S}, \mathscr{O}_{\mathfrak{S}}/\mathscr{K})$, then the morphism $f_0 : X_0 \to S_0$ induced by f* (**I**, 10.5.6) *is proper*.

It is immediate that this definition does not depend on the sheaf of ideals of definition $\mathscr{K}$ for $\mathscr{S}$ considered; indeed, if $\mathscr{K}'$ is a second sheaf of ideals of definition such that $\mathscr{K}' \subset \mathscr{K}$, and if we set $\mathscr{J}' = f^*(\mathscr{K}')\mathscr{O}_{\mathfrak{X}}$, $X'_0 = (\mathfrak{X}, \mathscr{O}_{\mathfrak{X}}/\mathscr{J}')$, $S'_0 = (\mathscr{S}, \mathscr{O}_{\mathfrak{S}}/\mathscr{K}')$, then the morphism $f'_0 : X'_0 \to S'_0$ induced by f is such that the diagram

$$\begin{tikzcd} X_0 \arrow[r, "f_0"] \arrow[d, "i"'] & S_0 \arrow[d, "j"] \\ X'_0 \arrow[r, "f'_0"] & S'_0 \end{tikzcd}$$

is commutative, i and j being surjective immersions; it is equivalent to say that f_0 or f'_0 is proper, by virtue of (**II**, 5.4.5).

We note that, for all $n \geqslant 0$, if we set $X_n = (\mathfrak{X}, \mathscr{O}_{\mathfrak{X}}/\mathscr{J}^{n+1})$, $S_n = (\mathfrak{S}, \mathscr{O}_{\mathfrak{S}}/\mathscr{K}^{n+1})$, then the morphism $f_n : X_n \to S_n$ induced by f (**I**, 10.5.6) is proper for all n whenever it is for $n = 0$ (**II**, 5.4.6).

If $g : Y \to Z$ is a proper morphism of locally Noetherian usual preschemes, Z' a closed subset of Z, Y' a closed subset of Y such that $g(Y') \subset Z'$, then the extension $\widehat{g} : Y_{/Y'} \to Z_{/Z'}$ of g to the completions (**I**, 10.9.1) is a proper morphism of formal preschemes, as it follows from the definition and from (**II**, 5.4.5).

Let $\mathfrak{X}$ and $\mathfrak{S}$ be two locally Noetherian formal preschemes, $f : \mathfrak{X} \to \mathfrak{S}$ a morphism *of finite type* (**I**, 10.13.3); the notation being the same as above, we say that a subset Z of the underlying space of $\mathfrak{X}$ is *proper* over $\mathfrak{S}$ (or proper for f) if, considered as a subset of X_0, Z is *proper over* S_0 (**II**, 5.4.10). All the properties of proper subsets of usual preschemes stated in (**II**, 5.4.10) are still true for the proper subsets of formal preschemes, as it follows immediately from the definitions.

Theorem (3.4.2). — *Let $\mathfrak{X}$ and $\mathfrak{Y}$ be locally Noetherian formal preschemes, $f : \mathfrak{X} \to \mathfrak{Y}$ a proper morphism. For every coherent $\mathscr{O}_{\mathfrak{X}}$-module $\mathscr{F}$, the $\mathscr{O}_{\mathfrak{Y}}$-modules $\mathrm{R}^q f_*(\mathscr{F})$ are coherent for all $q \geqslant 0$.*

Let $\mathscr{J}$ be a sheaf of ideals of definition for $\mathfrak{Y}$, $\mathscr{K} = f^*(\mathscr{J})\mathscr{O}_{\mathfrak{X}}$, and consider the $\mathscr{O}_{\mathfrak{X}}$-modules

$$\mathscr{F}_k = \mathscr{F} \otimes_{\mathscr{O}_{\mathfrak{Y}}} (\mathscr{O}_{\mathfrak{Y}} / \mathscr{J}^{k+1}) = \mathscr{F} / \mathscr{K}^{k+1}\mathscr{F} \quad (k \geqslant 0) \tag{3.4.2.1}$$

which evidently form a *projective system* of topological $\mathscr{O}_{\mathfrak{X}}$-modules, such that $\mathscr{F} = \varprojlim_k \mathscr{F}_k$ (**I**, 10.11.3). **III** | 120
On the other hand, it follows from Theorem (3.4.2) that each of the $\mathrm{R}^q f_*(\mathscr{F})$, being coherent, is naturally equipped with a topological $\mathscr{O}_{\mathfrak{Y}}$-module structure (**I**, 10.11.6), and so are the $\mathrm{R}^q f_*(\mathscr{F}_k)$. The canonical homomorphisms $\mathscr{F} \to \mathscr{F}_k = \mathscr{F} / \mathscr{K}^{k+1}\mathscr{F}$ canonically correspond to homomorphisms

$$\mathrm{R}^q f_*(\mathscr{F}) \longrightarrow \mathrm{R}^q f_*(\mathscr{F}_k),$$

which are necessarily continuous for the topological $\mathscr{O}_{\mathfrak{Y}}$-module structures above (**I**, 10.11.6), and form a projective system, giving the limit a canonical functorial homomorphism

$$\mathrm{R}^q f_*(\mathscr{F}) \longrightarrow \varprojlim_k \mathrm{R}^q f_*(\mathscr{F}_k), \tag{3.4.2.2}$$

which will be a continuous homomorphism of topological $\mathscr{O}_{\mathfrak{Y}}$-modules. We will prove along with Theorem (3.4.2) the

Corollary (3.4.3). — *Each of the homomorphisms (3.4.2.2) is a topological isomorphism. In addition, if $\mathfrak{Y}$ is Noetherian, then the projective system $(\mathrm{R}^q f_*(\mathscr{F} / \mathscr{K}^{k+1}\mathscr{F}))_{k \geqslant 0}$ satisfies the* (ML)*-condition* (**0**, 13.1.1).

We will begin by establishing Theorem (3.4.2) and Corollary (3.4.3) when Y is a Noetherian formal affine scheme (**I**, 10.4.1):

Corollary (3.4.4). — *Under the hypotheses of Theorem (3.4.2), suppose in addition that $\mathfrak{Y} = \mathrm{Spf}(A)$, where A is an adic Noetherian ring. Let $\mathfrak{J}$ be an ideal of definition for A, and set $\mathscr{F}_k = \mathscr{F} / \mathfrak{J}^{k+1}\mathscr{F}$ for $k \geqslant 0$. Then the $\mathrm{H}^n(\mathfrak{X}, \mathscr{F})$ are A-modules of finite type; the projective system $(\mathrm{H}^n(\mathfrak{X}, \mathscr{F}_k))_{k \geqslant 0}$ satisfies the* (ML)*-condition for all n; if we set*

$$N_{n,k} = \mathrm{Ker}\left(\mathrm{H}^n(\mathfrak{X}, \mathscr{F}) \longrightarrow \mathrm{H}^n(\mathfrak{X}, \mathscr{F}_k)\right) \tag{3.4.4.1}$$

(also equal to $\mathrm{Im}(\mathrm{H}^n(\mathfrak{X}, \mathfrak{J}^{k+1}\mathscr{F}) \to \mathrm{H}^n(\mathfrak{X}, \mathscr{F}))$ by the exact sequence in cohomology), then the $N_{n,k}$ define on $\mathrm{H}^n(\mathfrak{X}, \mathscr{F})$ a $\mathfrak{J}$-good filtration (**0**, 13.7.7)*; finally, the canonical homomorphism*

$$\mathrm{H}^n(\mathfrak{X}, \mathscr{F}) \longrightarrow \varprojlim_k \mathrm{H}^n(\mathfrak{X}, \mathscr{F}_k) \tag{3.4.4.2}$$

is a topological isomorphism for all n (the left hand side being equipped with the $\mathfrak{J}$-adic topology, the $\mathrm{H}^n(\mathfrak{X}, \mathscr{F}_k)$ with the discrete topology).

Set

$$S = \mathrm{gr}(A) = \bigoplus_{k \geqslant 0} \mathfrak{J}^k / \mathfrak{J}^{k+1}, \ \mathscr{M} = \mathrm{gr}(\mathscr{F}) = \bigoplus_{k \geqslant 0} \mathfrak{J}^k \mathscr{F} / \mathfrak{J}^{k+1}\mathscr{F}.$$

We know that $\mathfrak{J}^\Delta$ is a sheaf of ideals of definition for $\mathfrak{Y}$ (**I**, 10.3.1); let $\mathscr{K} = f^*(\mathfrak{J}^\Delta)\mathscr{O}_{\mathfrak{X}}$, $X_0 = (\mathfrak{X}, \mathscr{O}_{\mathfrak{X}} / \mathscr{K})$, $Y_0 = (\mathfrak{Y}, \mathscr{O}_{\mathfrak{Y}} / \mathfrak{J}^\Delta) = \mathrm{Spec}(A_0)$, with $A_0 = A/\mathfrak{J}$. It is clear that the $\mathscr{M}_k = \mathfrak{J}^k \mathscr{F} / \mathfrak{J}^{k+1}\mathscr{F}$ are coherent $\mathscr{O}_{X_0}$-modules (**I**, 10.11.3). Consider on the other hand the quasi-coherent graded $\mathscr{O}_{X_0}$-algebra

$$\mathscr{S} = \mathscr{O}_{X_0} \otimes_{A_0} S = \mathrm{gr}(\mathscr{O}_{\mathfrak{X}}) = \bigoplus_{k \geqslant 0} \mathscr{K}^k / \mathscr{K}^{k+1}. \tag{3.4.4.4}$$

The hypothesis that $\mathscr{F}$ is an $\mathscr{O}_{\mathfrak{X}}$-module of finite type implies first that $\mathscr{M}$ is a graded $\mathscr{S}$-module *of finite type*. Indeed, the question is local on $\mathfrak{X}$, and we can thus suppose that $\mathfrak{X} = \mathrm{Spf}(B)$, where B is an adic Noetherian ring, and $\mathscr{F} = N^\Delta$, where N is a B-module of finite type (**I**, 10.10.5); **III** | 121 we have in addition $X_0 = \mathrm{Spec}(B_0)$, where $B_0 = B/\mathfrak{J}B$, and the quasi-coherent $\mathcal{O}_{X_0}$-modules $\mathscr{S}$ and $\mathscr{M}$ are respectively equal to $\widetilde{S'}$ and $\widetilde{M'}$, where $S' = \bigoplus_{k\geqslant 0}((\mathfrak{J}^k/\mathfrak{J}^{k+1})\otimes_{A_0} B_0)$ and $M' = \bigoplus_{k\geqslant 0}((\mathfrak{J}^k/\mathfrak{J}^{k+1})\otimes_{A_0} N_0)$, with $N_0 = N/\mathfrak{J}N$; we then evidently have $M' = S'\otimes_{B_0} N_0$, and as N_0 is a B_0-module of finite type, M' is a S'-module of finite type, hence our assertion (**I**, 1.3.13).

As the morphism $f_0 : X_0 \to Y_0$ is *proper* by hypothesis, we can apply Corollary (3.3.2) to $\mathscr{S}$, $\mathscr{M}$, and the morphism f_0: taking into account Corollary (1.4.11), we conclude that for *all* $n \geqslant 0$, $\bigoplus_{k\geqslant 0}\mathrm{H}^n(X_0,\mathscr{M}_k)$ is a graded S-module *of finite type*. This proves that the condition (F_n) of (**0**, 13.7.7) is satisfied for *all* $n\geqslant 0$, when we consider the strictly projective system $(\mathscr{F}/\mathfrak{J}^k\mathscr{F})_{k\geqslant 0}$ of sheaves of abelian groups on X_0, each equipped with its natural "filtered A-module" structure. We can thus apply (**0**, 13.7.7), which proves that:

1st. The projective system $(\mathrm{H}^n(\mathfrak{X},\mathscr{F}_k))_{k\geqslant 0}$ satisfies the (ML)-condition.
2nd. If $\mathrm{H}'^n = \varprojlim_k \mathrm{H}^n(\mathfrak{X},\mathscr{F}_k)$, then H'^n is an A-module of finite type.
3rd. The filtration defined on H'^n by the kernels of the canonical homomorphisms $\mathrm{H}'^n \to \mathrm{H}^n(\mathfrak{X},\mathscr{F}_k)$ is $\mathfrak{J}$-good.

Note that on the other hand, if we set $X_k = (\mathfrak{X},\mathcal{O}_{\mathfrak{X}}/\mathscr{K}^{k+1})$, then $\mathscr{F}_k$ is a coherent $\mathcal{O}_{X_k}$-module (**I**, 10.11.3), and if U is an affine open set in X_0, then U is also an affine open set in each of the X_k (**I**, 5.1.9), so $\mathrm{H}^n(U,\mathscr{F}_k) = 0$ for all $n > 0$ and all k (1.3.1) and $\mathrm{H}^0(U,\mathscr{F}_k)\to\mathrm{H}^0(U,\mathscr{F}_h)$ is surjective for $h\leqslant k$ (**I**, 1.3.9). We are thus in the conditions of (**0**, 13.3.2) and applying (**0**, 13.3.1) proves that H'^n canonically identifies with $\mathrm{H}^n(\mathfrak{X},\varprojlim_k\mathscr{F}_k) = \mathrm{H}^n(\mathfrak{X},\mathscr{F})$; this finishes the proof of Corollary (3.4.4).

(3.4.5). We return to the proof of (3.4.2) and (3.4.3). We first prove the propositions for the case $\mathfrak{Y} = \mathrm{Spf}(A)$ envisaged in (3.4.4); for this, for all $g\in A$, apply (3.4.4) to the Noetherian affine formal scheme induced on the open set $\mathfrak{Y}_g = \mathfrak{D}(g)$ of $\mathfrak{Y}$, which is equal to $\mathrm{Spf}(A_{\{g\}})$, and to the formal prescheme induced by $\mathfrak{X}$ on $f^{-1}(\mathfrak{Y}_g)$; note that $\mathfrak{Y}_g$ is also an affine open set in the prescheme $Y_k = (\mathfrak{Y},\mathcal{O}_{\mathfrak{Y}}/(\mathfrak{J}^\Delta)^{k+1})$, and as $\mathscr{F}_k$ is a coherent $\mathcal{O}_{X_k}$-module, we have

$$\mathrm{H}^n(f^{-1}(\mathfrak{Y}_g),\mathscr{F}_k) = \Gamma(\mathfrak{Y}_g,\mathrm{R}^n f_*(\mathscr{F}_k))$$

for all $k\geqslant 0$ by virtue of Corollary (1.4.11). The canonical homomorphism

$$\mathrm{H}^n(f^{-1}(\mathfrak{Y}_g),\mathscr{F}) \longrightarrow \varprojlim_k \Gamma(\mathfrak{Y}_g,\mathrm{R}^n f_*(\mathscr{F}_k))$$

is an isomorphism; but we have (**0**, 3.2.6)

$$\varprojlim_k \Gamma(\mathfrak{Y}_g,\mathrm{R}^n f_*(\mathscr{F}_k)) = \Gamma(\mathfrak{Y}_g,\varprojlim_k \mathrm{R}^n f_*(\mathscr{F}_k)),$$

and as the sheaf $\mathrm{R}^n f_*(\mathscr{F})$ is the sheaf associated to the presheaf $\mathfrak{Y}_g\mapsto\mathrm{H}^n(f^{-1}(\mathfrak{Y}_g),\mathscr{F})$ on the **III** | 122
$\mathfrak{Y}_g$ (**0**, 3.2.1), we have shown that the homomorphism (3.4.2.2) is *bijective*. Let us now prove that $\mathrm{R}^n f_*(\mathscr{F})$ is a coherent $\mathcal{O}_{\mathfrak{Y}}$-module, and more precisely that we have

$$\mathrm{R}^n f_*(\mathscr{F}) = \big(\mathrm{H}^n(\mathfrak{X},\mathscr{F})\big)^\Delta. \tag{3.4.5.1}$$

With the above notation, we have, since $\mathscr{F}_k$ is a coherent $\mathcal{O}_{X_k}$-module (1.4.13),

$$\Gamma(\mathfrak{Y}_g,\mathrm{R}^n f_*(\mathscr{F}_k)) = (\Gamma(\mathfrak{Y},\mathrm{R}^n f_*(\mathscr{F}_k)))_g = (\mathrm{H}^n(\mathfrak{X},\mathscr{F}_k))_g.$$

Now the $\mathrm{H}^n(\mathfrak{X},\mathscr{F}_k)$ form a projective system satisfying (ML), and their projective limit $\mathrm{H}^n(\mathfrak{X},\mathscr{F})$ is an A-module of finite type. We conclude (**0**, 13.7.8) that we have

$$\varprojlim_k \big((\mathrm{H}^n(\mathfrak{X},\mathscr{F}_k))_g\big) = \mathrm{H}^n(\mathfrak{X},\mathscr{F})\otimes_A A_{\{g\}} = \Gamma(\mathfrak{Y}_g,(\mathrm{H}^n(\mathfrak{X},\mathscr{F}))^\Delta),$$

taking into account (**I**, 10.10.8) applied to A and $A_{\{g\}}$; this proves (3.4.5.1) since $\Gamma(\mathfrak{Y}_g,\mathrm{R}^n f_*(\mathscr{F})) = \varprojlim_k \Gamma(\mathfrak{Y}_g,\mathrm{R}^n f_*(\mathscr{F}_k))$.

As (3.4.2.2) is then an isomorphism of coherent $\mathcal{O}_{\mathfrak{Y}}$-modules, it is necessarily a *topological* isomorphism (**I**, 10.11.6). Finally, it follows from the relations $\mathrm{R}^n f_*(\mathscr{F}_k) = (\mathrm{H}^n(\mathfrak{X},\mathscr{F}_k))^\Delta$ that the projective system $(\mathrm{R}^n f_*(\mathscr{F}_k))_{k\geqslant 0}$ satisfies (ML) (**I**, 10.10.2).

Once (3.4.2) and (3.4.3) are proved in the case where the formal prescheme $\mathfrak{Y}$ is affine Noetherian, it is immediate to pass to the general case for (3.4.2) and the first assertion of (3.4.3), which are local on $\mathfrak{Y}$. As for the second assertion of (3.4.3), it suffices, $\mathfrak{Y}$ being Noetherian, to cover it by a finite number of Noetherian affine open sets U_i and to note that the restrictions of the projective system $(\mathrm{R}^q f_*(\mathscr{F}_k))$ to each of the U_i satisfies (ML).

Along the way, we have in addition proved:

Corollary (3.4.6). — *Under the hypotheses of Corollary (3.4.4), the canonical homomorphism*

(3.4.6.1) $$\mathrm{H}^q(\mathfrak{X}, \mathscr{F}) \longrightarrow \Gamma(\mathfrak{Y}, \mathrm{R}^q f_*(\mathscr{F}))$$

is bijective.

§4. The fundamental theorem of proper morphisms. Applications

4.1. The fundamental theorem

(4.1.1). Let X and Y be Noetherian usual preschemes, let $f : X \to Y$ be a proper morphism, let Y' be a closed subset of Y, and let

$$X' = f^{-1}(Y').$$

We denote by $\widehat{X}$ and $\widehat{Y}$ the formal preschemes $X_{/X'}$ and $Y_{/Y'}$, obtained by completing X and Y along X' and Y', respectively (**I**, 10.8.5). We denote by $\widehat{f} : \widehat{X} \to \widehat{Y}$ the extension of f to these completions (**I**, 10.9.1). For every coherent $\mathscr{O}_X$-module $\mathscr{F}$, we denote by

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$$\widehat{\mathscr{F}} = \mathscr{F}_{/X'}$$

its completion along X' (**I**, 10.8.4); this is a coherent $\mathscr{O}_{\widehat{X}}$-module (**I**, 10.8.8).

(4.1.2). Let $\mathscr{J}$ be a coherent ideal of $\mathscr{O}_Y$ such that

$$\mathrm{Supp}(\mathscr{O}_Y/\mathscr{J}) = Y'$$

(**I**, 5.2.1). We know (**I**, 4.4.5) that

$$\mathscr{K} = f^*(\mathscr{J})\mathscr{O}_X$$

is a coherent ideal of $\mathscr{O}_X$ satisfying

$$\mathrm{Supp}(\mathscr{O}_X/\mathscr{K}) = X'.$$

For every $k \geqslant 0$, consider the coherent $\mathscr{O}_X$-module

$$\mathscr{F}_k = \mathscr{F} \otimes_{\mathscr{O}_Y} (\mathscr{O}_Y/\mathscr{J}^{k+1}) = \mathscr{F}/\mathscr{K}^{k+1}\mathscr{F}.$$

The $\mathscr{O}_Y$-modules

$$\mathrm{R}^n f_*(\mathscr{F}) \quad \text{and} \quad \mathrm{R}^n f_*(\mathscr{F}_k)$$

are coherent for every n (3.2.1). For every $k \geqslant 0$ and every n, the canonical homomorphism $\mathscr{F} \to \mathscr{F}_k$ induces, by functoriality, a homomorphism

(4.1.2.1) $$\mathrm{R}^n f_*(\mathscr{F}) \longrightarrow \mathrm{R}^n f_*(\mathscr{F}_k).$$

Since $\mathscr{F}_k$ is an $\mathscr{O}_X/\mathscr{K}^{k+1}$-module, $\mathrm{R}^n f_*(\mathscr{F}_k)$ is an $\mathscr{O}_Y/\mathscr{J}^{k+1}$-module (**0**, 12.2.1). We therefore deduce from (4.1.2.1) a homomorphism

(4.1.2.2) $$\mathrm{R}^n f_*(\mathscr{F}) \otimes_{\mathscr{O}_Y} (\mathscr{O}_Y/\mathscr{J}^{k+1}) \longrightarrow \mathrm{R}^n f_*(\mathscr{F}_k).$$

The two sides of (4.1.2.2) form projective systems. The projective limit of the left-hand side is the completion

$$\big(\mathrm{R}^n f_*(\mathscr{F})\big)_{/Y'},$$

which we shall denote by

$$\widehat{\mathrm{R}^n f_*(\mathscr{F})}.$$

The homomorphisms (4.1.2.2) are compatible with the projective systems and hence define, on passing to the limit, a canonical homomorphism

(4.1.2.3) $$\phi_n : \widehat{\mathrm{R}^n f_*(\mathscr{F})} \longrightarrow \varprojlim_k \mathrm{R}^n f_*(\mathscr{F}_k).$$

Moreover, (4.1.2.2) is a homomorphism of $\mathscr{O}_Y/\mathscr{J}^{k+1}$-modules. By (**I**, 10.8.3), it may therefore be regarded as a continuous homomorphism of topological $\mathscr{O}_{\widehat{Y}}$-modules endowed with their pseudo-discrete topologies (**0**, 3.8.1). Consequently, ϕ_n is a continuous homomorphism of topological $\mathscr{O}_{\widehat{Y}}$-modules.

(4.1.3). Let $i : \widehat{X} \to X$ be the canonical morphism of ringed spaces defined in (**I**, 10.8.7). We have the commutative diagram

(4.1.3.1)
$$\begin{tikzcd} X_k \arrow[r, "h_k"] \arrow[dr, "i_k"'] & \widehat{X} \arrow[d, "i"] \\ & X \end{tikzcd}$$

where X_k is the closed subprescheme of X defined by $\mathscr{K}^{k+1}$, i_k is the canonical immersion, and h_k is the morphism of ringed spaces corresponding to the identity on the underlying spaces and to the canonical homomorphism

$$\mathscr{O}_{\widehat{X}} \longrightarrow \mathscr{O}_X/\mathscr{K}^{k+1}$$

(**I**, 10.5.2). Moreover,

$$\widehat{\mathscr{F}} = i^*(\mathscr{F})$$

up to canonical isomorphism (**I**, 10.8.8). We have

(4.1.3.2)
$$\mathrm{H}^n(X_k, i_k^*(\mathscr{F}_k)) = \mathrm{H}^n(X, \mathscr{F}_k)$$

up to canonical isomorphism.

Indeed, **III** | 124

$$\mathscr{F}_k = (i_k)_*(i_k^*(\mathscr{F}_k))$$

(G, II, 4.9.1). The canonical homomorphism

$$\mathrm{H}^n(\widehat{X}, \widehat{\mathscr{F}}) \longrightarrow \mathrm{H}^n(X_k, h_k^*(\widehat{\mathscr{F}}))$$

(**0**, 12.1.3.5) may thus also be written

(4.1.3.3)
$$\mathrm{H}^n(\widehat{X}, \widehat{\mathscr{F}}) \longrightarrow \mathrm{H}^n(X, \mathscr{F}_k).$$

These homomorphisms form a projective system and therefore define a canonical homomorphism

(4.1.3.4)
$$\psi_{n,X} : \mathrm{H}^n(\widehat{X}, \widehat{\mathscr{F}}) \longrightarrow \varprojlim_k \mathrm{H}^n(X, \mathscr{F}_k).$$

Replacing X by an open subset of the form $f^{-1}(V)$, where V is an affine open subset of Y, and taking (1.4.11) into account, we obtain homomorphisms

(4.1.3.5)
$$\psi_{n,V} : \mathrm{H}^n(\widehat{X} \cap f^{-1}(V), \widehat{\mathscr{F}}) \longrightarrow \varprojlim_k \Gamma(V, \mathrm{R}^n f_*(\mathscr{F}_k)).$$

These homomorphisms plainly commute with restriction from V to a smaller affine open subset. They therefore define a canonical homomorphism of sheaves

(4.1.3.6)
$$\psi_n : \mathrm{R}^n \widehat{f}_*(\widehat{\mathscr{F}}) \longrightarrow \varprojlim_k \mathrm{R}^n f_*(\mathscr{F}_k).$$

(4.1.4). Finally, let $j : \widehat{Y} \to Y$ be the canonical morphism of ringed spaces (**I**, 10.8.7). Since $\mathrm{R}^n f_*(\mathscr{F})$ is a coherent $\mathscr{O}_Y$-module (3.2.1), we have, up to canonical isomorphism,

$$j^*(\mathrm{R}^n f_*(\mathscr{F})) = \mathrm{R}^n \widehat{f_*(\mathscr{F})}$$

(**I**, 10.8.8). There is consequently a canonical homomorphism

(4.1.4.1)
$$\rho_n : \mathrm{R}^n \widehat{f_*(\mathscr{F})} = j^*(\mathrm{R}^n f_*(\mathscr{F})) \longrightarrow \mathrm{R}^n \widehat{f}_*(i^*(\mathscr{F})) = \mathrm{R}^n \widehat{f}_*(\widehat{\mathscr{F}})$$

defined for ringed spaces in general; see the proof of (1.4.15). We shall show that the diagram

(4.1.4.2)
$$\begin{tikzcd} \widehat{\mathrm{R}^n f_*(\mathscr{F})} \arrow[rr, "\rho_n"] \arrow[dr, "\phi_n"'] & & \mathrm{R}^n \widehat{f}_*(\widehat{\mathscr{F}}) \arrow[dl, "\psi_n"] \\ & \varprojlim_k \mathrm{R}^n f_*(\mathscr{F}_k) & \end{tikzcd}$$

is commutative. It is enough to prove the commutativity of the diagram formed by the corresponding homomorphisms of presheaves. We may therefore restrict to the case where Y is affine, and it is enough to prove that

(4.1.4.3)
$$\begin{tikzcd} \widehat{\mathrm{H}^n(X,\mathscr{F})} \arrow[rr, "\rho_n"] \arrow[dr, "\phi_n"'] & & \mathrm{H}^n(\widehat{X}, \widehat{\mathscr{F}}) \arrow[dl, "\psi_{n,X}"] \\ & \varprojlim_k \mathrm{H}^n(X, \mathscr{F}_k) & \end{tikzcd}$$

is commutative. The commutativity of (4.1.3.1), together with the relations between cohomology groups established in (4.1.3), gives the commutative diagram III | 125

$$\begin{tikzcd} \mathrm{H}^n(X,\mathscr{F}) \arrow[r] \arrow[dr] & \mathrm{H}^n(\widehat{X},\widehat{\mathscr{F}}) = \mathrm{H}^n(\widehat{X}, i^*(\mathscr{F})) \arrow[d] \\ & \mathrm{H}^n(X,\mathscr{F}_k) = \mathrm{H}^n(X_k, i_k^*(\mathscr{F})). \end{tikzcd}$$

The commutativity of (4.1.4.3) follows immediately.

Theorem (4.1.5). — *Let $f : X \to Y$ be a proper morphism of Noetherian preschemes, let Y' be a closed subset of Y, and put $X' = f^{-1}(Y')$. Then, for every coherent $\mathscr{O}_X$-module $\mathscr{F}$,*

$$\mathrm{R}^n \widehat{f}_*(\widehat{\mathscr{F}})$$

is a coherent $\mathscr{O}_{\widehat{Y}}$-module, and the homomorphisms ϕ_n, ψ_n, and ρ_n in diagram (4.1.4.2) *are topological isomorphisms.*

Proof. It is enough to prove that ϕ_n and ψ_n are isomorphisms. Indeed, $\mathrm{R}^n f_*(\mathscr{F})$ is coherent (3.2.1), so its completion $\widehat{\mathrm{R}^n f_*(\mathscr{F})}$ is coherent (**I**, 10.8.8); the bicontinuity of ϕ_n, ψ_n, and ρ_n is then automatic (**I**, 10.11.6). □

(4.1.6). *Remarks.*

(i) Put

$$\widehat{\mathscr{F}_k} = \widehat{\mathscr{F}} / \widehat{\mathscr{K}}^{k+1} \widehat{\mathscr{F}}.$$

Then, immediately,

$$\mathscr{F}_k = i_*(\widehat{\mathscr{F}_k}),$$

and the canonical homomorphism (4.1.3.6) is precisely the homomorphism already defined in (3.4.2.2):

(4.1.6.1)
$$\mathrm{R}^n \widehat{f}_*(\widehat{\mathscr{F}}) \longrightarrow \varprojlim_k \mathrm{R}^n \widehat{f}_*(\widehat{\mathscr{F}_k}).$$

Thus the fact that ψ_n is an isomorphism is a special case of (3.4.3). We shall nevertheless give a direct proof below, avoiding the delicate considerations concerning projective limits of spectral sequences (**0**, 13.7) on which the general theorem (3.4.3) rests.

(ii) Since the ψ_n are isomorphisms, saying that the ϕ_n are isomorphisms is equivalent to saying that
$$\rho_n = \psi_n^{-1} \circ \phi_n$$
are isomorphisms. Theorem (4.1.5) states, in particular, that the formation of $\mathrm{R}^n f_*$ commutes with completion; it may therefore be called the first comparison theorem between the "algebraic" and the "formal" theories.

We shall begin by proving the affine form of (4.1.5).

Corollary (4.1.7). — *Under the hypotheses of (4.1.5), suppose in addition that $Y = \mathrm{Spec}(A)$, where A is Noetherian, and that $\mathscr{J} = \widetilde{\mathfrak{J}}$, where $\mathfrak{J}$ is an ideal of A, so that*
$$\mathscr{F}_k = \mathscr{F}/\mathfrak{J}^{k+1}\mathscr{F}.$$
The canonical homomorphism
$$\phi_n : \widehat{\mathrm{H}^n(X,\mathscr{F})} \longrightarrow \varprojlim_k \mathrm{H}^n(X,\mathscr{F}_k) \tag{4.1.7.1}$$
where the first term is the separated completion of $\mathrm{H}^n(X,\mathscr{F})$ for the $\mathfrak{J}$-preadic topology, is an isomorphism. For every n, the projective system
$$\big(\mathrm{H}^n(X,\mathscr{F}_k)\big)_{k\geqslant 0}$$
satisfies condition (ML), *and the canonical homomorphism*
$$\psi_n : \mathrm{H}^n(\widehat{X},\widehat{\mathscr{F}}) \longrightarrow \varprojlim_k \mathrm{H}^n(X,\mathscr{F}_k) \tag{4.1.7.2}$$
is an isomorphism. Finally, the filtration on $\mathrm{H}^n(X,\mathscr{F})$ defined by the kernels of the canonical homomorphisms **III | 126**
$$\mathrm{H}^n(X,\mathscr{F}) \longrightarrow \mathrm{H}^n(X,\mathscr{F}_k)$$
is $\mathfrak{J}$-good (**0**, 13.7.7), *and ϕ_n is a topological isomorphism.*[2]

Proof. Fix $n \geqslant 0$ throughout this proof and, for simplicity, put
$$\mathrm{H} = \mathrm{H}^n(X,\mathscr{F}), \qquad \mathrm{H}_k = \mathrm{H}^n(X,\mathscr{F}_k). \tag{4.1.7.3}$$
Put also
$$\mathrm{R}_k = \mathrm{Ker}(\mathrm{H} \longrightarrow \mathrm{H}_k), \qquad \text{a sub-}A\text{-module of H}. \tag{4.1.7.4}$$
The cohomology exact sequence
$$\mathrm{H}^n(X,\mathfrak{J}^{k+1}\mathscr{F}) \longrightarrow \mathrm{H}^n(X,\mathscr{F}) \longrightarrow \mathrm{H}^n(X,\mathscr{F}_k) \xrightarrow{\partial} \mathrm{H}^{n+1}(X,\mathfrak{J}^{k+1}\mathscr{F}) \longrightarrow \mathrm{H}^{n+1}(X,\mathscr{F})$$
shows that
$$\mathrm{R}_k = \mathrm{Im}\big(\mathrm{H}^n(X,\mathfrak{J}^{k+1}\mathscr{F}) \longrightarrow \mathrm{H}^n(X,\mathscr{F})\big).$$
We put
$$\begin{aligned} \mathrm{Q}_k &= \mathrm{Ker}\big(\mathrm{H}^{n+1}(X,\mathfrak{J}^{k+1}\mathscr{F}) \longrightarrow \mathrm{H}^{n+1}(X,\mathscr{F})\big) \\ &= \mathrm{Im}\big(\mathrm{H}^n(X,\mathscr{F}_k) \longrightarrow \mathrm{H}^{n+1}(X,\mathfrak{J}^{k+1}\mathscr{F})\big). \end{aligned} \tag{4.1.7.5}$$
We therefore have the exact sequence
$$0 \longrightarrow \mathrm{R}_k \longrightarrow \mathrm{H} \longrightarrow \mathrm{H}_k \longrightarrow \mathrm{Q}_k \longrightarrow 0. \tag{4.1.7.6}$$
(4.1.7.7). Let $x \in \mathfrak{J}^m$, where $m \geqslant 0$. Multiplication by x on $\mathfrak{J}^k\mathscr{F}$ is a homomorphism
$$\mathfrak{J}^k\mathscr{F} \longrightarrow \mathfrak{J}^{k+m}\mathscr{F},$$
and hence gives a homomorphism
$$\mu_{x,m} : \mathrm{H}^n(X,\mathfrak{J}^k\mathscr{F}) \longrightarrow \mathrm{H}^n(X,\mathfrak{J}^{k+m}\mathscr{F}). \tag{4.1.7.8}$$

[2] The following proof, which is simpler than the original proof, and the supplement concerning the filtration of $\mathrm{H}^n(X,\mathscr{F})$, were communicated to us by J.-P. Serre.

Let

$$S = \bigoplus_{k\geqslant 0} \mathfrak{J}^k$$

be the graded A-algebra. We know that the multiplications $\mu_{x,m}$ endow

$$\mathrm{E} = \bigoplus_{k\geqslant 0} \mathrm{H}^n(X, \mathfrak{J}^k \mathscr{F})$$

with the structure of a graded S-module of finite type (3.3.2); the graded ring S is Noetherian (**II**, 2.1.5).

Lemma (4.1.7.9). — *The submodules* R_k *of* H *define a* $\mathfrak{J}$*-good filtration on* H*.*

Proof. First let us show that

(4.1.7.10) $$\mathfrak{J}^m \mathrm{R}_k \subset \mathrm{R}_{k+m},$$

where multiplication on $\mathrm{H} = \mathrm{H}^n(X, \mathscr{F})$ by an element $x \in \mathfrak{J}^m$ is the map $\mu_{x,0}$. For every $x \in \mathfrak{J}^m$, the diagram

(4.1.7.11)
$$\begin{tikzcd} \mathfrak{J}^{k+1}\mathscr{F} \arrow[r, "x"] \arrow[d] & \mathfrak{J}^{k+m+1}\mathscr{F} \arrow[d] \\ \mathscr{F} \arrow[r, "x"'] & \mathscr{F} \end{tikzcd}$$

is commutative; its horizontal arrows are multiplication by x, and its vertical arrows are the canonical injections. Consequently, the corresponding diagram **III** | 127

$$\begin{tikzcd} \mathrm{H}^n(X, \mathfrak{J}^{k+1}\mathscr{F}) \arrow[r, "\mu_{x,m}"] \arrow[d] & \mathrm{H}^n(X, \mathfrak{J}^{k+m+1}\mathscr{F}) \arrow[d] \\ \mathrm{H}^n(X, \mathscr{F}) \arrow[r, "\mu_{x,0}"'] & \mathrm{H}^n(X, \mathscr{F}) \end{tikzcd}$$

is commutative. The interpretation of R_k as the image of

$$\mathrm{H}^n(X, \mathfrak{J}^{k+1}\mathscr{F}) \longrightarrow \mathrm{H}^n(X, \mathscr{F})$$

therefore proves (4.1.7.10). It also shows that the graded S-module

$$\mathrm{R} = \bigoplus_{k\geqslant 0} \mathrm{R}_k$$

is a quotient of the sub-S-module

$$\mathrm{M} = \bigoplus_{k\geqslant 0} \mathrm{H}^n(X, \mathfrak{J}^{k+1}\mathscr{F})$$

of E. The preceding observation thus shows that R is an S-module of finite type. This is equivalent to the assertion of the lemma (Bourbaki, *Alg. comm.*, Chap. III, §3, no. 1, Th. 1). ☐

(4.1.7.12). Consider now the graded S-module

$$\mathrm{N} = \bigoplus_{k\geqslant 0} \mathrm{H}^{n+1}(X, \mathfrak{J}^{k+1}\mathscr{F}),$$

defined as in (4.1.7.8). It too is an S-module of finite type by (3.3.2). By (4.1.7.5), we have $\mathrm{Q}_k \subset \mathrm{N}_k$ for every k, and diagram (4.1.7.11), with n replaced by $n+1$, shows that

$$S_m \mathrm{Q}_k = \mathfrak{J}^m \mathrm{Q}_k \subset \mathrm{Q}_{k+m}.$$

In other words, Q is a graded sub-S-module of N and is therefore of finite type.

(4.1.7.13). Let α_m denote the canonical injection

$$\mathfrak{J}^m \longrightarrow A,$$

which may also be written $S_m \to S_0$. Since $\mathfrak{J}^{k+1}\mathscr{F}_k = 0$, the A-module $\mathrm{H}^n(X,\mathscr{F}_k)$ is annihilated by $\mathfrak{J}^{k+1}$. As Q_k is the image of the A-homomorphism

$$\mathrm{H}^n(X,\mathscr{F}_k) \longrightarrow \mathrm{H}^{n+1}(X,\mathfrak{J}^{k+1}\mathscr{F}),$$

Q_k is likewise annihilated by $\mathfrak{J}^{k+1}$. Equivalently, in the S-module Q,

$$\alpha_{k+1}(S_{k+1})\mathrm{Q}_k = 0. \tag{4.1.7.14}$$

Since Q is an S-module of finite type, there exist integers k_0 and h such that

$$\mathrm{Q}_{k+h} = S_h\mathrm{Q}_k \qquad (k \geqslant k_0)$$

(**II**, 2.1.6, ii). This relation and (4.1.7.14) imply that there exists an integer $r > 0$ such that

$$\alpha_r(S_r)\mathrm{Q} = 0. \tag{4.1.7.15}$$

(4.1.7.16). The canonical injection

$$\mathfrak{J}^{k+m}\mathscr{F} \longrightarrow \mathfrak{J}^k\mathscr{F}$$

induces in cohomology an A-homomorphism

$$\nu_m : \mathrm{H}^{n+1}(X,\mathfrak{J}^{k+m}\mathscr{F}) \longrightarrow \mathrm{H}^{n+1}(X,\mathfrak{J}^k\mathscr{F}). \tag{4.1.7.17}$$

For every $x \in \mathfrak{J}^m$, we plainly have the factorization

$$\mathrm{H}^{n+1}(X,\mathfrak{J}^k\mathscr{F}) \xrightarrow{\mu_{x,m}} \mathrm{H}^{n+1}(X,\mathfrak{J}^{k+m}\mathscr{F}) \xrightarrow{\nu_m} \mathrm{H}^{n+1}(X,\mathfrak{J}^k\mathscr{F}), \tag{4.1.7.18}$$

whose composite is $\mu_{x,0}$.

It follows that, for every sub-A-module P of $\mathrm{H}^{n+1}(X,\mathfrak{J}^k\mathscr{F})$, we have in the S-module N **III** | 128

$$\nu_m(S_m\mathrm{P}) = \alpha_m(S_m)\mathrm{P}. \tag{4.1.7.19}$$

Lemma (4.1.7.20). — *There exists an integer $m > 0$ such that*

$$\nu_m(\mathrm{Q}_{k+m}) = 0$$

for every $k \geqslant k_0$.

Proof. Take m to be a multiple of h with $m \geqslant r$. Since

$$\mathrm{Q}_{k+m} = S_m\mathrm{Q}_k \qquad (k \geqslant k_0),$$

relations (4.1.7.19) and (4.1.7.15) give

$$\nu_m(\mathrm{Q}_{k+m}) = \alpha_m(S_m)\mathrm{Q}_k \subset \alpha_r(S_r)\mathrm{Q}_k = 0.$$

□

(4.1.7.21). The commutative diagram

$$\begin{tikzcd}
\mathrm{H}^n(X,\mathscr{F}) \arrow[r] & \mathrm{H}^n(X,\mathscr{F}_k) \arrow[r, "\partial"] & \mathrm{H}^{n+1}(X,\mathfrak{J}^{k+1}\mathscr{F}) \arrow[r] & \mathrm{H}^{n+1}(X,\mathscr{F}) \\
\mathrm{H}^n(X,\mathscr{F}) \arrow[r] \arrow[u] & \mathrm{H}^n(X,\mathscr{F}_{k+m}) \arrow[r, "\partial"] \arrow[u] & \mathrm{H}^{n+1}(X,\mathfrak{J}^{k+m+1}\mathscr{F}) \arrow[r] \arrow[u] & \mathrm{H}^{n+1}(X,\mathscr{F}) \arrow[u]
\end{tikzcd}$$

itself comes from the commutative diagram

$$\begin{tikzcd}
0 \arrow[r] & \mathfrak{J}^{k+1}\mathscr{F} \arrow[r] & \mathscr{F} \arrow[r] & \mathscr{F}_k \arrow[r] & 0 \\
0 \arrow[r] & \mathfrak{J}^{k+m+1}\mathscr{F} \arrow[r] \arrow[u] & \mathscr{F} \arrow[r] \arrow[u] & \mathscr{F}_{k+m} \arrow[r] \arrow[u] & 0.
\end{tikzcd}$$

Here the vertical arrows are the canonical maps. We therefore obtain the commutative diagram

$$\begin{tikzcd} 0 \arrow[r] & R_k \arrow[r] & H \arrow[r] & H_k \arrow[r] & Q_k \arrow[r] & 0 \\ 0 \arrow[r] & R_{k+m} \arrow[r] \arrow[u] & H \arrow[r] \arrow[u, equal] & H_{k+m} \arrow[r] \arrow[u] & Q_{k+m} \arrow[r] \arrow[u, "v_m"] & 0, \end{tikzcd}$$

whose rows are exact. For $k \geqslant k_0$, the last vertical arrow is zero by (4.1.7.20). Thus the image of H_{k+m} in H_k is contained in

$$\mathrm{Ker}(H_k \longrightarrow Q_k) = \mathrm{Im}(H \longrightarrow H_k).$$

Conversely, commutativity shows that it contains $\mathrm{Im}(H \to H_k)$, and hence it is equal to that image. The same is therefore true of the images in H_k of $H_{k'}$ for $k' \geqslant k+m$. This proves condition *(ML)* for the projective system $(H_k)_{k \geqslant 0}$.

Moreover, for every affine open subset U of X,

$$H^i(U, \mathscr{F}_k) = 0 \qquad (i > 0)$$

by (1.3.1), and, for $m > 0$, the map

$$H^0(U, \mathscr{F}_{k+m}) \longrightarrow H^0(U, \mathscr{F}_k)$$

is surjective (**I**, 1.3.9). We may therefore
apply (**0**, 13.3.1); the canonical homomorphism III | 129

$$H^n(\widehat{X}, \widehat{\mathscr{F}}) \longrightarrow \varprojlim_k H^n(X, \mathscr{F}_k)$$

is bijective for every $n \geqslant 0$.

Since the projective system $(H/R_k)_{k \geqslant 0}$ is strict, we may pass to the projective limit in the exact sequences

(4.1.7.22) $$0 \longrightarrow H/R_k \longrightarrow H_k \longrightarrow Q_k \longrightarrow 0$$

by (**0**, 13.2.2). Since $v_m(Q_{k+m}) = 0$, we have

$$\varprojlim_k Q_k = 0,$$

and hence a topological isomorphism

$$\varprojlim_k (H/R_k) \simeq \varprojlim_k H_k.$$

But the filtration (R_k) of H is $\mathfrak{J}$-good, and therefore defines on H the $\mathfrak{J}$-preadic topology. Thus $\varprojlim_k (H/R_k)$ is the separated completion of H for this topology. This completes the proof of (4.1.7).

□

(4.1.8). We can now complete the proof of (4.1.5). For every affine open subset V of Y,

$$\Gamma(V, \widehat{R^n f_*(\mathscr{F})})$$

is the separated completion of $\Gamma(V, R^n f_*(\mathscr{F}))$ for the $\mathfrak{J}$-preadic topology, if $\mathscr{J}|V = \widetilde{\mathfrak{J}}$, because $R^n f_*(\mathscr{F})$ is a coherent $\mathscr{O}_Y$-module (**I**, 10.8.4). Furthermore,

$$\Gamma\left(V, \varprojlim_k R^n f_*(\mathscr{F}_k)\right) = \varprojlim_k \Gamma(V, R^n f_*(\mathscr{F}_k))$$

by (**0**, 3.2.6). It follows from (4.1.7) and (1.4.11) that ϕ_n is a topological isomorphism. Likewise, again by (1.4.11), (4.1.7) shows that the homomorphism $\psi_{n,V}$ of (4.1.3.5) is an isomorphism. By the definition of $R^n \widehat{f}_*(\widehat{\mathscr{F}})$, it follows that ψ_n is an isomorphism.

Corollary (4.1.9). — *Under the hypotheses of (4.1.4), for every affine open subset V of Y, the canonical homomorphism*

$$H^n(\widehat{X} \cap f^{-1}(V), \widehat{\mathscr{F}}) \longrightarrow \Gamma(\widehat{Y} \cap V, R^n \widehat{f}_*(\widehat{\mathscr{F}}))$$

is bijective.

(4.1.10). *Remark.* Let $f : X \to Y$ be a morphism of finite type between Noetherian usual preschemes, and let $\mathscr{F}$ be a coherent $\mathscr{O}_X$-module whose support is proper over Y (**II**, 5.4.10). We then know (3.2.4) that $\mathrm{R}^n f_*(\mathscr{F})$ is a coherent $\mathscr{O}_Y$-module for every $n \geqslant 0$. Moreover, we may always suppose that

$$\mathscr{F} = u_*(\mathscr{G}), \qquad \mathscr{G} = u^*(\mathscr{F}),$$

where $\mathscr{G}$ is a coherent $\mathscr{O}_Z$-module, Z is a suitable closed subprescheme of X whose underlying space is $\operatorname{Supp}(\mathscr{F})$, and $u : Z \to X$ is the canonical injection (**I**, 9.3.5). If

$$\mathscr{G}_k = \mathscr{G} \otimes_{\mathscr{O}_Y} (\mathscr{O}_Y / \mathscr{J}^{k+1}),$$

then

$$\mathscr{G}_k = u^*(\mathscr{F}_k), \qquad \mathrm{R}^n f_*(\mathscr{F}_k) = \mathrm{R}^n (f \circ u)_*(\mathscr{G}_k)$$

by (1.3.4), and, taking (**I**, 10.9.5) into account,

$$\mathrm{R}^n \widehat{f}_*(\widehat{\mathscr{F}}) = \mathrm{R}^n \widehat{(f \circ u)}_*(\widehat{\mathscr{G}}).$$

We may therefore apply (4.1.5) to $\mathscr{G}$ and to the proper morphism $f \circ u$. It follows that, under these hypotheses, the results of (4.1.5) remain valid for $\mathscr{F}$ and f.

4.2. Particular cases and variants

The most useful form of the comparison theorem (4.1.5) is the following.

Proposition (4.2.1). — *Let Y be a locally Noetherian prescheme, let $f : X \to Y$ be a proper morphism, and let $\mathscr{F}$ be a coherent $\mathscr{O}_X$-module. Then, for every $y \in Y$ and every p,*

$$\left(\mathrm{R}^p f_*(\mathscr{F})\right)_y$$

is an $\mathscr{O}_y$-module of finite type and is therefore separated for the $\mathfrak{m}_y$-preadic topology. Moreover, there is a canonical topological isomorphism **III** | 130

(4.2.1.1)
$$\left(\widehat{\mathrm{R}^p f_*(\mathscr{F})}\right)_y \simeq \varprojlim_k \mathrm{H}^p\left(f^{-1}(y), \mathscr{F} \otimes_{\mathscr{O}_Y} (\mathscr{O}_y / \mathfrak{m}_y^k)\right).$$

Here the first term is the completion of $\left(\mathrm{R}^p f_*(\mathscr{F})\right)_y$ for the $\mathfrak{m}_y$-preadic topology. In the second term, for every $k \geqslant 0$, $f^{-1}(y)$ is regarded as the underlying space of the prescheme

$$X \times_Y \operatorname{Spec}(\mathscr{O}_y / \mathfrak{m}_y^k)$$

(**I**, 3.6.1).

Proof. Since $\mathscr{O}_y$ is a Noetherian local ring and $\left(\mathrm{R}^p f_*(\mathscr{F})\right)_y$ is an $\mathscr{O}_y$-module of finite type (3.2.1), the $\mathfrak{m}_y$-preadic topology on this module is separated (**0**, 7.3.5).

When Y is Noetherian and y is closed, the other assertions follow from (4.1.7), after replacing Y by an affine neighbourhood of y and taking $Y' = \{y\}$; here we use (G, II, 4.9.1).

In general, put

$$Y_1 = \operatorname{Spec}(\mathscr{O}_y), \qquad X_1 = X \times_Y Y_1, \qquad \mathscr{F}_1 = \mathscr{F} \otimes_{\mathscr{O}_Y} \mathscr{O}_{Y_1},$$

and let

$$f_1 = f \times 1_{Y_1} : X_1 \longrightarrow Y_1.$$

The prescheme Y_1 is Noetherian, the morphism f_1 is proper (**II**, 5.4.2, iii), and $\mathscr{F}_1$ is coherent (**0**, 5.3.11). Let y_1 be the unique closed point of Y_1. The proposition already proved applies to f_1, $\mathscr{F}_1$, and y_1. We have

$$\mathscr{O}_{y_1} = \mathscr{O}_y, \qquad f_1^{-1}(y_1) = f^{-1}(y)$$

(**I**, 3.6.5). Furthermore, the preschemes

$$X \times_Y \operatorname{Spec}(\mathscr{O}_y / \mathfrak{m}_y^k) \quad \text{and} \quad X_1 \times_{Y_1} \operatorname{Spec}(\mathscr{O}_y / \mathfrak{m}_y^k)$$

are canonically identified (**I**, 3.3.9), as are

$$\mathscr{F}_1 \otimes_{\mathscr{O}_{Y_1}} (\mathscr{O}_y / \mathfrak{m}_y^k) \quad \text{and} \quad \mathscr{F} \otimes_{\mathscr{O}_Y} (\mathscr{O}_y / \mathfrak{m}_y^k)$$

(**I**, 9.1.6). It remains only to observe that

$$\mathrm{R}^p f_{1*}(\mathscr{F}_1) \simeq \mathrm{R}^p f_*(\mathscr{F}) \otimes_{\mathscr{O}_Y} \mathscr{O}_{Y_1}.$$

This follows from (1.4.15), since the local morphism $\operatorname{Spec}(\mathscr{O}_y) \to Y$ is flat (**0**, 6.7.1), (**I**, 2.4.2). □

The following corollary uses the terminology of dimension theory (Chapter IV) and will not be applied before that chapter.

Corollary (4.2.2). — *Let Y be a locally Noetherian prescheme, let $f : X \to Y$ be a proper morphism, let $y \in Y$, and let r be the dimension of $f^{-1}(y)$. Then, for every coherent $\mathscr{O}_X$-module $\mathscr{F}$, the sheaves $\mathrm{R}^p f_*(\mathscr{F})$ vanish in a neighbourhood of y for every $p > r$.*

Proof. For every k, we have

$$\mathrm{H}^p\left(f^{-1}(y), \mathscr{F} \otimes_{\mathscr{O}_Y} (\mathscr{O}_y/\mathfrak{m}_y^k)\right) = 0$$

by (G, II, 4.15.2). It follows from (4.2.1) that the separated completion of $\left(\mathrm{R}^p f_*(\mathscr{F})\right)_y$ for the $\mathfrak{m}_y$-preadic topology is zero. Since this topology is separated, we also have

$$\left(\mathrm{R}^p f_*(\mathscr{F})\right)_y = 0.$$

The conclusion follows because $\mathrm{R}^p f_*(\mathscr{F})$ is coherent (**0**, 5.2.2). □

(4.2.3). The result (4.2.1) is used especially when $p = 0$. We obtain the following corollary.

Corollary (4.2.4). — *Under the hypotheses of (4.2.1), there is a canonical topological isomorphism*

$$\left(\widehat{f_*(\mathscr{F})}\right)_y \simeq \varprojlim_k \Gamma\left(f^{-1}(y), \mathscr{F}_y/\mathfrak{m}_y^k \mathscr{F}_y\right).$$

4.3. Zariski's connectedness theorem The results of this section and the next generalize well-known theorems of Zariski, and may all be deduced from (4.2.4). They are consequences of the following theorem.

Theorem (4.3.1). — (Connectedness theorem). *Let Y be a locally Noetherian prescheme and let $f : X \to Y$ be a proper morphism. Then*

$$\mathscr{A}(X) = f_*(\mathscr{O}_X)$$

is a coherent $\mathscr{O}_Y$-algebra.

Let Y' be the finite Y-scheme such that $\mathscr{A}(Y') = \mathscr{A}(X)$; it is determined up to a Y-isomorphism (**II**, 1.3.1), **III** | 131
(**II**, 6.1.3). *If $f' = \mathscr{A}(e)$ is the Y-morphism $X \to Y'$ deduced from the identity isomorphism*

$$e : \mathscr{A}(Y') \simeq \mathscr{A}(X)$$

(**II**, 1.2.7), *then f' is proper, $f'_*(\mathscr{O}_X)$ is isomorphic to $\mathscr{O}_{Y'}$, and every fibre $f'^{-1}(y')$, $y' \in Y'$, is connected and nonempty.*

Proof. Let $g : Y' \to Y$ be the structural morphism. To prove that the homomorphism

$$\theta : \mathscr{O}_{Y'} \longrightarrow f'_*(\mathscr{O}_X)$$

entering into the definition of f' is bijective, it is enough, since Y' is affine over Y, to prove that

$$g_*(\theta) : g_*(\mathscr{O}_{Y'}) \longrightarrow g_*\left(f'_*(\mathscr{O}_X)\right) = f_*(\mathscr{O}_X)$$

is the identity (**II**, 1.4.2). This follows from the definitions, since $g_*(\mathscr{O}_{Y'}) = \mathscr{A}(Y')$ and $f_*(\mathscr{O}_X) = \mathscr{A}(X)$. The coherence of $\mathscr{A}(X)$ is a special case of the finiteness theorem (3.2.1). Since f is proper and g is separated, f' is proper (**II**, 5.4.3). It remains only to prove (4.3.2). □

Corollary (4.3.2). — *Under the hypotheses of (4.3.1), suppose in addition that $f_*(\mathscr{O}_X)$ is isomorphic to $\mathscr{O}_Y$. Then every fibre $f^{-1}(y)$ of f is connected and nonempty.*

Proof. The hypothesis that $f_*(\mathscr{O}_X)$ is isomorphic to $\mathscr{O}_Y$ already implies that f is dominant, hence surjective because f is a closed map. As in (4.2.1), we may reduce to the case where y is closed in Y. Since $f^{-1}(y)$ is a Noetherian space, it has finitely many connected components; these are also the connected components of the underlying space of the completion $\widehat{X}$ along $f^{-1}(y)$.

If Z_i, $1 \leqslant i \leqslant n$, are these connected components, then

$$\Gamma(\widehat{X}, \mathscr{O}_{\widehat{X}}) = \prod_{i=1}^n \Gamma(Z_i, \mathscr{O}_{\widehat{X}}),$$

and none of the rings on the right is zero, since the unit section is nonzero at every point of $\widehat{X}$. Applying (4.1.5) to $\mathscr{F} = \mathscr{O}_X$, whose completion along $f^{-1}(y)$ is $\mathscr{O}_{\widehat{X}}$, shows that $\Gamma(\widehat{X}, \mathscr{O}_{\widehat{X}})$ is isomorphic to the separated $\mathfrak{m}_y$-adic completion $\widehat{\mathscr{O}}_y$ of the local ring $\mathscr{O}_y$. It is therefore a local ring and cannot be a direct product of several nonzero rings. Thus $n = 1$. □

Corollary (4.3.3). — *Under the hypotheses of (4.3.1), for every $y \in Y$ the set of connected components of the fibre $f^{-1}(y)$ is in bijective correspondence with the finite set of points of the fibre $g^{-1}(y)$, where $g : Y' \to Y$ is the structural morphism; in other words, it corresponds to the set of maximal ideals of $\big(f_*(\mathscr{O}_X)\big)_y$.*

Proof. Since Y' is finite over Y, the space $g^{-1}(y)$ is finite and discrete (**II**, 6.1.7). As

$$f^{-1}(y) = f'^{-1}\big(g^{-1}(y)\big),$$

the assertion follows from this observation and (4.3.1). □

This gives a remarkable interpretation of the Y-prescheme Y' defined in (4.3.1). The factorization $f = g \circ f'$ of the proper morphism f is analogous to the factorization obtained by K. Stein for holomorphic maps of analytic spaces, and will henceforth be called the *Stein factorization* of f.

Remark (4.3.4). — Let k be an extension of the field $k(y)$. If

$$f^{-1}(y) \otimes_{k(y)} k = X \times_Y \operatorname{Spec}(k)$$

is connected, then $f^{-1}(y)$ is connected as well, since it is the image of the former under a projection morphism (**I**, 3.4.7). For a morphism $f : X \to Y$ of preschemes and a point $y \in Y$, we shall say that the fibre $f^{-1}(y)$ is *geometrically connected* if, for every extension k of $k(y)$, the prescheme $f^{-1}(y) \otimes_{k(y)} k = X \times_Y \operatorname{Spec}(k)$ is connected.

Under the hypotheses of (4.3.2), the conclusion can be strengthened: the fibres $f^{-1}(y)$ are geometrically connected. Indeed, for every extension k of $k(y)$ there exist a local Noetherian ring A and a local homomorphism **III | 132**

$$\varphi : \mathscr{O}_y \longrightarrow A$$

such that A is a flat $\mathscr{O}_y$-module and the residue field of A is $k(y)$-isomorphic to k (**0**, 10.3.1). Let $Y_1 = \operatorname{Spec}(A)$, and let $h : Y_1 \to Y$ be the local morphism corresponding to φ, taking the unique closed point y_1 of Y_1 to y (**I**, 2.4.1). Set

$$X_1 = X \times_Y Y_1, \qquad f_1 = f \times 1_{Y_1}.$$

The prescheme Y_1 is Noetherian, f_1 is proper (**II**, 5.4.2), and $f_1^{-1}(y_1)$ is a $k(y_1)$-prescheme isomorphic to $X \times_Y \operatorname{Spec}(k)$. It is therefore enough to prove that

$$f_{1*}(\mathscr{O}_{X_1}) = \mathscr{O}_{Y_1},$$

so that (4.3.2) may be applied to f_1. Now h is flat by (**I**, 2.4.2); hence, by (1.4.15) with $q = 0$,

$$f_{1*}(\mathscr{O}_{X_1}) = h^*\big(f_*(\mathscr{O}_X)\big) = h^*(\mathscr{O}_Y) = \mathscr{O}_{Y_1}.$$

In the general case of (4.3.1), the same argument shows, with the notation of that theorem, that

$$f_{1*}(\mathscr{O}_{X_1}) = h^*\big(g_*(\mathscr{O}_{Y'})\big),$$

and that the Stein factorization $f_1 = g_1 \circ f_1'$ of f_1 satisfies

$$g_1 = g \times 1_{Y_1}$$

(**II**, 1.5.2), the corresponding finite Y_1-scheme being $Y_1' = Y' \times_Y Y_1$. Taking the transitivity of fibres into account (**I**, 3.6.4), it follows from (4.3.3) that the number of connected components of $f_1^{-1}(y_1)$ is the number of elements of

$$g_1^{-1}(y_1) = g^{-1}(y) \otimes_{k(y)} k.$$

If k is chosen algebraically closed, this number is independent of the algebraically closed extension chosen and is equal to the *geometric number of points* of $g^{-1}(y)$ (**I**, 6.4.7), or equivalently to

$$\sum_i [k(y_i') : k(y)]_s,$$

where the y_i' range over the finite set $g^{-1}(y)$. This number is called the *geometric number of connected components* of $f^{-1}(y)$. The fields $k(y_i')$ are precisely the residue fields of the semilocal ring $\big(f_*(\mathscr{O}_X)\big)_y$.

Proposition (4.3.5). — *Let X and Y be locally Noetherian integral preschemes, and let $f : X \to Y$ be a proper dominant morphism. For every $y \in Y$, the number of connected components of $f^{-1}(y)$ is at most the number of maximal ideals of the integral closure $\mathscr{O}'_y$ of $\mathscr{O}_y$ in the field of rational functions* $\mathrm{R}(X)$.

Proof. For every open set U of Y,

$$\Gamma\big(U, f_*(\mathscr{O}_X)\big) = \Gamma\big(f^{-1}(U), \mathscr{O}_X\big)$$

is the intersection of the local rings $\mathscr{O}_x$ for $x \in f^{-1}(U)$ (**I**, 8.2.1.1). It follows at once that $\big(f_*(\mathscr{O}_X)\big)_y$ is a subring of $\mathrm{R}(X)$ containing $\mathscr{O}_y$. Since $f_*(\mathscr{O}_X)$ is a coherent $\mathscr{O}_Y$-module, $\big(f_*(\mathscr{O}_X)\big)_y$ is a finite $\mathscr{O}_y$-module, and hence is contained in $\mathscr{O}'_y$. Every maximal ideal of such a ring is the intersection of that ring with a maximal ideal of $\mathscr{O}'_y$ [SZ60, vol. I, pp. 257 and 259], proving the proposition. □

Definition (4.3.6). — An integral local ring is said to be *unibranch* if its integral closure is again a local ring. A point y of an integral prescheme Y is said to be *unibranch* if the local ring $\mathscr{O}_y$ is unibranch; this is in particular the case when Y is normal at y.

Let A be an integral local ring and let K be its field of fractions. The ring A is unibranch if and only if every subring A_1 of K that contains A and is a finite A-algebra is local. Indeed, let A' be the integral closure of A. By the first Cohen–Seidenberg theorem (Bourbaki, *Algèbre commutative*, Chapter V, §2, no. 1, Theorem 1), every maximal ideal of A_1 is the contraction to A_1 of a maximal ideal of A'; thus, if A' is local, so is A_1.

Conversely, A' is the inductive limit of the directed family of finite A-subalgebras A_α of A'. If **III** | 133
each A_α is local, then for $A_\alpha \subset A_\beta$ the maximal ideal of A_α is the contraction of the maximal ideal of A_β, by the same argument; hence A' is local (**0**, 10.3.1.3).

Notice also that if the completion of a Noetherian local ring A is integral—a property expressed by saying that A is *analytically integral*—then A is unibranch. Let $\mathfrak{m}$ be the maximal ideal of A, let K be its field of fractions, and let K' be the field of fractions of $\widehat{A}$; then

$$K' = K \otimes_A \widehat{A}.$$

Let A'_F be a finite A-subalgebra of K. The subring B_F of K' generated by $\widehat{A}$ and A'_F is isomorphic to

$$A'_F \otimes_A \widehat{A};$$

it is a finite $\widehat{A}$-module and is the completion of A'_F for the $\mathfrak{m}$-adic topology (**0**, 7.3.3), (**0**, 7.3.6). Since A'_F is semilocal (Bourbaki, *Algèbre commutative*, Chapter IV, §2, no. 5, Corollary 3 to Proposition 9), while its completion is integral, A'_F can have only one maximal ideal $\mathfrak{m}'_F$, and $\mathfrak{m}'_F \cap A = \mathfrak{m}$. This proves the assertion.

Corollary (4.3.7). — *Under the hypotheses of (4.3.5), suppose that the algebraic closure of* $\mathrm{R}(Y)$ *in* $\mathrm{R}(X)$ *has separable degree n and that $y \in Y$ is unibranch. Then the fibre $f^{-1}(y)$ has at most n connected components. In particular, if the algebraic closure of* $\mathrm{R}(Y)$ *in* $\mathrm{R}(X)$ *is radicial over* $\mathrm{R}(Y)$, *then $f^{-1}(y)$ is connected.*

Proof. Let $\mathscr{O}'_y$ be the integral closure of $\mathscr{O}_y$, and let $\mathscr{O}''_y$ be the integral closure of $\mathscr{O}_y$ in $\mathrm{R}(X)$. The latter is also the integral closure of $\mathscr{O}'_y$ in $\mathrm{R}(X)$. If $\mathscr{O}'_y$ is local, then $\mathscr{O}''_y$ is a semilocal ring having at most n maximal ideals [SZ60, vol. I, p. 289, Theorem 22].

This corollary is essentially the form in which Zariski states his "connectedness theorem" for algebraic schemes. □

Remark (4.3.8). — If, in addition to the hypotheses of (4.3.7), Y is normal at y, then the fibre $f^{-1}(y)$ is geometrically connected: with the notation of (4.3.4), $g^{-1}(y)$ is a single point y', and $k(y')$ is radicial over $k(y)$.

Definition (4.3.9). — Let Y be a locally Noetherian prescheme. A morphism of finite type $f : X \to Y$ is said to be *universally open* if, for every irreducible locally Noetherian prescheme Y' and every dominant morphism $g : Y' \to Y$, every irreducible component of $X' = X \times_Y Y'$ dominates Y'.

If Y is irreducible, this says that if η and η' are the generic points of Y and Y', respectively, and if $f' = f \times 1_{Y'}$, every irreducible component of X' has its generic point in $f'^{-1}(\eta')$. Consequently, for every open set U of Y, the restricted morphism $f^{-1}(U) \to U$ is universally open.

Corollary (4.3.10). — *Let X and Y be locally Noetherian integral preschemes, and let $f : X \to Y$ be proper, dominant, and universally open. If the algebraic closure of* $\mathrm{R}(Y)$ *in* $\mathrm{R}(X)$ *is radicial over* $\mathrm{R}(Y)$*, then every fibre $f^{-1}(y)$, $y \in Y$, is geometrically connected.*

Proof. We may reduce to the case $Y = \operatorname{Spec}(B)$, where B is an integral Noetherian ring. By (**II**, 7.1.7), there exists an integrally closed Noetherian local ring A that dominates $\mathscr{O}_y$ and has $\mathrm{R}(Y)$ as its field of fractions. Put $Y' = \operatorname{Spec}(A)$, and let $h : Y' \to Y$ be the morphism corresponding to the canonical injection $B \to A$. It is birational, hence dominant; moreover, if y' is the unique closed point of Y', then $h(y') = y$.

Set

$$X' = X \times_Y Y', \qquad f' = f \times 1_{Y'}.$$

Let η, η', and ξ be the generic points of Y, Y', and X, respectively. Then $f(\xi) = \eta$ and $h^{-1}(\eta) = \{\eta'\}$; moreover $k(\eta) = k(\eta') = \mathrm{R}(Y)$. Thus $f'^{-1}(\eta')$ is isomorphic to $f^{-1}(\eta)$ (**I**, 3.6.4), and it has a single generic point, since ξ is the generic point of $f^{-1}(\eta)$ (**0**, 2.1.8). By universal openness, every irreducible component of X' has its generic point in $f'^{-1}(\eta')$. Hence X' is irreducible; its generic point ξ' is that of $f'^{-1}(\eta')$, and $k(\xi') = k(\xi)$.

Put $X'' = X'_{\mathrm{red}}$ and $f'' = f'_{\mathrm{red}}$. Then X'' is integral and Noetherian, f'' is proper (**II**, 5.4.6), and the underlying spaces of the fibres $f'^{-1}(y')$ and $f''^{-1}(y')$ are the same. Moreover, **III | 134**

$$\mathrm{R}(X'') = k(\xi') = \mathrm{R}(X).$$

Thus f'' satisfies the hypotheses of (4.3.8), and $f''^{-1}(y')$ is geometrically connected.

Let k now be any extension of $k(y)$. There exists an extension k_1 of $k(y)$ in which both $k(y')$ and k may be regarded as subextensions (Bourbaki, *Algèbre*, Chapter V, §4, Proposition 2). By the preceding paragraph,

$$f''^{-1}(y') \times_{Y'} \operatorname{Spec}(k_1)$$

is connected. It has the same associated reduced prescheme as $f'^{-1}(y') \times_{Y'} \operatorname{Spec}(k_1)$ (**I**, 5.1.8), so the latter is connected; it is isomorphic to $f^{-1}(y) \times_Y \operatorname{Spec}(k_1)$ (**I**, 3.6.4). Therefore $f^{-1}(y) \times_Y \operatorname{Spec}(k)$ is connected, and the observation at the beginning of (4.3.4) completes the proof. □

Remark (4.3.11). — *(i)* The preceding argument is due in substance to Zariski [Zar51], except that in his setting one may take for A the integral closure of $\mathscr{O}_y$, since this is a Noetherian ring for the local rings of classical algebraic geometry. Zariski also proves that if Y is the Chow variety of a projective space $\mathbf{P}^r_k$ over a field k, and X is the closed subset of $\mathbf{P}^r_k \times_k Y$ defining the Chow correspondence between $\mathbf{P}^r_k$ and Y, then the projection $X \to Y$ is universally open (*loc. cit.*, lemma on p. 82). It appears that this is the only formal property of "Chow coordinates" used in certain applications. In such a situation it is therefore useful to replace the language of specialization of cycles in projective space by that of fibres of a proper morphism, possibly assumed universally open or subject to analogous local regularity conditions.

(ii) In Chapter IV we shall see that a universally open morphism $f : X \to Y$ may equivalently be defined by requiring that, for every morphism $Z \to Y$, the base-changed morphism $f_{(Z)} : X_{(Z)} \to Z$ be open. One can also show that if f satisfies the hypotheses of (4.3.10), and if y is a specialization of y', then the geometric number of connected components of $f^{-1}(y)$ is at most that of $f^{-1}(y')$.

Corollary (4.3.12). — *Under the hypotheses of (4.3.5), suppose in addition that* $\mathrm{R}(Y)$ *is algebraically closed in* $\mathrm{R}(X)$*, and let y be a normal point of Y. Then $f^{-1}(y)$ is geometrically connected, and there exists an open neighbourhood U of y in Y such that*

$$f_*(\mathscr{O}_X|_{f^{-1}(U)}) \simeq \mathscr{O}_Y|_U.$$

In particular, if Y is normal and $\mathrm{R}(Y)$ *is algebraically closed in* $\mathrm{R}(X)$*, then $f_*(\mathscr{O}_X) \simeq \mathscr{O}_Y$.*

Proof. The assertion concerning $f^{-1}(y)$ is a special case of (4.3.8).

If **III | 135**

$$X \xrightarrow{f'} Y' \xrightarrow{g} Y$$

is the Stein factorization of f (4.3.3), it follows that $g^{-1}(y)$ consists of a single point y'. Moreover,

$$\mathscr{O}_y \subset \mathscr{O}_{y'} = \big(f_*(\mathscr{O}_X)\big)_y \subset \mathrm{R}(X).$$

Since $\mathscr{O}_{y'}$ is finite over $\mathscr{O}_y$, and hence *a fortiori* algebraic over $\mathrm{R}(Y)$ inside the relevant fraction field, the hypothesis forces it to lie in $\mathrm{R}(Y)$. Since y is normal, necessarily $\mathscr{O}_{y'} = \mathscr{O}_y$. Thus g is a local isomorphism at y' (**I**, 6.5.4), which proves the first part. The second assertion follows from the first: the additional hypothesis makes g bijective and a local isomorphism in a neighbourhood of every point of Y', hence an isomorphism. □

The fact that (4.3.7) is available for schemes yields, for example, the following application.

Proposition (4.3.13). — *Let A be a unibranch Noetherian local ring, let $\mathfrak{a}$ be an ideal of definition of A, let*

$$A_0 = A/\mathfrak{a}, \qquad S = \mathrm{gr}_{\mathfrak{a}}(A)$$

be the graded ring associated to the $\mathfrak{a}$-preadic filtration. Suppose that S is a graded A_0-algebra generated by S_1, where S_1 is a finite A_0-module. Then $\mathrm{Proj}(S)$ is a connected A_0-scheme.

Proof. Let $\mathfrak{m}$ be the maximal ideal of A. The scheme $Y = \mathrm{Spec}(A)$ is integral, and the point y corresponding to $\mathfrak{m}$ is its unique closed point. For some integer p,

$$\mathfrak{m}^p \subset \mathfrak{a} \subset \mathfrak{m},$$

so $V(\mathfrak{a}) = \{\mathfrak{m}\}$. Put

$$S' = \bigoplus_{n \geqslant 0} \mathfrak{a}^n \qquad \text{and} \qquad X = \mathrm{Proj}(S').$$

This is the Y-scheme obtained by blowing up the ideal $\mathfrak{a}$. It is integral, and the structural morphism $f : X \to Y$ is birational (**II**, 8.1.4) and projective. Therefore (4.3.7) applies and shows that $f^{-1}(y)$ is connected. But the underlying space of $f^{-1}(y)$ is that of

$$\mathrm{Proj}(S' \otimes_A A_0)$$

(**I**, 3.6.1), (**II**, 2.8.10); since $S' \otimes_A A_0 = S$ by definition, the proposition follows. □

4.4. Zariski's "main theorem"

Proposition (4.4.1). — *Let Y be a locally Noetherian prescheme and let $f : X \to Y$ be a proper morphism. Let X' be the set of points $x \in X$ that are isolated in their fibre $f^{-1}(f(x))$. Then X' is open in X. If*

$$f = g \circ f'$$

is the Stein factorization of f (4.3.3), the restriction of f' to X' is an isomorphism of X' onto the subprescheme induced on an open subset U of Y', and $X' = f'^{-1}(U)$.

Proof. Since $g^{-1}(f(x))$ is finite and discrete (4.3.3), (**II**, 6.1.7), the point x is isolated in $f^{-1}(f(x))$ if and only if it is isolated in $f'^{-1}(f'(x))$. We may therefore reduce to the case $f' = f$, so that $f_*(\mathscr{O}_X) = \mathscr{O}_Y$. If $x \in X'$, the fibre $f^{-1}(f(x))$, being connected by (4.3.2), must then consist of the single point x.

Since f is closed, for every open neighbourhood V of x in X, the closed set $f(X - V)$ does not contain $y = f(x)$. If U is its complement in Y, then $f^{-1}(U) \subset V$. Thus the inverse images under f of a fundamental system of open neighbourhoods of y form a fundamental system of open neighbourhoods of x. The equality $f_*(\mathscr{O}_X) = \mathscr{O}_Y$ and the definition of the direct image of a sheaf (**0**, 3.4.1), (4.2.1) now imply that, if $f = (\psi, \theta)$, the homomorphism

$$\theta_y^\# : \mathscr{O}_y \longrightarrow \mathscr{O}_x$$

is an isomorphism. Hence there are open neighbourhoods V of x and U of y such that the restriction of f is an isomorphism $V \simeq U$ (**I**, 6.5.4).

By the preceding paragraph we may moreover arrange that $f^{-1}(U) = V$. It follows immediately **III** | 136
that $V \subset X'$, which completes the proof. □

The following proposition was proved by Chevalley for algebraic schemes.

Proposition (4.4.2). — *Let Y be a locally Noetherian prescheme and let $f : X \to Y$ be a morphism. The following conditions are equivalent:*

a) *f is finite;*
b) *f is affine and proper;*
c) *f is proper and, for every $y \in Y$, the set $f^{-1}(y)$ is finite.*

Proof. Condition a) implies b) by (**II**, 6.1.2) and (**II**, 6.1.11). If f is proper and affine, then so is

$$f^{-1}(y) \longrightarrow \operatorname{Spec}(k(y))$$

(**II**, 1.6.2), (**II**, 5.4.2). The finiteness theorem (3.2.1), applied to the structural sheaf of $f^{-1}(y)$, shows that $f^{-1}(y) = \operatorname{Spec}(A)$ for a finite $k(y)$-algebra A. Thus $f^{-1}(y)$ is a finite set (**II**, 6.1.7), and b) implies c).

Finally, if c) holds, the fibre $f^{-1}(y)$ is an algebraic prescheme over $k(y)$ whose underlying set is finite, and is therefore discrete (**I**, 6.4.4). With the notation of (4.4.1), we have $X' = X$, so $f' : X \to Y'$ is an isomorphism. Since g is finite, c) implies a). □

Theorem (4.4.3). — (Zariski's "main theorem"). *Let Y be a Noetherian prescheme, let $f : X \to Y$ be a quasi-projective morphism, and let X' be the set of points $x \in X$ that are isolated in their fibre $f^{-1}(f(x))$. Then X' is open in X, and the subprescheme induced by X on X' is isomorphic to a subprescheme induced on an open subset of a Y-prescheme Y' finite over Y.*

Proof. There is a projective Y-prescheme Z such that X is Y-isomorphic to the subprescheme induced on an open subset of Z (**II**, 5.3.2), (**II**, 5.5.1). We are thus reduced to the case where f is projective, hence proper (**II**, 5.5.3), and the theorem follows immediately from (4.4.1). □

Remark (4.4.4). — If X is reduced (respectively, if X is irreducible and X' is nonempty), one may require in (4.4.3) that Y' be reduced (respectively, irreducible). Indeed, Y' may be replaced by the schematic closure $\overline{X'}$ of X' in Y' (**I**, 9.5.11), (**II**, 6.1.5). If X' is reduced, so is $\overline{X'}$ (**I**, 9.5.9); if X' is nonempty and irreducible, then $\overline{X'}$ is irreducible as well.

Corollary (4.4.5). — *Let Y be a locally Noetherian scheme, let $f : X \to Y$ be a morphism of finite type, and let x be a point of X isolated in its fibre $f^{-1}(f(x))$. Then there is an open neighbourhood of x in X which is isomorphic to an open subset of a Y-prescheme finite over Y.*

Proof. Set $y = f(x)$, let U be an affine open neighbourhood of y in Y, and let V be an affine open neighbourhood of x in X, contained in $f^{-1}(U)$. Since Y is separated, the inclusion $U \to Y$ is affine (**II**, 1.6.3); and since V is affine over U (*ibid.*), the restriction of f to V is an affine morphism $V \to Y$ (**II**, 1.6.2). A fortiori, this restriction is quasi-projective, since it is of finite type (**I**, 6.3.5), (**II**, 5.3.4). It remains only to apply (4.4.3) to this restriction. □

Corollary (4.4.5) may be stated in the language of commutative algebra as follows.

Corollary (4.4.6). — *Let A be a Noetherian ring, let B be an A-algebra of finite type, let $\mathfrak{q}$ be a prime ideal of B, and let $\mathfrak{p}$ be its inverse image in A. Suppose that $\mathfrak{q}$ is both maximal and minimal in the set of prime ideals of B whose inverse image is $\mathfrak{p}$. Then there exist an element $g \in B - \mathfrak{q}$, a finite A-algebra A', and an element $f' \in A'$ such that the A-algebras B_g and $A'_{f'}$ are isomorphic.*

Proof. Apply (4.4.5) to $Y = \operatorname{Spec}(A)$ and $X = \operatorname{Spec}(B)$. The hypothesis on $\mathfrak{q}$ says precisely that $\mathfrak{q}$ is isolated in its fibre (**I**, 1.1.7). □

The following consequence appears less general.

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Corollary (4.4.7). — *Let A be a Noetherian local ring, let B be an A-algebra of finite type, and let $\mathfrak{n}$ be a prime ideal of B whose inverse image in A is the maximal ideal $\mathfrak{m}$. Suppose that $\mathfrak{n}$ is maximal in B and minimal in the set of prime ideals of B whose inverse image is $\mathfrak{m}$ (which also means that $\mathfrak{m}B$ is $\mathfrak{n}$-primary). Then there exist a finite A-algebra A' and a maximal ideal $\mathfrak{m}'$ of A' (whose inverse image in A is $\mathfrak{m}$) such that $B_{\mathfrak{n}}$ is isomorphic, as an A-algebra, to $A'_{\mathfrak{m}'}$.*

The following special case of (4.4.7) is also sometimes called the "main theorem".

Corollary (4.4.8). — *Under the hypotheses of (4.4.7), suppose moreover that A and B are integral domains with the same field of fractions K. If A is integrally closed, then $A = B$.*

Proof. Remark (4.4.4) shows that, in applying (4.4.7), we may suppose that A' is an integral domain with field of fractions K. The hypothesis on A then gives $A' = A$, hence $B_{\mathfrak{n}} = A$. Since $A \subset B \subset B_{\mathfrak{n}}$, it follows that $A = B$. □

Statement (4.4.8) is the form in which Zariski gave his "main theorem" (extended here to arbitrary Noetherian integral local rings).

The preceding corollaries were local variants of (4.4.3), which is a global result. Here is another consequence of a global nature.

Corollary (4.4.9). — *Let Y be an integral locally Noetherian prescheme, and let $f : X \to Y$ be a separated morphism of finite type which is birational. Suppose moreover that Y is normal and that every fibre $f^{-1}(y)$, for $y \in Y$, is finite. Then f is an open immersion. If in addition f is closed (in particular, if f is proper), then f is an isomorphism.*

Proof. Let $x \in X$ and put $y = f(x)$. Since $f^{-1}(y)$ is a scheme algebraic over $k(y)$, the assumption that it is finite implies that it is discrete (**I**, 6.4.4). Moreover, $\mathscr{O}_y$ is integrally closed, and $\mathscr{O}_x$ and $\mathscr{O}_y$ have the same field of fractions (**I**, 7.1.5). We may therefore apply (4.4.8); if $f = (\psi, \theta)$, the homomorphism

$$\theta_y^\# : \mathscr{O}_y \longrightarrow \mathscr{O}_x$$

is bijective. It follows from (**I**, 6.5.4) that f is a local isomorphism. Since f is separated and X is integral, f is an open immersion (**I**, 8.2.8). The last assertion follows from the fact that f is dominant. □

Proposition (4.4.10). — *Let Y be a locally Noetherian prescheme, and let $f : X \to Y$ be a morphism locally of finite type. The set X' of points $x \in X$ which are isolated in their fibre $f^{-1}(f(x))$ is open in X.*

Proof. The question is local on X and Y, so we may suppose that X and Y are affine Noetherian and that f is of finite type. The morphism f is then affine and of finite type, hence quasi-projective (**II**, 5.3.4); it suffices to apply (4.4.3). □

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Corollary (4.4.11). — *Let Y be a locally Noetherian prescheme, and let $f : X \to Y$ be a proper morphism. The set U of points $y \in Y$ for which $f^{-1}(y)$ is discrete is open in Y, and the restricted morphism $f^{-1}(U) \to U$ is finite. In particular, every proper quasi-finite morphism $X \to Y$ is finite.*

Proof. The complement of U in Y is the image under f of $X - X'$, which is closed in X by (4.4.10); since f is a closed map, U is open. Furthermore, (**II**, 6.2.2) shows that $f^{-1}(y)$ is finite for every $y \in U$. Since the restricted morphism $f^{-1}(U) \to U$ is proper (**II**, 5.4.1), it is finite by (4.4.2). □

Remark (4.4.12). — *(i)* As announced in (**II**, 6.2.7), we shall show in Chapter V that if Y is locally Noetherian, every separated quasi-finite morphism $f : X \to Y$ is quasi-affine, hence quasi-projective. It will follow that the conclusion of the main theorem (4.4.3) remains valid when f is assumed only to be separated and of finite type. Indeed, (4.4.10) shows that X' is open in X; and since X is locally Noetherian, the restriction of f to X' is still of finite type (**I**, 6.3.5), hence quasi-finite by the definition of X', and clearly separated. We may therefore apply (4.4.3) to this restriction.

(ii) In Chapter IV we shall give a more elementary proof of (4.4.10), using dimension theory.

4.5. Completions of modules of homomorphisms

Proposition (4.5.1). — *Let A be a Noetherian ring, let $\mathfrak{J}$ be an ideal of A, let X be an A-prescheme of finite type, and let $\mathscr{F}$ and $\mathscr{G}$ be two coherent $\mathscr{O}_X$-modules whose supports have proper intersection over $Y = \operatorname{Spec}(A)$* (**II**, 5.4.10)*. Then, for every integer $n \geqslant 0$,*

$$\operatorname{Ext}^n_{\mathscr{O}_X}(X; \mathscr{F}, \mathscr{G})$$

is a finite A-module, and its separated completion for the $\mathfrak{J}$-preadic topology is canonically identified, with the notation of (4.1.7), with

$$\operatorname{Ext}^n_{\mathscr{O}_{\widehat{X}}}(\widehat{X}; \widehat{\mathscr{F}}, \widehat{\mathscr{G}}).$$

Proof. We know from (T, 4.2) that there is a biregular spectral sequence $\mathrm{E}(\mathscr{F}, \mathscr{G})$ whose abutment is

$$\operatorname{Ext}_{\mathscr{O}_X}(X; \mathscr{F}, \mathscr{G})$$

and whose E_2-terms are

$$\mathrm{E}_2^{pq} = \mathrm{H}^p\big(X, \mathscr{E}xt^q_{\mathscr{O}_X}(\mathscr{F}, \mathscr{G})\big).$$

The $\mathscr{O}_X$-module $\mathscr{E}xt^q_{\mathscr{O}_X}(\mathscr{F},\mathscr{G})$ is coherent (**0**, 12.3.3); its support is contained in the intersection of the supports of $\mathscr{F}$ and $\mathscr{G}$ (T, 4.2.2), and is consequently proper over Y (**II**, 5.4.10). It follows from (3.2.4) that the E_2^{pq} are finite A-modules, and then from (**0**, 11.1.8) that the same is true of every term E_r^{pq} of the spectral sequence and of its abutment.

On the other hand, if $i : \widehat{X} \to X$ is the canonical morphism, then $\widehat{\mathscr{F}}$ and $\widehat{\mathscr{G}}$ are canonically identified with $i^*(\mathscr{F})$ and $i^*(\mathscr{G})$, and i is flat (**I**, 10.8.8), (**0**, 10.8.9). For every $q \geqslant 0$ there is therefore a canonical i-morphism

$$u_q : \mathscr{E}xt^q_{\mathscr{O}_X}(\mathscr{F},\mathscr{G}) \longrightarrow \mathscr{E}xt^q_{\mathscr{O}_{\widehat{X}}}(\widehat{\mathscr{F}},\widehat{\mathscr{G}})$$

(**0**, 12.3.4), and the corresponding $\mathscr{O}_{\widehat{X}}$-homomorphism

$$u_q^\# : i^*\big(\mathscr{E}xt^q_{\mathscr{O}_X}(\mathscr{F},\mathscr{G})\big) \longrightarrow \mathscr{E}xt^q_{\mathscr{O}_{\widehat{X}}}(\widehat{\mathscr{F}},\widehat{\mathscr{G}})$$

is an isomorphism (**0**, 12.3.5). In other words, (**I**, 10.8.8) identifies

$$\mathscr{E}xt^q_{\mathscr{O}_{\widehat{X}}}(\widehat{\mathscr{F}},\widehat{\mathscr{G}})$$

canonically with the completion of $\mathscr{E}xt^q_{\mathscr{O}_X}(\mathscr{F},\mathscr{G})$, in the notation of (4.1.7).

The comparison theorem (4.1.10) now shows, for every $p \geqslant 0$, that

$$\mathrm{H}^p\big(\widehat{X}, \mathscr{E}xt^q_{\mathscr{O}_{\widehat{X}}}(\widehat{\mathscr{F}},\widehat{\mathscr{G}})\big)$$

is canonically identified with the separated completion, for the $\mathfrak{J}$-preadic topology, of

$$\mathrm{H}^p\big(X, \mathscr{E}xt^q_{\mathscr{O}_X}(\mathscr{F},\mathscr{G})\big).$$

Let $\mathrm{E}(\widehat{\mathscr{F}},\widehat{\mathscr{G}})$ denote the biregular spectral sequence of (T, 4.2) associated with $\widehat{\mathscr{F}}$ and $\widehat{\mathscr{G}}$. If $\widehat{A}$ is the separated completion of A for the $\mathfrak{J}$-preadic topology, then, up to canonical isomorphism, **III** | 139

$$\mathrm{E}_2^{pq}(\widehat{\mathscr{F}},\widehat{\mathscr{G}}) = \mathrm{E}_2^{pq}(\mathscr{F},\mathscr{G}) \otimes_A \widehat{A}$$

(**0**, 7.3.3).

The flat morphism i defines a canonical homomorphism of spectral sequences

$$\varphi : \mathrm{E}(\mathscr{F},\mathscr{G}) \longrightarrow \mathrm{E}(\widehat{\mathscr{F}},\widehat{\mathscr{G}}) = \mathrm{E}(i^*(\mathscr{F}), i^*(\mathscr{G})).$$

On the E_2-terms and on the abutment, respectively, this is the homomorphism

$$\varphi_2^{pq} : \mathrm{H}^p\big(X, \mathscr{E}xt^q_{\mathscr{O}_X}(\mathscr{F},\mathscr{G})\big) \longrightarrow \mathrm{H}^p\big(\widehat{X}, \mathscr{E}xt^q_{\mathscr{O}_{\widehat{X}}}(\widehat{\mathscr{F}},\widehat{\mathscr{G}})\big)$$

and the homomorphism

$$\varphi^n : \mathrm{Ext}^n_{\mathscr{O}_X}(X; \mathscr{F},\mathscr{G}) \longrightarrow \mathrm{Ext}^n_{\mathscr{O}_{\widehat{X}}}(\widehat{X}; \widehat{\mathscr{F}},\widehat{\mathscr{G}}),$$

deduced from u_q and u_0, respectively, by functoriality (**0**, 12.3.4).

After tensoring with $\widehat{A}$, the φ_r^{pq} and φ^n give homomorphisms of $\widehat{A}$-modules

$$\psi_r^{pq} : \mathrm{E}_r^{pq}(\mathscr{F},\mathscr{G}) \otimes_A \widehat{A} \longrightarrow \mathrm{E}_r^{pq}(\widehat{\mathscr{F}},\widehat{\mathscr{G}})$$

and

$$\psi^n : \mathrm{Ext}^n_{\mathscr{O}_X}(X; \mathscr{F},\mathscr{G}) \otimes_A \widehat{A} \longrightarrow \mathrm{Ext}^n_{\mathscr{O}_{\widehat{X}}}(\widehat{X}; \widehat{\mathscr{F}},\widehat{\mathscr{G}}).$$

Since $\widehat{A}$ is a flat A-module (**0**, 7.3.3), the $\widehat{A}$-modules

$$\mathrm{E}_r^{pq}(\mathscr{F},\mathscr{G}) \otimes_A \widehat{A}$$

form a biregular spectral sequence with abutment

$$\mathrm{Ext}^n_{\mathscr{O}_X}(X; \mathscr{F},\mathscr{G}) \otimes_A \widehat{A},$$

and the ψ_r^{pq} and ψ^n form a morphism of spectral sequences. Since the ψ_2^{pq} are isomorphisms, the same is true of the ψ^n (**0**, 11.1.5). □

Corollary (4.5.2). — *Under the hypotheses of (4.5.1), suppose in addition that A is a Noetherian $\mathfrak{J}$-adic ring. Then, for every integer $n \geqslant 0$,*

$$\mathrm{Ext}^n_{\mathscr{O}_X}(X;\mathscr{F},\mathscr{G}) \simeq \mathrm{Ext}^n_{\mathscr{O}_{\widehat{X}}}(\widehat{X};\widehat{\mathscr{F}},\widehat{\mathscr{G}})$$

canonically.

Proof. The finite A-module $\mathrm{Ext}^n_{\mathscr{O}_X}(X;\mathscr{F},\mathscr{G})$ is separated and complete for the $\mathfrak{J}$-preadic topology (**0**, 7.3.6). □

The special case $n = 0$ of (4.5.1) may be stated as follows.

Corollary (4.5.3). — *Under the hypotheses of (4.5.1), for every homomorphism $u : \mathscr{F} \to \mathscr{G}$ let $\widehat{u} : \widehat{\mathscr{F}} \to \widehat{\mathscr{G}}$ denote the completed homomorphism* (**I**, 10.8.4)*. There is a canonical isomorphism*

(4.5.3.1) $$\widehat{\mathrm{Hom}_{\mathscr{O}_X}(\mathscr{F},\mathscr{G})} \simeq \mathrm{Hom}_{\mathscr{O}_{\widehat{X}}}(\widehat{\mathscr{F}},\widehat{\mathscr{G}}),$$

where the left-hand side is the separated completion, for the $\mathfrak{J}$-preadic topology, of the A-module $\mathrm{Hom}_{\mathscr{O}_X}(\mathscr{F},\mathscr{G})$. This isomorphism is obtained by passing to separated completions in the homomorphism $u \mapsto \widehat{u}$.

4.6. Relations between formal morphisms and ordinary morphisms

Proposition (4.6.1). — *Let Y be a locally Noetherian prescheme, let $f : X \to Y$ be a proper morphism, let $\mathscr{F}$ be a coherent $\mathscr{O}_X$-module which is flat over Y, and let $y \in Y$. Suppose that, for some integer n,*

$$\mathrm{H}^n\big(f^{-1}(y), \mathscr{F} \otimes_{\mathscr{O}_Y} k(y)\big) = 0.$$

Then there is a neighbourhood U of y in Y such that

$$\mathrm{R}^n f_*(\mathscr{F})|U = 0,$$

and, for every integer $p \geqslant 0$, the canonical homomorphism

$$\big(\mathrm{R}^{n-1} f_*(\mathscr{F})\big)_y \longrightarrow \mathrm{H}^{n-1}\big(f^{-1}(y), \mathscr{F} \otimes_{\mathscr{O}_Y} (\mathscr{O}_y/\mathfrak{m}_y^{p+1})\big)$$

(4.2.1) is surjective.

Proof. The $\mathscr{O}_Y$-module $\mathrm{R}^n f_*(\mathscr{F})$ is coherent (3.2.1); hence the first assertion will follow once we prove

$$\big(\mathrm{R}^n f_*(\mathscr{F})\big)_y = 0$$

(**0**, 5.2.2). By (4.2.1), it is enough to prove that

$$\mathrm{H}^n\big(f^{-1}(y), \mathscr{F} \otimes_{\mathscr{O}_Y} (\mathscr{O}_y/\mathfrak{m}_y^{p+1})\big) = 0$$

for every p. This is true by hypothesis for $p = 0$; we proceed by induction on p.

Put

$$X_p = X \times_Y \mathrm{Spec}(\mathscr{O}_y/\mathfrak{m}_y^{p+1}).$$

Then X_{p-1} is a closed subprescheme of X_p with the same underlying space (**I**, 3.6.1). The induction hypothesis

$$\mathrm{H}^n\big(X_{p-1}, \mathscr{F} \otimes_{\mathscr{O}_Y} (\mathscr{O}_y/\mathfrak{m}_y^{p})\big) = 0$$

therefore gives

$$\mathrm{H}^n\big(X_p, \mathscr{F} \otimes_{\mathscr{O}_Y} (\mathscr{O}_y/\mathfrak{m}_y^{p})\big) = 0.$$

On the other hand, the exact sequence of $\mathscr{O}_{X_p}$-modules

$$0 \longrightarrow \mathfrak{m}_y^p\mathscr{F}/\mathfrak{m}_y^{p+1}\mathscr{F} \longrightarrow \mathscr{F}/\mathfrak{m}_y^{p+1}\mathscr{F} \longrightarrow \mathscr{F}/\mathfrak{m}_y^{p}\mathscr{F} \longrightarrow 0$$

gives the exact cohomology sequence

$$\mathrm{H}^n\big(X_p, \mathfrak{m}_y^p\mathscr{F}/\mathfrak{m}_y^{p+1}\mathscr{F}\big) \longrightarrow \mathrm{H}^n\big(X_p, \mathscr{F}/\mathfrak{m}_y^{p+1}\mathscr{F}\big) \longrightarrow \mathrm{H}^n\big(X_p, \mathscr{F}/\mathfrak{m}_y^{p}\mathscr{F}\big).$$

It is thus enough to show that

(4.6.1.1) $$\mathrm{H}^n\big(X_p, \mathfrak{m}_y^p\mathscr{F}/\mathfrak{m}_y^{p+1}\mathscr{F}\big) = 0.$$

Indeed, the middle term will then be a submodule of the last term, which is zero by the induction hypothesis.

The fibre **III | 140**

$$Z = f^{-1}(y) = X \times_Y \operatorname{Spec}(k(y))$$

is a closed subprescheme of X_p, and $\mathfrak{m}_y^p \mathscr{F} / \mathfrak{m}_y^{p+1} \mathscr{F}$ is annihilated by $\mathfrak{m}_y$. It may therefore be regarded as an $\mathscr{O}_Z$-module, and

$$\mathrm{H}^n\big(Z, \mathfrak{m}_y^p \mathscr{F} / \mathfrak{m}_y^{p+1} \mathscr{F}\big) = \mathrm{H}^n\big(X_p, \mathfrak{m}_y^p \mathscr{F} / \mathfrak{m}_y^{p+1} \mathscr{F}\big).$$

We shall now prove that the canonical $\mathscr{O}_Z$-homomorphism

$$(\mathscr{F} / \mathfrak{m}_y \mathscr{F}) \otimes_{k(y)} (\mathfrak{m}_y^p / \mathfrak{m}_y^{p+1}) \longrightarrow \mathfrak{m}_y^p \mathscr{F} / \mathfrak{m}_y^{p+1} \mathscr{F} \tag{4.6.1.2}$$

is bijective. Once this is established, the fact that $\mathfrak{m}_y^p / \mathfrak{m}_y^{p+1}$ is a free $k(y)$-module gives

$$\begin{aligned} \mathrm{H}^n\big(Z, \mathfrak{m}_y^p \mathscr{F} / \mathfrak{m}_y^{p+1} \mathscr{F}\big) &= \mathrm{H}^n\big(Z, \mathscr{F} / \mathfrak{m}_y \mathscr{F}\big) \otimes_{k(y)} (\mathfrak{m}_y^p / \mathfrak{m}_y^{p+1}) \\ &= 0 \end{aligned}$$

(**0**, 12.2.3), since $\mathrm{H}^n(Z, \mathscr{F} / \mathfrak{m}_y \mathscr{F}) = 0$ by hypothesis. This proves (4.6.1.1).

To finish the proof of the first assertion it remains only to show that (4.6.1.2) is bijective. The question is pointwise on X, and $\mathscr{F}_x$ is a flat $\mathscr{O}_y$-module by hypothesis for every $x \in f^{-1}(y)$. It is therefore enough to apply (**0**, 10.2.1), c), since $\mathscr{F}_x / \mathfrak{m}_y \mathscr{F}_x$ is flat over the field $k(y) = \mathscr{O}_y / \mathfrak{m}_y$.

For the second assertion of (4.6.1), we reduce at once, as in (4.2.1), to the case in which Y is affine and y is closed. The same argument shows, for every $k > 0$, that

$$\mathrm{H}^n\big(X_{k+1}, \mathfrak{m}_y^k \mathscr{F} / \mathfrak{m}_y^{k+p+1} \mathscr{F}\big) = 0. \tag{4.6.1.3}$$

It follows from (4.2.1) that **III | 141**

$$\big(\mathrm{R}^n f_*(\mathfrak{m}_y^k \mathscr{F})\big)_y = 0. \tag{4.6.1.4}$$

The exact cohomology sequence now yields the exactness of

$$\big(\mathrm{R}^{n-1} f_*(\mathscr{F})\big)_y \longrightarrow \big(\mathrm{R}^{n-1} f_*(\mathscr{F} / \mathfrak{m}_y^p \mathscr{F})\big)_y \longrightarrow \big(\mathrm{R}^n f_*(\mathfrak{m}_y^p \mathscr{F})\big)_y = 0.$$

Since y is closed and Y is affine, we have (**I**, 1.4.11)

$$\mathrm{R}^{n-1} f_*(\mathscr{F} / \mathfrak{m}_y^p \mathscr{F}) = \big(\mathrm{H}^{n-1}(X, \mathscr{F} / \mathfrak{m}_y^p \mathscr{F})\big)^\sim = \big(\mathrm{H}^{n-1}(f^{-1}(y), \mathscr{F} / \mathfrak{m}_y^p \mathscr{F})\big)^\sim$$

(G, II, 4.9.1). The module

$$\mathrm{H}^{n-1}(f^{-1}(y), \mathscr{F} / \mathfrak{m}_y^p \mathscr{F})$$

is an $\mathscr{O}_y / \mathfrak{m}_y^p$-module, and hence

$$\big(\mathrm{R}^{n-1} f_*(\mathscr{F} / \mathfrak{m}_y^p \mathscr{F})\big)_y = \mathrm{H}^{n-1}(f^{-1}(y), \mathscr{F} / \mathfrak{m}_y^p \mathscr{F}).$$

This completes the proof. □

Corollary (4.6.2). — *Let Y be a locally Noetherian prescheme, let $f : X \to Y$ be a proper flat morphism, let $\mathscr{F}$ and $\mathscr{G}$ be two locally free $\mathscr{O}_X$-modules, and let y be a point of Y. Put*

$$X_y = f^{-1}(y) = X \times_Y \operatorname{Spec}(k(y)), \qquad \mathscr{F}_y = \mathscr{F} \otimes_{\mathscr{O}_Y} k(y), \qquad \mathscr{G}_y = \mathscr{G} \otimes_{\mathscr{O}_Y} k(y),$$

and suppose that

$$\mathrm{H}^1\big(X_y, \mathscr{H}om_{\mathscr{O}_{X_y}}(\mathscr{F}_y, \mathscr{G}_y)\big) = 0. \tag{4.6.2.1}$$

Then, for every homomorphism $u_0 : \mathscr{F}_y \to \mathscr{G}_y$, there exist an open neighbourhood U of y in Y and a homomorphism

$$u : \mathscr{F}|_{f^{-1}(U)} \longrightarrow \mathscr{G}|_{f^{-1}(U)}$$

such that $u_0 = u \otimes 1$.

Proof. The hypothesis permits us to apply (4.6.1), with $n = 1$ and $p = 0$, to the coherent $\mathscr{O}_X$-module

$$\mathscr{H} = \mathscr{H}om_{\mathscr{O}_X}(\mathscr{F}, \mathscr{G}).$$

Indeed, $\mathscr{H}$ is locally free and hence, a fortiori, flat over Y, while the $\mathscr{O}_{X_y}$-module $\mathscr{H} \otimes_{\mathscr{O}_Y} k(y)$ identifies with $\mathscr{H}om_{\mathscr{O}_{X_y}}(\mathscr{F}_y, \mathscr{G}_y)$ (**0**, 6.2.2). We may suppose that $Y = \operatorname{Spec}(A)$ is affine. Then (**I**, 1.4.11) gives

$$\mathrm{R}^0 f_*(\mathscr{H}) = \big(\operatorname{Hom}_{\mathscr{O}_X}(\mathscr{F}, \mathscr{G})\big)^{\sim},$$

and consequently

$$\big(\mathrm{R}^0 f_*(\mathscr{H})\big)_y = \operatorname{Hom}_{\mathscr{O}_X}(\mathscr{F}, \mathscr{G}) \otimes_A \mathscr{O}_y.$$

By (4.6.1), the canonical homomorphism

$$\operatorname{Hom}_{\mathscr{O}_X}(\mathscr{F}, \mathscr{G}) \otimes_A \mathscr{O}_y \longrightarrow \operatorname{Hom}_{\mathscr{O}_{X_y}}(\mathscr{F}_y, \mathscr{G}_y)$$

is surjective. This proves the corollary, since every element of the module on the left may be written as $u \otimes (1/s)$, where $s \in A \setminus \mathfrak{m}_y$. $\square$

Corollary (4.6.3). — *Under the hypotheses of (4.6.2), if u_0 is injective (respectively surjective, bijective), one may choose u to be injective (respectively surjective, bijective).*

Proof. We may restrict ourselves to the case $U = Y$. It is enough to prove that, if u_0 is injective (respectively surjective), then

$$\operatorname{Ker} u_x = 0 \quad (\text{respectively } \operatorname{Coker} u_x = 0)$$

for every $x \in f^{-1}(y)$. Indeed, $\operatorname{Ker} u$ and $\operatorname{Coker} u$ are coherent $\mathscr{O}_X$-modules (**0**, 5.3.4); hence there is a neighbourhood V of $f^{-1}(y)$ in X on which the restriction of $\operatorname{Ker} u$ (respectively $\operatorname{Coker} u$) is zero (**0**, 5.2.2). Since f is closed, there is an open neighbourhood $U' \subset U$ of y such that $f^{-1}(U') \subset V$, which proves the assertion after replacing U by U'.

By hypothesis, **III** | 142

$$u_x \otimes 1 : \mathscr{F}_x \otimes_{\mathscr{O}_y} k(y) \longrightarrow \mathscr{G}_x \otimes_{\mathscr{O}_y} k(y)$$

is injective (respectively surjective), $\mathscr{F}_x$ and $\mathscr{G}_x$ are finite free $\mathscr{O}_x$-modules, and $\mathscr{O}_x$ is flat over $\mathscr{O}_y$. If $u_x \otimes 1$ is injective, the injectivity of u_x follows from (**0**, 10.2.4). If $u_x \otimes 1$ is surjective, then, a fortiori, the induced homomorphism

$$\mathscr{F}_x/\mathfrak{m}_x\mathscr{F}_x \longrightarrow \mathscr{G}_x/\mathfrak{m}_x\mathscr{G}_x$$

is surjective. Since $\mathscr{G}_x$ is a finite $\mathscr{O}_x$-module and $\mathscr{O}_x$ is a local ring with maximal ideal $\mathfrak{m}_x$, the conclusion follows from Nakayama's lemma (Bourbaki, *Algèbre*, chap. VIII, §6, no. 3, cor. 4 to prop. 6). $\square$

Corollary (4.6.4). — *Let Y be a locally Noetherian prescheme, let $f : X \to Y$ be a proper flat morphism, let y be a point of Y, and put $X_y = X \times_Y \operatorname{Spec}(k(y))$. Let $\mathscr{E}_0$ be a locally free $\mathscr{O}_{X_y}$-module such that*

$$\mathrm{H}^1\big(X_y, \mathscr{H}om_{\mathscr{O}_{X_y}}(\mathscr{E}_0, \mathscr{E}_0)\big) = 0. \tag{4.6.4.1}$$

Let $\mathscr{F}$ and $\mathscr{G}$ be two locally free $\mathscr{O}_X$-modules such that $\mathscr{F}_y$ and $\mathscr{G}_y$, with the notation of (4.6.2), are isomorphic to $\mathscr{E}_0$. Then there is an open neighbourhood U of y such that $\mathscr{F}|_{f^{-1}(U)}$ and $\mathscr{G}|_{f^{-1}(U)}$ are isomorphic.

Corollary (4.6.5). — *Under the hypotheses of (4.6.4) on f, X, and Y, suppose that*

$$\mathrm{H}^1(X_y, \mathscr{O}_{X_y}) = 0.$$

If $\mathscr{F}$ and $\mathscr{G}$ are two invertible $\mathscr{O}_X$-modules such that $\mathscr{F}_y$ and $\mathscr{G}_y$ are isomorphic, then there is an open neighbourhood U of y such that $\mathscr{F}|_{f^{-1}(U)}$ and $\mathscr{G}|_{f^{-1}(U)}$ are isomorphic.

Proof. It is enough to apply (4.6.4) to the modules $\mathscr{F}^{-1} \otimes \mathscr{G}$ and $\mathscr{O}_X$. $\square$

Remarks (4.6.6). — (i) Using (4.6.5), we shall establish in Chapter V the classification of invertible sheaves on a projective bundle announced in (**II**, 4.2.7).

(ii) The result of (4.6.1) will appear in §7 as a consequence of more general propositions.

Proposition (4.6.7). — *Let Z be a locally Noetherian prescheme, and let X and Y be two Z-preschemes whose structural morphisms*

$$g : X \longrightarrow Z, \qquad h : Y \longrightarrow Z$$

are proper. Let $f : X \to Y$ be a Z-morphism, let z be a point of Z, and put

$$f_z = f \times_Z 1 : X \times_Z \operatorname{Spec}(k(z)) \longrightarrow Y \times_Z \operatorname{Spec}(k(z)).$$

(i) If f_z is finite (respectively a closed immersion), there is an open neighbourhood U of z such that the morphism

$$g^{-1}(U) \longrightarrow h^{-1}(U)$$

induced by f is finite (respectively a closed immersion).

(ii) Suppose in addition that g is flat. If f_z is an isomorphism, there is an open neighbourhood U of z such that the morphism $g^{-1}(U) \to h^{-1}(U)$ induced by f is an isomorphism.

Proof. In either case it is enough to prove that, for every $y \in h^{-1}(z)$, there is a neighbourhood V_y of y such that the restriction

$$f^{-1}(V_y) \longrightarrow V_y$$

of f is finite (respectively a closed immersion, an isomorphism). Indeed, if V is the union of the V_y, then $f^{-1}(V) \to V$ is finite (respectively a closed immersion, an isomorphism) (**II**, 6.1.1), (**I**, 4.2.4). Since h is closed, there is an open neighbourhood U of z such that $h^{-1}(U) \subset V$, and the proposition follows.

(i) First note that f is proper (**II**, 5.4.3). If f_z is finite, the required neighbourhood V_y, for each $y \in h^{-1}(z)$, exists by (4.4.11).

To treat the case in which f_z is a closed immersion, we may therefore already suppose that f is finite. Thus $X = \operatorname{Spec}(\mathscr{B})$, where $\mathscr{B}$ is a coherent $\mathscr{O}_Y$-algebra, and f corresponds (**II**, 1.2.7) to the canonical homomorphism **III | 143**

$$u : \mathscr{O}_Y \longrightarrow \mathscr{B}.$$

If, for every $y \in h^{-1}(z)$, the homomorphism $u_y : \mathscr{O}_y \to \mathscr{B}_y$ is surjective, then u is surjective on some neighbourhood V_y of y, since $\operatorname{Coker} u$ is coherent (**0**, 5.3.4), (**0**, 5.2.2). Now the finite morphism f_z corresponds to

$$v = u \otimes 1 : \mathscr{O}_Y \otimes_{\mathscr{O}_Z} k(z) \longrightarrow \mathscr{B} \otimes_{\mathscr{O}_Z} k(z),$$

and the hypothesis that f_z is a closed immersion implies that

$$u_y \otimes 1 : \mathscr{O}_y \otimes_{\mathscr{O}_z} k(z) \longrightarrow \mathscr{B}_y \otimes_{\mathscr{O}_z} k(z)$$

is surjective. Since $\mathscr{B}_y$ is a finite $\mathscr{O}_y$-module and $\mathscr{O}_y$ is a local Noetherian ring, the conclusion follows, as in (4.6.3), from Nakayama's lemma.

(ii) The same reasoning shows that it is now enough to prove that u_y is bijective, knowing that $u_y \otimes 1$ is bijective. The required injectivity follows from the lemma below; surjectivity follows from the preceding argument. □

Lemma (4.6.7.1). — *Let A and B be local Noetherian rings, let $\rho : A \to B$ be a local homomorphism, and let $u : N \to M$ be a homomorphism of B-modules. Suppose that M is flat over A, that N is a finite B-module, and that*

$$u \otimes 1 : N \otimes_A k \longrightarrow M \otimes_A k,$$

where k is the residue field of A, is injective. Then N is flat over A and u is injective.

Proof. To prove the first assertion, we must show that, for every pair of finite A-modules P, Q and every injective A-homomorphism $v : P \to Q$, the homomorphism

$$1_N \otimes v : N \otimes_A P \longrightarrow N \otimes_A Q$$

is injective. Consider the commutative diagram

$$\begin{tikzcd} N \otimes_A P \arrow[r, "1_N \otimes v"] \arrow[d, "u \otimes 1_P"'] & N \otimes_A Q \arrow[d, "u \otimes 1_Q"] \\ M \otimes_A P \arrow[r, "1_M \otimes v"] & M \otimes_A Q. \end{tikzcd}$$

Since $1_M \otimes v$ is injective by the flatness of M, it is enough to prove that $u \otimes 1_P$ is injective.

Let $\mathfrak{m}$ be the maximal ideal of A. The $\mathfrak{m}$-adic filtration on the A-module $N \otimes_A P$ is also its $\mathfrak{m}B$-adic filtration as a B-module. The topology defined by this filtration is separated: B is Noetherian, $\mathfrak{m}B$ is contained in the radical of B, and $N \otimes_A P$ is a finite B-module, since N is finite over B and P is finite over A (**0**, 7.3.5). It is therefore enough to prove that

$$\operatorname{gr}(u \otimes 1_P) : \operatorname{gr}_\bullet(N \otimes_A P) \longrightarrow \operatorname{gr}_\bullet(M \otimes_A P)$$

is injective, where the associated graded modules are taken with respect to the $\mathfrak{m}$-adic filtrations (Bourbaki, *Algèbre commutative*, chap. III, § 2, no. 8, cor. 1 to th. 1).

Since M is flat over A, the homomorphisms

$$M \otimes_A (\mathfrak{m}^n P) \longrightarrow \mathfrak{m}^n (M \otimes_A P)$$

are bijective. Consequently the canonical homomorphism

$$\phi_M : \operatorname{gr}_0(M) \otimes_A \operatorname{gr}_\bullet(P) \longrightarrow \operatorname{gr}_\bullet(M \otimes_A P)$$

is bijective.

We have the commutative diagram **III** | 144

$$\begin{tikzcd} \operatorname{gr}_0(N) \otimes_A \operatorname{gr}_\bullet(P) \arrow[r, "\operatorname{gr}_0(u)\otimes 1"] \arrow[d, "\phi_N"'] & \operatorname{gr}_0(M) \otimes_A \operatorname{gr}_\bullet(P) \arrow[d, "\phi_M"] \\ \operatorname{gr}_\bullet(N \otimes_A P) \arrow[r, "\operatorname{gr}(u\otimes 1)"] & \operatorname{gr}_\bullet(M \otimes_A P). \end{tikzcd}$$

Here ϕ_M is bijective and ϕ_N is surjective. Moreover, $\operatorname{gr}_0(u)$ is injective by hypothesis, and

$$\operatorname{gr}_0(N) \otimes_A \operatorname{gr}_\bullet(P) = \operatorname{gr}_0(N) \otimes_k \operatorname{gr}_\bullet(P),$$
$$\operatorname{gr}_0(M) \otimes_A \operatorname{gr}_\bullet(P) = \operatorname{gr}_0(M) \otimes_k \operatorname{gr}_\bullet(P).$$

Thus $\operatorname{gr}_0(u) \otimes 1$ is also injective. It follows from the diagram that $\operatorname{gr}(u \otimes 1_P)$ is injective, proving the first assertion. The second follows from the same argument by taking $P = A$. □

Proposition (4.6.8). — *Let Z be a locally Noetherian prescheme, and let X and Y be two Z-preschemes whose structural morphisms $g : X \to Z$ and $h : Y \to Z$ are proper. Let Z' be a closed subset of Z, put*

$$X' = g^{-1}(Z'), \qquad Y' = h^{-1}(Z'),$$

and denote by

$$\widehat{X} = X_{/X'}, \qquad \widehat{Y} = Y_{/Y'}, \qquad \widehat{Z} = Z_{/Z'}$$

the formal completions along these closed subsets. Let $f : X \to Y$ be a Z-morphism and let $\widehat{f} : \widehat{X} \to \widehat{Y}$ be its extension to the formal completions. In order that $\widehat{f}$ be an isomorphism (respectively a closed immersion), it is necessary and sufficient that there exist an open neighbourhood U of Z' such that the morphism

$$g^{-1}(U) \longrightarrow h^{-1}(U)$$

induced by f is an isomorphism (respectively a closed immersion).

Proof. The sufficiency of the condition is immediate (**I**, 10.14.7). To prove its necessity, it is enough, by the same reasoning as in (4.6.7), to show that for every $y \in Y'$ there is an open neighbourhood V_y of y such that $f^{-1}(V_y) \to V_y$ is an isomorphism (respectively a closed immersion). We are thus reduced to the case

$$Y = Z = \operatorname{Spec}(A),$$

where A is Noetherian.

By hypothesis (**I**, 10.9.1), (**I**, 10.14.2), the fibre $f^{-1}(y)$ consists of a single point for every $y \in Y'$. Since f is proper (**II**, 5.4.3), (4.4.11) gives an open neighbourhood U of y for which $f^{-1}(U) \to U$ is finite. We may therefore suppose that f is finite, so that

$$X = \operatorname{Spec}(B),$$

where B is a finite A-algebra.

If $Y' = V(\mathfrak{J})$, then

$$\widehat{Y} = \mathrm{Spf}(\widehat{A}), \qquad \widehat{X} = \mathrm{Spf}(\widehat{B}),$$

where $\widehat{A}$ is the separated completion of A for the $\mathfrak{J}$-adic topology, and $\widehat{B}$ is the separated completion of B for the $\mathfrak{J}B$-adic topology, or, equivalently, the separated completion of the A-module B for the $\mathfrak{J}$-adic topology. Furthermore, $\widehat{f}$ corresponds to the continuous extension

$$\widehat{\phi} : \widehat{A} \longrightarrow \widehat{B}$$

of the canonical ring homomorphism $\phi : A \to B$, and the hypothesis is that $\widehat{\phi}$ is surjective (respectively bijective) (**I**, 10.14.2). But $\widehat{\phi}$ is also the continuous extension of ϕ regarded as a homomorphism of A-modules. It follows from (**I**, 10.8.14) that there is an open neighbourhood U of Y' such that the restriction to U of the homomorphism of $\mathscr{O}_Y$-modules

$$\widetilde{\phi} : \widetilde{A} \longrightarrow \widetilde{B}$$

is surjective (respectively bijective), which completes the proof. □

4.7. An ampleness criterion

Theorem (4.7.1). — *Let Y be a locally Noetherian prescheme, let $f : X \to Y$ be a proper morphism, let $\mathscr{L}$ be an invertible $\mathscr{O}_X$-module, and let y be a point of Y. Put*

$$X_y = X \times_Y \mathrm{Spec}(k(y)) = f^{-1}(y),$$

and let $g : X_y \to X$ be the canonical morphism. If

$$\mathscr{L}_y = g^*(\mathscr{L}) = \mathscr{L} \otimes_{\mathscr{O}_Y} k(y)$$

is ample on X_y, then there is an open neighbourhood U of y in Y such that $\mathscr{L}|_{f^{-1}(U)}$ is ample for the restriction of f to $f^{-1}(U)$.

Proof. **III | 145**
I) Put

$$Y' = \mathrm{Spec}(\mathscr{O}_y), \qquad X' = X \times_Y Y', \qquad \mathscr{L}' = \mathscr{L} \otimes_{\mathscr{O}_Y} \mathscr{O}_y.$$

We first prove that $\mathscr{L}'$ is ample for $f' = f_{(Y')}$. We have the commutative diagram

$$\begin{tikzcd} X & X' \arrow[l] \arrow[d, "f'"] & X_y \arrow[l] \arrow[d, "f_y"] \\ Y \arrow[from=u, "f"'] & Y' \arrow[l] & \mathrm{Spec}(k(y)) \arrow[l] \end{tikzcd}$$

Since f' is proper (**II**, 5.4.2), (iii), and $\mathscr{O}_y$ is Noetherian, (**II**, 4.6.6) shows that we may reduce to the case

$$Y = Y' = \mathrm{Spec}(\mathscr{O}_y), \qquad X = X',$$

assume that $\mathscr{L}_y$ is ample for f_y, and prove that $\mathscr{L}$ is ample for f. We apply the criterion (2.6.1), c), and prove that for every coherent $\mathscr{O}_X$-module $\mathscr{F}$ there is an integer N such that

$$\mathrm{H}^1(X, \mathscr{F}(n)) = 0 \quad \text{for all } n \geqslant N, \qquad \mathscr{F}(n) = \mathscr{F} \otimes \mathscr{L}^{\otimes n}.$$

The point y is closed in Y and corresponds to the maximal ideal $\mathfrak{m}$ of $\mathscr{O}_y$. Hence X_y is the closed subprescheme of X defined by the coherent ideal

$$\mathscr{J} = f^*(\widetilde{\mathfrak{m}})\mathscr{O}_X = \mathfrak{m}\mathscr{O}_X$$

(**I**, 4.4.5), and g is its canonical injection. Consider the graded $k(y)$-algebra

$$S = \mathrm{gr}(\mathscr{O}_y) = \bigoplus_{j \geqslant 0} \mathfrak{m}^j / \mathfrak{m}^{j+1}.$$

It is of finite type because $\mathscr{O}_y$ is Noetherian. The $\mathscr{O}_X$-algebra

$$\mathscr{S} = f^*(\widetilde{S})$$

is quasi-coherent and of finite type, and is annihilated by $\mathscr{J}$. Thus, if $\mathscr{S}_y = g^*(\mathscr{S})$, then $\mathscr{S}_y$ is a quasi-coherent $\mathscr{O}_{X_y}$-algebra of finite type and

$$\mathscr{S} = g_*(\mathscr{S}_y).$$

Put

$$\mathscr{M}_j = \mathfrak{m}^j\mathscr{F}/\mathfrak{m}^{j+1}\mathscr{F}, \qquad \mathscr{M} = \bigoplus_{j\geqslant 0}\mathscr{M}_j = \operatorname{gr}(\mathscr{F}).$$

Since $\mathscr{F}$ is coherent, $\mathscr{M}$ is a quasi-coherent $\mathscr{S}$-module of finite type (**0**, 10.1.1); it too is annihilated by $\mathscr{J}$. Consequently, if

$$\mathscr{M}'_j = g^*(\mathscr{M}_j), \qquad \mathscr{M}' = g^*(\mathscr{M}) = \bigoplus_{j\geqslant 0}\mathscr{M}'_j,$$

then $\mathscr{M}'$ is a graded quasi-coherent $\mathscr{S}_y$-module of finite type and $\mathscr{M} = g_*(\mathscr{M}')$. Moreover, on putting

$$\mathscr{M}'_j(n) = \mathscr{M}'_j \otimes \mathscr{L}_y^{\otimes n},$$

we have $\mathscr{M}'_j(n) = g^*(\mathscr{M}_j(n))$.

The morphism f_y is proper (**II**, 5.4.2), (iii), and $\mathscr{L}_y$ is ample; hence f_y is projective (**II**, 5.5.4), (**II**, 4.6.11). We may therefore apply (2.4.1), (ii), to $\operatorname{Spec}(k(y))$, f_y, $\mathscr{S}_y$, $\mathscr{L}_y$, and $\mathscr{M}'$. There is an integer N such that, for $n \geqslant N$,

$$\mathrm{H}^q(X_y, \mathscr{M}'_j(n)) = 0$$

for every $q > 0$ and every j. It follows that

$$\mathrm{H}^q(X, \mathscr{M}_j(n)) = 0$$

for every $q > 0$ and every j (G, II, 4.9.1).

Now put

$$\mathscr{F}(n)_j = \mathscr{F}(n)/\mathfrak{m}^{j+1}\mathscr{F}(n).$$

For $j \geqslant 1$ we have

$$\mathscr{F}(n)_{j-1} = \mathscr{F}(n)_j/\mathscr{M}_j(n), \qquad \mathscr{F}(n)_0 = \mathscr{M}_0(n).$$

Thus $\mathrm{H}^1(X, \mathscr{F}(n)_0) = 0$, and the exact cohomology sequence gives, for every $j \geqslant 1$,

$$\mathrm{H}^1(X, \mathscr{F}(n)_j) = \mathrm{H}^1(X, \mathscr{F}(n)_{j-1}).$$

Hence $\mathrm{H}^1(X, \mathscr{F}(n)_j) = 0$ for every $j \geqslant 0$. By (4.2.1), it follows that

$$\mathrm{H}^1(X, \mathscr{F}(n)) = 0,$$

which proves the assertion.

II) Return to the notation at the beginning of the proof. We may suppose that $Y = \operatorname{Spec}(A)$ is affine. **III** | 146
Since f' is of finite type and $\mathscr{L}'$ is ample for f', there is an integer $m > 0$ such that $\mathscr{L}'^{\otimes m}$ is very ample for f' (**II**, 4.6.11). Replacing $\mathscr{L}$ by $\mathscr{L}^{\otimes m}$, it is enough to consider the case in which $\mathscr{L}'$ is very ample for f' and to prove that $\mathscr{L}|_{f^{-1}(U)}$ is very ample for the restriction of f.

Since f' is proper, there is, for a suitable integer $r > 0$, a closed Y'-immersion

$$j : X' \longrightarrow P = \mathbf{P}^r_{Y'}$$

such that $\mathscr{L}' \simeq j^*(\mathscr{O}_P(1))$ (**II**, 5.5.4), (ii). This closed immersion corresponds canonically to a surjective $\mathscr{O}_{X'}$-homomorphism

$$u : \mathscr{O}_{X'}^{r+1} \longrightarrow \mathscr{L}'$$

(**II**, 4.2.3), or equivalently (**0**, 5.1.1) to $r+1$ sections s'_i $(0 \leqslant i \leqslant r)$ of $\mathscr{L}'$ over X' which generate this $\mathscr{O}_{X'}$-module.

By definition these are also sections of $f'_*(\mathscr{L}')$ over Y', and

$$f'_*(\mathscr{L}') = f_*(\mathscr{L}) \otimes_{\mathscr{O}_Y} \mathscr{O}_{Y'}$$

(**0**, 5.4.10). Since Y is affine and $\mathscr{O}_y$ is the local ring of A at the prime ideal $\mathfrak{j}_y$, each s'_i has the form

$$s'_i = s''_i/t_i,$$

where s''_i is a section of $f_*(\mathscr{L})$ over Y and $t_i \in A \setminus \mathfrak{j}_y$. It follows that there are an open neighbourhood V of y in Y and sections s_i of $f_*(\mathscr{L})|_V$ such that $s'_i = s_i/1$ (the space Y' is contained in V; see (**I**, 2.4.2)).

The sections s_i are sections of $\mathscr{L}$ over $f^{-1}(V)$ and therefore define a homomorphism

$$v : \left(\mathscr{O}_X|_{f^{-1}(V)}\right)^{r+1} \longrightarrow \mathscr{L}|_{f^{-1}(V)}.$$

By construction it is surjective at every point of $f^{-1}(y)$. Since $\mathrm{Coker}(v)$ is coherent (**0**, 5.3.4), its support is closed (**0**, 5.2.2); hence there is an open neighbourhood $W \subset f^{-1}(V)$ of $f^{-1}(y)$ on which v is surjective. Because f is closed, after shrinking around y we may choose an open neighbourhood $U \subset V$ with $f^{-1}(U) \subset W$. The conclusion now follows from (**II**, 4.2.3). ☐

4.8. Finite morphisms of formal preschemes

Proposition (4.8.1). — *Let $\mathfrak{Y}$ be a locally Noetherian formal prescheme, $\mathscr{K}$ an ideal of definition of $\mathfrak{Y}$, and $f : \mathfrak{X} \to \mathfrak{Y}$ a morphism of formal preschemes. The following conditions are equivalent:*

a) *$\mathfrak{X}$ is locally Noetherian, f is an adic morphism* (**I**, 10.12.1)*, and, if $\mathscr{J} = f^*(\mathscr{K})\mathscr{O}_{\mathfrak{X}}$, the morphism*

$$f_0 : (\mathfrak{X}, \mathscr{O}_{\mathfrak{X}}/\mathscr{J}) \longrightarrow (\mathfrak{Y}, \mathscr{O}_{\mathfrak{Y}}/\mathscr{K})$$

induced by f is finite.

b) *$\mathfrak{X}$ is locally Noetherian and is the inductive limit of an adic inductive (Y_n)-system (X_n) such that the morphism $X_0 \to Y_0$ is finite.*

c) *Every point of $\mathfrak{Y}$ has a Noetherian formal affine open neighbourhood V such that $f^{-1}(V)$ is a formal affine open and $\Gamma(f^{-1}(V), \mathscr{O}_{\mathfrak{X}})$ is a $\Gamma(V, \mathscr{O}_{\mathfrak{Y}})$-module of finite type.*

Proof. It is immediate from (**I**, 10.12.3) that *a)* implies *b)*. To see that *b)* implies *c)*, we may suppose that

$$\mathfrak{Y} = \mathrm{Spf}(B), \qquad \mathfrak{X} = \mathrm{Spf}(A),$$

where B is an adic Noetherian ring and $\mathfrak{K}$ is an ideal of definition of B. By hypothesis, X_0 is an affine scheme whose ring A_0 is a $B/\mathfrak{K}$-module of finite type (**II**, 6.1.3). By (**I**, 5.1.9), each X_n is an affine scheme; if A_n denotes its ring, condition *b)* implies that, for $m \leqslant n$,

$$A_m \simeq A_n/\mathfrak{K}^{m+1}A_n.$$

It follows that $\mathfrak{X} \simeq \mathrm{Spf}(A)$, where $A = \varprojlim A_n$, and the conclusion follows from (**0**, 7.2.9).

Finally, to prove that *c)* implies *a)*, we may again restrict to the case

$$\mathfrak{Y} = \mathrm{Spf}(B), \qquad \mathfrak{X} = \mathrm{Spf}(A),$$

where A is a finite B-algebra. Since $A/\mathfrak{K}A$ is then a finite $B/\mathfrak{K}$-algebra, (**I**, 10.10.9) shows that the conditions in *a)* are satisfied. ☐

Definition (4.8.2). — When the equivalent conditions *a)*, *b)*, and *c)* of (4.8.1) are satisfied, the morphism f is said to be *finite*; one also says that $\mathfrak{X}$ is a *finite formal $\mathfrak{Y}$-prescheme*, or a formal prescheme *finite over* $\mathfrak{Y}$.

Proposition (4.8.3). — (i) *A closed immersion of locally Noetherian formal preschemes is a finite morphism.*

(ii) *The composite of two finite morphisms of locally Noetherian formal preschemes is a finite morphism.*

(iii) *Let $\mathfrak{X}$, $\mathfrak{Y}$, and $\mathfrak{S}$ be locally Noetherian formal preschemes, $f : \mathfrak{X} \to \mathfrak{S}$ a finite morphism, and $g : \mathfrak{Y} \to \mathfrak{S}$ a morphism. Then the projection*

$$\mathfrak{X} \times_{\mathfrak{S}} \mathfrak{Y} \longrightarrow \mathfrak{Y}$$

is finite.

(iv) *Let $\mathfrak{S}$ be a locally Noetherian formal prescheme, and let $\mathfrak{X}'$ and $\mathfrak{Y}'$ be locally Noetherian formal preschemes such that $\mathfrak{X}' \times_{\mathfrak{S}} \mathfrak{Y}'$ is locally Noetherian. If $\mathfrak{X}$ and $\mathfrak{Y}$ are locally Noetherian formal $\mathfrak{S}$-preschemes and*

$$f : \mathfrak{X} \longrightarrow \mathfrak{X}', \qquad g : \mathfrak{Y} \longrightarrow \mathfrak{Y}'$$

are finite $\mathfrak{S}$-morphisms, then $f \times_{\mathfrak{S}} g$ is finite.

(v) *Let*

$$f : \mathfrak{X} \longrightarrow \mathfrak{Y}, \qquad g : \mathfrak{Y} \longrightarrow \mathfrak{Z}$$

be morphisms of locally Noetherian formal preschemes such that g is of finite type and separated. If $g \circ f$ is finite, then f is finite.

Proof. Assertion *(i)* is immediate. Using criterion *a)* of (4.8.1), the other assertions reduce at once to the corresponding propositions for morphisms of ordinary preschemes (**II**, 6.1.5). We leave the details to the reader, following the model of (**I**, 10.13.5). □

Corollary (4.8.4). — *Under the hypotheses of* (**I**, 10.9.9), *if f is a finite morphism, then so is its extension $\widehat{f}$ to the completions.*

Corollary (4.8.5). — *Let $\mathfrak{X}$ be a formal prescheme finite over $\mathfrak{Y}$, and let $f : \mathfrak{X} \to \mathfrak{Y}$ be the structural morphism. For every open subset $U \subset \mathfrak{Y}$, the formal prescheme $f^{-1}(U)$ is finite over U.*

Proposition (4.8.6). — *If $f : \mathfrak{X} \to \mathfrak{Y}$ is a finite morphism of locally Noetherian formal preschemes, then $f_*(\mathscr{O}_\mathfrak{X})$ is a coherent $\mathscr{O}_\mathfrak{Y}$-algebra.*

Proof. We may regard f as the inductive limit of an inductive system (f_n) of morphisms $f_n : X_n \to Y_n$. Let us show that the f_n are finite and that $f_*(\mathscr{O}_\mathfrak{X})$ is isomorphic to the projective limit of the $(f_n)_*(\mathscr{O}_{X_n})$; this will prove the assertion (**I**, 10.10.5). It is enough to restrict to the case

$$\mathfrak{Y} = \operatorname{Spf}(B), \qquad \mathfrak{X} = \operatorname{Spf}(A),$$

and to observe that, if $\mathfrak{K}$ is an ideal of definition of B and A is a B-module of finite type, then $A/\mathfrak{K}^{n+1}A$ is a module of finite type over $B/\mathfrak{K}^{n+1}B$, and A is the projective limit of the $A/\mathfrak{K}^{n+1}A$. □

Conversely:

Proposition (4.8.7). — *Let $\mathfrak{Y}$ be a locally Noetherian formal prescheme and $\mathscr{A}$ a coherent $\mathscr{O}_\mathfrak{Y}$-algebra. There exists a formal prescheme $\mathfrak{X}$ finite over $\mathfrak{Y}$, unique up to a unique $\mathfrak{Y}$-isomorphism, such that*

$$f_*(\mathscr{O}_\mathfrak{X}) = \mathscr{A},$$

where $f : \mathfrak{X} \to \mathfrak{Y}$ is the structural morphism.

Proof. Let $\mathscr{K}$ be an ideal of definition of $\mathfrak{Y}$, and set III | 148

$$Y_n = (\mathfrak{Y}, \mathscr{O}_\mathfrak{Y}/\mathscr{K}^{n+1}), \qquad \mathscr{A}_n = \mathscr{A}/\mathscr{K}^{n+1}\mathscr{A}.$$

Clearly $\mathscr{A}_n$ is a finite $\mathscr{O}_{Y_n}$-algebra, and hence defines a finite Y_n-prescheme

$$X_n = \operatorname{Spec}(\mathscr{A}_n)$$

(**II**, 6.1.3). For $m \leqslant n$, the canonical surjective homomorphism

$$h_{mn} : \mathscr{A}_n \longrightarrow \mathscr{A}_m$$

defines a morphism $u_{mn} : X_m \to X_n$ for which the diagram

$$\begin{tikzcd} X_n \arrow[d, "f_n"'] & X_m \arrow[l, "u_{mn}"'] \arrow[d, "f_m"] \\ Y_n & Y_m \arrow[l] \end{tikzcd}$$

is commutative, where f_n denotes the structural morphism; it also identifies X_m with $X_n \times_{Y_n} Y_m$, as follows at once from (**II**, 1.4.6). The formal prescheme $\mathfrak{X}$ obtained as the inductive limit of the inductive system (X_n) is therefore locally Noetherian, and its structural morphism

$$f : \mathfrak{X} \longrightarrow \mathfrak{Y},$$

the inductive limit of the system (f_n), is finite (4.8.1) (**I**, 10.12.3.1). Moreover, the proof of (4.8.6) shows that $f_*(\mathscr{O}_\mathfrak{X})$ is the projective limit of the $\mathscr{A}_n$, and is therefore equal to $\mathscr{A}$ (**I**, 10.10.6). The uniqueness assertion follows from the more general result below. □

Proposition (4.8.8). — *Let $\mathfrak{Y}$ be a locally Noetherian formal prescheme, and let $\mathfrak{X}$ and $\mathfrak{X}'$ be formal $\mathfrak{Y}$-preschemes finite over $\mathfrak{Y}$, with structural morphisms*

$$f : \mathfrak{X} \longrightarrow \mathfrak{Y}, \qquad f' : \mathfrak{X}' \longrightarrow \mathfrak{Y}.$$

There is a canonical bijection

$$\operatorname{Hom}_\mathfrak{Y}(\mathfrak{X}, \mathfrak{X}') \xrightarrow{\sim} \operatorname{Hom}_{\mathscr{O}_\mathfrak{Y}\text{-}alg}\big(f'_*(\mathscr{O}_{\mathfrak{X}'}), f_*(\mathscr{O}_\mathfrak{X})\big).$$

Proof. The map $h \mapsto \mathscr{A}(h)$ is defined as in (**II**, 1.1.2). To prove that it is bijective, we reduce at once to the case in which $\mathfrak{Y} = \mathrm{Spf}(B)$ is a Noetherian formal affine prescheme. Then

$$\mathfrak{X} = \mathrm{Spf}(A), \qquad \mathfrak{X}' = \mathrm{Spf}(A'),$$

where A and A' are finite B-algebras, and

$$f_*(\mathscr{O}_{\mathfrak{X}}) = A^{\Delta}, \qquad f'_*(\mathscr{O}_{\mathfrak{X}'}) = A'^{\Delta}.$$

The conclusion follows from the bijective correspondence between $\mathfrak{Y}$-morphisms $\mathfrak{X} \to \mathfrak{X}'$ and continuous B-algebra homomorphisms $A' \to A$ (**I**, 10.2.2), together with the bijective correspondence between homomorphisms of B-modules $A' \to A$ and homomorphisms of $\mathscr{O}_{\mathfrak{Y}}$-modules $A'^{\Delta} \to A^{\Delta}$ (**I**, 10.10.2.3). ☐

Corollary (4.8.9). — *Under the canonical bijection of (4.8.8), closed immersions $\mathfrak{X} \to \mathfrak{X}'$ correspond to surjective homomorphisms of $\mathscr{O}_{\mathfrak{Y}}$-algebras*

$$f'_*(\mathscr{O}_{\mathfrak{X}'}) \longrightarrow f_*(\mathscr{O}_{\mathfrak{X}}).$$

Proof. The question is local on $\mathfrak{Y}$, and hence reduces to the definition of closed immersions of locally Noetherian formal preschemes (**I**, 10.14.2). ☐

Corollary (4.8.10). — *With the notation and hypotheses of (4.8.1), an adic morphism f is a closed immersion if and only if f_0 is a closed immersion of ordinary preschemes.*

Proof. This follows at once from (4.8.9) and the surjectivity criterion for a homomorphism of coherent $\mathscr{O}_{\mathfrak{Y}}$-modules (**I**, 10.11.5). ☐

Proposition (4.8.11). — *A morphism $f : \mathfrak{X} \to \mathfrak{Y}$ of locally Noetherian formal preschemes is finite if and only if it is proper and all its fibres $f^{-1}(y)$, for $y \in \mathfrak{Y}$, are finite.* **III** | 149

Proof. By the definitions (3.4.1) and (4.8.2), the assertion reduces immediately to the same assertion for f_0 (with the notation of (4.8.1)), which is precisely (4.4.2). ☐

§5. An existence theorem for coherent algebraic sheaves

5.1. Statement of the theorem

(5.1.1). Let A be an adic Noetherian ring and $\mathfrak{J}$ an ideal of definition of A, so that A is separated and complete for the $\mathfrak{J}$-adic topology. If $Y = \mathrm{Spec}(A)$, the affine formal prescheme $\mathrm{Spf}(A)$ is identified with the completion $\widehat{Y}$ of Y along the closed subset $Y' = \mathrm{V}(\mathfrak{J})$ (**I**, 10.10.1). Let X be an ordinary Y-prescheme of finite type, with structural morphism $f : X \to Y$. We denote by $\widehat{X}$ the completion of X along the closed subset $X' = f^{-1}(Y')$, or equivalently the formal $\widehat{Y}$-prescheme $X \times_Y \widehat{Y}$, and by $\widehat{f} : \widehat{X} \to \widehat{Y}$ the extension of f to the completions. Finally, for every coherent $\mathscr{O}_X$-module $\mathscr{F}$, we denote by $\widehat{\mathscr{F}}$ its completion $\mathscr{F}_{/X'}$, which is a coherent $\mathscr{O}_{\widehat{X}}$-module.

Proposition (5.1.2). — *With the hypotheses and notation of (5.1.1), let $\mathscr{F}$ be a coherent $\mathscr{O}_X$-module whose support is proper over Y* (**II**, 5.4.10). *The canonical homomorphisms (4.1.4)*

$$\rho_i : \mathrm{H}^i(X, \mathscr{F}) \longrightarrow \mathrm{H}^i(\widehat{X}, \widehat{\mathscr{F}})$$

are isomorphisms.

Proof. The A-module $\mathrm{H}^i(X, \mathscr{F})$ is of finite type (3.2.4), and is therefore identical with its separated completion for the $\mathfrak{J}$-preadic topology (**0**, 7.3.6). The proposition is consequently a special case of (4.1.10). ☐

Recall that the canonical isomorphisms ρ_i commute with the connecting homomorphisms associated with every exact sequence

$$0 \longrightarrow \mathscr{F}' \longrightarrow \mathscr{F} \longrightarrow \mathscr{F}'' \longrightarrow 0$$

of coherent $\mathscr{O}_X$-modules (**0**, 12.1.6) (**I**, 10.8.9).

Corollary (5.1.3). — *Let $\mathscr{F}$ and $\mathscr{G}$ be coherent $\mathscr{O}_X$-modules such that the intersection of their supports is proper over Y. The canonical homomorphism*

$$\text{(5.1.3.1)} \qquad \operatorname{Hom}_{\mathscr{O}_X}(\mathscr{F},\mathscr{G}) \longrightarrow \operatorname{Hom}_{\mathscr{O}_{\widehat{X}}}(\widehat{\mathscr{F}},\widehat{\mathscr{G}})$$

which sends a homomorphism $u : \mathscr{F} \to \mathscr{G}$ to its completion $\widehat{u} : \widehat{\mathscr{F}} \to \widehat{\mathscr{G}}$, is an isomorphism. Moreover, when f is closed, $\widehat{u}$ is injective (respectively, surjective) if and only if u is injective (respectively, surjective).

Proof. The first assertion is a special case of (4.5.3); it also follows from the fact that the source of (5.1.3.1) is an A-module of finite type and is therefore identical with its separated completion. For the second assertion, note that, by (**I**, 10.8.14), $\widehat{u}$ is injective (respectively, surjective) if and only if there is a neighbourhood of X' in X on which u is injective (respectively, surjective). The conclusion therefore follows from the lemma below. □

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Lemma (5.1.3.1). — *Under the hypotheses of (5.1.1), suppose in addition that f is closed. Every neighbourhood of X' in X is then equal to X.*

Proof. We first reduce to the case $f(X) = Y$. Indeed, by hypothesis, $f(X)$ is a closed subset Z of Y. We may replace f by f_{red} (**I**, 6.3.4), and hence suppose that X and Y are reduced; we may then replace Y by the reduced closed subprescheme having Z as its underlying space (**I**, 5.2.2), since every ideal of A is closed and every quotient ring of A is therefore adic and Noetherian.

We then have $f(X') = Y'$. If V is an open neighbourhood of X' in X, the set $f(X - V)$ is closed in Y and does not meet Y'. This is impossible unless $X - V$ is empty, because $\mathfrak{J}$ is contained in the radical of A (**I**, 1.1.15) (**0**, 7.1.10). Thus $V = X$. □

If we restrict attention to coherent $\mathscr{O}_X$-modules whose support is proper over Y, (5.1.3) says, in categorical language, that the functor

$$\mathscr{F} \longmapsto \widehat{\mathscr{F}}$$

is fully faithful from the category of such $\mathscr{O}_X$-modules to the category of coherent $\mathscr{O}_{\widehat{X}}$-modules. It thus gives an equivalence of the former category with a full subcategory of the latter (**0**, 8.1.6). The existence theorem will show that, when X is proper over Y, this subcategory is in fact the category of all coherent $\mathscr{O}_{\widehat{X}}$-modules. More precisely:

Theorem (5.1.4). — *Let A be an adic Noetherian ring, let $Y = \operatorname{Spec}(A)$, let $\mathfrak{J}$ be an ideal of definition of A, and put $Y' = \mathrm{V}(\mathfrak{J})$. Let $f : X \to Y$ be a separated morphism of finite type and put $X' = f^{-1}(Y')$. Let*

$$\widehat{Y} = Y_{/Y'} = \operatorname{Spf}(A), \qquad \widehat{X} = X_{/X'}$$

be the completions of Y and X along Y' and X', and let $\widehat{f} : \widehat{X} \to \widehat{Y}$ be the extension of f to the completions. Then the functor

$$\mathscr{F} \longmapsto \mathscr{F}_{/X'} = \widehat{\mathscr{F}}$$

is an equivalence from the category of coherent $\mathscr{O}_X$-modules whose support is proper over $\operatorname{Spec}(A)$ to the category of coherent $\mathscr{O}_{\widehat{X}}$-modules whose support is proper over $\operatorname{Spf}(A)$.

In other words, in view of (5.1.3):

Corollary (5.1.5). — *An $\mathscr{O}_{\widehat{X}}$-module is isomorphic to the completion of a coherent $\mathscr{O}_X$-module whose support is proper over $\operatorname{Spec}(A)$ if and only if it is coherent and its support is proper over $\operatorname{Spf}(A)$.*

The most important case is the following:

Corollary (5.1.6). — *Suppose that X is proper over $Y = \operatorname{Spec}(A)$. Then the functor $\mathscr{F} \mapsto \widehat{\mathscr{F}}$ is an equivalence between the category of coherent $\mathscr{O}_X$-modules and the category of coherent $\mathscr{O}_{\widehat{X}}$-modules.*

(5.1.7). *Scholium.* Taking account of the characterization of coherent sheaves on formal preschemes (**I**, 10.11.3), we see that, under the conditions of (5.1.1), giving a coherent $\mathscr{O}_X$-module whose support is proper over $\operatorname{Spec}(A)$ is equivalent—if

$$Y_n = \operatorname{Spec}(A/\mathfrak{J}^{n+1}), \qquad X_n = X \times_Y Y_n$$

—to giving a projective system $(\mathscr{F}_n)$ of coherent $\mathscr{O}_{X_n}$-modules such that, for $m \leqslant n$,

$$\mathscr{F}_m = \mathscr{F}_n \otimes_{\mathscr{O}_{Y_n}} \mathscr{O}_{Y_m}, \qquad \text{or equivalently} \qquad \mathscr{F}_m = \mathscr{F}_n / \mathfrak{J}^{m+1} \mathscr{F}_n,$$

and such that the support of $\mathscr{F}_0$ is a subset of X_0 proper over Y_0. By (**I**, 10.11.4), homomorphisms of coherent $\mathscr{O}_X$-modules are interpreted similarly as homomorphisms of projective systems of coherent $\mathscr{O}_{X_n}$-modules.

In all known applications, A is in fact a local adic Noetherian ring. Thus the Y_n are spectra of local Artinian rings, and the results of this subsection and of the preceding ones reduce, to a large extent, algebraic geometry over a local adic Noetherian ring to algebraic geometry over local Artinian rings. **III** | 151

Corollary (5.1.8). — *Under the hypotheses of (5.1.4), the map*

$$Z \longmapsto \widehat{Z} = Z_{/(Z \cap X')}$$

is a bijection from the set of closed subpreschemes Z of X which are proper over Y onto the set of closed formal subpreschemes of $\widehat{X}$ which are proper over $\widehat{Y}$.

Proof. A closed formal subprescheme of $\widehat{X}$ has the form

$$\left(T, (\mathscr{O}_{\widehat{X}}/\mathscr{A})|_T\right),$$

where $\mathscr{A}$ is a coherent ideal of $\mathscr{O}_{\widehat{X}}$ (**I**, 10.14.2). If T is proper over $\widehat{Y}$, (5.1.4) shows that $\mathscr{O}_{\widehat{X}}/\mathscr{A}$ is isomorphic to an $\mathscr{O}_{\widehat{X}}$-module of the form $\widehat{\mathscr{F}}$, where $\mathscr{F}$ is a coherent $\mathscr{O}_X$-module whose support is proper over Y. Moreover, (5.1.3) shows that the canonical homomorphism

$$\mathscr{O}_{\widehat{X}} \longrightarrow \mathscr{O}_{\widehat{X}}/\mathscr{A}$$

has the form $\widehat{u}$, where $u : \mathscr{O}_X \to \mathscr{F}$ is a surjective homomorphism of $\mathscr{O}_X$-modules. Consequently $\mathscr{F} = \mathscr{O}_X/\mathscr{N}$ for a coherent ideal $\mathscr{N}$ of $\mathscr{O}_X$, and $\mathscr{A} = \widehat{\mathscr{N}}$ (**I**, 10.8.8). The conclusion now follows from (**I**, 10.14.7). □

5.2. Proof of the existence theorem: the projective and quasi-projective case

(5.2.1). Under the conditions of (5.1.4), we shall say provisionally that a coherent $\mathscr{O}_{\widehat{X}}$-module is *algebraizable* if it is isomorphic to the completion $\widehat{\mathscr{F}}$ of a coherent $\mathscr{O}_X$-module $\mathscr{F}$ whose support is proper over Y.

Lemma (5.2.2). — *Let $\mathscr{F}'$ and $\mathscr{G}'$ be algebraizable $\mathscr{O}_{\widehat{X}}$-modules. For every homomorphism $u : \mathscr{F}' \to \mathscr{G}'$, the modules* $\operatorname{Ker}(u)$, $\operatorname{Im}(u)$, *and* $\operatorname{Coker}(u)$ *are algebraizable.*

Proof. Write

$$\mathscr{F}' = \widehat{\mathscr{F}}, \qquad \mathscr{G}' = \widehat{\mathscr{G}},$$

where $\mathscr{F}$ and $\mathscr{G}$ are coherent $\mathscr{O}_X$-modules whose supports are proper over Y. By (5.1.3), $u = \widehat{v}$ for a homomorphism $v : \mathscr{F} \to \mathscr{G}$. Since completion is exact, $\operatorname{Ker}(\widehat{v}) \simeq \widehat{\operatorname{Ker}(v)}$; and because the support of $\operatorname{Ker}(v)$ is contained in that of $\mathscr{F}$, it follows that $\operatorname{Ker}(u)$ is algebraizable. The arguments for $\operatorname{Im}(u)$ and $\operatorname{Coker}(u)$ are the same. □

Proposition (5.2.3). — *Let A be an adic Noetherian ring, $\mathfrak{J}$ an ideal of definition of A, $\mathfrak{Y} = \operatorname{Spf}(A)$, and $f : \mathfrak{X} \to \mathfrak{Y}$ a proper morphism of formal preschemes. Put*

$$Y_k = \operatorname{Spec}(A/\mathfrak{J}^{k+1}), \qquad X_k = \mathfrak{X} \times_{\mathfrak{Y}} Y_k,$$

and, for every $\mathscr{O}_{\mathfrak{X}}$-module $\mathscr{F}$, put

$$\mathscr{F}_k = \mathscr{F} \otimes_{\mathscr{O}_{\mathfrak{X}}} \mathscr{O}_{X_k} = \mathscr{F}/\mathfrak{J}^{k+1}\mathscr{F}.$$

Let $\mathscr{L}$ be an invertible $\mathscr{O}_{\mathfrak{X}}$-module, and suppose that $\mathscr{L}_0 = \mathscr{L}/\mathfrak{J}\mathscr{L}$ is an ample $\mathscr{O}_{X_0}$-module. For every $\mathscr{O}_{\mathfrak{X}}$-module $\mathscr{F}$ and every integer n, set

$$\mathscr{F}(n) = \mathscr{F} \otimes \mathscr{L}^{\otimes n}.$$

Then, for every coherent $\mathscr{O}_{\mathfrak{X}}$-module $\mathscr{F}$, there is an integer n_0 such that, for every $n \geqslant n_0$:

(i) *the canonical homomorphism*

$$\mathrm{H}^0(\mathfrak{X}, \mathscr{F}(n)) \longrightarrow \mathrm{H}^0(\mathfrak{X}, \mathscr{F}_k(n))$$

is surjective for every $k \geqslant 0$*;*

(ii) $\mathrm{H}^q(\mathfrak{X}, \mathscr{F}(n)) = 0$ *for every* $q > 0$.

Proof. The underlying spaces of $\mathfrak{X}$ and X_0 are the same. The sheaves

$$\mathscr{M}_k = \mathfrak{J}^k \mathscr{F} / \mathfrak{J}^{k+1} \mathscr{F},$$

being annihilated by $\mathfrak{J}$, may be regarded as coherent $\mathscr{O}_{X_0}$-modules (**0**, 5.3.10). Moreover, if

$$\mathscr{M}_k(n) = \mathscr{M}_k \otimes_{\mathscr{O}_{X_0}} \mathscr{L}_0^{\otimes n},$$

then

$$\mathscr{M}_k(n) = \mathfrak{J}^k \mathscr{F}(n) / \mathfrak{J}^{k+1} \mathscr{F}(n).$$

Since $\mathscr{L}_0$ is ample for $f_0 : X_0 \to Y_0$ and f_0 is proper, f_0 is projective (**II**, 5.5.4). Let **III | 152**

$$S = \bigoplus_{k \geqslant 0} \mathfrak{J}^k / \mathfrak{J}^{k+1}$$

be the graded A-algebra associated with the $\mathfrak{J}$-adic filtration of A. It is of finite type because A is Noetherian. Put

$$\mathscr{S}' = f_0^*(\widetilde{S}), \qquad \mathscr{M} = \bigoplus_{k \geqslant 0} \mathscr{M}_k.$$

Then $\mathscr{S}'$ is a quasi-coherent $\mathscr{O}_{X_0}$-algebra of finite type, and $\mathscr{M}$ is a quasi-coherent graded $\mathscr{S}'$-module of finite type, since $\mathscr{F}_0$ is coherent and generates $\mathscr{M}$ as an $\mathscr{S}'$-module. Theorem (2.4.1), *(ii)*, therefore gives an integer n_0 such that, for $n \geqslant n_0$ and every k,

(5.2.3.1) $$\mathrm{H}^q(X_0, \mathscr{M}_k(n)) = 0 \qquad (q > 0).$$

Regarded now as an $\mathscr{O}_{\mathfrak{X}}$-module, $\mathscr{M}_k(n)$ consequently also satisfies

$$\mathrm{H}^q(\mathfrak{X}, \mathscr{M}_k(n)) = 0 \qquad (q > 0,\ n \geqslant n_0).$$

Apply the exact cohomology sequence to

$$0 \longrightarrow \frac{\mathfrak{J}^k \mathscr{F}(n)}{\mathfrak{J}^{k+1} \mathscr{F}(n)} \longrightarrow \frac{\mathfrak{J}^h \mathscr{F}(n)}{\mathfrak{J}^{k+1} \mathscr{F}(n)} \longrightarrow \frac{\mathfrak{J}^h \mathscr{F}(n)}{\mathfrak{J}^{k} \mathscr{F}(n)} \longrightarrow 0.$$

For $0 \leqslant h < k$, $n \geqslant n_0$, and $q > 0$, induction on $k - h$ gives

(5.2.3.2) $$\mathrm{H}^q\left(\mathfrak{X}, \frac{\mathfrak{J}^h \mathscr{F}(n)}{\mathfrak{J}^k \mathscr{F}(n)}\right) = 0,$$

and in particular, for $h = 0$,

(5.2.3.3) $$\mathrm{H}^q(\mathfrak{X}, \mathscr{F}_k(n)) = 0 \qquad (n \geqslant n_0,\ k \geqslant 0,\ q > 0).$$

Another part of the exact cohomology sequence, with $h = 0$, gives

(5.2.3.4) $$\mathrm{H}^0(\mathfrak{X}, \mathscr{F}_{k+1}(n)) \longrightarrow \mathrm{H}^0(\mathfrak{X}, \mathscr{F}_k(n)) \longrightarrow \mathrm{H}^1\left(\mathfrak{X}, \frac{\mathfrak{J}^k \mathscr{F}(n)}{\mathfrak{J}^{k+1} \mathscr{F}(n)}\right) = 0.$$

It follows that, for $h \leqslant k$, the canonical map

(5.2.3.5) $$\mathrm{H}^0(\mathfrak{X}, \mathscr{F}_k(n)) \longrightarrow \mathrm{H}^0(\mathfrak{X}, \mathscr{F}_h(n))$$

is surjective. Thus, for every q, the projective system

$$\left(\mathrm{H}^q(\mathfrak{X}, \mathscr{F}_k(n))\right)_{k \geqslant 0}$$

satisfies condition (ML) when $n \geqslant n_0$.

Furthermore, every formal affine open U of $\mathfrak{X}$ is an affine open in each X_k (**I**, 10.5.2). Hence $\mathrm{H}^q(U, \mathscr{F}_k(n)) = 0$ for $q > 0$ (**I**, 1.3.1), and

$$\mathrm{H}^0(U, \mathscr{F}_k(n)) \longrightarrow \mathrm{H}^0(U, \mathscr{F}_h(n))$$

is surjective for $h \leqslant k$ (**I**, 1.3.9). The hypotheses of (**0**, 13.3.1) are therefore satisfied. For $n \geqslant n_0$ we obtain:

1° for every $q > 0$, the map

$$\mathrm{H}^q(\mathfrak{X}, \mathscr{F}(n)) \longrightarrow \varprojlim_k \mathrm{H}^q(\mathfrak{X}, \mathscr{F}_k(n))$$

is bijective; hence, by (5.2.3.3), $\mathrm{H}^q(\mathfrak{X}, \mathscr{F}(n)) = 0$;

2° the homomorphism

$$\mathrm{H}^0(\mathfrak{X}, \mathscr{F}(n)) \longrightarrow \varprojlim_k \mathrm{H}^0(\mathfrak{X}, \mathscr{F}_k(n))$$

is bijective. Since the homomorphisms (5.2.3.5) are surjective, so is each homomorphism

$$\varprojlim_k \mathrm{H}^0(\mathfrak{X}, \mathscr{F}_k(n)) \longrightarrow \mathrm{H}^0(\mathfrak{X}, \mathscr{F}_h(n)).$$

This proves both assertions. □

Corollary (5.2.4). — *Under the hypotheses of (5.2.3), for every coherent $\mathscr{O}_{\mathfrak{X}}$-module $\mathscr{F}$ there is an integer N such that, for $n \geqslant N$, $\mathscr{F}(n)$ is generated by its sections over $\mathfrak{X}$. Equivalently, $\mathscr{F}$ is isomorphic to a quotient* **III | 153** *of an $\mathscr{O}_{\mathfrak{X}}$-module of the form $(\mathscr{O}_{\mathfrak{X}}(-n))^k$.*

Proof. Since X_0 is Noetherian, the hypothesis on $\mathscr{L}_0$ and (**II**, 4.5.5) give an integer n_0 such that $\mathscr{F}_0(n)$ is generated by its sections over $\mathfrak{X}$ for $n \geqslant n_0$. We may take n_0 large enough that

$$\Gamma(\mathfrak{X}, \mathscr{F}(n)) \longrightarrow \Gamma(\mathfrak{X}, \mathscr{F}_0(n))$$

is also surjective for $n \geqslant n_0$ (5.2.3). There are therefore finitely many sections $s_i \in \Gamma(\mathfrak{X}, \mathscr{F}(n))$ whose images generate $\mathscr{F}_0(n)$ (**0**, 5.2.3). Since $\mathfrak{J}$ is contained in the maximal ideal of the local ring at every point of $\mathfrak{X}$, Nakayama's lemma applied to those local rings shows that the s_i generate $\mathscr{F}(n)$ (**0**, 5.1.1). □

(5.2.5). *Proof of the existence theorem: the projective case.* With the notation of (5.1.4), suppose that f is projective, so that there is an ample $\mathscr{O}_X$-module $\mathscr{L}$ (**I**, 5.5.4). By definition,

$$X_n = \widehat{X} \times_{\widehat{Y}} Y_n$$

is the closed subprescheme $X \times_Y Y_n = f^{-1}(Y_n)$ of X. If

$$\mathscr{L}' = \mathscr{L} \otimes_{\mathscr{O}_X} \mathscr{O}_{\widehat{X}}$$

is the completion of $\mathscr{L}$, then $\mathscr{L}'_0 = \mathscr{L}/\mathfrak{J}\mathscr{L}$, regarded as an $\mathscr{O}_{X_0}$-module; this module is ample (**II**, 4.6.13, (i *bis*)).

We may therefore apply (5.2.4) to $\mathscr{L}'$ and to any coherent $\mathscr{O}_{\widehat{X}}$-module $\mathscr{F}$. It follows that $\mathscr{F}$ is isomorphic to a quotient of

$$\mathscr{G} = \left(\mathscr{L}'^{\otimes(-n)}\right)^k$$

for suitable integers $n > 0$ and $k > 0$. The module $\mathscr{G}$ is the completion of $(\mathscr{L}^{\otimes(-n)})^k$ (**I**, 10.8.10), and is therefore algebraizable. Let $u : \mathscr{G} \to \mathscr{F}$ be the canonical surjection, and put $\mathscr{H} = \operatorname{Ker}(u)$; this is a coherent $\mathscr{O}_{\widehat{X}}$-module (**0**, 5.3.4). Applying the same argument to $\mathscr{H}$, we find an algebraizable $\mathscr{O}_{\widehat{X}}$-module $\mathscr{K}$ and a homomorphism $v : \mathscr{K} \to \mathscr{G}$ such that $\mathscr{H} = \operatorname{Im}(v)$. Thus $\mathscr{F} = \operatorname{Coker}(v)$, and $\mathscr{F}$ is algebraizable by (5.2.2).

(5.2.6). *Proof of the existence theorem: the quasi-projective case.* Retain the notation of (5.1.4), and now suppose that f is quasi-projective. There is a projective morphism $g : Z \to Y$ such that X is identified with the Y-prescheme induced on an open subset of Z (**II**, 5.3.2). If $Z' = g^{-1}(Y')$, then $X' = X \cap Z'$. Consequently the completion $\widehat{X} = X_{/X'}$ is identified with the formal prescheme induced by $\widehat{Z} = Z_{/Z'}$ on the open subset $X \cap Z'$ of $\widehat{Z}$ (**I**, 10.8.5).

Let $\mathscr{F}'$ be a coherent $\mathscr{O}_{\widehat{X}}$-module whose support T' is proper over $\widehat{Y}$. By definition, there is a closed subprescheme of X' with underlying space $T' \subset X'$ such that the restriction $T' \to Y'$ of f is proper. Hence T' is proper over Y, and is therefore closed in Z' (**II**, 5.4.10). It follows that $\mathscr{F}'$ is induced on $\widehat{X}$ by the $\mathscr{O}_{\widehat{Z}}$-module $\mathscr{G}'$ obtained by gluing $\mathscr{F}'$, on the open $\widehat{X}$, with the zero sheaf on the open $\widehat{Z} - T'$; the two sheaves agree on their intersection $\widehat{X} - T'$.

The module $\mathscr{G}'$ is coherent. By (5.2.5), there is a coherent $\mathscr{O}_Z$-module $\mathscr{G}$ such that $\mathscr{G}' = \widehat{\mathscr{G}}$. Let T be the support of $\mathscr{G}$; then $T' = T \cap Z'$ (**I**, 10.8.12). If h is the restriction of g to the reduced closed subprescheme of Z having T as its underlying space, then

$$T' = h^{-1}(Y') = T \cap g^{-1}(Y').$$

Thus $X \cap T$ is an open subset of T containing T'.

Since h is proper (**II**, 5.4.2), and hence closed, (5.1.3.1) gives $X \cap T = T$. Thus $T \subset X$; and since T is closed in Z, it is proper over Y. If $\mathscr{F}$ is the sheaf induced on X by $\mathscr{G}$, its completion $\widehat{\mathscr{F}}$ is induced on $\widehat{X}$ by $\widehat{\mathscr{G}}$ (**I**, 10.8.4), and is therefore equal to $\mathscr{F}'$. This completes the proof. **III** | 154

5.3. Proof of the existence theorem: the general case

Lemma (5.3.1). — *Under the conditions of (5.1.4), let*

$$0 \longrightarrow \mathscr{H} \longrightarrow \mathscr{F} \longrightarrow \mathscr{G} \longrightarrow 0$$

be an exact sequence of coherent $\mathscr{O}_{\widehat{X}}$-modules. If $\mathscr{G}$ and $\mathscr{H}$ are algebraizable, then $\mathscr{F}$ is algebraizable.

Proof. Suppose that

$$\mathscr{G} = \widehat{\mathscr{B}}, \qquad \mathscr{H} = \widehat{\mathscr{C}},$$

where $\mathscr{B}$ and $\mathscr{C}$ are coherent $\mathscr{O}_X$-modules whose supports are proper over Y. The module $\mathscr{F}$ canonically defines an element of the A-module

$$\operatorname{Ext}^1_{\mathscr{O}_{\widehat{X}}}(\widehat{X}; \widehat{\mathscr{B}}, \widehat{\mathscr{C}})$$

(**0**, 12.3.2). The hypotheses imply that this A-module is canonically isomorphic to

$$\operatorname{Ext}^1_{\mathscr{O}_X}(X; \mathscr{B}, \mathscr{C})$$

(4.5.2). There is therefore an exact sequence

$$0 \longrightarrow \mathscr{C} \longrightarrow \mathscr{A} \longrightarrow \mathscr{B} \longrightarrow 0$$

of coherent $\mathscr{O}_X$-modules whose canonical class maps to the class corresponding to $\mathscr{F}$. By definition, taking account of (**I**, 10.8.8), (ii), this says that $\mathscr{F}$ is isomorphic to $\widehat{\mathscr{A}}$. The lemma follows, since $\operatorname{Supp}(\mathscr{A})$ is contained in $\operatorname{Supp}(\mathscr{B}) \cup \operatorname{Supp}(\mathscr{C})$ and is therefore proper over Y. □

Corollary (5.3.2). — *Under the conditions of (5.1.1), let $u : \mathscr{F} \to \mathscr{G}$ be a homomorphism of coherent $\mathscr{O}_{\widehat{X}}$-modules. If $\mathscr{G}$, $\operatorname{Ker}(u)$, and $\operatorname{Coker}(u)$ are algebraizable, then $\mathscr{F}$ is algebraizable.*

Proof. Lemma (5.2.2), applied to $\mathscr{G} \to \operatorname{Coker}(u)$, shows that $\operatorname{Im}(u)$ is algebraizable. It then suffices to apply Lemma (5.3.1) to the exact sequence

$$0 \longrightarrow \operatorname{Ker}(u) \longrightarrow \mathscr{F} \longrightarrow \operatorname{Im}(u) \longrightarrow 0.$$

□

Lemma (5.3.3). — *Under the conditions of (5.1.1), let $h : Z \to Y$ be a morphism of finite type, let $\widehat{Z}$ be the completion of Z along $Z' = h^{-1}(Y')$, let $g : Z \to X$ be a proper Y-morphism, and let $\widehat{g} : \widehat{Z} \to \widehat{X}$ be its extension to the completions. For every algebraizable $\mathscr{O}_{\widehat{Z}}$-module $\mathscr{F}'$, the module $\widehat{g}_*(\mathscr{F}')$ is an algebraizable $\mathscr{O}_{\widehat{X}}$-module.*

Proof. If $\mathscr{F}$ is a coherent $\mathscr{O}_Z$-module such that $\mathscr{F}' = \widehat{\mathscr{F}}$, the first comparison theorem (4.1.5) identifies $\widehat{g}_*(\widehat{\mathscr{F}})$ with the completion of $g_*(\mathscr{F})$. □

Lemma (5.3.4). — *Let X be an ordinary Noetherian scheme, let X' be a closed subset of X, let $f : Z \to X$ be a proper morphism, and put*

$$Z' = f^{-1}(X'), \qquad \widehat{X} = X_{/X'}, \qquad \widehat{Z} = Z_{/Z'}.$$

Let $\widehat{f} : \widehat{Z} \to \widehat{X}$ be the extension of f to the completions. Let $\mathscr{M}$ be a coherent ideal of $\mathscr{O}_X$ such that, if

$$U = X - \operatorname{Supp}(\mathscr{O}_X/\mathscr{M}),$$

the restriction $f^{-1}(U) \to U$ of f is an isomorphism. Then, for every coherent $\mathscr{O}_{\widehat{X}}$-module $\mathscr{F}$, there is an integer $n > 0$ such that the kernel and cokernel of the canonical homomorphism

$$\rho_{\mathscr{F}} : \mathscr{F} \longrightarrow \widehat{f}_*(\widehat{f}^*(\mathscr{F}))$$

(**0**, 4.4.3) *are annihilated by* $\widehat{\mathscr{M}}^n$.

Proof. We may restrict to the case $X = \operatorname{Spec}(B)$, where B is a Noetherian ring; then $X' = \mathrm{V}(\mathfrak{K})$ for an ideal $\mathfrak{K}$ of B. We first reduce to the case in which B is adic Noetherian and $\mathfrak{K}$ is an ideal of definition.

Let B_1 be the separated completion of B for the $\mathfrak{K}$-preadic topology. If $\mathfrak{K}_1 = \mathfrak{K}B_1$, then B_1 is an adic Noetherian ring and $\mathfrak{K}_1$ is an ideal of definition. Put $X_1 = \operatorname{Spec}(B_1)$, and let $h : X_1 \to X$ be the morphism corresponding to the canonical homomorphism $B \to B_1$. If $X_1' = h^{-1}(X')$, then $X_1' = \mathrm{V}(\mathfrak{K}_1)$. Finally put III | 155

$$Z_1 = Z \times_X X_1, \qquad f_1 : Z_1 \longrightarrow X_1.$$

The morphism f_1 is proper (**II**, 5.4.2). Denote by $\widehat{X}_1$ the completion of X_1 along X_1', by

$$\widehat{Z}_1 = Z_1 \times_{X_1} \widehat{X}_1$$

the completion of Z_1 along $Z_1' = f_1^{-1}(X_1')$, and by $\widehat{f}_1$ the extension of f_1 to the completions.

The extension $\widehat{h} : \widehat{X}_1 \to \widehat{X}$ corresponds to the identity map of B_1 and is therefore an isomorphism (**I**, 10.9.1). The corresponding morphism $\widehat{Z}_1 \to \widehat{Z}$ is consequently also an isomorphism, and these isomorphisms identify $\widehat{f}_1$ with $\widehat{f}$. Moreover, $\mathscr{M}_1 = h^*(\mathscr{M})$ is a coherent ideal of $\mathscr{O}_{X_1}$ and

$$\operatorname{Supp}(\mathscr{O}_{X_1}/\mathscr{M}_1) = h^{-1}\big(\operatorname{Supp}(\mathscr{O}_X/\mathscr{M})\big)$$

(**I**, 9.1.13). Thus, if $U_1 = X_1 - \operatorname{Supp}(\mathscr{O}_{X_1}/\mathscr{M}_1)$, then $U_1 = h^{-1}(U)$, and the restriction $f_1^{-1}(U_1) \to U_1$ is an isomorphism (**I**, 3.2.7). Finally, $\widehat{\mathscr{M}}$ and $\widehat{\mathscr{M}_1}$ are identified by $\widehat{h}$ (**I**, 10.9.5). All the hypotheses of the lemma are therefore satisfied by X_1, X_1', f_1, and $\mathscr{M}_1$; henceforth we may suppose that B is adic Noetherian and that $\mathfrak{K}$ is an ideal of definition.

We then have $\widehat{X} = \operatorname{Spf}(B)$ and $\mathscr{F} = N^{\Delta}$, where N is a finite-type B-module. Thus $\mathscr{F} = \widehat{\mathscr{G}}$, where $\mathscr{G}$ is the coherent $\mathscr{O}_X$-module $\widetilde{N}$ (**I**, 10.10.5), and

$$\widehat{f}^*(\mathscr{F}) = \widehat{f^*(\mathscr{G})}$$

(**I**, 10.9.5). By the first comparison theorem (4.1.5),

$$\widehat{f}_*\big(\widehat{f^*(\mathscr{G})}\big) \simeq f_*\big(\widehat{f^*(\mathscr{G})}\big),$$

and, by (5.1.3), the canonical homomorphism $\rho_{\mathscr{F}}$ is precisely $\widehat{\rho}_{\mathscr{G}}$.

The kernel $\mathscr{P}$ and cokernel $\mathscr{R}$ of

$$\rho_{\mathscr{G}} : \mathscr{G} \longrightarrow f_*\big(f^*(\mathscr{G})\big)$$

are coherent $\mathscr{O}_X$-modules, and their restrictions to U are zero. There is therefore an integer $n > 0$ such that

$$\mathscr{M}^n\mathscr{P} = \mathscr{M}^n\mathscr{R} = 0$$

(**I**, 9.3.4). It follows that

$$\widehat{\mathscr{M}}^n\widehat{\mathscr{P}} = \widehat{\mathscr{M}}^n\widehat{\mathscr{R}} = 0$$

(**I**, 10.8.8) (**I**, 10.8.10), which proves the lemma. □

(5.3.5). *End of the proof of the existence theorem.* Retain the hypotheses of (5.1.4). We use Noetherian induction (**0**, 2.2.2), assuming the theorem known for every closed subprescheme T of X whose underlying space is distinct from X; here $\widehat{T}$ denotes the completion of T along $T \cap X'$. We may suppose that X is nonempty.

Since f is separated and of finite type, Chow's lemma (**II**, 5.6.1) gives a Y-scheme Z and a Y-morphism $g : Z \to X$ such that the structural morphism $h : Z \to Y$ is quasi-projective, g is projective and surjective, and there is a nonempty open subset U of X for which $g^{-1}(U) \to U$ is an isomorphism. Let $\mathscr{M}$ be a coherent ideal of $\mathscr{O}_X$ defining a closed subprescheme with underlying space $X - U$ (**I**, 5.2.2), and let $\mathscr{F}$ be a coherent $\mathscr{O}_{\widehat{X}}$-module whose support E is proper over Y. Denote by $\widehat{Z}$ the completion of Z along $h^{-1}(Y')$, and by $\widehat{g} : \widehat{Z} \to \widehat{X}$ the extension of g to the completions.

The module $\widehat{g}^*(\mathscr{F})$ is a coherent $\mathscr{O}_{\widehat{Z}}$-module whose support is contained in $g^{-1}(E)$ and is therefore proper over Y, since g is projective and hence proper (**II**, 5.4.6). Since h is quasi-projective, $\widehat{g}^*(\mathscr{F})$ is algebraizable by (5.2.6). It follows from (5.3.3), since g is proper, that III | 156

$$\widehat{g}_*\big(\widehat{g}^*(\mathscr{F})\big)$$

is an algebraizable $\mathscr{O}_{\widehat{X}}$-module.

Apply (5.3.4) to $\mathscr{F}$ and g. The kernel $\mathscr{P}$ and cokernel $\mathscr{R}$ of

$$\rho_{\mathscr{F}} : \mathscr{F} \longrightarrow \widehat{g}_*(\widehat{g}^*(\mathscr{F}))$$

are annihilated by a power $\widehat{\mathscr{M}}^n$. Let T be the closed subprescheme of X defined by $\mathscr{M}^n$, whose underlying space is $X - U$, and let $j : T \to X$ be the canonical injection. Its extension $\widehat{j} : \widehat{T} \to \widehat{X}$ is the canonical injection (**I**, 10.14.7), and we may write

$$\mathscr{P} = \widehat{j}_*(\widehat{j}^*(\mathscr{P})), \qquad \mathscr{R} = \widehat{j}_*(\widehat{j}^*(\mathscr{R})).$$

Since U is nonempty, the induction hypothesis shows that $\widehat{j}^*(\mathscr{P})$ and $\widehat{j}^*(\mathscr{R})$ are algebraizable $\mathscr{O}_{\widehat{T}}$-modules. Lemma (5.3.3) then shows that $\mathscr{P}$ and $\mathscr{R}$ are algebraizable, and (5.3.2) finally proves that $\mathscr{F}$ is algebraizable.

5.4. Application: comparison between morphisms of ordinary preschemes and morphisms of formal preschemes. Algebraizable formal preschemes

Theorem (5.4.1). — *Let A be an adic Noetherian ring, let $\mathfrak{J}$ be an ideal of definition of A, and put*

$$S = \operatorname{Spec}(A), \qquad S' = \mathrm{V}(\mathfrak{J}).$$

Let $u : X \to S$ be a proper morphism and let $v : Y \to S$ be a separated morphism of finite type. Let $\widehat{S}$, $\widehat{X}$, and $\widehat{Y}$ be the completions of S, X, and Y along S', $u^{-1}(S')$, and $v^{-1}(S')$, respectively. For every S-morphism $f : X \to Y$, let $\widehat{f} : \widehat{X} \to \widehat{Y}$ be its extension to the completions. Then the map $f \mapsto \widehat{f}$ is a bijection

$$\operatorname{Hom}_S(X, Y) \xrightarrow{\sim} \operatorname{Hom}_{\widehat{S}}(\widehat{X}, \widehat{Y}).$$

Proof. We first prove injectivity. Suppose that $f, g : X \to Y$ are S-morphisms such that $\widehat{f} = \widehat{g}$. There is then an open neighbourhood V of $X' = u^{-1}(S')$ on which f and g coincide (**I**, 10.9.4). Since u is closed, (5.1.3.1) gives $V = X$, and hence $f = g$.

We now prove surjectivity. Let $h : \widehat{X} \to \widehat{Y}$ be an $\widehat{S}$-morphism. Put

$$Z = X \times_S Y,$$

and denote the canonical projections by $p : Z \to X$ and $q : Z \to Y$. The prescheme Z is of finite type over S (**I**, 6.3.4), and is therefore Noetherian. Let $\widehat{Z}$ be its completion along

$$Z' = p^{-1}(u^{-1}(S')).$$

The formal prescheme $\widehat{Z}$ is canonically identified with $\widehat{X} \times_{\widehat{S}} \widehat{Y}$, and its projections onto $\widehat{X}$ and $\widehat{Y}$ are identified with $\widehat{p}$ and $\widehat{q}$ (**I**, 10.9.7).

Since Y is separated over S, $\widehat{Y}$ is separated over $\widehat{S}$ (**I**, 10.15.7). The graph morphism

$$\Gamma_h = (1_{\widehat{X}}, h) : \widehat{X} \longrightarrow \widehat{Z}$$

is consequently a closed immersion (**I**, 10.15.4). Let $\mathfrak{T}$ be the closed formal subprescheme of $\widehat{Z}$ associated with this immersion, and let $j : \mathfrak{T} \to \widehat{Z}$ be the canonical injection. Thus

$$\Gamma_h = j \circ w,$$

where $w : \widehat{X} \to \mathfrak{T}$ is an isomorphism (**I**, 10.14.3) whose inverse is $\widehat{p} \circ j$. Moreover, $\mathfrak{T}$ is proper over $\widehat{S}$ because $\widehat{X}$ is. It follows from (5.1.8) that there is a closed subprescheme T of Z such that

$$\mathfrak{T} = \widehat{T} = T_{/(T \cap Z')},$$

and that $j = \widehat{i}$, where $i : T \to Z$ is the canonical injection (**I**, 10.14.7).

The morphism $p \circ i : T \to X$ is an isomorphism: its extension to the completions is the isomorphism $\widehat{p} \circ \widehat{i}$, and it suffices to apply (4.6.8), noting as above that S is the only neighbourhood of S' in S. Let $g : X \to T$ be the inverse of $p \circ i$, and put **III | 157**

$$f = q \circ i \circ g.$$

Then $f : X \to Y$ has graph $\Gamma_f = i \circ g$. Since $\widehat{g}$ is the inverse of $\widehat{p \circ i}$, we have

$$\widehat{\Gamma_f} = \widehat{i} \circ \widehat{g} = j \circ w = \Gamma_h.$$

But $\widehat{\Gamma_f} = \Gamma_{\widehat{f}}$ (**I**, 10.9.8); hence $h = \widehat{f}$, which completes the proof. □

In categorical language, the functor $X \mapsto \widehat{X}$ is therefore fully faithful (**0**, 8.1.6) from the category of preschemes proper over $\operatorname{Spec}(A)$ to the category of formal preschemes proper over $\operatorname{Spf}(A)$, for every adic Noetherian ring A. It consequently gives an equivalence between the first category and a full subcategory of the second. The objects of this subcategory will henceforth be called *algebraizable formal preschemes*. For such a formal prescheme $\mathfrak{X}$, there is an ordinary prescheme X, proper over $\operatorname{Spec}(A)$ and determined up to unique isomorphism, such that $\mathfrak{X} \simeq \widehat{X}$.

(5.4.2). *Scholium.* With the notation of (5.4.1), put

$$S_n = \operatorname{Spec}(A/\mathfrak{J}^{n+1}), \qquad X_n = X \times_S S_n, \qquad Y_n = Y \times_S S_n.$$

It follows from (5.4.1) and (**I**, 10.12.3) that giving an S-morphism $f : X \to Y$ is equivalent to giving an adic inductive (S_n)-system (**I**, 10.12.2) of S_n-morphisms $f_n : X_n \to Y_n$.

Remark (5.4.3). — Contrary to what the existence theorem (5.1.6) might suggest, there are formal preschemes proper over $\operatorname{Spf}(A)$ which are *not algebraizable*, just as there are compact analytic spaces which do not arise from complex algebraic varieties. We shall encounter such preschemes later in the "theory of moduli," which, when the base field is $\mathbb{C}$, deals precisely with infinitesimal variations of the complex structure of a complete algebraic variety; it is well known that such variations can give rise to analytic varieties which are not algebraic.

Proposition (5.4.4). — *Let A be an adic Noetherian ring, put $\mathfrak{S} = \operatorname{Spf}(A)$, and let*

$$g : \mathfrak{X} \longrightarrow \mathfrak{S}, \qquad h : \mathfrak{Y} \longrightarrow \mathfrak{S}$$

be two proper morphisms of formal preschemes. Let $f : \mathfrak{X} \to \mathfrak{Y}$ be an $\mathfrak{S}$-morphism. If f is finite and $\mathfrak{Y}$ is algebraizable, then $\mathfrak{X}$ is algebraizable.

Proof. The hypotheses on g and h already imply that f is proper (3.4.1); for f to be finite, it is enough that the fibre $f^{-1}(y)$ be finite for every $y \in \mathfrak{Y}$ (**0**, 4.8.11). By hypothesis,

$$\mathscr{B} = f_*(\mathscr{O}_{\mathfrak{X}})$$

is a coherent $\mathscr{O}_{\mathfrak{Y}}$-algebra (**0**, 4.8.6). Write $\mathfrak{Y} = \widehat{Y}$ and $h = \widehat{w}$, where $w : Y \to \operatorname{Spec}(A)$ is a proper morphism of ordinary preschemes. The existence theorem gives a coherent $\mathscr{O}_Y$-algebra $\mathscr{C}$ such that $\mathscr{B} = \widehat{\mathscr{C}}$. Put

$$X = \operatorname{Spec}(\mathscr{C}),$$

and let $u : X \to Y$ be the structural morphism. The definition of $\mathfrak{X}$ from $\mathscr{B}$ (**0**, 4.8.7) then shows that $\mathfrak{X}$ is canonically isomorphic to $\widehat{X}$ and that f is identified with $\widehat{u}$; it is enough to check this when Y is affine. □

Proposition (5.1.8) is a special case of (5.4.4).

Theorem (5.4.5). — *Let A be an adic Noetherian ring, let $\mathfrak{J}$ be an ideal of definition of A, put*

$$S = \operatorname{Spec}(A), \qquad \mathfrak{S} = \widehat{S} = \operatorname{Spf}(A),$$

and let $f : \mathfrak{X} \to \mathfrak{S}$ be a proper morphism of formal preschemes. Put

$$S_k = \operatorname{Spec}(A/\mathfrak{J}^{k+1}), \qquad X_k = \mathfrak{X} \times_{\mathfrak{S}} S_k,$$

and, for every $\mathscr{O}_{\mathfrak{X}}$-module $\mathscr{F}$, put

$$\mathscr{F}_k = \mathscr{F} \otimes_{\mathscr{O}_{\mathfrak{X}}} \mathscr{O}_{X_k} = \mathscr{F}/\mathfrak{J}^{k+1}\mathscr{F}.$$

Let $\mathscr{L}$ be an invertible $\mathscr{O}_{\mathfrak{X}}$-module, and suppose that **III** | 158

$$\mathscr{L}_0 = \mathscr{L}/\mathfrak{J}\mathscr{L}$$

is an ample $\mathscr{O}_{X_0}$-module. Then $\mathfrak{X}$ is algebraizable. Moreover, if X is a proper S-prescheme such that $\mathfrak{X} \simeq \widehat{X}$, there is an ample $\mathscr{O}_X$-module $\mathscr{M}$ such that $\mathscr{L} \simeq \widehat{\mathscr{M}}$; in particular, X is projective over S.

Proof. Apply (5.2.3) with $\mathscr{F} = \mathscr{O}_{\mathfrak{X}}$. There is an integer n_0 such that, for $n \geqslant n_0$, the canonical homomorphism

$$\Gamma(\mathfrak{X}, \mathscr{L}^{\otimes n}) \longrightarrow \Gamma(X_0, \mathscr{L}_0^{\otimes n})$$

is surjective. Choose $n \geqslant n_0$ large enough that $\mathscr{L}_0^{\otimes n}$ is very ample for S_0 (**II**, 4.5.10). Since $f_0 : X_0 \to S_0$ is proper, $\Gamma(X_0, \mathscr{L}_0^{\otimes n})$ is a finite-type A-module (3.2.1). There is consequently a finite-type A-submodule

$$E \subset \Gamma(\mathfrak{X}, \mathscr{L}^{\otimes n})$$

whose image in $\Gamma(X_0, \mathscr{L}_0^{\otimes n})$ is the whole module.

For every $k \geqslant 0$, consider the homomorphism

$$u_k : E/\mathfrak{J}^{k+1}E \longrightarrow \Gamma(X_k, \mathscr{L}_k^{\otimes n})$$

induced by the canonical injection $E \to \Gamma(\mathfrak{X}, \mathscr{L}^{\otimes n})$. The module $(f_k)_*(\mathscr{L}_k^{\otimes n})$ is quasi-coherent, and

$$\Gamma\big(S_k, (f_k)_*(\mathscr{L}_k^{\otimes n})\big) = \Gamma(X_k, \mathscr{L}_k^{\otimes n}).$$

Thus u_k defines a homomorphism

$$\widetilde{u}_k : \widetilde{E/\mathfrak{J}^{k+1}E} \longrightarrow (f_k)_*(\mathscr{L}_k^{\otimes n}),$$

and hence, by adjunction, a homomorphism

$$\widetilde{u}_k^\sharp : f_k^*\big(\widetilde{E/\mathfrak{J}^{k+1}E}\big) \longrightarrow \mathscr{L}_k^{\otimes n}.$$

Put

$$\mathscr{G}_k = f_k^*\big(\widetilde{E/\mathfrak{J}^{k+1}E}\big).$$

We have $\mathscr{G}_0 = \mathscr{G}_k/\mathfrak{J}\mathscr{G}_k$ (**I**, 9.1.5), and

$$\widetilde{u}_0^\sharp : \mathscr{G}_0/\mathfrak{J}\mathscr{G}_0 \longrightarrow \mathscr{L}_0^{\otimes n}/\mathfrak{J}\mathscr{L}_0^{\otimes n}$$

is obtained from $\widetilde{u}_k^\sharp$ by passage to the quotients.

By the definition of E, $\widetilde{u}_0^\sharp$ is the canonical homomorphism

$$\sigma : f_0^*\big((f_0)_*(\mathscr{L}_0^{\otimes n})\big) \longrightarrow \mathscr{L}_0^{\otimes n}.$$

Since $\mathscr{L}_0^{\otimes n}$ is very ample, $\widetilde{u}_0^\sharp$ is surjective (**II**, 4.4.3). Nakayama's lemma then shows that every $\widetilde{u}_k^\sharp$ is surjective.

Each $\widetilde{u}_k^\sharp$ therefore defines (**II**, 4.2.2) an S_k-morphism

$$g_k : X_k \longrightarrow P_k = \mathbf{P}(E/\mathfrak{J}^{k+1}E).$$

For $h \leqslant k$,

$$P_h = P_k \times_{S_k} S_h$$

by (**II**, 4.1.3). The relations among the $\widetilde{u}_k^\sharp$ and (**II**, 4.2.10) show that (g_k) is an adic inductive (S_k)-system (**I**, 10.12.2). The g_k consequently define an $\mathfrak{S}$-morphism of formal preschemes

$$g : \mathfrak{X} \longrightarrow \mathfrak{P},$$

where $\mathfrak{P}$ is the inductive limit of the system (P_k), or equivalently the completion $\widehat{P}$ of $P = \mathbf{P}(E)$. The very ampleness of $\mathscr{L}_0^{\otimes n}$ implies that g_0 is a closed immersion (**II**, 4.4.3). Hence g is a closed immersion of formal preschemes (**0**, 4.8.10), and $\mathfrak{X}$ is algebraizable by (5.1.8).

The existence theorem (5.1.6) now shows that $\mathscr{L}$ is the completion $\widehat{\mathscr{M}}$ of an invertible $\mathscr{O}_X$-module $\mathscr{M}$. Moreover, $\mathscr{L}^{\otimes n}$ is the completion of $\mathscr{M}^{\otimes n}$ (**I**, 10.8.10), and the homomorphisms $\widetilde{u}_k^\sharp$ define a unique homomorphism

$$v : f^*(\widetilde{E}) \longrightarrow \mathscr{M}^{\otimes n}$$

(5.1.7). Since the $\widetilde{u}_k^\sharp$ are surjective, $\widehat{v}$ is surjective (**I**, 10.11.5), and therefore so is v (5.1.3). Let $r : X \to P$ be the morphism defined by v (**II**, 4.2.2). Its extension to the completions is g. Since g is a closed immersion, so is r, by (5.1.8) and (5.4.1). It follows that $\mathscr{M}^{\otimes n}$ is very ample (**II**, 4.4.6), and hence that $\mathscr{M}$ is ample (**II**, 4.5.10). □

Remark (5.4.6). — Let A be an adic Noetherian ring, put $\mathfrak{S} = \mathrm{Spf}(A)$, and let $f : \mathfrak{X} \to \mathfrak{S}$ be a proper morphism of formal preschemes. Let $\mathscr{N}$ be the coherent ideal of $\mathscr{O}_{\mathfrak{X}}$ such that, for every affine formal open subset U of $\mathfrak{X}$,

$$\Gamma(U, \mathscr{N})$$

is the nilradical of $\Gamma(U, \mathscr{O}_{\mathfrak{X}})$. The existence of this ideal follows readily from (**I**, 10.10.2) and from the fact that every ring homomorphism $B \to C$ sends the nilradical of B into that of C. Let $\mathfrak{X}'$ be the closed formal subprescheme of $\mathfrak{X}$ defined by $\mathscr{N}$ (**I**, 10.14.2). It would be interesting to know whether algebraizability of $\mathfrak{X}'$ implies algebraizability of $\mathfrak{X}$ itself.

One would doubtless arrive at a solution of this problem if one knew how to classify, for example by invariants of a cohomological nature, the *extensions* of a structural sheaf $\mathscr{O}_{\mathfrak{X}}$—for an ordinary or a formal prescheme—by a square-zero ideal. Equivalently, these are the $\mathscr{O}_{\mathfrak{X}}$-algebras $\mathscr{A}$ such that $\mathscr{O}_{\mathfrak{X}} \simeq \mathscr{A} / \mathscr{J}$, where $\mathscr{J}$ is a square-zero ideal of $\mathscr{A}$. **III** | 159

5.5. A decomposition of certain preschemes

Proposition (5.5.1). — *Let A be an adic Noetherian ring, let $\mathfrak{J}$ be an ideal of definition of A, and put $Y = \mathrm{Spec}(A)$. Let $f : X \to Y$ be a separated morphism of finite type, and put*

$$Y_0 = \mathrm{Spec}(A/\mathfrak{J}), \qquad X_0 = X \times_Y Y_0 = f^{-1}(Y_0).$$

Let Z_0 be an open subset of X_0 which is proper over Y_0. Then there is an open and closed subset Z of X, proper over Y, such that

$$Z \cap X_0 = Z_0.$$

Proof. By hypothesis, there is an open subset T of X such that $T \cap X_0 = Z_0$. Let $\widehat{T}$ be the completion along Z_0 of the prescheme induced by X on T. The support of $\mathscr{O}_{\widehat{T}}$ is Z_0, which is proper over Y_0; hence $\widehat{T}$ is proper over $\widehat{Y} = \mathrm{Spf}(A)$ (3.4.1).

It follows from (5.1.8) that there is a closed subprescheme Z of T, proper over Y, such that, if $i : Z \to T$ is the canonical injection, then

$$\widehat{i} : \widehat{Z} \longrightarrow \widehat{T}$$

is an isomorphism; here $\widehat{Z}$ is the completion of Z along Z_0. By (4.6.8), there is an open neighbourhood V of Z_0 in T such that $i^{-1}(V) \to V$ is an isomorphism. But $i^{-1}(V)$ is a neighbourhood of Z_0 in Z, and is therefore equal to Z by (5.1.3.1). Thus Z is open in T, and hence in X. It is also closed in X, since Z is proper over Y and X is separated over Y. This proves the proposition. □

Corollary (5.5.2). — *If X_0 is proper over Y_0, then X is the union of two disjoint open subsets Z and Z' such that Z is proper over Y and contains X_0. Moreover, every closed subset P of X which is proper over Y is contained in Z.*

Proof. Apply (5.5.1) with $Z_0 = X_0$, and put $Z' = X - Z$. For the last assertion, $P \cap Z'$ is closed in P and is therefore proper over Y. If it were nonempty, $f(P \cap Z')$ would be a nonempty closed subset of Y, and hence would meet Y_0 (5.1.3.1). This contradicts $Z' \cap X_0 = \varnothing$, and proves that $P \subset Z$. □

§6. Local and global Tor functors; Künneth formula

6.1. Introduction **III** | 137

(6.1.1). Let $f : X \to Y$ be a morphism of preschemes, and let $\mathscr{F}$ be a quasi-coherent $\mathscr{O}_X$-module. In studying the higher direct images $\mathrm{R}^n f_*(\mathscr{F})$, one is led to consider the following general problem. Given a base-change morphism $g : Y' \to Y$, put

$$X' = X_{(Y')} = X \times_Y Y', \qquad \mathscr{F}' = \mathscr{F} \otimes_{\mathscr{O}_Y} \mathscr{O}_{Y'}, \qquad f' = f_{(Y')} : X' \longrightarrow Y'.$$

We seek information about the higher direct images $\mathrm{R}^n f'_*(\mathscr{F}')$, assuming that the $\mathrm{R}^n f_*(\mathscr{F})$ are known. One sees readily (cf. for example (7.7.2)) that this reduces to studying the variation of

$$\mathrm{R}^n f_*(\mathscr{F} \otimes_{\mathscr{O}_Y} \mathscr{G})$$

as $\mathscr{G}$ ranges over the quasi-coherent $\mathscr{O}_Y$-modules; in other words, to studying the functor

$$\mathscr{G} \longmapsto \mathrm{R}^n f_*(\mathscr{F} \otimes_{\mathscr{O}_Y} \mathscr{G}).$$

If $\mathscr{F}$ is flat over Y, the functor $\mathscr{G} \mapsto \mathscr{F} \otimes_{\mathscr{O}_Y} \mathscr{G}$ is exact (**0**, 6.7.4), and consequently the composite functors

$$\mathscr{G} \longmapsto \mathrm{R}^i f_*(\mathscr{F} \otimes_{\mathscr{O}_Y} \mathscr{G})$$

again form a cohomological functor. This is no longer true in general. In order to apply cohomological methods, one is therefore led to replace $\mathscr{G} \mapsto \mathrm{R}^n f_*(\mathscr{F} \otimes_{\mathscr{O}_Y} \mathscr{G})$ by other functors which are always cohomological.

These functors, which generalize the Tor functors of module theory, are defined in Sections 6.3–6.7. There are in fact two such generalizations, one local and the other global, related by spectral sequences discussed in Section 6.7. As an application of those spectral sequences one obtains, under suitable hypotheses, a Künneth formula expressing

$$\mathrm{R}^n(f_1 \times f_2)_*(\mathscr{F}_1 \otimes_Y \mathscr{F}_2)$$

in terms of the higher direct images $\mathrm{R}^p(f_1)_*(\mathscr{F}_1)$ and $\mathrm{R}^q(f_2)_*(\mathscr{F}_2)$. Other spectral sequences (6.8) generalize the associativity spectral sequences for the Tor functor of modules; finally, the base-change problem also leads to spectral sequences (6.9).

(6.1.2). Moreover, one observes in (6.10) that the cohomological functors

$$\mathscr{G} \longmapsto \mathscr{T}_\bullet(\mathscr{G})$$

so defined are locally on Y of the form

$$\mathscr{G} \longmapsto \mathscr{H}_\bullet(\mathscr{L}_\bullet \otimes_Y \mathscr{G}),$$

where $\mathscr{L}_\bullet$ is a complex of locally free $\mathscr{O}_Y$-modules, defined up to homotopy, and $\mathscr{H}_\bullet$ denotes homology. It is therefore useful to set aside the particular situation which produced $\mathscr{T}_\bullet$ and to study, in general, functors of the preceding form, imposing when necessary appropriate finiteness hypotheses on the $\mathscr{L}_i$ or on the $\mathscr{H}_i(\mathscr{L}_\bullet)$. This is the subject of Section 7, which can for the most part be read independently of Section 6. **III** | 138

The most important properties of these functors concern exactness of a component $\mathscr{T}_i$ of $\mathscr{T}_\bullet$. We shall give various criteria for establishing such properties. As an application, we shall obtain conditions under which, with the notation of (6.1.1), the functor

$$\mathscr{G} \longmapsto \mathrm{R}^n f_*(\mathscr{F} \otimes_{\mathscr{O}_Y} \mathscr{G})$$

is exact; we shall express this by saying that $\mathscr{F}$ is *cohomologically flat over Y in dimension n*. Another important property of the components $\mathscr{T}_i$ of

$$\mathscr{T}_\bullet(\mathscr{G}) = \mathscr{H}_\bullet(\mathscr{L}_\bullet \otimes_Y \mathscr{G})$$

is semicontinuity of the function

$$y \longmapsto \dim_{k(y)}(T_i(k(y))).$$

When $\mathscr{T}_i$ is exact, semicontinuity is replaced by continuity; by a theorem of Grauert the converse also holds when Y is reduced (7.8.4).

(6.1.3). In Sections 6 and 7 we systematically use hypercohomology, taking complexes of sheaves rather than sheaves as arguments throughout, although the necessity of this point of view will become apparent only in later chapters. The cohomological formalism developed here will moreover become more transparent in the chapter of this Treatise devoted to an algebra of cohomological functors of coherent sheaves, including the formalism of duality. This, however, will require developments beyond the scope of the present chapter.

(6.1.4). For brevity, let $K^\bullet$ and $K'^\bullet$ be two complexes in an abelian category C. We shall call a morphism of complexes

$$f : K^\bullet \longrightarrow K'^\bullet$$

a *homotopism* if there is a morphism $g : K'^\bullet \to K^\bullet$ such that both composites $f \circ g$ and $g \circ f$ are homotopic to the identity. By abuse of language, when such a homotopism exists we shall also say that $K^\bullet$ and $K'^\bullet$ are homotopic.

Suppose that the hypercohomology of a covariant additive functor $T : \mathsf{C} \to \mathsf{C}'$, with values in an abelian category C', can be defined relative to a complex of C (**0**, 11.4.3). It is then immediate that a homotopism $K^\bullet \to K'^\bullet$ canonically defines an isomorphism

$$\mathbf{R}^\bullet T(K^\bullet) \xrightarrow{\sim} \mathbf{R}^\bullet T(K'^\bullet)$$

on hypercohomology *(loc. cit.)*.

6.2. Hypercohomology of complexes of modules on a prescheme

(6.2.1). Let X be a prescheme, and let $\mathscr{K}^\bullet = (\mathscr{K}^i)_{i\in\mathbf{Z}}$ be a complex of $\mathscr{O}_X$-modules whose differential has degree $+1$. Recall that, for every morphism $f : X \to Y$ of preschemes, the $\mathscr{O}_Y$-modules of hypercohomology $\mathscr{H}^n(f, \mathscr{K}^\bullet)$, also denoted $\mathscr{H}^n_f(\mathscr{K}^\bullet)$ or $\mathrm{R}^n f_*(\mathscr{K}^\bullet)$, have been defined for every $n \in \mathbf{Z}$ (**0**, 12.4.1). The hypercohomology $\mathscr{H}^\bullet(f, \mathscr{K}^\bullet)$ is the common abutment of the two spectral functors $'\mathscr{E}(f, \mathscr{K}^\bullet)$ and $''\mathscr{E}(f, \mathscr{K}^\bullet)$ whose E_2-terms are

$$'\mathscr{E}_2^{pq} = \mathscr{H}^p\big(\mathscr{H}^q(f, \mathscr{K}^\bullet)\big), \tag{6.2.1.1}$$

and

$$''\mathscr{E}_2^{pq} = \mathscr{H}^p\big(f, \mathscr{H}^q(\mathscr{K}^\bullet)\big) = \mathrm{R}^p f_*\big(\mathscr{H}^q(\mathscr{K}^\bullet)\big). \tag{6.2.1.2}$$

Here $\mathscr{H}^q(f, \mathscr{K}^\bullet)$ is the complex whose component of degree i is **III** | 139

$$\mathscr{H}^q(f, \mathscr{K}^i) = \mathrm{R}^q f_*(\mathscr{K}^i)$$

(loc. cit.). Recall also that, when Y is reduced to a point, the corresponding hypercohomology is denoted $\mathbf{H}^\bullet(X, \mathscr{K}^\bullet)$; it consists of modules over $\Gamma(X, \mathscr{O}_X)$ and is independent of the one-point prescheme Y chosen. When $Y = X$ and $f = 1_X$,

$$\mathscr{H}^n(f, \mathscr{K}^\bullet) = \mathscr{H}^n(\mathscr{K}^\bullet),$$

the cohomology of the complex $\mathscr{K}^\bullet$. When $\mathscr{K}^i = 0$ except for $i = i_0$, one has

$$\mathscr{H}^n(f, \mathscr{K}^\bullet) = \mathrm{R}^{n-i_0} f_*(\mathscr{K}^{i_0}).$$

The spectral sequence $'\mathscr{E}(f, \mathscr{K}^\bullet)$ is always regular; both spectral sequences are biregular when $\mathscr{K}^\bullet$ is bounded below (**0**, 12.4.1). Every homotopism $h : \mathscr{K}^\bullet \to \mathscr{K}'^\bullet$ of complexes of $\mathscr{O}_X$-modules (6.1.4) induces an isomorphism

$$\mathscr{H}^\bullet(f, \mathscr{K}^\bullet) \xrightarrow{\sim} \mathscr{H}^\bullet(f, \mathscr{K}'^\bullet)$$

on hypercohomology. The same conclusion holds if one assumes only that

$$\mathscr{H}^\bullet(h) : \mathscr{H}^\bullet(\mathscr{K}^\bullet) \xrightarrow{\sim} \mathscr{H}^\bullet(\mathscr{K}'^\bullet)$$

is an isomorphism and that both complexes are bounded below. This follows immediately from (**0**, 11.1.5), applied to the spectral sequence (6.2.1.2) and its analogue for $\mathscr{K}'^\bullet$.

Finally, for every open covering $\mathfrak{U} = (U_\alpha)$ of X, the hypercohomology $\mathbf{H}^\bullet(\mathfrak{U}, \mathscr{K}^\bullet)$ has also been defined (**0**, 12.4.5) as the cohomology of the bicomplex $\mathrm{C}^\bullet(\mathfrak{U}, \mathscr{K}^\bullet)$, whose component of bidegree (i, j) is by definition $\mathrm{C}^i(\mathfrak{U}, \mathscr{K}^j)$. The modules $\mathbf{H}^n(\mathfrak{U}, \mathscr{K}^\bullet)$ are again modules over $\Gamma(X, \mathscr{O}_X)$.

Proposition (6.2.2). — *Let X be a scheme, and let $\mathfrak{U} = (U_\alpha)$ be a covering of X by affine open subsets. For every complex $\mathscr{K}^\bullet$ of quasi-coherent $\mathscr{O}_X$-modules, the hypercohomology modules*

$$\mathbf{H}^\bullet(X, \mathscr{K}^\bullet) \quad \textit{and} \quad \mathbf{H}^\bullet(\mathfrak{U}, \mathscr{K}^\bullet)$$

are canonically isomorphic.

Proof. Every finite intersection V of members of $\mathfrak{U}$ is affine (**I**, 5.5.6); hence $\mathrm{H}^q(V, \mathscr{K}^i) = 0$ for every i and every $q > 0$ (1.3.1). The proposition is therefore a special case of (**0**, 12.4.7). □

Proposition (6.2.3). — *Let $f : X \to Y$ be a quasi-compact separated morphism of preschemes. For every complex $\mathscr{K}^\bullet$ of quasi-coherent $\mathscr{O}_X$-modules, the $\mathscr{O}_Y$-modules $\mathscr{H}^n(f, \mathscr{K}^\bullet)$ are quasi-coherent.*

Proof. The modules

$$\mathscr{H}^q(f,\mathscr{K}^i) = \mathrm{R}^q f_*(\mathscr{K}^i)$$

are quasi-coherent $\mathscr{O}_Y$-modules (1.4.10). Hence so is ${}'\mathscr{E}_2^{pq}$: by (6.2.1.1), it is the quotient of the kernel of a homomorphism of quasi-coherent modules by the image of such a homomorphism (**I**, 4.1.1). For the same reason, all terms, cycles, and boundaries of the first spectral sequence are quasi-coherent. Regularity implies that $\mathrm{Z}_\infty({}'\mathscr{E}_2^{pq})$ equals one of the $\mathrm{Z}_k({}'\mathscr{E}_2^{pq})$ and is therefore quasi-coherent. Likewise,

$$\mathrm{B}_\infty({}'\mathscr{E}_2^{pq}) = \varinjlim_k \mathrm{B}_k({}'\mathscr{E}_2^{pq})$$

is quasi-coherent (**0**, 11.2.4), (**I**, 4.1.1); thus the ${}'\mathscr{E}_\infty^{pq}$ are quasi-coherent as well.

Since the spectral sequence is regular, the filtration $F^\bullet\mathscr{H}^n(f,\mathscr{K}^\bullet)$ is discrete and exhaustive. Equivalently, $\mathscr{H}^n(f,\mathscr{K}^\bullet)$ is the union of an increasing sequence $(\mathscr{G}_k)_{k\geqslant 0}$ of $\mathscr{O}_Y$-modules such that $\mathscr{G}_0 = 0$ and every quotient $\mathscr{G}_k/\mathscr{G}_{k-1}$ is one of the ${}'\mathscr{E}_\infty^{pq}$, and hence is quasi-coherent. It follows inductively that the $\mathscr{G}_k$ are quasi-coherent (1.4.17). Since

$$\mathscr{H}^n(f,\mathscr{K}^\bullet) = \varinjlim_k \mathscr{G}_k,$$

the proposition follows from (**I**, 4.1.1). □

Corollary (6.2.4). — *Under the hypotheses of (6.2.3), for every affine open subset V of Y, the canonical homomorphism* **III | 140**

$$\mathbf{H}^n\big(f^{-1}(V),\mathscr{K}^\bullet\big) \longrightarrow \Gamma\big(V,\mathscr{H}^n(f,\mathscr{K}^\bullet)\big) \tag{6.2.4.1}$$

is bijective for every $n \in \mathbf{Z}$.

Proof. The proof is the same as that of (1.4.11), using (6.2.2): one replaces the sheaf there by $\mathscr{K}^\bullet$, uses the Cech bicomplex $f_*\mathrm{C}^\bullet(\mathfrak{U},\mathscr{K}^\bullet)$, and replaces ordinary cohomology by $\mathscr{H}^\bullet(f,\mathscr{K}^\bullet)$. The latter is a quasi-coherent $\mathscr{O}_Y$-module by (6.2.3). □

Proposition (6.2.5). — *Let Y be a locally Noetherian prescheme, let $f : X \to Y$ be proper, and let $\mathscr{K}^\bullet$ be a complex of $\mathscr{O}_X$-modules such that the $\mathscr{O}_X$-modules $\mathscr{H}^q(\mathscr{K}^\bullet)$ are coherent. Then the $\mathscr{O}_Y$-modules $\mathscr{H}^n(f,\mathscr{K}^\bullet)$ are coherent.*

Proof. The question is local on Y, so we may suppose that Y is Noetherian and affine. By (6.2.4), it is enough to prove that the $\mathbf{H}^n(X,\mathscr{K}^\bullet)$ are finite-type modules over $\Gamma(Y,\mathscr{O}_Y)$ (3.2.1). By (6.2.2),

$$\mathbf{H}^\bullet(X,\mathscr{K}^\bullet) = \mathbf{H}^\bullet(\mathfrak{U},\mathscr{K}^\bullet),$$

where $\mathfrak{U}$ may be taken finite, since X is quasi-compact. The cochains in each complex $\mathrm{C}^\bullet(\mathfrak{U},\mathscr{K}^j)$ are alternating by definition. Hence there is an integer $r > 0$ such that

$$\mathrm{C}^i(\mathfrak{U},\mathscr{K}^j) = 0 \qquad \text{for } i < 0 \text{ and } i > r.$$

It follows from (**0**, 11.3.3) that both spectral sequences of the bicomplex $\mathrm{C}^\bullet(\mathfrak{U},\mathscr{K}^\bullet)$ are biregular.

Intersections of members of $\mathfrak{U}$ are affine open subsets (**I**, 5.5.6), so each functor

$$\mathscr{F} \longmapsto \mathrm{C}^i(\mathfrak{U},\mathscr{F})$$

is exact in the category of quasi-coherent $\mathscr{O}_X$-modules. Consequently the E_2-terms of the second spectral sequence are

$$''E_2^{pq} = \mathrm{H}^p\big(\mathrm{C}^\bullet(\mathfrak{U},\mathscr{H}^q(\mathscr{K}^\bullet))\big) = \mathbf{H}^p\big(\mathfrak{U},\mathscr{H}^q(\mathscr{K}^\bullet)\big) = \mathrm{H}^p\big(X,\mathscr{H}^q(\mathscr{K}^\bullet)\big)$$

(**0**, 11.3.2), (1.4.1). Since f is proper, these are finite-type $\Gamma(Y,\mathscr{O}_Y)$-modules (3.2.1). Biregularity now implies that the $\mathbf{H}^n(X,\mathscr{K}^\bullet)$ are finite-type $\Gamma(Y,\mathscr{O}_Y)$-modules (**0**, 11.1.8).

A fortiori, if $\mathscr{K}^\bullet$ is a complex of coherent $\mathscr{O}_X$-modules, then the $\mathscr{H}^n(f,\mathscr{K}^\bullet)$ are coherent under the hypotheses on Y and f in (6.2.5) (**0**, 5.3.4). □

(6.2.6). The hypercohomology $\mathscr{H}^\bullet(f, \mathscr{K}^\bullet)$ is a cohomological functor on the category of bounded-below complexes of $\mathscr{O}_X$-modules (**0**, 12.4.4). It is a cohomological functor on the category of all complexes of $\mathscr{O}_X$-modules when f is quasi-compact and the underlying space of X is locally Noetherian. Indeed, f_* then commutes with inductive limits *(G, II, 3.10.1)*—the question being local on Y—and (**0**, 11.5.2) applies.

Finally, if f is separated, $\mathscr{H}^\bullet(f, \mathscr{K}^\bullet)$ is a cohomological functor on the category of complexes of quasi-coherent $\mathscr{O}_X$-modules. This is immediate when Y is affine, for then X is a scheme and the canonical isomorphism (6.2.2) reduces the assertion to the fact that

$$\mathscr{K}^\bullet \longmapsto \mathbf{H}^\bullet(\mathfrak{U}, \mathscr{K}^\bullet)$$

is cohomological. This in turn is immediate because

$$\mathscr{K}^\bullet \longmapsto \mathbf{C}^\bullet(\mathfrak{U}, \mathscr{K}^\bullet)$$

is exact in this category (**I**, 1.3.7).

In general, for every affine open subset V of Y, $f^{-1}(V)$ is a scheme. To apply the preceding **III** | 141
argument, it is enough to verify that, for an exact sequence

$$0 \longrightarrow \mathscr{K}'^\bullet \longrightarrow \mathscr{K}^\bullet \longrightarrow \mathscr{K}''^\bullet \longrightarrow 0$$

of complexes of quasi-coherent $\mathscr{O}_X$-modules, the connecting homomorphism

$$\partial : \mathscr{H}^n(f, \mathscr{K}''^\bullet)|V \longrightarrow \mathscr{H}^{n+1}(f, \mathscr{K}'^\bullet)|V$$

does not depend on the affine open covering $\mathfrak{U}$ of $f^{-1}(V)$ used to define it. If $\mathfrak{U}'$ is an affine open refinement of $\mathfrak{U}$, this follows from commutativity of the triangle of canonical isomorphisms

$$\begin{tikzcd}
\mathbf{H}^\bullet(\mathfrak{U}, \mathscr{K}^\bullet) \arrow[r, "\sim"] \arrow[d, "\iota"'] & \mathbf{H}^\bullet(f^{-1}(V), \mathscr{K}^\bullet) \\
\mathbf{H}^\bullet(\mathfrak{U}', \mathscr{K}^\bullet) \arrow[ur, "\sim"'] &
\end{tikzcd}$$

and of the diagram

$$\begin{tikzcd}
\mathbf{H}^n(\mathfrak{U}, \mathscr{K}''^\bullet) \arrow[r, "\partial"] \arrow[d, "\iota"'] & \mathbf{H}^{n+1}(\mathfrak{U}, \mathscr{K}'^\bullet) \arrow[d, "\iota"] \\
\mathbf{H}^n(\mathfrak{U}', \mathscr{K}''^\bullet) \arrow[r, "\partial"'] & \mathbf{H}^{n+1}(\mathfrak{U}', \mathscr{K}'^\bullet).
\end{tikzcd}$$

Whenever one of the preceding conditions holds and $\mathscr{K}^\bullet \to \mathscr{K}'^\bullet$ is a homotopism (6.1.4), the resulting isomorphism

$$\mathscr{H}^\bullet(f, \mathscr{K}^\bullet) \xrightarrow{\sim} \mathscr{H}^\bullet(f, \mathscr{K}'^\bullet)$$

is an isomorphism of ∂-functors (**0**, 11.4.4).

(6.2.7). Everything above applies naturally, with only a change of notation, to a complex $\mathscr{K}_\bullet$ of quasi-coherent $\mathscr{O}_X$-modules whose differential has degree -1: it is enough to consider the complex $\mathscr{K}^\bullet = (\mathscr{K}^i)$ defined by

$$\mathscr{K}^i = \mathscr{K}_{-i} \qquad (i \in \mathbf{Z}).$$

6.3. Hypertor of two complexes of modules

(6.3.1). Let A be a commutative ring, and let $P_\bullet$ and $Q_\bullet$ be two complexes of A-modules whose differentials have degree -1. Let $L_{\bullet\bullet}$ and $M_{\bullet\bullet}$ be projective Cartan–Eilenberg resolutions of $P_\bullet$ and $Q_\bullet$, respectively (**0**, 11.6.1). Taking the sum of the first degrees and the sum of the second degrees,

$$L_{\bullet\bullet} \otimes_A M_{\bullet\bullet}$$

is a bicomplex whose differentials have degree -1. Its homology

$$\mathrm{H}_\bullet(L_{\bullet\bullet} \otimes_A M_{\bullet\bullet})$$

does not depend on the chosen Cartan–Eilenberg resolutions and is, by definition, the hyperhomology of the bifunctor $P_\bullet \otimes_A Q_\bullet$ in $P_\bullet$ and $Q_\bullet$ (**0**, 11.6.5). We put

(6.3.1.1) $$\mathbf{Tor}_n^A(P_\bullet, Q_\bullet) = \mathrm{H}_n(L_{\bullet\bullet} \otimes_A M_{\bullet\bullet})$$

and call this A-module the *hypertor of index n* of the two complexes $P_\bullet$ and $Q_\bullet$. In the category of bounded-below complexes of A-modules, the modules $\mathbf{Tor}_n^A(P_\bullet, Q_\bullet)$ form a homological bifunctor in $P_\bullet$ and $Q_\bullet$ (**0**, 11.6.5). Moreover: **III** | 142

Proposition (6.3.2). — *The bifunctor $\mathbf{Tor}_\bullet^A(P_\bullet, Q_\bullet)$ is the common abutment of two bifunctorial spectral sequences $'E(P_\bullet, Q_\bullet)$ and $''E(P_\bullet, Q_\bullet)$ whose E_2-terms are*

(6.3.2.1) $$'E^2_{pq} = \mathrm{H}_p\big(\mathrm{Tor}_q^A(P_\bullet, Q_\bullet)\big),$$

and

(6.3.2.2) $$''E^2_{pq} = \bigoplus_{q'+q''=q} \mathrm{Tor}_p^A\big(\mathrm{H}_{q'}(P_\bullet), \mathrm{H}_{q''}(Q_\bullet)\big).$$

In (6.3.2.1), $\mathrm{Tor}_q^A(P_\bullet, Q_\bullet)$ denotes the bicomplex formed by the A-modules $\mathrm{Tor}_q^A(P_i, Q_j)$. The spectral sequence (6.3.2.2) is always regular. If $P_\bullet$ and $Q_\bullet$ are bounded below, or if A has finite cohomological dimension, then both spectral sequences (6.3.2.1) and (6.3.2.2) are biregular.

Proof. This follows from (**0**, 11.6.5). Indeed, if A has finite cohomological dimension n, every A-module has a projective resolution of length n *(M, VI, 2.1).* □

Corollary (6.3.3). — *Let $P'_\bullet$ and $Q'_\bullet$ be complexes of A-modules, and let*

$$u : P_\bullet \longrightarrow P'_\bullet, \qquad v : Q_\bullet \longrightarrow Q'_\bullet$$

be homomorphisms of complexes. If the induced homomorphisms

$$\mathrm{H}_\bullet(u) : \mathrm{H}_\bullet(P_\bullet) \longrightarrow \mathrm{H}_\bullet(P'_\bullet), \qquad \mathrm{H}_\bullet(v) : \mathrm{H}_\bullet(Q_\bullet) \longrightarrow \mathrm{H}_\bullet(Q'_\bullet)$$

are bijective, then the homomorphism

$$\mathbf{Tor}_n^A(P_\bullet, Q_\bullet) \longrightarrow \mathbf{Tor}_n^A(P'_\bullet, Q'_\bullet)$$

induced by u and v is bijective.

Proof. The induced homomorphism

$$''E(P_\bullet, Q_\bullet) \longrightarrow'' E(P'_\bullet, Q'_\bullet)$$

is an isomorphism on the E_2-terms. The conclusion follows from regularity of these spectral sequences (6.3.2) and (**0**, 11.1.5). □

Proposition (6.3.4). — *Let $P_\bullet$ and $Q_\bullet$ be bounded-below complexes of A-modules. Let $L_{\bullet\bullet}$ and $M_{\bullet\bullet}$ be bicomplexes of flat A-modules such that, for every i, $L_{i,\bullet}$ is a resolution of P_i and $M_{i,\bullet}$ is a resolution of Q_i. Then there are canonical isomorphisms*

(6.3.4.1) $$\begin{aligned}\mathbf{Tor}_n^A(P_\bullet, Q_\bullet) &\xrightarrow{\sim} \mathrm{H}_n(L_{\bullet\bullet} \otimes_A Q_\bullet)\\ &\xrightarrow{\sim} \mathrm{H}_n(P_\bullet \otimes_A M_{\bullet\bullet})\\ &\xrightarrow{\sim} \mathrm{H}_n(L_{\bullet\bullet} \otimes_A M_{\bullet\bullet}).\end{aligned}$$

Proof. This follows from (**0**, 11.6.5), parts (ii) and (iii), and from the definition of flat A-modules. □

Remarks (6.3.5). — *(i)* With the notation of (6.3.1), the bicomplexes $L_{\bullet\bullet} \otimes_A M_{\bullet\bullet}$ and $M_{\bullet\bullet} \otimes_A L_{\bullet\bullet}$ are canonically isomorphic. Hence there is a canonical isomorphism

$$\mathbf{Tor}_n^A(P_\bullet, Q_\bullet) \xrightarrow{\sim} \mathbf{Tor}_n^A(Q_\bullet, P_\bullet).$$

(ii) Let F and G be A-modules, and let $P_\bullet$ and $Q_\bullet$ be the complexes concentrated in degree 0 with values F and G, respectively. Projective resolutions $L_\bullet$ of F and $M_\bullet$ of G may be regarded, after adjoining zeros, as Cartan–Eilenberg resolutions of $P_\bullet$ and $Q_\bullet$. Thus in this case

$$\mathbf{Tor}_n^A(P_\bullet, Q_\bullet) = \mathrm{Tor}_n^A(F, G).$$

Proposition (6.3.6). — *Let $(P_\bullet^\lambda)$ and $(Q_\bullet^\mu)$ be two filtered inductive systems of complexes of A-modules. There is a canonical isomorphism*

$$\varinjlim_{\lambda,\mu} \mathbf{Tor}_n^A(P_\bullet^\lambda, Q_\bullet^\mu) \xrightarrow{\sim} \mathbf{Tor}_n^A\left(\varinjlim_\lambda P_\bullet^\lambda, \varinjlim_\mu Q_\bullet^\mu\right). \tag{6.3.6.1}$$

Proof. Put

$$P_\bullet = \varinjlim_\lambda P_\bullet^\lambda, \qquad Q_\bullet = \varinjlim_\mu Q_\bullet^\mu.$$

By functoriality, the modules $\mathbf{Tor}_n^A(P_\bullet^\lambda, Q_\bullet^\mu)$ form an inductive system, and the canonical maps to $\mathbf{Tor}_n^A(P_\bullet, Q_\bullet)$ give the homomorphism (6.3.6.1). More generally, they give a canonical homomorphism of spectral sequences

$$\varinjlim_{\lambda,\mu}{}'' E(P_\bullet^\lambda, Q_\bullet^\mu) \longrightarrow {}'' E(P_\bullet, Q_\bullet),$$

whose homomorphism on abutments is (6.3.6.1).

The spectral sequence ${}''E(P_\bullet, Q_\bullet)$ is regular by (6.3.2). So is $\varinjlim''_{\lambda,\mu} E(P_\bullet^\lambda, Q_\bullet^\mu)$, as follows from the definitions (**0**, 11.1.7) and the proof of (**0**, 11.3.3). By (**0**, 11.1.5), it is therefore enough to prove that

$$\varinjlim_{\lambda,\mu}{}'' E(P_\bullet^\lambda, Q_\bullet^\mu) \longrightarrow {}'' E(P_\bullet, Q_\bullet) \tag{6.3.6.2}$$

is bijective on the E^2-terms.

Since homology commutes with inductive limits of complexes of modules, it remains only to prove that, for filtered inductive systems (F^λ) and (G^μ) of A-modules, the canonical homomorphism

$$\varinjlim_{\lambda,\mu} \mathrm{Tor}_n^A(F^\lambda, G^\mu) \longrightarrow \mathrm{Tor}_n^A\left(\varinjlim_\lambda F^\lambda, \varinjlim_\mu G^\mu\right)$$

is bijective.

For each F^λ, consider its canonical free resolution

$$L_\bullet^\lambda : \quad \cdots \longrightarrow L_{i+1}^\lambda \longrightarrow L_i^\lambda \longrightarrow \cdots \longrightarrow L_1^\lambda \longrightarrow L_0^\lambda \longrightarrow 0,$$

where L_0^λ is the A-module of formal linear combinations of elements of F^λ, and L_{i+1}^λ is the A-module of formal linear combinations of elements of $\mathrm{Ker}(L_i^\lambda \to L_{i-1}^\lambda)$. The $L_\bullet^\lambda$ form an inductive system of complexes. Put

$$F = \varinjlim_\lambda F^\lambda, \qquad L_i = \varinjlim_\lambda L_i^\lambda.$$

Since filtered inductive limits are exact, the L_i form a resolution $L_\bullet$ of F; moreover, each L_i is flat, being an inductive limit of free A-modules (**0**, 6.1.2).

Likewise, let $M_\bullet^\mu$ be the canonical free resolution of G^μ, and put

$$M_\bullet = \varinjlim_\mu M_\bullet^\mu, \qquad G = \varinjlim_\mu G^\mu.$$

Then $M_\bullet$ is a flat resolution of G. By (6.3.5) and (6.3.4),

$$\mathrm{Tor}_n^A(F, G) = \mathrm{H}_n(L_\bullet \otimes_A M_\bullet).$$

Since homology commutes with inductive limits of complexes of modules,

$$\mathrm{H}_n(L_\bullet \otimes_A M_\bullet) = \varinjlim_{\lambda,\mu} \mathrm{H}_n(L^\lambda_\bullet \otimes_A M^\mu_\bullet).$$

The last modules are $\mathrm{Tor}^A_n(F^\lambda, G^\mu)$, which proves (6.3.6.1).

If, in addition, there is an i_0 such that $P^\lambda_i = Q^\mu_i = 0$ for $i < i_0$, for all λ and μ, the same argument proves that the canonical homomorphism

$$\text{(6.3.6.3)} \qquad \varinjlim_{\lambda,\mu}{}' E(P^\lambda_\bullet, Q^\mu_\bullet) \longrightarrow {}' E(P_\bullet, Q_\bullet)$$

is bijective. ☐

Proposition (6.3.7). — *Suppose that $P_\bullet$ and $Q_\bullet$ are bounded below. If $P_\bullet$ consists of flat A-modules, there is a canonical A-linear isomorphism of ∂-functors in $Q_\bullet$*

$$\text{(6.3.7.1)} \qquad \mathbf{Tor}^A_n(P_\bullet, Q_\bullet) \xrightarrow{\sim} \mathrm{H}_n(P_\bullet \otimes_A Q_\bullet).$$

Proof. The spectral sequence (6.3.2.1) is biregular and degenerate, so existence of (6.3.7.1) follows from (**0**, 11.1.6). Computing hypertor from a projective Cartan–Eilenberg resolution of $P_\bullet$ as in (6.3.4) shows immediately that this isomorphism is an isomorphism of ∂-functors in $Q_\bullet$. ☐

(6.3.8). Let $\rho : A \longrightarrow A'$ be a homomorphism of rings. We shall define a functorial A-linear homomorphism of degree zero, canonically associated with ρ, **III** | 144

$$\text{(6.3.8.1)} \qquad \rho_{P_\bullet,Q_\bullet} : \mathbf{Tor}^A_n(P_\bullet, Q_\bullet) \longrightarrow \mathbf{Tor}^{A'}_n(P_\bullet \otimes_A A', Q_\bullet \otimes_A A').$$

For this purpose, let $L_{\bullet\bullet}$ be a projective Cartan–Eilenberg resolution of $P_\bullet$, and let $L'_{\bullet\bullet}$ be a projective Cartan–Eilenberg resolution of $P_\bullet \otimes_A A'$. We first show that there is an A'-linear homomorphism of complexes

$$L_{\bullet\bullet} \otimes_A A' \longrightarrow L'_{\bullet\bullet},$$

determined up to homotopy. Indeed, the construction of $L_{\bullet\bullet}$ is completely determined once one chooses, arbitrarily for each i, a projective resolution $(X^{\mathrm{B}}_{ij})_{j\geqslant 0}$ of $\mathrm{B}_i(P_\bullet)$ and a projective resolution $(X^{\mathrm{H}}_{ij})_{j\geqslant 0}$ of $\mathrm{H}_i(P_\bullet)$. These are respectively $\mathrm{B}^{\mathrm{I}}_i(L_{\bullet\bullet})$ and $\mathrm{H}^{\mathrm{I}}_i(L_{\bullet\bullet})$. One then obtains successively

$$\mathrm{Z}^{\mathrm{I}}_i(L_{\bullet\bullet}) = \mathrm{H}^{\mathrm{I}}_i(L_{\bullet\bullet}) \oplus \mathrm{B}^{\mathrm{I}}_i(L_{\bullet\bullet}), \qquad L_{i,\bullet} = \mathrm{Z}^{\mathrm{I}}_i(L_{\bullet\bullet}) \oplus \mathrm{B}^{\mathrm{I}}_{i-1}(L_{\bullet\bullet}).$$

Now $X^{\mathrm{B}}_{i,\bullet} \otimes_A A'$ is not in general a resolution of $\mathrm{B}_i(P_\bullet) \otimes_A A'$, but it is still a complex of projective A'-modules. Hence there is an A'-linear homomorphism

$$X^{\mathrm{B}}_{i,\bullet} \otimes_A A' \longrightarrow \mathrm{B}^{\mathrm{I}}_i(L'_{\bullet\bullet})$$

compatible with the augmentations and determined up to homotopy (M, V, 1.1). Likewise there is an A'-linear homomorphism

$$X^{\mathrm{H}}_{i,\bullet} \otimes_A A' \longrightarrow \mathrm{H}^{\mathrm{I}}_i(L'_{\bullet\bullet})$$

determined up to homotopy. By the construction just recalled, these maps give, for every i, an A'-linear homomorphism

$$L_{i,\bullet} \otimes_A A' \longrightarrow L'_{i,\bullet}.$$

For $i \in \mathbb{Z}$ these homomorphisms are compatible with the differentials $L_{i,\bullet} \longrightarrow L_{i-1,\bullet}$ and their analogues for $L'_{\bullet\bullet}$; together they therefore give the required homomorphism $L_{\bullet\bullet} \otimes_A A' \longrightarrow L'_{\bullet\bullet}$.

To define (6.3.8.1), take in the same way projective Cartan–Eilenberg resolutions $M_{\bullet\bullet}$ and $M'_{\bullet\bullet}$ of $Q_\bullet$ and $Q_\bullet \otimes_A A'$, respectively, and an A'-linear homomorphism $M_{\bullet\bullet} \otimes_A A' \longrightarrow M'_{\bullet\bullet}$. The resulting homomorphism

$$(L_{\bullet\bullet} \otimes_A A') \otimes_{A'} (M_{\bullet\bullet} \otimes_A A') \longrightarrow L'_{\bullet\bullet} \otimes_{A'} M'_{\bullet\bullet}$$

gives, by composition, an A-linear homomorphism of bicomplexes

$$L_{\bullet\bullet} \otimes_A M_{\bullet\bullet} \longrightarrow L'_{\bullet\bullet} \otimes_{A'} M'_{\bullet\bullet}.$$

Passing to homology yields (6.3.8.1); it is well defined because it is induced by a homomorphism of complexes determined up to homotopy.

If $\rho' : A' \longrightarrow A''$ is a second homomorphism of rings and $\rho'' = \rho' \circ \rho : A \longrightarrow A''$, then

$$\rho''_{P_\bullet,Q_\bullet} = \rho'_{P'_\bullet,Q'_\bullet} \circ \rho_{P_\bullet,Q_\bullet}, \qquad P'_\bullet = P_\bullet \otimes_A A', \quad Q'_\bullet = Q_\bullet \otimes_A A'.$$

The preceding homomorphism of bicomplexes also defines functorial homomorphisms of spectral sequences

$${}'E^r_{pq}(P_\bullet, Q_\bullet) \longrightarrow {}' E^r_{pq}(P_\bullet \otimes_A A', Q_\bullet \otimes_A A')$$

and

$${}''E^r_{pq}(P_\bullet, Q_\bullet) \longrightarrow {}'' E^r_{pq}(P_\bullet \otimes_A A', Q_\bullet \otimes_A A').$$

They are independent of the Cartan–Eilenberg resolutions chosen and satisfy the same transitivity property.

Proposition (6.3.9). — *Let $\rho : A \longrightarrow A'$ be a homomorphism of rings such that A' is flat as an A-module. There are canonical functorial isomorphisms*

(6.3.9.1)
$$\mathbf{Tor}^{A'}_n(P_\bullet \otimes_A A', Q_\bullet \otimes_A A') \xrightarrow{\sim} \mathbf{Tor}^A_n(P_\bullet, Q_\bullet) \otimes_A A',$$

(6.3.9.2)
$${}'E(P_\bullet \otimes_A A', Q_\bullet \otimes_A A') \xrightarrow{\sim} {}' E(P_\bullet, Q_\bullet) \otimes_A A',$$

and

(6.3.9.3)
$${}''E(P_\bullet \otimes_A A', Q_\bullet \otimes_A A') \xrightarrow{\sim} {}'' E(P_\bullet, Q_\bullet) \otimes_A A'.$$

Proof. Since the functor $M \longmapsto M \otimes_A A'$ is exact, $L_{\bullet\bullet} \otimes_A A'$ and $M_{\bullet\bullet} \otimes_A A'$ are projective Cartan– **III | 145**
Eilenberg resolutions of $P_\bullet \otimes_A A'$ and $Q_\bullet \otimes_A A'$, respectively. The assertion follows. □

(6.3.10). Let $\rho : A \longrightarrow A'$ be a homomorphism of rings. For every complex $P'_\bullet$ of A'-modules, let $P'_{\bullet[\rho]}$ denote the same complex regarded as a complex of A-modules. The identity map $P'_{\bullet[\rho]} \longrightarrow P'_\bullet$ is the composite of the canonical maps

$$P'_{\bullet[\rho]} \longrightarrow P'_{\bullet[\rho]} \otimes_A A' \xrightarrow{\mu} P'_\bullet,$$

where μ is the A'-linear homomorphism $\mu(x \otimes a') = a'x$. If $Q'_\bullet$ is a second complex of A'-modules, there are therefore canonical functorial homomorphisms of degree zero

(6.3.10.1)
$$\begin{aligned}\mathbf{Tor}^A_n(P'_{\bullet[\rho]}, Q'_{\bullet[\rho]}) &\longrightarrow \mathbf{Tor}^{A'}_n(P'_{\bullet[\rho]} \otimes_A A', Q'_{\bullet[\rho]} \otimes_A A') \\ &\longrightarrow \mathbf{Tor}^{A'}_n(P'_\bullet, Q'_\bullet).\end{aligned}$$

The first arrow is the A-linear homomorphism defined in (6.3.8); the second follows by functoriality from the A'-linear homomorphisms

$$P'_{\bullet[\rho]} \otimes_A A' \longrightarrow P'_\bullet, \qquad Q'_{\bullet[\rho]} \otimes_A A' \longrightarrow Q'_\bullet.$$

There are analogous homomorphisms for the spectral sequences of (6.3.2), together with the evident transitivity properties, whose formulation is left to the reader.

Proposition (6.3.11). — *Let $\rho : A \longrightarrow A'$ be a homomorphism of rings such that A' is flat as an A-module. For every complex $P'_\bullet$ of A'-modules and every bounded-below complex $Q_\bullet$ of A-modules, there is a canonical functorial isomorphism*

(6.3.11.1)
$$\mathbf{Tor}^A_n(P'_{\bullet[\rho]}, Q_\bullet) \xrightarrow{\sim} \mathbf{Tor}^{A'}_n(P'_\bullet, Q_\bullet \otimes_A A').$$

Proof. If $M_{\bullet\bullet}$ is a projective Cartan–Eilenberg resolution of $Q_\bullet$, then $M_{\bullet\bullet} \otimes_A A'$ is a projective Cartan–Eilenberg resolution of $Q_\bullet \otimes_A A'$. Moreover, up to canonical isomorphism,

$$P'_{\bullet[\rho]} \otimes_A M_{\bullet\bullet} = P'_\bullet \otimes_{A'} (M_{\bullet\bullet} \otimes_A A').$$

The assertion follows from (6.3.4). □

Remark (6.3.12). — Let (A^λ) be a filtered inductive system of rings, and let $(P_\bullet^\lambda)$ and $(Q_\bullet^\lambda)$ be two inductive systems of complexes of A^λ-modules. There is a canonical isomorphism generalizing (6.3.6.1),

(6.3.12.1) $$\varinjlim_\lambda \mathbf{Tor}_n^{A^\lambda}(P_\bullet^\lambda, Q_\bullet^\lambda) \xrightarrow{\sim} \mathbf{Tor}_n^A(P_\bullet, Q_\bullet),$$

where

$$A = \varinjlim_\lambda A^\lambda, \qquad P_\bullet = \varinjlim_\lambda P_\bullet^\lambda, \qquad Q_\bullet = \varinjlim_\lambda Q_\bullet^\lambda.$$

Once the homomorphisms

$$\mathbf{Tor}_n^{A^\lambda}(P_\bullet^\lambda, Q_\bullet^\lambda) \longrightarrow \mathbf{Tor}_n^{A^\mu}(P_\bullet^\mu, Q_\bullet^\mu) \qquad (\lambda \leqslant \mu)$$

have been defined by (6.3.10), the proof is the same as that of (6.3.6).

Proposition (6.3.13). — *Let S be a multiplicative subset of A, and let $P_\bullet$ and $Q_\bullet$ be complexes of A-modules on which multiplication by every element of S is bijective. Thus, if $A' = S^{-1}A$, both complexes consist of A'-modules. Then there is a canonical isomorphism*

$$\mathbf{Tor}_n^A(P_\bullet, Q_\bullet) \xrightarrow{\sim} \mathbf{Tor}_n^{A'}(P_\bullet, Q_\bullet).$$

Proof. The hypothesis implies that the canonical homomorphisms

$$P_\bullet \longrightarrow P_\bullet \otimes_A A', \qquad Q_\bullet \longrightarrow Q_\bullet \otimes_A A'$$

are bijective. Functoriality of hypertor also shows that each $s \in S$ acts bijectively on $\mathbf{Tor}_n^A(P_\bullet, Q_\bullet)$; hence

$$\mathbf{Tor}_n^A(P_\bullet, Q_\bullet) \longrightarrow \mathbf{Tor}_n^A(P_\bullet, Q_\bullet) \otimes_A A'$$

is bijective. Since A' is flat over A, the assertion follows from (6.3.9). The same argument gives **III** | 146
canonical isomorphisms of the associated spectral sequences. □

6.4. Local hypertor functors of complexes of quasi-coherent modules: the affine scheme case

(6.4.1). Let S be an affine scheme with ring A, and let X and Y be affine S-schemes with rings B and C, respectively; thus B and C are A-algebras. Every complex $\mathscr{P}_\bullet$ of quasi-coherent $\mathscr{O}_X$-modules is of the form $\widetilde{P}_\bullet$ for a complex $P_\bullet$ of B-modules, and similarly every complex $\mathscr{Q}_\bullet$ of quasi-coherent $\mathscr{O}_Y$-modules is of the form $\widetilde{Q_\bullet}$ for a complex $Q_\bullet$ of C-modules ((**I**, 1.3.7) and (**I**, 1.3.8)).

Regarded as complexes of A-modules, $P_\bullet$ and $Q_\bullet$ have hypertor modules $\mathbf{Tor}_n^A(P_\bullet, Q_\bullet)$. By bifunctoriality, the A-algebras B and C act on this A-module; these actions make it a (B, C)-bimodule, equivalently a module over

$$B \otimes_A C = A(X \times_S Y).$$

We have thus defined a quasi-coherent $\mathscr{O}_{X \times_S Y}$-module

(6.4.1.1) $$\mathscr{T}or_n^{\mathscr{O}_S}(\mathscr{P}_\bullet, \mathscr{Q}_\bullet) = \big(\mathbf{Tor}_n^A(P_\bullet, Q_\bullet)\big)^\sim.$$

It is called the *local hypertor of index n* of $\mathscr{P}_\bullet$ and $\mathscr{Q}_\bullet$, and is also denoted $\mathscr{T}or_n^S(\mathscr{P}_\bullet, \mathscr{Q}_\bullet)$.

Lemma (6.4.2). — *With the notation of (6.4.1), suppose that $A = R^{-1}A'$, where A' is a ring and R is a multiplicative subset of A'. Let $S' = \operatorname{Spec}(A')$. Then X and Y may be regarded as S'-schemes and $X \times_{S'} Y = X \times_S Y$ ((**I**, 1.6.2) and (3.2.4)); moreover,*

$$\mathscr{T}or_n^{S'}(\mathscr{P}_\bullet, \mathscr{Q}_\bullet) = \mathscr{T}or_n^S(\mathscr{P}_\bullet, \mathscr{Q}_\bullet).$$

Proof. This follows from (6.4.1.1) and (6.3.13). □

(6.4.3). With the notation and hypotheses of (6.4.1), let $\mathscr{F} = \widetilde{F}$ be a quasi-coherent $\mathscr{O}_X$-module and $\mathscr{G} = \widetilde{G}$ a quasi-coherent $\mathscr{O}_Y$-module. Regarding $\mathscr{F}$ and $\mathscr{G}$ as complexes concentrated in degree zero, we write $\mathscr{T}or_n^{\mathscr{O}_S}(\mathscr{F}, \mathscr{G})$, or $\mathscr{T}or_n^S(\mathscr{F}, \mathscr{G})$, for their local hypertor of index n. By (6.3.5)(ii),

(6.4.3.1) $$\mathscr{T}or_n^{\mathscr{O}_S}(\mathscr{F}, \mathscr{G}) = \big(\mathrm{Tor}_n^A(F, G)\big)^\sim.$$

Return now to two arbitrary complexes of quasi-coherent modules $\mathscr{P}_\bullet$ and $\mathscr{Q}_\bullet$. Formulas (6.4.1.1) and (6.4.3.1), together with (6.3.2), show that $\mathscr{T}or_n^S(\mathscr{P}_\bullet, \mathscr{Q}_\bullet)$ is the common abutment of two spectral sequences ${}'\mathscr{E}(\mathscr{P}_\bullet, \mathscr{Q}_\bullet)$ and ${}''\mathscr{E}(\mathscr{P}_\bullet, \mathscr{Q}_\bullet)$ whose E_2-terms are

(6.4.3.2) $$ {}'\mathscr{E}_{pq}^2 = \mathscr{H}_p\big(\mathscr{T}or_q^S(\mathscr{P}_\bullet, \mathscr{Q}_\bullet)\big)$$

and

(6.4.3.3) $$ {}''\mathscr{E}_{pq}^2 = \bigoplus_{q'+q''=q} \mathscr{T}or_p^S\big(\mathscr{H}_{q'}(\mathscr{P}_\bullet), \mathscr{H}_{q''}(\mathscr{Q}_\bullet)\big).$$

Here $\mathscr{T}or_q^S(\mathscr{P}_\bullet, \mathscr{Q}_\bullet)$ in (6.4.3.2) denotes the bicomplex of quasi-coherent $\mathscr{O}_{X\times_S Y}$-modules $\mathscr{T}or_q^S(\mathscr{P}_i, \mathscr{Q}_j)$.

(6.4.4). Let

$$X^{(1)} = \operatorname{Spec}(B^{(1)}), \qquad Y^{(1)} = \operatorname{Spec}(C^{(1)})$$

be two further affine schemes, where $B^{(1)}$ and $C^{(1)}$ are A-algebras, and let **III** | 147

$$u : X^{(1)} \longrightarrow X, \qquad v : Y^{(1)} \longrightarrow Y$$

be S-morphisms corresponding to the A-homomorphisms $\phi : B \longrightarrow B^{(1)}$ and $\psi : C \longrightarrow C^{(1)}$. Consider the complexes

$$u^*(\mathscr{P}_\bullet) = (P_\bullet \otimes_B B^{(1)})^\sim, \qquad v^*(\mathscr{Q}_\bullet) = (Q_\bullet \otimes_C C^{(1)})^\sim.$$

The canonical A-linear homomorphisms

$$P_\bullet \longrightarrow P_\bullet \otimes_B B^{(1)}, \qquad Q_\bullet \longrightarrow Q_\bullet \otimes_C C^{(1)}$$

give, by functoriality, an A-linear homomorphism

$$\mathbf{Tor}_n^A(P_\bullet, Q_\bullet) \longrightarrow \mathbf{Tor}_n^A(P_\bullet \otimes_B B^{(1)}, Q_\bullet \otimes_C C^{(1)}).$$

Again by functoriality, this is in fact a homomorphism of $(B \otimes_A C)$-modules. It therefore defines a $(u \times_S v)$-morphism

(6.4.4.1) $$\theta : \mathscr{T}or_n^S(\mathscr{P}_\bullet, \mathscr{Q}_\bullet) \longrightarrow \mathscr{T}or_n^S(u^*\mathscr{P}_\bullet, v^*\mathscr{Q}_\bullet),$$

and hence a homomorphism of $\mathscr{O}_{X^{(1)}\times_S Y^{(1)}}$-modules

(6.4.4.2) $$\theta^\sharp : (u\times_S v)^*\mathscr{T}or_n^S(\mathscr{P}_\bullet, \mathscr{Q}_\bullet) \longrightarrow \mathscr{T}or_n^S(u^*\mathscr{P}_\bullet, v^*\mathscr{Q}_\bullet).$$

This is plainly a morphism of bi-∂-functors on the categories of bounded-below quasi-coherent modules. The homomorphism (6.4.4.2) need not be bijective; nevertheless:

Lemma (6.4.5). — *With the notation of (6.4.4), if u and v are open immersions, then (6.4.4.2) is bijective.*

Proof. Identify $X^{(1)}$ and $Y^{(1)}$ with open subsets of X and Y. They are unions of open subsets of the form $\mathrm{D}(f)$ and $\mathrm{D}(g)$, where $f \in B$ and $g \in C$; the schemes induced on $\mathrm{D}(\phi(f))$ and $\mathrm{D}(f)$, and likewise on $\mathrm{D}(\psi(g))$ and $\mathrm{D}(g)$, are isomorphic.

It is enough to prove the lemma when $X^{(1)} = \mathrm{D}(f)$ and $Y^{(1)} = \mathrm{D}(g)$. Indeed, once this is known, in the general case it suffices to show that the restriction of $\theta^\sharp$ to every $\mathrm{D}(f)\times_S \mathrm{D}(g)$ is an isomorphism. If

$$u_1 : \mathrm{D}(f) \longrightarrow X^{(1)}, \qquad v_1 : \mathrm{D}(g) \longrightarrow Y^{(1)}$$

are the canonical inclusions, this restriction is $(u_1 \times_S v_1)^*(\theta^\sharp)$. From (6.4.4) and (**0**, 4.4.8), it is immediate that composing it with the canonical homomorphism

(6.4.5.1) $$(u_1\times_S v_1)^*\mathscr{T}or_n^S(u^*\mathscr{P}_\bullet, v^*\mathscr{Q}_\bullet) \longrightarrow \mathscr{T}or_n^S((u')^*\mathscr{P}_\bullet, (v')^*\mathscr{Q}_\bullet),$$

where $u' = u \circ u_1$ and $v' = v \circ v_1$, gives the canonical homomorphism

(6.4.5.2) $$(u'\times_S v')^*\mathscr{T}or_n^S(\mathscr{P}_\bullet, \mathscr{Q}_\bullet) \longrightarrow \mathscr{T}or_n^S((u')^*\mathscr{P}_\bullet, (v')^*\mathscr{Q}_\bullet).$$

If (6.4.5.1) and (6.4.5.2) are isomorphisms, then so is $(u_1\times_S v_1)^*(\theta^\sharp)$.

Suppose therefore that $X^{(1)} = \mathrm{D}(f)$ and $Y^{(1)} = \mathrm{D}(g)$, so that $B^{(1)} = B_f$ and $C^{(1)} = C_g$. Then $u^*(\mathscr{P}_\bullet)$ and $v^*(\mathscr{Q}_\bullet)$ identify with $((P_\bullet)_f)^\sim$ and $((Q_\bullet)_g)^\sim$, respectively. Moreover, $X^{(1)}\times_S Y^{(1)}$ identifies with the open subscheme $\mathrm{D}(f\otimes g)$ of

$$X \times_S Y = \operatorname{Spec}(B \otimes_A C)$$

((**II**, 4.3.2.4)). It remains to prove that the homomorphism

$$(6.4.5.3)\qquad \big(\mathbf{Tor}_n^A(P_\bullet, Q_\bullet)\big)_{f\otimes g} \longrightarrow \mathbf{Tor}_n^A((P_\bullet)_f, (Q_\bullet)_g),$$

induced by the canonical homomorphisms $P_\bullet \longrightarrow (P_\bullet)_f$ and $Q_\bullet \longrightarrow (Q_\bullet)_g$, is bijective. **III** | 148

By (**0**, 1.6.1),

$$(P_\bullet)_f = \varinjlim_m P_\bullet^{(m)},$$

where every $P_\bullet^{(m)}$ is the same complex of B-modules as $P_\bullet$, and for $m \leqslant n$ the map $P_\bullet^{(m)} \longrightarrow P_\bullet^{(n)}$ is multiplication by f^{n-m}. The analogous statement holds for $Q_\bullet$, with g in place of f. The homomorphism

$$\mathbf{Tor}_n^A(P_\bullet^{(m)}, Q_\bullet^{(m)}) \longrightarrow \mathbf{Tor}_n^A(P_\bullet^{(n)}, Q_\bullet^{(n)})$$

corresponding to these transition maps is, by definition, multiplication by $(f\otimes g)^{n-m}$. The conclusion follows from (**0**, 1.6.1), applied to the left-hand side of (6.4.5.3), and from (6.3.6). □

(6.4.6). With the notation of (6.4.4), one defines in the same way canonical homomorphisms of spectral functors

$$(6.4.6.1)\qquad \begin{aligned}(u\times_S v)^*\,{}'\mathscr{E}(\mathscr{P}_\bullet, \mathscr{Q}_\bullet) &\longrightarrow {}'\mathscr{E}(u^*\mathscr{P}_\bullet, v^*\mathscr{Q}_\bullet),\\ (u\times_S v)^*\,{}''\mathscr{E}(\mathscr{P}_\bullet, \mathscr{Q}_\bullet) &\longrightarrow {}''\mathscr{E}(u^*\mathscr{P}_\bullet, v^*\mathscr{Q}_\bullet).\end{aligned}$$

The argument of (6.4.5) shows that when u and v are open immersions, these homomorphisms are bijective. Taking (6.3.6.2) and (6.3.6.3) into account, the argument proves that they are isomorphisms on the E^2-terms, while (6.4.5) proves that they are isomorphisms on the abutments. The assertion therefore follows from (**0**, 11.1.2) and (**0**, 11.2.4).

6.5. Local hypertor functors of complexes of quasi-coherent modules: the general case

(6.5.1). Let S now be an arbitrary scheme and let X and Y be two S-schemes. Let $\mathscr{P}_\bullet$ be a complex of quasi-coherent $\mathscr{O}_X$-modules and $\mathscr{Q}_\bullet$ a complex of quasi-coherent $\mathscr{O}_Y$-modules. Set $Z = X\times_S Y$. We shall define quasi-coherent $\mathscr{O}_Z$-modules

$$\mathscr{T}\!or_n^S(\mathscr{P}_\bullet, \mathscr{Q}_\bullet),$$

called the *local hypertors* of $\mathscr{P}_\bullet$ and $\mathscr{Q}_\bullet$, which reduce to those of (6.4) when S, X, and Y are affine.

When $\mathscr{P}_\bullet$ and $\mathscr{Q}_\bullet$ are concentrated in degree zero, with respective terms $\mathscr{F}$ and $\mathscr{G}$, we write $\mathscr{T}\!or_n^S(\mathscr{F},\mathscr{G})$ instead of $\mathscr{T}\!or_n^S(\mathscr{P}_\bullet, \mathscr{Q}_\bullet)$.

(6.5.2). Suppose first that S is affine, and let (X_λ) and (Y_μ) be coverings of X and Y by affine open subsets. Then

$$Z_{\lambda\mu} = X_\lambda \times_S Y_\mu$$

form an affine open covering of Z. Put

$$\mathscr{P}_{\lambda,\bullet} = \mathscr{P}_\bullet|X_\lambda, \qquad \mathscr{Q}_{\mu,\bullet} = \mathscr{Q}_\bullet|Y_\mu.$$

For each pair (λ,μ) we thus have a quasi-coherent $\mathscr{O}_{Z_{\lambda\mu}}$-module

$$\mathscr{F}_{\lambda\mu} = \mathscr{T}\!or_n^S(\mathscr{P}_{\lambda,\bullet}, \mathscr{Q}_{\mu,\bullet}).$$

We must show that the $\mathscr{F}_{\lambda\mu}$ satisfy the gluing condition (**0**, 3.3.1). It is enough to verify that, for every affine open

$$U \subset X_\lambda \cap X_{\lambda'}, \qquad V \subset Y_\mu \cap Y_{\mu'},$$

the restrictions of $\mathscr{F}_{\lambda\mu}$ and $\mathscr{F}_{\lambda'\mu'}$ to $U\times_S V$ are canonically isomorphic. This follows at once from their canonical isomorphisms with

$$\mathscr{T}\!or_n^S(\mathscr{P}_\bullet|U, \mathscr{Q}_\bullet|V)$$

given by (6.4.5).

The definition and (6.4.5) also show that the resulting $\mathscr{O}_Z$-module is independent, up to a unique isomorphism compatible with the local identifications, of the chosen open coverings. We denote it by **III** | 149

$$\mathscr{T}\!or_n^S(\mathscr{P}_\bullet, \mathscr{Q}_\bullet).$$

Finally, for every open subset U of X and every open subset V of Y, its restriction to $U \times_S V$ is canonically isomorphic to

$$\mathcal{T}or_n^S(\mathcal{P}_\bullet|U, \mathcal{Q}_\bullet|V).$$

(6.5.3). Pass now to the general case, where S is arbitrary. Let (S_α) be a covering of S by affine open subsets, and let X_α and Y_α be the inverse images of S_α in X and Y. We must show that the sheaves

$$\mathcal{G}_\alpha = \mathcal{T}or_n^{S_\alpha}(\mathcal{P}_\bullet|X_\alpha, \mathcal{Q}_\bullet|Y_\alpha)$$

satisfy the gluing condition.

For every affine open subset $T \subset S_\alpha \cap S_\beta$, let U and V be its inverse images in X and Y. It is enough to define canonical isomorphisms between the restrictions of $\mathcal{G}_\alpha$ and $\mathcal{G}_\beta$ to $U \times_S V$ and

$$\mathcal{T}or_n^T(\mathcal{P}_\bullet|U, \mathcal{Q}_\bullet|V).$$

We may further restrict to the case in which $T = \mathrm{D}(f_\alpha) = \mathrm{D}(f_\beta)$, where f_α and f_β are sections of $\mathcal{O}_S$ over S_α and S_β, respectively. By (6.4.2),

$$\mathcal{T}or_n^T(\mathcal{P}_\bullet|U, \mathcal{Q}_\bullet|V)$$

is canonically isomorphic both to $\mathcal{T}or_n^{S_\alpha}(\mathcal{P}_\bullet|U, \mathcal{Q}_\bullet|V)$ and to $\mathcal{T}or_n^{S_\beta}(\mathcal{P}_\bullet|U, \mathcal{Q}_\bullet|V)$. Combining these isomorphisms with the local identifications of (6.5.2) completes the definition of the $\mathcal{O}_Z$-module

$$\mathcal{T}or_n^S(\mathcal{P}_\bullet, \mathcal{Q}_\bullet).$$

For every open subset U of X and every open subset V of Y, the restriction of this module to $U \times_S V$ is canonically isomorphic to

$$\mathcal{T}or_n^S(\mathcal{P}_\bullet|U, \mathcal{Q}_\bullet|V).$$

This construction plainly defines, on the categories of bounded-below complexes of quasi-coherent modules, a bi-∂-functor

$$\mathcal{T}or_\bullet^S(\mathcal{P}_\bullet, \mathcal{Q}_\bullet)$$

with values in $\mathcal{O}_Z$-modules. Indeed the assertion is local on X, Y, and S, by (6.4.5) and by the fact that (6.4.4.2) is a morphism of bi-∂-functors.

If $\mathcal{P}_\bullet$ and $\mathcal{Q}_\bullet$ are concentrated in degree zero, with respective terms $\mathcal{F}$ and $\mathcal{G}$, then

$$\mathcal{T}or_0^S(\mathcal{F}, \mathcal{G}) = \mathcal{F} \otimes_S \mathcal{G},$$

the external tensor product defined in (**I**, 9.1.2); this follows from (6.4.1.1) and (**I**, 9.1.3).

(6.5.4). The preceding construction and (6.4.6) show that $\mathcal{T}or_n^S(\mathcal{P}_\bullet, \mathcal{Q}_\bullet)$ is the common abutment of two spectral functors $'\mathcal{E}(\mathcal{P}_\bullet, \mathcal{Q}_\bullet)$ and $''\mathcal{E}(\mathcal{P}_\bullet, \mathcal{Q}_\bullet)$, whose E_2-terms are

$$'\mathcal{E}_{pq}^2 = \mathcal{H}_p\big(\mathcal{T}or_q^S(\mathcal{P}_\bullet, \mathcal{Q}_\bullet)\big) \tag{6.5.4.1}$$

and

$$''\mathcal{E}_{pq}^2 = \bigoplus_{q'+q''=q} \mathcal{T}or_p^S\big(\mathcal{H}_{q'}(\mathcal{P}_\bullet), \mathcal{H}_{q''}(\mathcal{Q}_\bullet)\big). \tag{6.5.4.2}$$

The spectral sequence (6.5.4.2) is always regular. Both spectral sequences are biregular when $\mathcal{P}_\bullet$ and $\mathcal{Q}_\bullet$ are bounded below. A further case in which they are biregular is described next.

(6.5.5). Let T be a topological space. A sheaf of rings $\mathcal{A}$ on T is said to have *cohomological dimension at most n* if the ring $\mathcal{A}_t$ has cohomological dimension at most n for every $t \in T$. In this case the ringed space $(T, \mathcal{A})$ is also said to have cohomological dimension at most n. A sheaf of rings, or a ringed space, has *finite cohomological dimension* if this holds for some integer n.

If the $\mathcal{A}_t$ are Noetherian commutative local rings, saying that they have cohomological dimension at most n means that they are regular and have Krull dimension at most n ((**0**, 17.3.1)). In the terminology of dimension theory to be introduced in Chapter IV, a locally Noetherian scheme T has cohomological dimension at most n exactly when it is regular ((**0**, 4.1.4)) and has dimension at most n. Equivalently, for every affine open subset U of T, the ring $\Gamma(U, \mathcal{O}_T)$ has cohomological dimension at most n ((**0**, 17.2.6)). **III** | 150

Consequently, if S is locally Noetherian and has finite cohomological dimension, then (6.3.2) shows that both $'\mathcal{E}(\mathcal{P}_\bullet, \mathcal{Q}_\bullet)$ and $''\mathcal{E}(\mathcal{P}_\bullet, \mathcal{Q}_\bullet)$ are biregular.

Finally, under the canonical isomorphism $X \times_S Y \xrightarrow{\sim} Y \times_S X$, $\mathscr{T}\!or_n^S(\mathscr{P}_\bullet, \mathscr{Q}_\bullet)$ corresponds canonically to $\mathscr{T}\!or_n^S(\mathscr{Q}_\bullet, \mathscr{P}_\bullet)$.

Proposition (6.5.6). — *Let $(\mathscr{P}_{\alpha\bullet})$ be a filtered direct system of complexes of quasi-coherent $\mathscr{O}_X$-modules. There is a canonical isomorphism*

(6.5.6.1)
$$\varinjlim_{\alpha} \mathscr{T}\!or_n^S(\mathscr{P}_{\alpha\bullet}, \mathscr{Q}_\bullet) \xrightarrow{\sim} \mathscr{T}\!or_n^S\left(\varinjlim_{\alpha} \mathscr{P}_{\alpha\bullet}, \mathscr{Q}_\bullet\right).$$

Proof. The question is local on S, X, and Y. We may therefore suppose that S, X, and Y are affine, in which case the proposition reduces to (6.3.6). ☐

(6.5.7). *Remarks.*

(i) Consider in particular the case $S = X = Y$, with $\mathscr{P}_\bullet$ and $\mathscr{Q}_\bullet$ two complexes of quasi-coherent $\mathscr{O}_S$-modules. Then the $\mathscr{T}\!or_n^S(\mathscr{P}_\bullet, \mathscr{Q}_\bullet)$ are quasi-coherent $\mathscr{O}_S$-modules. Moreover, for every point $z \in S$, (6.5.6) gives a canonical isomorphism

(6.5.7.1)
$$\left(\mathscr{T}\!or_n^S(\mathscr{P}_\bullet, \mathscr{Q}_\bullet)\right)_z \xrightarrow{\sim} \mathbf{Tor}_n^{\mathscr{O}_z}((\mathscr{P}_\bullet)_z, (\mathscr{Q}_\bullet)_z),$$

because the question is local and, by (6.4.1.1), reduces to the case of modules.

(ii) The definition of hypertor can be generalized to two complexes $\mathscr{P}_\bullet, \mathscr{Q}_\bullet$ of $\mathscr{O}_X$-modules on the same ringed space $(X, \mathscr{O}_X)$. For every open subset U of X, put

$$A(U) = \Gamma(U, \mathscr{O}_X), \qquad P_\bullet(U) = \Gamma(U, \mathscr{P}_\bullet), \qquad Q_\bullet(U) = \Gamma(U, \mathscr{Q}_\bullet).$$

The $A(U)$-modules

$$\mathbf{Tor}_n^{A(U)}(P_\bullet(U), Q_\bullet(U))$$

then form a presheaf on X; denote the associated $\mathscr{O}_X$-module by $\mathscr{T}\!or_n^X(\mathscr{P}_\bullet, \mathscr{Q}_\bullet)$. When X is a prescheme, (6.3.12) shows that this module is canonically isomorphic to the hypertor defined above. We shall not pursue this generalization further.

Proposition (6.5.8). — *Let X and Y be two S-preschemes, and let $\mathscr{F}$ (respectively $\mathscr{G}$) be a quasi-coherent $\mathscr{O}_X$-module (respectively $\mathscr{O}_Y$-module). If $\mathscr{F}$ or $\mathscr{G}$ is S-flat, then*

$$\mathscr{T}\!or_n^S(\mathscr{F}, \mathscr{G}) = 0 \qquad (n \neq 0).$$

Proof. The question is local on X and Y, so we may suppose that X, Y, and S are affine, with respective rings B, C, and A, and write $\mathscr{F} = \widetilde{M}$ and $\mathscr{G} = \widetilde{N}$, where M is a B-module and N a C-module. Suppose, for example, that $\mathscr{F}$ is S-flat. This means that, for every $s \in S$, M_s is a flat A_s-module ((**0**, 6.7.1)); hence M is a flat A-module ((**0**, 6.3.3)). Thus $\mathrm{Tor}_n^A(M, N) = 0$ for $n > 0$ and every C-module N ((**0**, 6.1.1)), and the conclusion follows from (6.4.1.1). ☐

Corollary (6.5.9). — *Let X and Y be two S-preschemes, and let $\mathscr{P}_\bullet$ (respectively $\mathscr{Q}_\bullet$) be a bounded-below complex of quasi-coherent $\mathscr{O}_X$-modules (respectively $\mathscr{O}_Y$-modules). Suppose that every $\mathscr{P}_i$ is S-flat. Then there is a canonical isomorphism of ∂-functors in $\mathscr{Q}_\bullet$* III | 151

(6.5.9.1)
$$\mathscr{T}\!or_\bullet^S(\mathscr{P}_\bullet, \mathscr{Q}_\bullet) \xrightarrow{\sim} \mathscr{H}_\bullet(\mathscr{P}_\bullet \otimes_S \mathscr{Q}_\bullet).$$

Proof. When S, X, and Y are affine, this is precisely (6.3.7); the general case follows by the arguments of (6.5.2) and (6.5.3). ☐

(6.5.10). *Corollary.* Suppose that X is flat over S ((**0**, 6.7.1)), that $\mathscr{P}_\bullet$ and $\mathscr{Q}_\bullet$ are bounded below, and that every $\mathscr{P}_i$ is a locally free $\mathscr{O}_X$-module, not necessarily of finite type. Then the homomorphism (6.5.9.1) is bijective.

Indeed, the hypothesis of (6.5.9) is satisfied: flatness is, by definition, a pointwise property on X, and every direct sum of flat modules is flat ((**0**, 6.1.2)).

Proposition (6.5.11). — *Let X' and Y' be two S-preschemes, and let $f : X \to X'$ and $g : Y \to Y'$ be two affine S-morphisms. For a complex $\mathscr{P}_\bullet$ of quasi-coherent $\mathscr{O}_X$-modules and a complex $\mathscr{Q}_\bullet$ of quasi-coherent $\mathscr{O}_Y$-modules, there is a canonical functorial isomorphism*

(6.5.11.1)
$$(f \times_S g)_*(\mathscr{T}\!or_n^S(\mathscr{P}_\bullet, \mathscr{Q}_\bullet)) \xrightarrow{\sim} \mathscr{T}\!or_n^S(f_*\mathscr{P}_\bullet, g_*\mathscr{Q}_\bullet).$$

Proof. Since f and g are affine, $f_*\mathscr{P}_\bullet$ and $g_*\mathscr{Q}_\bullet$ are complexes of quasi-coherent modules ((**II**, 1.2.6)). If $Z' = X' \times_S Y'$, both sides of (6.5.11.1) are quasi-coherent $\mathscr{O}_{Z'}$-modules by (6.5.1). We readily reduce to the case in which S, X', and Y' are affine; then X and Y are affine as well, and the assertion follows immediately from (6.4.1.1) and (**I**, 1.6.3). ☐

(6.5.12). *Remark.* Let X' and Y' be two S-preschemes, with X' flat over S. Let X be a closed subprescheme of X', and let $i : X \to X'$ be the canonical immersion. Let $\mathscr{P}_\bullet$ (respectively $\mathscr{Q}_\bullet$) be a bounded-below complex of quasi-coherent $\mathscr{O}_X$-modules (respectively $\mathscr{O}_{Y'}$-modules). Finally, let $\mathscr{L}'_{\bullet\bullet}$ be a resolution of $i_*\mathscr{P}_\bullet$ consisting of locally free $\mathscr{O}_{X'}$-modules, such that every point of X' has an affine open neighborhood U for which $\mathscr{L}'_{j\bullet}|U$ is a free resolution of $i_*\mathscr{P}_j|U$, for every j. There is then a canonical isomorphism

$$(6.5.12.1) \qquad (i \times_S 1)_*\big(\mathscr{T}or^S_\bullet(\mathscr{P}_\bullet, \mathscr{Q}_\bullet)\big) \xrightarrow{\sim} \mathscr{H}_\bullet(\mathscr{L}'_{\bullet\bullet} \otimes_S \mathscr{Q}_\bullet).$$

If S, X', and Y' are affine, and if $\mathscr{L}'_{j\bullet}$ is a free resolution of $i_*\mathscr{P}_j$ for every j, then (6.5.11) reduces us to the case $X' = X$, with X flat over S, and it is enough to apply (6.3.4). In general, (6.5.12.1) is defined locally, and it remains to verify that these local definitions give a global isomorphism.

For this, recall the definition of the first isomorphism (6.3.4.1). It comes from an isomorphism of spectral sequences ((**0**, 11.6.5) and (**0**, 11.5.3)), itself obtained from a morphism of bicomplexes

$$L_{\bullet\bullet} \longrightarrow L''_{\bullet\bullet},$$

where $L_{\bullet\bullet}$ is the given resolution of $P_\bullet$, and $L''_{\bullet\bullet}$ is a projective resolution of $P_\bullet$ in the category of **III** | 152
bounded-below complexes of A-modules ((**0**, 11.5.2.2)). The assertion follows from the fact that (6.3.4.1) is independent of the chosen projective resolution $L''_{\bullet\bullet}$, since any two such resolutions are connected by a homotopy (M, V, 1.2).

Proposition (6.5.13). — *Let X and Y be two S-preschemes, and suppose that one of the following conditions holds:*

(i) X and $Z = X \times_S Y$ are locally Noetherian, and X is flat over S;
(ii) S and X are locally Noetherian, and Y is of finite type over S.

Let $\mathscr{P}_\bullet$ (respectively $\mathscr{Q}_\bullet$) be a bounded-below complex of quasi-coherent $\mathscr{O}_X$-modules (respectively $\mathscr{O}_Y$-modules). Suppose in addition that, for every n, $\mathscr{H}_n(\mathscr{P}_\bullet)$ (respectively $\mathscr{H}_n(\mathscr{Q}_\bullet)$) is an $\mathscr{O}_X$-module (respectively $\mathscr{O}_Y$-module) of finite type. Then

$$\mathscr{T}or^S_n(\mathscr{P}_\bullet, \mathscr{Q}_\bullet)$$

is a coherent $\mathscr{O}_Z$-module for every n.

Proof. Since $\mathscr{P}_\bullet$ and $\mathscr{Q}_\bullet$ are bounded below, the spectral sequence ${}''\mathscr{E}(\mathscr{P}_\bullet, \mathscr{Q}_\bullet)$ is biregular by (6.5.4). In either case (i) or (ii), Z is locally Noetherian, so (**0**, 11.1.8) reduces the assertion to the coherence of the terms ${}''\mathscr{E}^2_{pq}$. The finite-type hypothesis on the homology modules and formula (6.5.4.2) reduce the proposition to the special case in which $\mathscr{P}_\bullet$ and $\mathscr{Q}_\bullet$ are concentrated in degree zero, namely the following corollary. ☐

Corollary (6.5.14). — *Suppose that one of conditions* (i) *and* (ii) *of* (6.5.13) *holds, and let $\mathscr{F}$ (respectively $\mathscr{G}$) be a quasi-coherent $\mathscr{O}_X$-module (respectively $\mathscr{O}_Y$-module) of finite type. Then $\mathscr{T}or^S_n(\mathscr{F}, \mathscr{G})$ is a coherent $\mathscr{O}_Z$-module for every n.*

Proof. The question is local on X and Y, so we may suppose that S, X, and Y are affine.

(i) Under the hypotheses of (i), S, X, and Z are Noetherian. There is therefore a locally free resolution $\mathscr{L}_\bullet$ of $\mathscr{F}$ consisting of $\mathscr{O}_X$-modules of finite type ((**I**, 1.3.7)). Since X is flat over S, (6.3.4) gives

$$\mathscr{T}or^S_n(\mathscr{F}, \mathscr{G}) = \mathscr{H}_n(\mathscr{L}_\bullet \otimes_S \mathscr{G}).$$

The modules $\mathscr{L}_i \otimes_S \mathscr{G}$ are quasi-coherent $\mathscr{O}_Z$-modules of finite type ((**I**, 9.1.1)), hence are coherent. It follows that $\mathscr{H}_n(\mathscr{L}_\bullet \otimes_S \mathscr{G})$ is coherent ((**0**, 5.3.4)).

(ii) Now suppose that condition (ii) holds. Since $A(Y)$ is a quotient of a polynomial $A(S)$-algebra in finitely many indeterminates ((**I**, 6.3.3)), Y is a closed subprescheme of an affine S-scheme Y' that is flat and of finite type over S. The scheme Y' is Noetherian ((**I**, 6.3.7)), so there exists a locally free resolution $\mathscr{M}_\bullet$ of $j_*\mathscr{G}$ by $\mathscr{O}_{Y'}$-modules of finite type, where $j : Y \to Y'$ is the canonical immersion. By (6.5.12), $\mathscr{T}or_n^S(\mathscr{F},\mathscr{G})$ is the inverse image under $1 \times j$ of the $\mathscr{O}_{Z'}$-module

$$\mathscr{H}_n(\mathscr{F} \otimes_S \mathscr{M}_\bullet), \qquad Z' = X \times_S Y'.$$

As in (i), the $\mathscr{F} \otimes_S \mathscr{M}_i$ are coherent $\mathscr{O}_{Z'}$-modules, and (**0**, 5.3.4) again gives the conclusion. □

(6.5.15). The theory developed above for two complexes $\mathscr{P}_\bullet$, $\mathscr{Q}_\bullet$ of quasi-coherent sheaves on two S-preschemes X and Y extends without difficulty to an arbitrary finite family of S-preschemes $X^{(i)}$ $(1 \leqslant i \leqslant m)$, each equipped with a complex $\mathscr{P}_\bullet^{(i)}$ of quasi-coherent $\mathscr{O}_{X^{(i)}}$-modules. If

$$Z = X^{(1)} \times_S X^{(2)} \times_S \cdots \times_S X^{(m)},$$

one thereby defines a quasi-coherent $\mathscr{O}_Z$-module

$$\mathscr{T}or_n^S(\mathscr{P}_\bullet^{(1)}, \mathscr{P}_\bullet^{(2)}, \ldots, \mathscr{P}_\bullet^{(m)}).$$

We leave the development of the theory in this general case to the reader, and record only, for later reference, the E_2-term of the second, regular spectral sequence whose abutment is this hypertor:

$$''\mathscr{E}_{pq}^2 = \bigoplus_{q_1+\cdots+q_m=q} \mathscr{T}or_p^S(\mathscr{H}_{q_1}(\mathscr{P}_\bullet^{(1)}), \ldots, \mathscr{H}_{q_m}(\mathscr{P}_\bullet^{(m)})). \tag{6.5.15.1}$$

In (6.8) we shall study the associativity spectral sequences to which these hypertor functors of an arbitrary finite number of complexes give rise.

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6.6. Global hypertor functors of complexes of quasi-coherent modules and Künneth spectral sequences: the affine-base case

(6.6.1). Let $S = \mathrm{Spec}(A)$ be an affine scheme, and let $X^{(i)}$ $(i = 1, 2)$ be two quasi-compact S-schemes. For $i = 1, 2$, let $\mathscr{P}_\bullet^{(i)}$ be a bounded-below complex of quasi-coherent $\mathscr{O}_{X^{(i)}}$-modules whose differential has degree -1. Choose a finite covering

$$\mathfrak{U}^{(i)} = (U_\alpha^{(i)})$$

of $X^{(i)}$ by affine open subsets. Put

$$X = X^{(1)} \times_S X^{(2)}.$$

This is a quasi-compact S-scheme ((**I**, 5.5.1) and (**I**, 6.6.4)), and $\mathfrak{U} = \mathfrak{U}^{(1)} \times_S \mathfrak{U}^{(2)}$ is the covering of X by the affine open subsets $U_\alpha^{(1)} \times_S U_\beta^{(2)}$.

For $p \leqslant 0$ and $q \in \mathbf{Z}$, the group of $(-p)$-alternating cochains

$$C^{-p}(\mathfrak{U}^{(i)}, \mathscr{P}_q^{(i)})$$

of the covering $\mathfrak{U}^{(i)}$ with coefficients in $\mathscr{P}_q^{(i)}$ (G, II, 5.1) is an A-module. For $p > 0$, put $C^p(\mathfrak{U}^{(i)}, \mathscr{P}_q^{(i)}) = 0$. This defines a bicomplex of A-modules

$$C^\bullet(\mathfrak{U}^{(i)}, \mathscr{P}_\bullet^{(i)})$$

whose two differentials have degree -1.

It follows from the definitions ((**0**, 12.1.2)) that the homology module

$$H_n(C^\bullet(\mathfrak{U}^{(i)}, \mathscr{P}_\bullet^{(i)}))$$

of this bicomplex, viewed in the usual way as a simple complex for the total degree, is the hypercohomology module

$$\mathrm{H}^{-n}(\mathfrak{U}^{(i)}, \mathscr{P}^{(i)\bullet}),$$

where $\mathscr{P}^{(i)\bullet}$ is the complex with differential of degree $+1$ whose component of degree q is $\mathscr{P}_{-q}^{(i)}$. By abuse of notation we write this as

$$\mathrm{H}^{-n}(\mathfrak{U}^{(i)}, \mathscr{P}_\bullet^{(i)}).$$

By (6.2.2), this module is canonically isomorphic to

$$\mathrm{H}^{-n}(X^{(i)}, \mathscr{P}^{(i)\bullet}),$$

also written $\mathrm{H}^{-n}(X^{(i)}, \mathscr{P}_\bullet^{(i)})$. It is therefore independent of the chosen finite covering $\mathfrak{U}^{(i)}$.

(6.6.2). Apply the general theory of hyperhomology of functors with respect to bicomplexes ((**0**, 11.7.4)) to the two bicomplexes of A-modules

$$L_{\bullet\bullet}^{(i)} = C^\bullet(\mathfrak{U}^{(i)}, \mathscr{P}_\bullet^{(i)}) \qquad (i = 1, 2)$$

and to the covariant bifunctor $L_{\bullet\bullet}^{(1)} \otimes_A L_{\bullet\bullet}^{(2)}$. Because the cochains are alternating and the coverings $\mathfrak{U}^{(i)}$ are finite, the modules

$$C^{-p}(\mathfrak{U}^{(i)}, \mathscr{P}_q^{(i)})$$

are nonzero for only finitely many values of p, independently of q. In particular, both degrees of each $L_{\bullet\bullet}^{(i)}$ are bounded below.

Denote by

$$\mathbf{Tor}_n^A(L_{\bullet\bullet}^{(1)}, L_{\bullet\bullet}^{(2)})$$

or by

$$\mathbf{Tor}_n^S(\mathfrak{U}^{(1)}, \mathfrak{U}^{(2)}; \mathscr{P}_\bullet^{(1)}, \mathscr{P}_\bullet^{(2)})$$

the n-th hyperhomology module of $L_{\bullet\bullet}^{(1)} \otimes_A L_{\bullet\bullet}^{(2)}$. We call it the hypertor of index n of $\mathscr{P}_\bullet^{(1)}$ and $\mathscr{P}_\bullet^{(2)}$ relative to the coverings $\mathfrak{U}^{(1)}$ and $\mathfrak{U}^{(2)}$. When these complexes are concentrated in degree zero, with terms $\mathscr{F}^{(1)}$ and $\mathscr{F}^{(2)}$, write

$$\mathbf{Tor}_n^S(\mathfrak{U}^{(1)}, \mathfrak{U}^{(2)}; \mathscr{F}^{(1)}, \mathscr{F}^{(2)})$$

for their hypertor. Following the general conventions, the same notation

$$\mathbf{Tor}_n^S(\mathfrak{U}^{(1)}, \mathfrak{U}^{(2)}; \mathscr{P}_\bullet^{(1)}, \mathscr{P}_\bullet^{(2)})$$

also denotes the bicomplex whose component of bidegree (j, k) is

$$\mathbf{Tor}_n^S(\mathfrak{U}^{(1)}, \mathfrak{U}^{(2)}; \mathscr{P}_j^{(1)}, \mathscr{P}_k^{(2)}).$$

The construction $L_{\bullet\bullet}^{(i)}$ is exact in $\mathscr{P}_\bullet^{(i)}$, since all intersections of members of $\mathfrak{U}^{(i)}$ are affine ((**I**, 5.5.6) and (**I**, 1.3.11)). Consequently

$$\mathbf{Tor}_\bullet^S(\mathfrak{U}^{(1)}, \mathfrak{U}^{(2)}; \mathscr{P}_\bullet^{(1)}, \mathscr{P}_\bullet^{(2)})$$

is a covariant bi-∂-functor in $\mathscr{P}_\bullet^{(1)}$ and $\mathscr{P}_\bullet^{(2)}$, with values in the category of A-modules ((**0**, 11.7.3)). By (**0**, 11.7.2), it is the common abutment of six biregular spectral functors, denoted

$${}^{(t)}E^S(\mathfrak{U}^{(1)}, \mathfrak{U}^{(2)}; \mathscr{P}_\bullet^{(1)}, \mathscr{P}_\bullet^{(2)}),$$

or simply

$${}^{(t)}E(\mathfrak{U}^{(1)}, \mathfrak{U}^{(2)}; \mathscr{P}_\bullet^{(1)}, \mathscr{P}_\bullet^{(2)}),$$

where t is one of a, b, a', b', c, d. Their E_2-terms are

$$\begin{aligned}
{}^{(a)}E_{pq}^2 &= \bigoplus_{q_1+q_2=q} \mathrm{Tor}_p^A\big(H_{q_1}(L_{\bullet\bullet}^{(1)}), H_{q_2}(L_{\bullet\bullet}^{(2)})\big), \\
{}^{(b)}E_{pq}^2 &= H_p\big(\mathbf{Tor}_q^{A,\mathrm{II}}(L_{\bullet\bullet}^{(1)}, L_{\bullet\bullet}^{(2)})\big), \\
{}^{(a')}E_{pq}^2 &= \bigoplus_{q_1+q_2=q} \mathrm{Tor}_p^A\big(H_{q_1}^{\mathrm{I}}(L_{\bullet\bullet}^{(1)}), H_{q_2}^{\mathrm{I}}(L_{\bullet\bullet}^{(2)})\big), \\
{}^{(b')}E_{pq}^2 &= H_p\big(\mathbf{Tor}_q^{A}(L_{\bullet\bullet}^{(1)}, L_{\bullet\bullet}^{(2)})\big), \\
{}^{(c)}E_{pq}^2 &= \bigoplus_{q_1+q_2=q} \mathrm{Tor}_p^A\big(H_{q_1}^{\mathrm{II}}(L_{\bullet\bullet}^{(1)}), H_{q_2}^{\mathrm{II}}(L_{\bullet\bullet}^{(2)})\big), \\
{}^{(d)}E_{pq}^2 &= H_p\big(\mathbf{Tor}_q^{A,\mathrm{I}}(L_{\bullet\bullet}^{(1)}, L_{\bullet\bullet}^{(2)})\big).
\end{aligned}$$

These notations agree with those of the general theory of hyperhomology. We now make the initial terms more explicit.

(6.6.3). *Spectral sequences* (a) *and* (a′). By (6.6.1),

$$H_n(L^{(i)}_{\bullet\bullet}) = \mathrm{H}^{-n}(X^{(i)}, \mathscr{P}^{(i)}_\bullet),$$

and hence

$${}^{(a)}E^2_{pq} = \bigoplus_{q_1+q_2=q} \mathrm{Tor}^A_p\big(\mathrm{H}^{-q_1}(X^{(1)}, \mathscr{P}^{(1)}_\bullet), \mathrm{H}^{-q_2}(X^{(2)}, \mathscr{P}^{(2)}_\bullet)\big).$$

By definition, the component of degree k of the complex $H^{\mathrm{I}}_n(L^{(i)}_{\bullet\bullet})$ is

$$H_n\big(C^\bullet(\mathfrak{U}^{(i)}, \mathscr{P}^{(i)}_k)\big) = \mathrm{H}^{-n}(\mathfrak{U}^{(i)}, \mathscr{P}^{(i)}_k).$$

By (6.1.4.1), this module is canonically isomorphic to $\mathrm{H}^{-n}(X^{(i)}, \mathscr{P}^{(i)}_k)$. Therefore

$${}^{(a')}E^2_{pq} = \bigoplus_{q_1+q_2=q} \mathrm{Tor}^A_p\big(\mathrm{H}^{-q_1}(X^{(1)}, \mathscr{P}^{(1)}_\bullet), \mathrm{H}^{-q_2}(X^{(2)}, \mathscr{P}^{(2)}_\bullet)\big),$$

where each $\mathrm{H}^{-q_i}(X^{(i)}, \mathscr{P}^{(i)}_\bullet)$ in this last formula denotes the complex obtained by taking cohomology term by term.

(6.6.4). *Spectral sequences* (b) *and* (b′). By definition,

$$\mathbf{Tor}^{A,\mathrm{II}}_q(L^{(1)}_{\bullet\bullet}, L^{(2)}_{\bullet\bullet})$$

is a bicomplex whose component of bidegree (h,k) is

$$\mathrm{Tor}^A_q\big(C^{-h}(\mathfrak{U}^{(1)}, \mathscr{P}^{(1)}_\bullet), C^{-k}(\mathfrak{U}^{(2)}, \mathscr{P}^{(2)}_\bullet)\big).$$

Let $\Phi^{(i)}$ be the index set of $\mathfrak{U}^{(i)}$. For $r \geqslant 0$, the complex $C^r(\mathfrak{U}^{(i)}, \mathscr{P}^{(i)}_\bullet)$ is the direct sum of the complexes

$$\Gamma(U^{(i)}_\rho, \mathscr{P}^{(i)}_\bullet),$$

where $U^{(i)}_\rho$ is the intersection of the $U^{(i)}_\xi$ for $\xi \in \rho$, and ρ ranges over the appropriate elements of $\mathfrak{P}(\Phi^{(i)})$. Consequently

$$\mathrm{Tor}^A_q\big(C^{-h}(\mathfrak{U}^{(1)}, \mathscr{P}^{(1)}_\bullet), C^{-k}(\mathfrak{U}^{(2)}, \mathscr{P}^{(2)}_\bullet)\big)$$

is the direct sum of the A-modules

$$\mathrm{Tor}^A_q\big(\Gamma(U^{(1)}_\sigma, \mathscr{P}^{(1)}_\bullet), \Gamma(U^{(2)}_\tau, \mathscr{P}^{(2)}_\bullet)\big),$$

where σ and τ range over $\mathfrak{P}(\Phi^{(1)})$ and $\mathfrak{P}(\Phi^{(2)})$ with

$$\mathrm{Card}(\sigma) = -(h+1), \qquad \mathrm{Card}(\tau) = -(k+1),$$

respectively. Since $X^{(1)}$ and $X^{(2)}$ are schemes, the $U^{(i)}_\rho$ are affine. Thus (6.4.1.1) gives

$$\mathrm{Tor}^A_q\big(\Gamma(U^{(1)}_\sigma, \mathscr{P}^{(1)}_\bullet), \Gamma(U^{(2)}_\tau, \mathscr{P}^{(2)}_\bullet)\big) = \Gamma\big(U^{(1)}_\sigma \times_S U^{(2)}_\tau, \mathscr{T}\!or^S_q(\mathscr{P}^{(1)}_\bullet, \mathscr{P}^{(2)}_\bullet)\big).$$

It follows that ${}^{(b)}E^2_{pq}$ is the $(-p)$-th cohomology module of the complex

$$L^\bullet(\Phi^{(1)}, \Phi^{(2)}; \mathscr{S})$$

of bialternating cochains on $\Phi^{(1)}$ and $\Phi^{(2)}$ with values in the coefficient system

$$\mathscr{S} : (\sigma,\tau) \longmapsto \Gamma\big(U^{(1)}_\sigma \times_S U^{(2)}_\tau, \mathscr{T}\!or^S_q(\mathscr{P}^{(1)}_\bullet, \mathscr{P}^{(2)}_\bullet)\big) \tag{6.6.4.1}$$

((**0**, 11.8.4)). By (**0**, 11.8.5) and (**0**, 11.8.6), this complex has the same cohomology as the complex of all cochains

$$C^\bullet(\Phi^{(1)}, \Phi^{(2)}; \mathscr{S})$$

and as the complex

$$P^\bullet(\Phi^{(1)}, \Phi^{(2)}; \mathscr{S}),$$

whose elements are linear combinations of

$$\lambda(\sigma,\tau) \in \Gamma\big(U^{(1)}_\sigma \times_S U^{(2)}_\tau, \mathscr{T}\!or^S_q(\mathscr{P}^{(1)}_\bullet, \mathscr{P}^{(2)}_\bullet)\big),$$

where

$$\sigma = (\alpha_0, \ldots, \alpha_h), \qquad \tau = (\beta_0, \ldots, \beta_h)$$

are sequences with the same number of terms. Now

$$U_\sigma^{(1)} \times_S U_\tau^{(2)} = (U_{\alpha_0}^{(1)} \times_S U_{\beta_0}^{(2)}) \cap \cdots \cap (U_{\alpha_h}^{(1)} \times_S U_{\beta_h}^{(2)})$$

by (**I**, 3.2.7). Let $\mathfrak{U}$ be the covering of $Z = X^{(1)} \times_S X^{(2)}$ by the affine open subsets $U_\alpha^{(1)} \times_S U_\beta^{(2)}$. Since Z is a scheme, (6.1.3.1) yields

$$ {}^{(b)}E_{pq}^2 = \mathrm{H}^{-p}\big(Z, \mathscr{T}\!or_q^S(\mathscr{P}_\bullet^{(1)}, \mathscr{P}_\bullet^{(2)})\big). \tag{6.6.4.2}$$

Next,

$$\mathbf{Tor}_q^A(L_{\bullet\bullet}^{(1)}, L_{\bullet\bullet}^{(2)})$$

is a bicomplex whose component of bidegree (h,k) is the direct sum of the A-modules

$$\mathrm{Tor}_q^A\big(C^{-h_1}(\mathfrak{U}^{(1)}, \mathscr{P}_{k_1}^{(1)}), C^{-h_2}(\mathfrak{U}^{(2)}, \mathscr{P}_{k_2}^{(2)})\big),$$

where $h_1 + h_2 = h$ and $k_1 + k_2 = k$. Expanding the cochain modules as above shows that this component is also the direct sum of

$$\Gamma\big(U_\sigma^{(1)} \times_S U_\tau^{(2)}, \mathscr{T}\!or_q^S(\mathscr{P}_{k_1}^{(1)}, \mathscr{P}_{k_2}^{(2)})\big),$$

where $k_1 + k_2 = k$ and

$$\mathrm{Card}(\sigma) + \mathrm{Card}(\tau) = -h - 2.$$

The term ${}^{(b')}E_{pq}^2$ is the $(-p)$-th cohomology module of a bicomplex $N^{\bullet\bullet} = (N^{hk})$. For fixed k, the simple complex $N^{\bullet k}$ consists of bialternating cochains on $\Phi^{(1)}$ and $\Phi^{(2)}$ with values in the coefficient system

$$\mathscr{S}_k : (\sigma, \tau) \longmapsto \Gamma\left(U_\sigma^{(1)} \times_S U_\tau^{(2)}, \bigoplus_{k_1+k_2=k} \mathscr{T}\!or_q^S(\mathscr{P}_{k_1}^{(1)}, \mathscr{P}_{k_2}^{(2)})\right).$$

These coefficient systems form a complex $\mathscr{S}^\bullet$, whose differential is induced by that of the simple complex associated to the bicomplex

$$\mathscr{T}\!or_q^S(\mathscr{P}^{(1)\bullet}, \mathscr{P}^{(2)\bullet}).$$

By (**0**, 11.8.9), $N^{\bullet\bullet}$ has the same cohomology as

$$C^\bullet(\Phi^{(1)}, \Phi^{(2)}; \mathscr{S}^\bullet),$$

and also the same cohomology as

$$P^\bullet(\Phi^{(1)}, \Phi^{(2)}; \mathscr{S}^\bullet),$$

whose elements of bidegree (h,k) are linear combinations of

$$\lambda(\sigma, \tau) \in \Gamma\left(U_\sigma^{(1)} \times_S U_\tau^{(2)}, \bigoplus_{k_1+k_2=k} \mathscr{T}\!or_q^S(\mathscr{P}_{k_1}^{(1)}, \mathscr{P}_{k_2}^{(2)})\right),$$

with $\sigma = (\alpha_0, \ldots, \alpha_h)$ and $\tau = (\beta_0, \ldots, \beta_h)$ having the same number of terms ((**0**, 11.8.10)). As above, ${}^{(b')}E_{pq}^2$ is therefore the $(-p)$-th cohomology module of

$$C^\bullet(\mathfrak{U}, \mathscr{Q}^\bullet),$$

where $\mathscr{Q}^\bullet$ is the simple complex associated to the bicomplex

$$\mathscr{T}\!or_q^S(\mathscr{P}^{(1)\bullet}, \mathscr{P}^{(2)\bullet})$$

of $\mathscr{O}_Z$-modules. With the conventions of (6.6.1), we obtain

$$ {}^{(b')}E_{pq}^2 = \mathrm{H}^{-p}\big(Z, \mathscr{T}\!or_q^S(\mathscr{P}_\bullet^{(1)}, \mathscr{P}_\bullet^{(2)})\big). \tag{6.6.4.3}$$

(6.6.5). *Spectral sequences* (c) *and* (d). By definition, the component of degree h of the complex $H_n^{\mathrm{II}}(L_{\bullet\bullet}^{(i)})$ is the A-module

$$C^{-h}(\mathfrak{U}^{(i)}, \mathscr{H}_n(\mathscr{P}_\bullet^{(i)})),$$

because the functor C^{-h} is exact. Hence, by the definition of the hypertor of two modules relative to two coverings ((6.6.2)),

$$^{(c)}E_{pq}^2 = \bigoplus_{q_1+q_2=q} \mathbf{Tor}_p^S(\mathfrak{U}^{(1)}, \mathfrak{U}^{(2)}; \mathscr{H}_{q_1}(\mathscr{P}_\bullet^{(1)}), \mathscr{H}_{q_2}(\mathscr{P}_\bullet^{(2)})).$$

Finally, by definition,

$$\mathbf{Tor}_q^{A,\mathrm{I}}(L_{\bullet\bullet}^{(1)}, L_{\bullet\bullet}^{(2)})$$

is a bicomplex whose component of bidegree (h,k) is the A-module

$$\mathbf{Tor}_q^S(\mathfrak{U}^{(1)}, \mathfrak{U}^{(2)}; \mathscr{P}_h^{(1)}, \mathscr{P}_k^{(2)}).$$

Consequently

$$^{(d)}E_{pq}^2 = H_p\left(\mathbf{Tor}_q^S(\mathfrak{U}^{(1)}, \mathfrak{U}^{(2)}; \mathscr{P}_\bullet^{(1)}, \mathscr{P}_\bullet^{(2)})\right).$$

(6.6.6). The theory of hyperhomology for functors of bicomplexes ((**0**, 11.7.3)) shows, as in (6.3.4), that for every *flat* Cartan–Eilenberg resolution $M_{\bullet\bullet}^{(i)}$ of $L_{\bullet\bullet}^{(i)}$ in the category of complexes of modules bounded below $(i = 1, 2)$, there are canonical isomorphisms of bi-∂-functors

(6.6.6.1).

$$\begin{aligned} \mathbf{Tor}_\bullet^S(\mathfrak{U}^{(1)}, \mathfrak{U}^{(2)}; \mathscr{P}_\bullet^{(1)}, \mathscr{P}_\bullet^{(2)}) &\xrightarrow{\sim} H_\bullet(M_{\bullet\bullet}^{(1)} \otimes_A M_{\bullet\bullet}^{(2)}) \\ &\xrightarrow{\sim} H_\bullet(M_{\bullet\bullet}^{(1)} \otimes_A L_{\bullet\bullet}^{(2)}) \\ &\xrightarrow{\sim} H_\bullet(L_{\bullet\bullet}^{(1)} \otimes_A M_{\bullet\bullet}^{(2)}). \end{aligned} \tag{6.6.6.1}$$

(6.6.7). We shall now show that the global hypertor defined in (6.6.2), and the six corresponding spectral sequences, do not depend, up to canonical isomorphisms, on the finite affine open coverings $\mathfrak{U}^{(i)}$ used to define them. It is enough to show that if $\mathfrak{V}^{(i)}$ are two other coverings of the same kind, with $\mathfrak{V}^{(i)}$ finer than $\mathfrak{U}^{(i)}$ for $i = 1, 2$, then there are canonical isomorphisms of spectral functors

(6.6.7, *t*).

$$^{(t)}E(\mathfrak{U}^{(1)}, \mathfrak{U}^{(2)}; \mathscr{P}_\bullet^{(1)}, \mathscr{P}_\bullet^{(2)}) \xrightarrow{\sim} {}^{(t)}E(\mathfrak{V}^{(1)}, \mathfrak{V}^{(2)}; \mathscr{P}_\bullet^{(1)}, \mathscr{P}_\bullet^{(2)}), \tag{6.6.7, t}$$

where t is one of a, b, a', b', c, d.

For $i = 1, 2$ there are homomorphisms of bicomplexes

$$C^\bullet(\mathfrak{U}^{(i)}, \mathscr{P}_\bullet^{(i)}) \longrightarrow C^\bullet(\mathfrak{V}^{(i)}, \mathscr{P}_\bullet^{(i)})$$

that are well defined up to homotopy ((**G**, II.5.7.1)). It follows at once that the homomorphisms (6.6.7, *t*) are canonically defined and compatible with the boundary operators in their abutments ((**0**, 11.3.2)). Moreover, the calculation of the E_2-terms of the spectral sequences (a), (b), (a′), and (b′) shows that (6.6.7, *t*) is an isomorphism on their E_2-terms. Since these spectral sequences are biregular, it follows that (6.6.7, *t*) is an isomorphism for these spectral functors, and hence an isomorphism of bi-∂-functors on their common abutment ((**0**, 11.1.5)).

In particular, if $\mathscr{F}^{(i)}$ is a quasi-coherent $\mathscr{O}_{X^{(i)}}$-module $(i = 1, 2)$, the canonical homomorphism

$$\mathbf{Tor}_n^S(\mathfrak{U}^{(1)}, \mathfrak{U}^{(2)}; \mathscr{F}^{(1)}, \mathscr{F}^{(2)}) \longrightarrow \mathbf{Tor}_n^S(\mathfrak{V}^{(1)}, \mathfrak{V}^{(2)}; \mathscr{F}^{(1)}, \mathscr{F}^{(2)})$$

is bijective. In view of the calculation in (6.6.5), the homomorphism (6.6.7, *t*) is therefore also an isomorphism on E_2-terms for $t = c$ and $t = d$. As above, we conclude that it is an isomorphism of spectral sequences for these two values of t.

The isomorphisms (6.6.7, *t*) may be regarded as defining inductive systems of spectral functors **III** | 157
on the filtering set of pairs $(\mathfrak{U}^{(1)}, \mathfrak{U}^{(2)})$ of finite affine open coverings of $X^{(1)}$ and $X^{(2)}$. We denote the inductive limit by

$$^{(t)}E(X^{(1)}, X^{(2)}; \mathscr{P}_\bullet^{(1)}, \mathscr{P}_\bullet^{(2)}) \quad \text{or} \quad {}^{(t)}E^S(X^{(1)}, X^{(2)}; \mathscr{P}_\bullet^{(1)}, \mathscr{P}_\bullet^{(2)}).$$

Its abutment is denoted
$$\mathbf{Tor}^S_\bullet(X^{(1)},X^{(2)};\mathscr{P}^{(1)}_\bullet,\mathscr{P}^{(2)}_\bullet)$$
and is called the *global hypertor* of the two complexes $\mathscr{P}^{(1)}_\bullet$ and $\mathscr{P}^{(2)}_\bullet$. If the two complexes are concentrated in degree zero, with terms $\mathscr{F}^{(1)}$ and $\mathscr{F}^{(2)}$, we write
$$\mathbf{Tor}^S_\bullet(X^{(1)},X^{(2)};\mathscr{F}^{(1)},\mathscr{F}^{(2)}).$$
In accordance with the general conventions,
$$\mathbf{Tor}^S_q(X^{(1)},X^{(2)};\mathscr{P}^{(1)}_\bullet,\mathscr{P}^{(2)}_\bullet)$$
will denote the bicomplex whose component of bidegree (h,k) is
$$\mathbf{Tor}^S_q(X^{(1)},X^{(2)};\mathscr{P}^{(1)}_h,\mathscr{P}^{(2)}_k).$$
(6.6.8). Under the hypotheses of (6.6.1), consider two S-morphisms
$$f_i : X^{(i)} \longrightarrow Y^{(i)},$$
where $Y^{(i)} = \operatorname{Spec}(B_i)$ is an affine S-scheme, so that B_i is an A-algebra $(i = 1,2)$. This defines an A-homomorphism
$$B_i \longrightarrow \Gamma(X^{(i)},\mathscr{O}_{X^{(i)}})$$
((**I**, 2.2.4)), and consequently each $L^{(i)}_{\bullet\bullet}$ defined in (6.6.2) is a bicomplex of B_i-modules. Thus
$$L^{(1)}_{\bullet\bullet} \otimes_A L^{(2)}_{\bullet\bullet}$$
is a quadricomplex of $(B_1 \otimes_A B_2)$-modules, and its six hyperhomology spectral functors may be regarded as taking values in the category of spectral sequences of $(B_1 \otimes_A B_2)$-modules.

Put
$$Y = Y^{(1)} \times_S Y^{(2)} = \operatorname{Spec}(B_1 \otimes_A B_2).$$
We may then consider the quasi-coherent $\mathscr{O}_Y$-modules associated with these modules ((**I**, 1.3.4)). For $t = a,b,a',b',c,d$, write
$${}^{(t)}\mathscr{E}(f_1,f_2;\mathscr{P}^{(1)}_\bullet,\mathscr{P}^{(2)}_\bullet)$$
for the six spectral sequences of $\mathscr{O}_Y$-modules
$$\left({}^{(t)}E(X^{(1)},X^{(2)};\mathscr{P}^{(1)}_\bullet,\mathscr{P}^{(2)}_\bullet)\right)^\sim,$$
and write
$$\mathscr{T}\!or^S_\bullet(f_1,f_2;\mathscr{P}^{(1)}_\bullet,\mathscr{P}^{(2)}_\bullet)$$
for their common abutment
$$\left(\mathbf{Tor}^S_\bullet(X^{(1)},X^{(2)};\mathscr{P}^{(1)}_\bullet,\mathscr{P}^{(2)}_\bullet)\right)^\sim.$$
When $\mathscr{P}^{(i)}_\bullet$ is concentrated in degree zero, with term $\mathscr{F}^{(i)}$ $(i = 1,2)$, we write this abutment as
$$\mathscr{T}\!or^S_\bullet(f_1,f_2;\mathscr{F}^{(1)},\mathscr{F}^{(2)}).$$

6.7. Global hypertor functors of complexes of quasi-coherent modules and Künneth spectral sequences: the general case

(6.7.1). We now generalize the definitions of (6.6.8) to the case where S is an arbitrary prescheme, $X^{(i)}$ and $Y^{(i)}$ are S-preschemes, and

$$f_i : X^{(i)} \longrightarrow Y^{(i)}$$

are separated and quasi-compact S-morphisms. For every pair of bounded-below complexes $\mathscr{P}_\bullet^{(i)}$ of quasi-coherent $\mathscr{O}_{X^{(i)}}$-modules $(i = 1, 2)$, the aim is to define a quasi-coherent $\mathscr{O}_Y$-module

$$\mathfrak{Tor}_n^S(f_1, f_2; \mathscr{P}_\bullet^{(1)}, \mathscr{P}_\bullet^{(2)})$$

for every n, together with six spectral functors, in such a way that these definitions reduce to those of (6.6.8) when S, $Y^{(1)}$, and $Y^{(2)}$ are affine; here $Y = Y^{(1)} \times_S Y^{(2)}$.

First suppose that $S = \operatorname{Spec}(A)$ is affine, while $Y^{(1)}$ and $Y^{(2)}$ are arbitrary. Let $W^{(i)}$ be an affine open subset of $Y^{(i)}$. Then $f_i^{-1}(W^{(i)})$ is a quasi-compact S-scheme, and

$$W = W^{(1)} \times_S W^{(2)}$$

is an affine open subset of Y. Let

$$f_i' : f_i^{-1}(W^{(i)}) \longrightarrow W^{(i)}$$

be the restriction of f_i, and let

$$\mathscr{P}_\bullet'^{(i)} = \mathscr{P}_\bullet^{(i)} |_{f_i^{-1}(W^{(i)})} \qquad (i = 1, 2).$$

By (6.6.8) we then have the spectral sequences of quasi-coherent $(\mathscr{O}_Y|W)$-modules

$$^{(t)}\mathscr{E}(f_1', f_2'; \mathscr{P}_\bullet'^{(1)}, \mathscr{P}_\bullet'^{(2)}),$$

and it remains to verify that they satisfy the gluing conditions of (**0**, 3.3.1).

We are immediately reduced to the case in which

$$Y^{(i)} = \operatorname{Spec}(B_i), \qquad W^{(i)} = D(g_i), \qquad g_i \in B_i,$$

so that

$$W = D(g_1 \otimes g_2)$$

in $Y = \operatorname{Spec}(B_1 \otimes_A B_2)$ ((**II**, 4.3.2.1)). Put $X'^{(i)} = f_i^{-1}(W^{(i)})$. We must establish an isomorphism of spectral functors

(6.7.1.1).

$$^{(t)}E(X'^{(1)}, X'^{(2)}; \mathscr{P}_\bullet'^{(1)}, \mathscr{P}_\bullet'^{(2)}) \xrightarrow{\sim} {}^{(t)}E(X^{(1)}, X^{(2)}; \mathscr{P}_\bullet^{(1)}, \mathscr{P}_\bullet^{(2)}) \otimes_B B_g, \tag{6.7.1.1}$$

where

$$B = B_1 \otimes_A B_2, \qquad g = g_1 \otimes g_2.$$

Choose finite affine open coverings $\mathfrak{U}^{(i)}$ of $X^{(i)}$, and let $\mathfrak{U}'^{(i)}$ be their traces on $X'^{(i)}$. The latter are again coverings by affine open subsets ((**I**, 5.5.10)), and more precisely **III | 158**

$$C^\bullet(\mathfrak{U}'^{(i)}, \mathscr{P}_\bullet'^{(i)}) = C^\bullet(\mathfrak{U}^{(i)}, \mathscr{P}_\bullet^{(i)}) \otimes_{B_i} (B_i)_{g_i}.$$

Put

$$L_{\bullet\bullet}^{(i)} = C^\bullet(\mathfrak{U}^{(i)}, \mathscr{P}_\bullet^{(i)}), \qquad L_{\bullet\bullet}'^{(i)} = C^\bullet(\mathfrak{U}'^{(i)}, \mathscr{P}_\bullet'^{(i)}).$$

Then

$$L_{\bullet\bullet}'^{(1)} \otimes_A L_{\bullet\bullet}'^{(2)} = (L_{\bullet\bullet}^{(1)} \otimes_{B_1} (B_1)_{g_1}) \otimes_A (L_{\bullet\bullet}^{(2)} \otimes_{B_2} (B_2)_{g_2}).$$

Since

$$B_g = (B_1)_{g_1} \otimes_A (B_2)_{g_2}$$

up to canonical isomorphism, we likewise have, up to canonical isomorphism,

$$L_{\bullet\bullet}'^{(1)} \otimes_A L_{\bullet\bullet}'^{(2)} = (L_{\bullet\bullet}^{(1)} \otimes_A L_{\bullet\bullet}^{(2)}) \otimes_B B_g.$$

Let $M^{(i)}_{\bullet\bullet}$ be a projective Cartan–Eilenberg resolution of $L^{(i)}_{\bullet\bullet}$, chosen to consist of B_i-modules. Because $(B_i)_{g_i}$ is flat over B_i,

$$M'^{(i)}_{\bullet\bullet} = M^{(i)}_{\bullet\bullet} \otimes_{B_i} (B_i)_{g_i}$$

is a projective Cartan–Eilenberg resolution of $L'^{(i)}_{\bullet\bullet}$. Moreover,

$$M'^{(1)}_{\bullet\bullet} \otimes_A M'^{(2)}_{\bullet\bullet} = \left(M^{(1)}_{\bullet\bullet} \otimes_A M^{(2)}_{\bullet\bullet}\right) \otimes_B B_g.$$

The desired isomorphism (6.7.1.1) now follows immediately from the definitions of the hyperhomology of a bicomplex and from the exactness of $G \longmapsto G \otimes_B B_g$ on B-modules.

(6.7.2). Now suppose that S is arbitrary, and let

$$u_i : Y^{(i)} \longrightarrow S \qquad (i = 1,2)$$

be the structural morphisms. Choose an affine open covering (S_α) of S, put

$$Y^{(i)}_\alpha = u_i^{-1}(S_\alpha), \qquad X^{(i)}_\alpha = f_i^{-1}(Y^{(i)}_\alpha),$$

and let

$$f_{i\alpha} : X^{(i)}_\alpha \longrightarrow Y^{(i)}_\alpha$$

be the restriction of f_i, which is separated and quasi-compact. The schemes

$$Y_\alpha = Y^{(1)}_\alpha \times_{S_\alpha} Y^{(2)}_\alpha$$

form an open covering of Y, and on every Y_α, (6.7.1) defines the spectral functors

$$^{(t)}\mathscr{E}^{S_\alpha}\left(f_{1\alpha}, f_{2\alpha}; \mathscr{P}^{(1)}_\bullet|X^{(1)}_\alpha, \mathscr{P}^{(2)}_\bullet|X^{(2)}_\alpha\right).$$

It remains again to show that these functors satisfy the gluing conditions.

We reduce immediately to the following situation: $S = \operatorname{Spec}(A)$ is affine, $S' = D(h)$ with $h \in A$, and $u_i(Y^{(i)}) \subset S'$. We may moreover assume that $Y^{(i)} = \operatorname{Spec}(B_i)$ is affine. It is then enough to define canonical isomorphisms

(6.7.2.1).

$$^{(t)}E^S\left(X^{(1)}, X^{(2)}; \mathscr{P}^{(1)}_\bullet, \mathscr{P}^{(2)}_\bullet\right) \xrightarrow{\sim} {}^{(t)}E^{S'}\left(X^{(1)}, X^{(2)}; \mathscr{P}^{(1)}_\bullet, \mathscr{P}^{(2)}_\bullet\right). \tag{6.7.2.1}$$

With the notation of (6.6.2), the $L^{(i)}_{\bullet\bullet}$ consist of A_h-modules, and hence, up to canonical isomorphism,

$$L^{(1)}_{\bullet\bullet} \otimes_{A_h} L^{(2)}_{\bullet\bullet} = L^{(1)}_{\bullet\bullet} \otimes_A L^{(2)}_{\bullet\bullet}.$$

Since a projective Cartan–Eilenberg resolution $M^{(i)}_{\bullet\bullet}$ of $L^{(i)}_{\bullet\bullet}$ may be chosen to consist of A_h-modules, the desired canonical isomorphism follows immediately.

Theorem (6.7.3). — *Let S be a prescheme, let $f_i : X^{(i)} \to Y^{(i)}$ be a separated and quasi-compact S-morphism of S-preschemes, and let $\mathscr{P}^{(i)}_\bullet$ be a bounded-below complex of quasi-coherent $\mathscr{O}_{X^{(i)}}$-modules $(i = 1,2)$. Put*

$$Y = Y^{(1)} \times_S Y^{(2)}.$$

There exists a bi-∂-functor

$$\mathfrak{Tor}^S_\bullet\left(f_1, f_2; \mathscr{P}^{(1)}_\bullet, \mathscr{P}^{(2)}_\bullet\right)$$

with values in the category of quasi-coherent $\mathscr{O}_Y$-modules, such that if $V^{(i)}$ is an affine open subset of $Y^{(i)}$ $(i = 1,2)$ and $V = V^{(1)} \times_S V^{(2)}$, then

$$\begin{aligned}&\mathfrak{Tor}^S_\bullet\left(f_1, f_2; \mathscr{P}^{(1)}_\bullet, \mathscr{P}^{(2)}_\bullet\right)|_V \\ &\quad = \left(\mathbf{Tor}^S_\bullet\left(f_1^{-1}(V^{(1)}), f_2^{-1}(V^{(2)}); \mathscr{P}^{(1)}_\bullet|f_1^{-1}(V^{(1)}), \mathscr{P}^{(2)}_\bullet|f_2^{-1}(V^{(2)})\right)\right)^\sim.\end{aligned}$$

This bifunctor is the common abutment of six biregular spectral functors **III | 159**

$$^{(t)}\mathscr{E}\left(f_1, f_2; \mathscr{P}^{(1)}_\bullet, \mathscr{P}^{(2)}_\bullet\right) \qquad (t = a, b, a', b', c, d),$$

whose E_2-terms are

$$
\begin{aligned}
&{}^{(a)}\mathscr{E}^2_{pq} = \bigoplus_{q_1+q_2=q} \mathscr{T}\!or^S_p\big(\mathscr{H}^{-q_1}(f_1,\mathscr{P}^{(1)}_\bullet),\mathscr{H}^{-q_2}(f_2,\mathscr{P}^{(2)}_\bullet)\big),\\
&{}^{(b)}\mathscr{E}^2_{pq} = \mathscr{H}^{-p}\big(f_1\times_S f_2,\mathfrak{Tor}^S_q(\mathscr{P}^{(1)}_\bullet,\mathscr{P}^{(2)}_\bullet)\big),\\
&{}^{(a')}\mathscr{E}^2_{pq} = \bigoplus_{q_1+q_2=q} \mathfrak{Tor}^S_p\big(\mathscr{H}^{-q_1}(f_1,\mathscr{P}^{(1)}_\bullet),\mathscr{H}^{-q_2}(f_2,\mathscr{P}^{(2)}_\bullet)\big),\\
&{}^{(b')}\mathscr{E}^2_{pq} = \mathscr{H}^{-p}\big(f_1\times_S f_2,\mathscr{T}\!or^S_q(\mathscr{P}^{(1)}_\bullet,\mathscr{P}^{(2)}_\bullet)\big),\\
&{}^{(c)}\mathscr{E}^2_{pq} = \bigoplus_{q_1+q_2=q} \mathfrak{Tor}^S_p\big(f_1,f_2;\mathscr{H}_{q_1}(\mathscr{P}^{(1)}_\bullet),\mathscr{H}_{q_2}(\mathscr{P}^{(2)}_\bullet)\big),\\
&{}^{(d)}\mathscr{E}^2_{pq} = \mathscr{H}_p\left(\mathfrak{Tor}^S_q(f_1,f_2;\mathscr{P}^{(1)}_\bullet,\mathscr{P}^{(2)}_\bullet)\right).
\end{aligned}
$$

The spectral sequences (a) *and* (b) *are called the* Künneth spectral sequences. *Notice that* (a) *and* (a′), *respectively* (b) *and* (b′), *are identical when* $\mathscr{P}^{(1)}_\bullet$ *and* $\mathscr{P}^{(2)}_\bullet$ *are concentrated in degree zero. In that case,* (c) *and* (d) *are degenerate and are therefore of no interest.*

Remark (6.7.4). — The global hypertors just defined include as special cases both the hypercohomology modules defined in (6.2.1) and the local hypertors defined in (6.5.3). Indeed, for every quasi-compact and separated morphism $f : X \to Y$ and every bounded-below complex $\mathscr{P}_\bullet$ of quasi-coherent $\mathscr{O}_X$-modules, there is a canonical isomorphism of ∂-functors in $\mathscr{P}_\bullet$:

(6.7.4.1).

$$\mathfrak{Tor}^Y_n(f,1_Y;\mathscr{P}_\bullet,\mathscr{O}_Y) \xrightarrow{\sim} \mathscr{H}^{-n}(f,\mathscr{P}_\bullet) \qquad (n\in\mathbf{Z}). \tag{6.7.4.1}$$

The gluing methods of (6.7.2) immediately reduce the proof to the case in which Y is affine. By (6.2.2), both sides of (6.7.4.1) may then be computed using the same finite covering $\mathfrak{U}$ of X by affine open subsets and, for the first side, the covering of Y consisting of Y itself. With the notation of (6.6.2), the bicomplex $L^{(2)}_{\bullet\bullet}$ is then concentrated in bidegree $(0,0)$, where it is equal to A, and the conclusion follows from (**0**, 11.7.5).

For a generalization, see (6.7.7). Notice, however, that if $\mathscr{O}_Y$ on the left side of (6.7.4.1) is replaced by an arbitrary quasi-coherent $\mathscr{O}_Y$-module $\mathscr{F}$, there is in general no longer an isomorphism with

$$\mathscr{H}^{-n}(f,\mathscr{P}_\bullet\otimes_{\mathscr{O}_Y}\mathscr{F}),$$

although the bicomplex $L^{(1)}_{\bullet\bullet}\otimes_A L^{(2)}_{\bullet\bullet}$ occurring in the preceding calculation still identifies with

$$C^\bullet(\mathfrak{U},\mathscr{P}_\bullet\otimes_Y\mathscr{F}).$$

On the other hand, there is a canonical isomorphism of bi-∂-functors

(6.7.4.2).

$$\mathfrak{Tor}^S_n(1_{X^{(1)}},1_{X^{(2)}};\mathscr{P}^{(1)}_\bullet,\mathscr{P}^{(2)}_\bullet) \xrightarrow{\sim} \mathscr{T}\!or^S_n(\mathscr{P}^{(1)}_\bullet,\mathscr{P}^{(2)}_\bullet). \tag{6.7.4.2}$$

Indeed, (6.7.1) and (6.7.2) reduce us again to the case in which S and the $X^{(i)}$ are affine. In computing the first side of (6.7.4.2), we may then take $\mathfrak{U}^{(i)}$ to consist of the single member $X^{(i)}$.

With the notation of (6.6.2), $L^{(i)}_{\bullet\bullet}$ then reduces to **III** | 160

$$\Gamma(X^{(i)},\mathscr{P}^{(i)}_\bullet),$$

regarded as a bicomplex whose components of first degree different from zero vanish. The equality of the two sides of (6.7.4.2) follows from (6.4.1.1) and (6.3.1).

Proposition (6.7.5). — *Let*

$$u:\mathscr{P}^{(1)}_\bullet\longrightarrow\mathscr{Q}^{(1)}_\bullet$$

be a homomorphism of bounded-below complexes of quasi-coherent $\mathscr{O}_{X^{(1)}}$-modules such that the induced homomorphism

$$\mathscr{H}_\bullet(u):\mathscr{H}_\bullet(\mathscr{P}^{(1)}_\bullet)\longrightarrow\mathscr{H}_\bullet(\mathscr{Q}^{(1)}_\bullet)$$

is an isomorphism. Then the homomorphisms

$$^{(t)}\mathscr{E}\big(f_1, f_2; \mathscr{P}_\bullet^{(1)}, \mathscr{P}_\bullet^{(2)}\big) \longrightarrow {}^{(t)}\mathscr{E}\big(f_1, f_2; \mathscr{Q}_\bullet^{(1)}, \mathscr{P}_\bullet^{(2)}\big)$$

induced by u are isomorphisms for $t = a, b, c$.

Proof. For the spectral sequence (c), the assertion follows from biregularity and from the fact that, by hypothesis, the homomorphism under consideration is an isomorphism on E_2-terms ((**0**, 11.1.5)). This already shows that

$$\mathfrak{Tor}_n^S\big(f_1, f_2; \mathscr{P}_\bullet^{(1)}, \mathscr{P}_\bullet^{(2)}\big) \longrightarrow \mathfrak{Tor}_n^S\big(f_1, f_2; \mathscr{Q}_\bullet^{(1)}, \mathscr{P}_\bullet^{(2)}\big)$$

is an isomorphism. Applying (6.7.4.1) and (6.7.4.2), we first see that the homomorphisms

$$\mathscr{H}^{-n}\big(f_1, \mathscr{P}_\bullet^{(1)}\big) \longrightarrow \mathscr{H}^{-n}\big(f_1, \mathscr{Q}_\bullet^{(1)}\big)$$

and

$$\mathscr{T}\!or_n^S\big(\mathscr{P}_\bullet^{(1)}, \mathscr{P}_\bullet^{(2)}\big) \longrightarrow \mathscr{T}\!or_n^S\big(\mathscr{Q}_\bullet^{(1)}, \mathscr{P}_\bullet^{(2)}\big)$$

induced by u are isomorphisms. The assertion for (a) and (b) now follows from the biregularity of these spectral sequences ((6.7.3)) and the bijectivity of the induced homomorphisms on E_2-terms ((**0**, 11.1.5)).

Notice also that if $u : \mathscr{P}_\bullet^{(1)} \to \mathscr{Q}_\bullet^{(1)}$ is a homotopy equivalence, then it induces canonical isomorphisms

$$^{(t)}\mathscr{E}\big(f_1, f_2; \mathscr{P}_\bullet^{(1)}, \mathscr{P}_\bullet^{(2)}\big) \xrightarrow{\sim} {}^{(t)}\mathscr{E}\big(f_1, f_2; \mathscr{Q}_\bullet^{(1)}, \mathscr{P}_\bullet^{(2)}\big)$$

for all six spectral sequences. Indeed, when S and the $Y^{(i)}$ are affine, u induces a homotopy equivalence of bicomplexes

$$C^\bullet\big(\mathfrak{U}^{(1)}, \mathscr{P}_\bullet^{(1)}\big) \longrightarrow C^\bullet\big(\mathfrak{U}^{(1)}, \mathscr{Q}_\bullet^{(1)}\big),$$

and the assertion follows from the general theory of hyperhomology ((**0**, 11.3.2)). The passage to the general case is made by gluing, using the fact that a homotopy equivalence of complexes induces one between projective Cartan–Eilenberg resolutions of those complexes (M, XVII, 1.2). □

Proposition (6.7.6). — *Suppose that either $\mathscr{P}_\bullet^{(1)}$ or $\mathscr{P}_\bullet^{(2)}$ consists of S-flat modules, both complexes being bounded below. There is then a canonical isomorphism of bi-∂-functors*

(6.7.6.1).

$$\mathfrak{Tor}_n^S\big(f_1, f_2; \mathscr{P}_\bullet^{(1)}, \mathscr{P}_\bullet^{(2)}\big) \xrightarrow{\sim} \mathscr{H}^{-n}\big(f_1 \times_S f_2, \mathscr{P}_\bullet^{(1)} \otimes_S \mathscr{P}_\bullet^{(2)}\big). \tag{6.7.6.1}$$

Proof. First suppose that S, $Y^{(1)}$, and $Y^{(2)}$ are affine, so that we are in the situation of (6.6.2), whose notation we retain. Suppose, for example, that $\mathscr{P}_\bullet^{(1)}$ consists of S-flat modules. Using (6.6.6), the hypertor is the homology of

$$L_{\bullet\bullet}^{(1)} \otimes_A M_{\bullet\bullet\bullet}^{(2)},$$

where $M_{\bullet\bullet\bullet}^{(2)}$ is a projective Cartan–Eilenberg resolution of $L_{\bullet\bullet}^{(2)}$ in the sense of (**0**, 11.7.1). On the other hand, the modules $L_{ij}^{(1)}$ are flat over A by the hypothesis ((**I**, 1.4.15.1)). It follows from (**0**, 11.7.5) that there is a canonical isomorphism

(6.7.6.2).

$$\mathbf{Tor}_\bullet^S\big(\mathfrak{U}^{(1)}, \mathfrak{U}^{(2)}; \mathscr{P}_\bullet^{(1)}, \mathscr{P}_\bullet^{(2)}\big) \xrightarrow{\sim} H_\bullet\big(L_{\bullet\bullet}^{(1)} \otimes_A L_{\bullet\bullet}^{(2)}\big). \tag{6.7.6.2}$$

There is also a natural homomorphism of bicomplexes

$$L_{\bullet\bullet}^{(1)} \otimes_A L_{\bullet\bullet}^{(2)} \longrightarrow C^\bullet(\mathfrak{U}, \mathscr{Q}_\bullet),$$

where $\mathfrak{U}$ is the covering of

$$Z = X^{(1)} \times_S X^{(2)}$$

by the affine open subsets $U_\alpha^{(1)} \times_S U_\beta^{(2)}$, and

$$\mathscr{Q}_\bullet = \mathscr{P}_\bullet^{(1)} \otimes_S \mathscr{P}_\bullet^{(2)}$$

is regarded as the simple complex for the total degree.

The substance of this homomorphism was already given in the calculation of spectral sequence (b′) in (6.6.4), with $q = 0$. Retaining the notation used there, one need only observe that there is, first, a natural homomorphism from $N^{\bullet k}$ to III | 161

$$C^\bullet\big(\Phi^{(1)}, \Phi^{(2)}; \mathscr{S}_k\big)$$

((**0**, 11.8.5)); second, a natural homomorphism from the latter complex to

$$P^\bullet\big(\Phi^{(1)}, \Phi^{(2)}; \mathscr{S}_k\big)$$

((**0**, 11.8.6)); and finally, a natural homomorphism from this last cochain complex to its subcomplex of alternating cochains ((**0**, 11.8.7)). The resulting homomorphism of bicomplexes induces an isomorphism in homology, as was seen in (6.6.4). Composing with (6.7.6.2) therefore gives an isomorphism

(6.7.6.3).

$$\mathbf{Tor}^S_n\big(\mathfrak{U}^{(1)}, \mathfrak{U}^{(2)}; \mathscr{P}^{(1)}_\bullet, \mathscr{P}^{(2)}_\bullet\big) \xrightarrow{\sim} \mathrm{H}^{-n}\big(\mathfrak{U}, \mathscr{P}^{(1)}_\bullet \otimes_S \mathscr{P}^{(2)}_\bullet\big). \tag{6.7.6.3}$$

It remains to prove that the isomorphism just defined is independent of the chosen open coverings. The right side of (6.7.6.3) is canonically isomorphic, by (6.2.2), to

$$\mathrm{H}^{-n}\big(X^{(1)} \times_S X^{(2)}, \mathscr{P}^{(1)}_\bullet \otimes_S \mathscr{P}^{(2)}_\bullet\big).$$

For the left side, use (6.6.7) and note, with the notation used there, that the following diagram commutes up to homotopy:

$$\begin{tikzcd}
C^\bullet(\mathfrak{U}^{(1)}, \mathscr{P}^{(1)}_\bullet) \otimes_A C^\bullet(\mathfrak{U}^{(2)}, \mathscr{P}^{(2)}_\bullet) \arrow[r] \arrow[d] & C^\bullet(\mathfrak{U}, \mathscr{Q}_\bullet) \arrow[d] \\
C^\bullet(\mathfrak{V}^{(1)}, \mathscr{P}^{(1)}_\bullet) \otimes_A C^\bullet(\mathfrak{V}^{(2)}, \mathscr{P}^{(2)}_\bullet) \arrow[r] & C^\bullet(\mathfrak{V}, \mathscr{Q}_\bullet).
\end{tikzcd}$$

The horizontal arrows are those defined above. Finally, one passes to the general case by gluing, exactly as in (6.7.1) and (6.7.2); the details are left to the reader. □

Proposition (6.7.7). — *Suppose that $\mathscr{P}^{(1)}_\bullet$ and $\mathscr{P}^{(2)}_\bullet$ are bounded below, and that either all the modules*

$$\mathscr{H}^{-n}(f_1, \mathscr{P}^{(1)}_\bullet)$$

or all the modules

$$\mathscr{H}^{-n}(f_2, \mathscr{P}^{(2)}_\bullet)$$

are S-flat. There is then a canonical isomorphism of bi-∂-functors, for $n \in \mathbf{Z}$,

(6.7.7.1).

$$\mathfrak{Tor}^S_n\big(f_1, f_2; \mathscr{P}^{(1)}_\bullet, \mathscr{P}^{(2)}_\bullet\big) \xrightarrow{\sim} \bigoplus_{q_1+q_2=n} \mathscr{H}^{-q_1}(f_1, \mathscr{P}^{(1)}_\bullet) \otimes_S \mathscr{H}^{-q_2}(f_2, \mathscr{P}^{(2)}_\bullet). \tag{6.7.7.1}$$

Proof. By (6.5.8), spectral sequence (a) of (6.7.3) degenerates. The assertion therefore follows immediately from (**0**, 11.1.6), since that spectral sequence is biregular ((6.7.3)). □

Theorem (6.7.8). — *Suppose that the following conditions hold:*

1° *The complexes $\mathscr{P}^{(1)}_\bullet$ and $\mathscr{P}^{(2)}_\bullet$ are bounded below.*

2° *Either $\mathscr{P}^{(1)}_\bullet$ or $\mathscr{P}^{(2)}_\bullet$ consists of S-flat modules.*

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3° *Either all the modules $\mathscr{H}^{-n}(f_1, \mathscr{P}^{(1)}_\bullet)$ or all the modules $\mathscr{H}^{-n}(f_2, \mathscr{P}^{(2)}_\bullet)$ are S-flat.*

There is then a canonical isomorphism of bi-∂-functors, for $n \in \mathbf{Z}$,

(6.7.8.1).

$$\mathscr{H}^n\big(f_1 \times_S f_2, \mathscr{P}^{(1)}_\bullet \otimes_S \mathscr{P}^{(2)}_\bullet\big) \xrightarrow{\sim} \bigoplus_{n_1+n_2=n} \mathscr{H}^{n_1}(f_1, \mathscr{P}^{(1)}_\bullet) \otimes_S \mathscr{H}^{n_2}(f_2, \mathscr{P}^{(2)}_\bullet). \tag{6.7.8.1}$$

This is the Künneth formula.

Proof. The formula follows from (6.7.6) and (6.7.7).

When S, $Y^{(1)}$, and $Y^{(2)}$ are affine, the inverse of (6.7.8.1) is obtained, with the notation of (6.7.6), from the homomorphism of bicomplexes

$$C^\bullet(\mathfrak{U}^{(1)}, \mathscr{P}_\bullet^{(1)}) \otimes_A C^\bullet(\mathfrak{U}^{(2)}, \mathscr{P}_\bullet^{(2)}) \longrightarrow C^\bullet(\mathfrak{U}, \mathscr{Q}_\bullet)$$

by the procedure defined in (G, I, 2.7), as follows from (G, I, 5.5). □

Proposition (6.7.9). — *Suppose that the following three conditions hold:*

1° *The schemes S, $Y^{(1)}$, and $Y^{(2)}$ are locally noetherian; f_1 and f_2 are proper; and either $Y^{(1)}$ or $Y^{(2)}$ is of finite type over S.*

2° *The complexes $\mathscr{P}_\bullet^{(1)}$ and $\mathscr{P}_\bullet^{(2)}$ are bounded below.*

3° *For every $n \in \mathbf{Z}$, the module $H_n(\mathscr{P}_\bullet^{(i)})$ is coherent, for $i = 1, 2$.*

Then

$$\mathfrak{Tor}_n^S(f_1, f_2; \mathscr{P}_\bullet^{(1)}, \mathscr{P}_\bullet^{(2)})$$

is a coherent $\mathscr{O}_Y$-module, where $Y = Y^{(1)} \times_S Y^{(2)}$.

Proof. It follows from (6.5.13) that the local hypertor modules

$$\mathscr{T}\!or_n^S(\mathscr{P}_\bullet^{(1)}, \mathscr{P}_\bullet^{(2)})$$

are coherent $\mathscr{O}_X$-modules, where $X = X^{(1)} \times_S X^{(2)}$. Indeed, each $X^{(i)}$ is locally noetherian, and one of them is, by hypothesis, of finite type over S ((**I**, 6.3.4) and (6.3.8)); hence X is locally noetherian.

Since Y is locally noetherian and $f_1 \times_S f_2$ is proper ((**II**, 5.4.2)), it follows from (6.2.5) that the terms ${}^{(b)}\mathscr{E}_{pq}^2$ of (6.7.3) are coherent $\mathscr{O}_Y$-modules. All the spectral sequences of (6.7.3) are biregular by hypothesis 2°, so the conclusion follows from (**0**, 11.1.8). □

(6.7.10). Let $Y'^{(i)}$ be two S-preschemes and let

$$v_i : Y'^{(i)} \longrightarrow Y^{(i)} \qquad (i = 1, 2)$$

be two S-morphisms. Write

$$v = v_1 \times_S v_2 : Y' \longrightarrow Y, \qquad Y' = Y'^{(1)} \times_S Y'^{(2)}.$$

For $i = 1, 2$, let $X'^{(i)}$ be an S-prescheme, and let

$$u_i : X'^{(i)} \longrightarrow X^{(i)}, \qquad f'_i : X'^{(i)} \longrightarrow Y'^{(i)}$$

be S-morphisms such that the diagrams

(6.7.10.1).

$$\begin{tikzcd} X'^{(i)} \arrow[r, "u_i"] \arrow[d, "f'_i"'] & X^{(i)} \arrow[d, "f_i"] \\ Y'^{(i)} \arrow[r, "v_i"'] & Y^{(i)} \end{tikzcd} \tag{6.7.10.1}$$

commute, the morphisms f'_i being separated and quasi-compact. There are then canonical $\mathscr{O}_{Y'}$-linear homomorphisms of spectral functors

(6.7.10.2).

$$v^{*\,(t)}\mathscr{E}(f_1, f_2; \mathscr{P}_\bullet^{(1)}, \mathscr{P}_\bullet^{(2)}) \longrightarrow {}^{(t)}\mathscr{E}(f'_1, f'_2; u_1^*(\mathscr{P}_\bullet^{(1)}), u_2^*(\mathscr{P}_\bullet^{(2)})) \tag{6.7.10.2}$$

for $t = a, a', b, b', c, d$.

To define them, first suppose that

$$S = \mathrm{Spec}(A), \qquad Y^{(i)} = \mathrm{Spec}(B_i), \qquad Y'^{(i)} = \mathrm{Spec}(B'_i)$$

are affine. The schemes $X^{(i)}$ and $X'^{(i)}$ are then quasi-compact. To compute

$${}^{(t)}\mathscr{E}(f_1, f_2; \mathscr{P}_\bullet^{(1)}, \mathscr{P}_\bullet^{(2)}),$$

choose, as in (6.6.1), finite affine open coverings $\mathfrak{U}^{(i)}$ of $X^{(i)}$. To compute

$${}^{(t)}\mathscr{E}\big(f_1', f_2'; u_1^*(\mathscr{P}_\bullet^{(1)}), u_2^*(\mathscr{P}_\bullet^{(2)})\big),$$

choose finite affine open coverings $\mathfrak{U}'^{(i)}$ of $X'^{(i)}$, respectively finer than the inverse-image coverings $u_i^{-1}(\mathfrak{U}^{(i)})$.

The bicomplex III | 163

$$C^\bullet\big(\mathfrak{U}^{(i)}, \mathscr{P}_\bullet^{(i)}\big) = L_{\bullet\bullet}^{(i)}$$

may canonically be regarded as a sub-bicomplex of

$$C^\bullet\big(u_i^{-1}(\mathfrak{U}^{(i)}), u_i^*(\mathscr{P}_\bullet^{(i)})\big)$$

((**0**, 4.4.3.2)). A simplicial map (G, II, 5.7) from $\mathfrak{U}'^{(i)}$ to $u_i^{-1}(\mathfrak{U}^{(i)})$ defines a homomorphism of bicomplexes

$$C^\bullet\big(u_i^{-1}(\mathfrak{U}^{(i)}), u_i^*(\mathscr{P}_\bullet^{(i)})\big) \longrightarrow C^\bullet\big(\mathfrak{U}'^{(i)}, u_i^*(\mathscr{P}_\bullet^{(i)})\big).$$

By composition one obtains a homomorphism

$$L_{\bullet\bullet}^{(i)} \longrightarrow L_{\bullet\bullet}'^{(i)} = C^\bullet\big(\mathfrak{U}'^{(i)}, u_i^*(\mathscr{P}_\bullet^{(i)})\big).$$

Changing the simplicial map replaces this homomorphism by a homotopic one (G, II, 5.7.1). Thus one obtains a well-defined homomorphism of spectral functors

(6.7.10.3).

$$\text{(6.7.10.3)} \qquad {}^{(t)}\mathscr{E}\big(\mathfrak{U}^{(1)}, \mathfrak{U}^{(2)}; \mathscr{P}_\bullet^{(1)}, \mathscr{P}_\bullet^{(2)}\big) \longrightarrow {}^{(t)}\mathscr{E}\big(\mathfrak{U}'^{(1)}, \mathfrak{U}'^{(2)}; u_1^*(\mathscr{P}_\bullet^{(1)}), u_2^*(\mathscr{P}_\bullet^{(2)})\big).$$

Suppose that $\mathfrak{V}^{(i)}$ is a finite affine open covering of $X^{(i)}$ finer than $\mathfrak{U}^{(i)}$, and that $\mathfrak{V}'^{(i)}$ is a finite affine open covering of $X'^{(i)}$ finer than both $\mathfrak{U}'^{(i)}$ and $u_i^{-1}(\mathfrak{V}^{(i)})$. One verifies immediately that the diagram

$$\begin{tikzcd}
C^\bullet(\mathfrak{U}^{(i)}, \mathscr{P}_\bullet^{(i)}) \arrow[r] \arrow[d] & C^\bullet(\mathfrak{V}^{(i)}, \mathscr{P}_\bullet^{(i)}) \arrow[d] \\
C^\bullet\big(\mathfrak{U}'^{(i)}, u_i^*(\mathscr{P}_\bullet^{(i)})\big) \arrow[r] & C^\bullet\big(\mathfrak{V}'^{(i)}, u_i^*(\mathscr{P}_\bullet^{(i)})\big)
\end{tikzcd}$$

commutes. Consequently, (6.7.10.3) is essentially independent of the coverings used. We have therefore defined a homomorphism of A-modules

(6.7.10.4).

$$\text{(6.7.10.4)} \qquad {}^{(t)}E\big(X^{(1)}, X^{(2)}; \mathscr{P}_\bullet^{(1)}, \mathscr{P}_\bullet^{(2)}\big) \longrightarrow {}^{(t)}E\big(X'^{(1)}, X'^{(2)}; u_1^*(\mathscr{P}_\bullet^{(1)}), u_2^*(\mathscr{P}_\bullet^{(2)})\big).$$

By the definition of the $u_i^*(\mathscr{P}_\bullet^{(i)})$ and the commutativity of (6.7.10.1), this is also a homomorphism of $(B_1 \otimes_A B_2)$-modules. Since the right-hand side of (6.7.10.4) consists of $(B_1' \otimes_A B_2')$-modules, it canonically induces a homomorphism of $(B_1' \otimes_A B_2')$-modules

(6.7.10.5).

$$\text{(6.7.10.5)} \qquad {}^{(t)}E\big(X^{(1)}, X^{(2)}; \mathscr{P}_\bullet^{(1)}, \mathscr{P}_\bullet^{(2)}\big) \otimes_{B_1 \otimes_A B_2} (B_1' \otimes_A B_2') \longrightarrow {}^{(t)}E\big(X'^{(1)}, X'^{(2)}; u_1^*(\mathscr{P}_\bullet^{(1)}), u_2^*(\mathscr{P}_\bullet^{(2)})\big).$$

In view of (**I**, 1.6.5), this is precisely (6.7.10.2) in the affine case.

It remains to pass to the general case by the gluing procedures of (6.7.1) and (6.7.2). The second passage is immediate. For the first, take, as in (6.7.1), elements $g_i \in B_i$ and their images $g_i' \in B_i'$, and set

$$g = g_1 \otimes g_2 \in B = B_1 \otimes_A B_2, \qquad g' = g_1' \otimes g_2' \in B' = B_1' \otimes_A B_2'.$$

Everything then reduces to the canonical isomorphism III | 164

$$(M \otimes_B B')_{g'} \xrightarrow{\sim} M_g \otimes_{B_g} B'_{g'}$$

((**0**, 1.5.4)). The details are left to the reader.

(6.7.11). The theory of global hypertor developed above for two S-morphisms $X^{(i)} \to Y^{(i)}$ and two bounded-below complexes $\mathscr{P}_\bullet^{(i)}$ of quasi-coherent modules extends immediately to the following general situation. Let S be a prescheme, let $(Y^{(i)})_{i\in I}$ be a finite family of S-preschemes, let

$$f_i : X^{(i)} \longrightarrow Y^{(i)} \qquad (i \in I)$$

be a finite family of separated and quasi-compact S-morphisms, and for each i let $\mathscr{P}_\bullet^{(i)}$ be a bounded-below complex of quasi-coherent $\mathscr{O}_{X^{(i)}}$-modules. If Y is the product of the S-preschemes $Y^{(i)}$, then for every $n \in \mathbf{Z}$ one defines a quasi-coherent $\mathscr{O}_Y$-module

$$\mathfrak{Tor}_n^S\big((f_i)_{i\in I}; (\mathscr{P}_\bullet^{(i)})_{i\in I}\big).$$

These modules form a covariant ∂-functor in each complex $\mathscr{P}_\bullet^{(i)}$; moreover, this functor is the common abutment of six spectral functors

$$^{(t)}\mathscr{E}\big((f_i)_{i\in I}; (\mathscr{P}_\bullet^{(i)})_{i\in I}\big).$$

We leave to the reader the repetition, in this general case, of the definitions and arguments given above for $I = \{1,2\}$.

When I has a single element, this construction recovers the hypercohomology $\mathscr{H}^\bullet(f, \mathscr{P}_\bullet)$ defined in (6.2.7), as already noted in (6.7.4). When I is the interval $1 \leqslant i \leqslant m$ in $\mathbf{N}$, we write

$$\mathfrak{Tor}_n^S(f_1, \ldots, f_m; \mathscr{P}_\bullet^{(1)}, \ldots, \mathscr{P}_\bullet^{(m)})$$

for

$$\mathfrak{Tor}_n^S\big((f_i)_{i\in I}; (\mathscr{P}_\bullet^{(i)})_{i\in I}\big).$$

Proposition (6.7.12). — *With the notation of (6.7.11), let J be a subset of I such that, for $i \in I - J$,*

$$X^{(i)} = Y^{(i)} = S,$$

the morphism f_i is the identity, and $\mathscr{P}_\bullet^{(i)}$ is the complex concentrated in degree 0 with value $\mathscr{O}_S$. There is then a canonical isomorphism of ∂-functors

(6.7.12.1).

$$\mathfrak{Tor}_n^S\big((f_i)_{i\in I}; (\mathscr{P}_\bullet^{(i)})_{i\in I}\big) \xrightarrow{\sim} \mathfrak{Tor}_n^S\big((f_i)_{i\in J}; (\mathscr{P}_\bullet^{(i)})_{i\in J}\big). \tag{6.7.12.1}$$

Proof. It is enough to define this isomorphism when S and the $Y^{(i)}$ are affine, since the gluing is carried out as usual. For $i \in I - J$, take $\mathfrak{U}^{(i)}$ to be the covering consisting of the single set S. Then

$$L_{\bullet\bullet}^{(i)} = C^\bullet(\mathfrak{U}^{(i)}, \mathscr{P}_\bullet^{(i)})$$

has only one nonzero term, in bidegree $(0,0)$, equal to

$$\Gamma(S, \mathscr{O}_S) = A(S).$$

The isomorphism (6.7.12.1) is therefore immediate. □

Remark (6.7.13). — With the notation of (6.7.3), consider the canonical S-isomorphism

$$Y^{(1)} \times_S Y^{(2)} \longrightarrow Y^{(2)} \times_S Y^{(1)}$$

of (**I**, 3.3.5). Under this isomorphism, the image of

$$\mathfrak{Tor}_n^S(f_1, f_2; \mathscr{P}_\bullet^{(1)}, \mathscr{P}_\bullet^{(2)})$$

is

$$\mathfrak{Tor}_n^S(f_2, f_1; \mathscr{P}_\bullet^{(2)}, \mathscr{P}_\bullet^{(1)}).$$

This is a local assertion, so it reduces to the case considered in (6.6.2). If $M_{\bullet\bullet\bullet}^{(i)}$ is a projective Cartan–Eilenberg resolution of $L_{\bullet\bullet}^{(i)}$ for $i = 1,2$, the isomorphism in question interchanges

$$M_{\bullet\bullet\bullet}^{(1)} \otimes_A M_{\bullet\bullet\bullet}^{(2)} \qquad \text{and} \qquad M_{\bullet\bullet\bullet}^{(2)} \otimes_A M_{\bullet\bullet\bullet}^{(1)}.$$

The assertion follows by taking the homology of the simple complexes associated with these tricomplexes.

6.8. The associativity spectral sequences of global hypertor

(6.8.1). Retain the hypotheses and notation of (6.7.11); in particular, the complexes $\mathscr{P}_\bullet^{(i)}$ are bounded below. Let $(I_j)_{j\in J}$ be a partition of the index set I. We seek an "associativity" relation between

$$\mathfrak{Tor}_n^S\big((f_i)_{i\in I};(\mathscr{P}_\bullet^{(i)})_{i\in I}\big)$$

and the "partial" hypertors

$$\mathscr{T}_{\bullet j} = \mathfrak{Tor}_\bullet^S\big((f_i)_{i\in I_j};(\mathscr{P}_\bullet^{(i)})_{i\in I_j}\big).$$

To simplify the notation, we shall restrict attention to the case where I is the interval $1\leqslant i\leqslant m$ and the partition consists of the two intervals

$$\{1,2,\ldots,r\}\qquad\text{and}\qquad\{r+1,\ldots,m\}.$$

Proposition (6.8.2). — *There is a canonical biregular spectral functor, called the* associativity spectral functor, *denoted*

$$^{(e)}\mathscr{E}^S\big(f_1,\ldots,f_m;\mathscr{P}_\bullet^{(1)},\ldots,\mathscr{P}_\bullet^{(m)}\big),$$

or simply $^{(e)}\mathscr{E}$ *with the same arguments, whose abutment is*

$$\mathfrak{Tor}_\bullet^S\big(f_1,\ldots,f_m;\mathscr{P}_\bullet^{(1)},\ldots,\mathscr{P}_\bullet^{(m)}\big)$$

and whose E_2*-term is*

$$^{(e)}\mathscr{E}^2_{pq} = \bigoplus_{q_1+q_2=q}\mathscr{T}\!or_p^S\Big(\mathfrak{Tor}_{q_1}^S\big(f_1,\ldots,f_r;\mathscr{P}_\bullet^{(1)},\ldots,\mathscr{P}_\bullet^{(r)}\big),\\ \mathfrak{Tor}_{q_2}^S\big(f_{r+1},\ldots,f_m;\mathscr{P}_\bullet^{(r+1)},\ldots,\mathscr{P}_\bullet^{(m)}\big)\Big).$$

Proof. Here Y is canonically identified with $Z^{(1)}\times_S Z^{(2)}$, where

$$Z^{(1)} = Y^{(1)}\times_S\cdots\times_S Y^{(r)}\quad\text{and}\quad Z^{(2)} = Y^{(r+1)}\times_S\cdots\times_S Y^{(m)}.$$

It is enough to treat the case in which S and all the $Y^{(i)}$ are affine. The passage from this case to the general one uses the methods of (6.7.1) and (6.7.2); the details are left to the reader. Thus it is enough to prove the following affine statement. □

Corollary (6.8.3). — *Let A be a ring, let $S=\operatorname{Spec}(A)$, and let $X^{(i)}$, $1\leqslant i\leqslant m$, be quasi-compact S-schemes. For each i, let $\mathscr{P}_\bullet^{(i)}$ be a bounded-below complex of quasi-coherent $\mathscr{O}_{X^{(i)}}$-modules. There is a canonical biregular spectral functor with abutment*

$$\mathbf{Tor}_\bullet^S\big(X^{(1)},\ldots,X^{(m)};\mathscr{P}_\bullet^{(1)},\ldots,\mathscr{P}_\bullet^{(m)}\big)$$

and E_2*-term*

$$^{(e)}E^2_{pq} = \bigoplus_{q_1+q_2=q}\operatorname{Tor}_p^A\Big(\mathbf{Tor}_{q_1}^S\big(X^{(1)},\ldots,X^{(r)};\mathscr{P}_\bullet^{(1)},\ldots,\mathscr{P}_\bullet^{(r)}\big),\\ \mathbf{Tor}_{q_2}^S\big(X^{(r+1)},\ldots,X^{(m)};\mathscr{P}_\bullet^{(r+1)},\ldots,\mathscr{P}_\bullet^{(m)}\big)\Big).$$

Proof. **III | 165**

Following the definition of (6.6.2), choose for each i a finite affine open covering $\mathfrak{U}^{(i)}$ of $X^{(i)}$, form the bicomplex

$$L_{\bullet\bullet}^{(i)} = C^\bullet\big(\mathfrak{U}^{(i)},\mathscr{P}_\bullet^{(i)}\big),$$

and choose a projective Cartan–Eilenberg resolution $M_{\bullet\bullet\bullet}^{(i)}$ of this bicomplex in the sense of (**0**, 11.7.1). The hypertor in question is the homology of the tensor product

$$M_{\bullet\bullet\bullet} = \bigotimes_{i=1}^m M_{\bullet\bullet\bullet}^{(i)}.$$

Regard $M_{\bullet\bullet\bullet}$ as a simple complex $N_\bullet$, the tensor product of the two simple complexes

$$N'_\bullet = \bigotimes_{i=1}^r M_{\bullet\bullet\bullet}^{(i)},\qquad N''_\bullet = \bigotimes_{i=r+1}^m M_{\bullet\bullet\bullet}^{(i)},$$

where $N'_\bullet$ and $N''_\bullet$ are graded by the sums of the total degrees of the $M^{(i)}_{\bullet\bullet\bullet}$.

The A-modules forming $N'_\bullet$ and $N''_\bullet$ are projective. Hence (6.5.9) gives

$$H_\bullet(M_{\bullet\bullet\bullet}) = \mathrm{Tor}^A_\bullet(N'_\bullet, N''_\bullet).$$

The required spectral sequence is therefore precisely (6.3.2.2) applied to $N'_\bullet$ and $N''_\bullet$, using the preceding interpretation of their homology modules for the two partial products

$$X^{(1)} \times_S \cdots \times_S X^{(r)} \quad \text{and} \quad X^{(r+1)} \times_S \cdots \times_S X^{(m)}.$$

Finally, the regularity assertions follow from (6.3.2). Indeed, since the $\mathscr{P}^{(i)}_\bullet$ are bounded below, so are the $M^{(i)}_{\bullet\bullet\bullet}$, and therefore so are $N'_\bullet$ and $N''_\bullet$. □

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(6.9.1). Retain the hypotheses and notation of (6.7.11); in particular, the complexes $\mathscr{P}^{(i)}_\bullet$ are bounded below. Let $g : S' \to S$ be a morphism of preschemes, and set

$$Y'^{(i)} = Y^{(i)}_{(S')}, \qquad X'^{(i)} = X^{(i)}_{(S')}, \qquad \mathscr{P}'^{(i)}_\bullet = \mathscr{P}^{(i)}_\bullet \otimes_{\mathscr{O}_S} \mathscr{O}_{S'}.$$

Thus $\mathscr{P}'^{(i)}_\bullet$ is a complex of quasi-coherent $\mathscr{O}_{X'^{(i)}}$-modules. Let

$$f'_i = (f_i)_{(S')} : X'^{(i)} \longrightarrow Y'^{(i)};$$

this morphism is separated and quasi-compact ((**I**, 5.5.1) and (6.6.4)).

We shall study the relation between the quasi-coherent $\mathscr{O}_{Y'}$-modules

$$\mathfrak{Tor}^{S'}_n\big((f'_i)_{i\in I}; (\mathscr{P}'^{(i)}_\bullet)_{i\in I}\big)$$

and

$$\mathfrak{Tor}^{S}_n\big((f_i)_{i\in I}; (\mathscr{P}^{(i)}_\bullet)_{i\in I}\big) \otimes_{\mathscr{O}_S} \mathscr{O}_{S'},$$

where

$$Y' = Y \times_S S' = Y_{(S')}.$$

The following particularly simple case reduces to (**I**, 1.4.15) when I has a single element and $\mathscr{P}_\bullet$ consists of a single module.

Proposition (6.9.2). — *If $g : S' \to S$ is flat, there is a canonical isomorphism of ∂-functors in the complexes $\mathscr{P}^{(i)}_\bullet$:*

(6.9.2.1).

$$\mathfrak{Tor}^{S}_n\big((f_i)_{i\in I}; (\mathscr{P}^{(i)}_\bullet)_{i\in I}\big) \otimes_{\mathscr{O}_S} \mathscr{O}_{S'} \xrightarrow{\sim} \mathfrak{Tor}^{S'}_n\big((f'_i)_{i\in I}; (\mathscr{P}'^{(i)}_\bullet)_{i\in I}\big). \tag{6.9.2.1}$$

Proof. It is enough to consider the case where S, S', and all the $Y^{(i)}$ are affine; the gluing is carried out by the methods of (6.7.1) and (6.7.2). Write

$$S = \mathrm{Spec}(A), \qquad S' = \mathrm{Spec}(A'),$$

and choose for each i a finite affine open covering $\mathfrak{U}^{(i)}$ of $X^{(i)}$. If $u_i : X'^{(i)} \to X^{(i)}$ is the canonical projection, then

$$\mathfrak{U}'^{(i)} = u_i^{-1}(\mathfrak{U}^{(i)})$$

is a finite affine open covering of $X'^{(i)}$ ((**II**, 1.5.5)), and

$$C^\bullet(\mathfrak{U}'^{(i)}, \mathscr{P}'^{(i)}_\bullet) = C^\bullet(\mathfrak{U}^{(i)}, \mathscr{P}^{(i)}_\bullet) \otimes_A A'.$$

Let $M^{(i)}_{\bullet\bullet\bullet}$ be a projective Cartan–Eilenberg resolution, in the sense of (**0**, 11.7.1), of

$$L^{(i)}_{\bullet\bullet} = C^\bullet(\mathfrak{U}^{(i)}, \mathscr{P}^{(i)}_\bullet)$$

by A-modules. Since A' is flat over A,

$$M^{(i)}_{\bullet\bullet\bullet} \otimes_A A'$$

is a projective Cartan–Eilenberg resolution, over A', of $L^{(i)}_{\bullet\bullet} \otimes_A A'$. The same flatness gives

$$H_\bullet\left(\bigotimes_{i=1}^{m}(M^{(i)}_{\bullet\bullet\bullet} \otimes_A A')\right) = H_\bullet\left(\bigotimes_{i=1}^{m} M^{(i)}_{\bullet\bullet\bullet}\right) \otimes_A A',$$

which proves (6.9.2.1). □

(6.9.2.2). When I consists of the single element 1, (6.9.2.1) also follows directly from (6.7.7). Indeed, take

$$Y_2 = X_2 = S', \qquad f_2 = 1_{S'},$$

and let $\mathscr{P}^{(2)}_\bullet$ be concentrated in degree 0 with value $\mathscr{O}_{S'}$. Then

$$\mathscr{H}^n(1_{S'}, \mathscr{O}_{S'})$$

vanishes for $n \neq 0$ and equals $\mathscr{O}_{S'}$ for $n = 0$ ((6.2.1)).

In the general case, replace $\mathscr{O}_{S'}$ by a bounded-below complex $\mathscr{Q}'_\bullet$ of quasi-coherent $\mathscr{O}_{S'}$-modules. Taking, for simplicity, $I = \{1, 2, \ldots, m\}$, one can then consider the ∂-functor

$$\mathfrak{Tor}^S_\bullet(f_1, \ldots, f_m, 1_{S'}; \mathscr{P}^{(1)}_\bullet, \ldots, \mathscr{P}^{(m)}_\bullet, \mathscr{Q}'_\bullet).$$

Proposition (6.9.3). — *There are three canonical biregular spectral functors, denoted*

$$^{(t)}\mathscr{E}(f_1, \ldots, f_m; \mathscr{P}^{(1)}_\bullet, \ldots, \mathscr{P}^{(m)}_\bullet, \mathscr{Q}'_\bullet), \qquad t = e, f, f',$$

with common abutment

$$\mathfrak{Tor}^S_\bullet(f_1, \ldots, f_m, 1_{S'}; \mathscr{P}^{(1)}_\bullet, \ldots, \mathscr{P}^{(m)}_\bullet, \mathscr{Q}'_\bullet),$$

and whose E_2-terms are respectively as follows.

(6.9.3.1).

$$^{(e)}\mathscr{E}^2_{pq} = \bigoplus_{q'+q''=q} \mathscr{T}\!or^S_p\Big(\mathfrak{Tor}^S_{q'}(f_1, \ldots, f_m; \mathscr{P}^{(1)}_\bullet, \ldots, \mathscr{P}^{(m)}_\bullet), \mathscr{H}_{q''}(\mathscr{Q}'_\bullet)\Big). \tag{6.9.3.1}$$

(6.9.3.2).

$$^{(f)}\mathscr{E}^2_{pq} = \bigoplus_{q_1+\cdots+q_{m+1}=q} \mathscr{T}\!or^{S'}_p\Big(\mathfrak{Tor}^S_{q_1}(f_1, 1_{S'}; \mathscr{P}^{(1)}_\bullet, \mathscr{O}_{S'}), \ldots, \mathfrak{Tor}^S_{q_m}(f_m, 1_{S'}; \mathscr{P}^{(m)}_\bullet, \mathscr{O}_{S'}), \mathscr{H}_{q_{m+1}}(\mathscr{Q}'_\bullet)\Big). \tag{6.9.3.2}$$

(6.9.3.3).

$$^{(f')}\mathscr{E}^2_{pq} = \bigoplus_{q_1+\cdots+q_m=q} \mathfrak{Tor}^{S'}_p\Big(f'_1, \ldots, f'_m, 1_{S'}; \mathscr{T}\!or^S_{q_1}(\mathscr{P}^{(1)}_\bullet, \mathscr{O}_{S'}), \ldots, \mathscr{T}\!or^S_{q_m}(\mathscr{P}^{(m)}_\bullet, \mathscr{O}_{S'}), \mathscr{Q}'_\bullet\Big). \tag{6.9.3.3}$$

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Proof. The sequence (e) is precisely the associativity sequence of (6.8.2), with $r = m$. To define the other two spectral sequences, it is again enough to treat the case in which S, S', and the $Y^{(i)}$ are affine. The passage to the general case uses the methods of (6.7.1) and (6.7.2) and is left to the reader. Thus it remains to prove the following affine statement. □

Corollary (6.9.4). — *Let A be a ring, let A' be an A-algebra, and put*

$$S = \operatorname{Spec}(A), \qquad S' = \operatorname{Spec}(A').$$

Let $X^{(i)}$, $1 \leqslant i \leqslant m$, be quasi-compact S-schemes, and for each i put

$$X'^{(i)} = X^{(i)}_{(S')};$$

this is a quasi-compact S'-scheme. For each i, let $\mathscr{P}_\bullet^{(i)}$ be a bounded-below complex of quasi-coherent $\mathscr{O}_{X^{(i)}}$-modules. Finally, let $Q'_\bullet$ be a bounded-below complex of A'-modules, and let $\widetilde{Q}'_\bullet$ be the associated complex.

There are three biregular spectral functors in the $\mathscr{P}_\bullet^{(i)}$ and in $Q'_\bullet$, with common abutment

$$\mathbf{Tor}_\bullet^S(X^{(1)},\ldots,X^{(m)},S';\mathscr{P}_\bullet^{(1)},\ldots,\mathscr{P}_\bullet^{(m)},\widetilde{Q}'_\bullet),$$

and whose E_2-terms are respectively

$$^{(e)}E^2_{pq} = \bigoplus_{q'+q''=q} \operatorname{Tor}_p^A\Big(\mathbf{Tor}_{q'}^S(X^{(1)},\ldots,X^{(m)};\mathscr{P}_\bullet^{(1)},\ldots,\mathscr{P}_\bullet^{(m)}),\ H_{q''}(Q'_\bullet)\Big),$$

$$^{(f)}E^2_{pq} = \bigoplus_{q_1+\cdots+q_{m+1}=q} \operatorname{Tor}_p^{A'}\Big(\mathbf{Tor}_{q_1}^S(X^{(1)},S';\mathscr{P}_\bullet^{(1)},\mathscr{O}_{S'}),\ldots,\ \mathbf{Tor}_{q_m}^S(X^{(m)},S';\mathscr{P}_\bullet^{(m)},\mathscr{O}_{S'}),H_{q_{m+1}}(Q'_\bullet)\Big),$$

and

$$^{(f')}E^2_{pq} = \bigoplus_{q_1+\cdots+q_m=q} \mathbf{Tor}_p^{S'}\Big(X'^{(1)},\ldots,X'^{(m)},S';\ \mathscr{T}or_{q_1}^S(\mathscr{P}_\bullet^{(1)},\mathscr{O}_{S'}),\ldots,\mathscr{T}or_{q_m}^S(\mathscr{P}_\bullet^{(m)},\mathscr{O}_{S'}),\widetilde{Q}'_\bullet\Big).$$

Proof. We do not return to the first of these spectral functors, which was treated in (6.8.3) and is included here only for reference. To define the other two, choose for each i a finite affine open covering $\mathfrak{U}^{(i)}$ of $X^{(i)}$. If

$$u_i : X'^{(i)} \longrightarrow X^{(i)}$$

is the canonical projection, put

$$\mathfrak{U}'^{(i)} = u_i^{-1}(\mathfrak{U}^{(i)}).$$

By (6.6.6), the common abutment is obtained by choosing, for $1 \leqslant i \leqslant m$, a projective Cartan–Eilenberg resolution $M_{\bullet\bullet\bullet}^{(i)}$ of

$$L_{\bullet\bullet}^{(i)} = C^\bullet(\mathfrak{U}^{(i)},\mathscr{P}_\bullet^{(i)})$$

in the sense of (**0**, 11.7.1), and then taking the homology of the tricomplex

$$M_{\bullet\bullet\bullet} = M_{\bullet\bullet\bullet}^{(1)} \otimes_A \cdots \otimes_A M_{\bullet\bullet\bullet}^{(m)} \otimes_A Q'_\bullet,$$

where $Q'_\bullet$ is regarded as a tricomplex whose last two degrees are zero. Set

$$M'^{(i)}_{\bullet\bullet\bullet} = M^{(i)}_{\bullet\bullet\bullet} \otimes_A A'.$$

Since $Q'_\bullet$ is a complex of A'-modules, we may rewrite

$$M_{\bullet\bullet\bullet} = M'^{(1)}_{\bullet\bullet\bullet} \otimes_{A'} \cdots \otimes_{A'} M'^{(m)}_{\bullet\bullet\bullet} \otimes_{A'} Q'_\bullet.$$

Regard each $M'^{(i)}_{\bullet\bullet\bullet}$ as a simple complex for its total degree. Its terms are projective A'-modules; hence (6.3.7), extended to any finite number of complexes, identifies the homology of $M_{\bullet\bullet\bullet}$ with

$$\operatorname{Tor}_\bullet^{A'}(M'^{(1)}_{\bullet\bullet\bullet},\ldots,M'^{(m)}_{\bullet\bullet\bullet},Q'_\bullet).$$

By (6.5.15), this is the abutment of a spectral sequence having the required regularity properties and E_2-term

$$E^2_{pq} = \bigoplus_{q_1+\cdots+q_{m+1}=q} \operatorname{Tor}_p^{A'}\Big(H_{q_1}(M'^{(1)}_{\bullet\bullet\bullet}),\ldots,H_{q_m}(M'^{(m)}_{\bullet\bullet\bullet}),\ H_{q_{m+1}}(Q'_\bullet)\Big).$$

By the definition of global hypertors,

$$H_{q_i}(M'^{(i)}_{\bullet\bullet\bullet}) = \mathbf{Tor}_{q_i}^S(X^{(i)},S';\mathscr{P}_\bullet^{(i)},\mathscr{O}_{S'}),$$

which gives the sequence (f).

On the other hand, regard $M'^{(i)}_{\bullet\bullet\bullet}$ as a bicomplex whose first degree is the sum of the first two degrees of the tricomplex and whose second degree is its third degree. Since its terms are projective A'-modules, the general theory of hyperhomology shows that the homology of **III** | 168

$$M'^{(1)}_{\bullet\bullet\bullet} \otimes_{A'} \cdots \otimes_{A'} M'^{(m)}_{\bullet\bullet\bullet} \otimes_{A'} Q'_\bullet$$

is canonically isomorphic to its hyperhomology ((**0**, 11.6.5)). It is therefore the abutment of a spectral sequence with E_2-term

$$E^2_{pq} = \bigoplus_{q_1+\cdots+q_{m+1}=q} \operatorname{Tor}^{A'}_p \Big(H^{\mathrm{II}}_{q_1}(M'^{(1)}_{\bullet\bullet\bullet}), \ldots, H^{\mathrm{II}}_{q_m}(M'^{(m)}_{\bullet\bullet\bullet}), H^{\mathrm{II}}_{q_{m+1}}(Q'_\bullet) \Big).$$

The second degree of $Q'_\bullet$ is zero, so

$$H^{\mathrm{II}}_n(Q'_\bullet) = 0 \quad (n \neq 0), \qquad H^{\mathrm{II}}_0(Q'_\bullet) = Q'_\bullet,$$

and the preceding formula becomes

$$E^2_{pq} = \bigoplus_{q_1+\cdots+q_m=q} \operatorname{Tor}^{A'}_p \Big(H^{\mathrm{II}}_{q_1}(M'^{(1)}_{\bullet\bullet\bullet}), \ldots, H^{\mathrm{II}}_{q_m}(M'^{(m)}_{\bullet\bullet\bullet}), Q'_\bullet \Big).$$

Moreover, by (6.3.4),

$$\begin{aligned} H^{\mathrm{II}}_{q_i}(M'^{(i)}_{\bullet\bullet\bullet}) &= H^{\mathrm{II}}_{q_i}(M^{(i)}_{\bullet\bullet\bullet} \otimes_A A') \\ &= \operatorname{Tor}^A_{q_i}(L^{(i)}_{\bullet\bullet}, A'). \end{aligned}$$

Now $L^{(i)}_{j,k}$ is the direct sum of the modules $\Gamma(V, \mathscr{P}^{(i)}_k)$, where V runs through the affine intersections of $j+1$ members of $\mathfrak{U}^{(i)}$. If $V' = u_i^{-1}(V)$, then V' is affine in $X'^{(i)}$, and (6.4.1.1) gives

$$\Gamma\big(V', \mathscr{T}\!or^S_{q_i}(\mathscr{P}^{(i)}_k, \mathscr{O}_{S'})\big) = \operatorname{Tor}^A_{q_i}\big(\Gamma(V, \mathscr{P}^{(i)}_k), A'\big).$$

Consequently, the bicomplex $H^{\mathrm{II}}_{q_i}(M'^{(i)}_{\bullet\bullet\bullet})$ is

$$C^\bullet\big(\mathfrak{U}'^{(i)}, \mathscr{T}\!or^S_{q_i}(\mathscr{P}^{(i)}_\bullet, \mathscr{O}_{S'})\big).$$

This gives the asserted E_2-term of the sequence (f'). Its biregularity under the stated hypotheses follows in the usual way, since, when $\mathscr{P}^{(i)}_\bullet$ is bounded below, every degree of $M'^{(i)}_{\bullet\bullet\bullet}$ is bounded below. □

Remark (6.9.5). — As in (6.7.6), replacing the complexes $\mathscr{P}^{(i)}_\bullet$ and $\mathscr{Q}'_\bullet$ by respectively homotopic complexes does not change the spectral sequences (e), (f), and (f'), up to canonical isomorphism.

Moreover, for the sequence (f), homomorphisms of complexes

$$\mathscr{P}^{(i)}_\bullet \longrightarrow \mathscr{R}^{(i)}_\bullet, \qquad \mathscr{Q}'_\bullet \longrightarrow \mathscr{T}'_\bullet$$

which induce isomorphisms in homology

$$\mathscr{H}_\bullet(\mathscr{P}^{(i)}_\bullet) \xrightarrow{\sim} \mathscr{H}_\bullet(\mathscr{R}^{(i)}_\bullet), \qquad \mathscr{H}_\bullet(\mathscr{Q}'_\bullet) \xrightarrow{\sim} \mathscr{H}_\bullet(\mathscr{T}'_\bullet)$$

also induce an isomorphism of spectral sequences

$$^{(f)}\mathscr{E}(f_1, \ldots, f_m; \mathscr{P}^{(1)}_\bullet, \ldots, \mathscr{P}^{(m)}_\bullet, \mathscr{Q}'_\bullet) \xrightarrow{\sim} {}^{(f)}\mathscr{E}(f_1, \ldots, f_m; \mathscr{R}^{(1)}_\bullet, \ldots, \mathscr{R}^{(m)}_\bullet, \mathscr{T}'_\bullet).$$

The proof is the same as that of (6.7.6), taking into account the result of that section and the regularity of the sequence (f).

Corollary (6.9.6). — *Under the hypotheses of (6.9.1), suppose that:*

1° *the complexes* $\mathscr{P}_\bullet^{(i)}$ *consist of S-flat modules, and the* $\mathscr{O}_Y$*-modules*

$$\mathfrak{Tor}_n^S(f_1,\ldots,f_m;\mathscr{P}_\bullet^{(1)},\ldots,\mathscr{P}_\bullet^{(m)})$$

are S-flat;

2° *the complexes* $\mathscr{P}_\bullet^{(i)}$ *and* $\mathscr{Q}'_\bullet$ *are bounded below.*

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$$\mathscr{P}'^{(i)}_\bullet = \mathscr{P}_\bullet^{(i)} \otimes_{\mathscr{O}_S} \mathscr{O}_{S'}.$$

There are canonical functorial isomorphisms

(6.9.6.1).

$$\begin{aligned}\mathfrak{Tor}_n^{S'}(f'_1,\ldots,f'_m,1_{S'};\mathscr{P}'^{(1)}_\bullet,\ldots,\mathscr{P}'^{(m)}_\bullet,\mathscr{Q}'_\bullet) &\xrightarrow{\sim} \\ \bigoplus_{n'+n''=n} \mathfrak{Tor}_{n'}^S(f_1,\ldots,f_m;\mathscr{P}_\bullet^{(1)},\ldots,\mathscr{P}_\bullet^{(m)}) &\otimes_{\mathscr{O}_S} \mathscr{H}_{n''}(\mathscr{Q}'_\bullet).\end{aligned} \tag{6.9.6.1}$$

In particular, when $\mathscr{Q}'_\bullet$ *consists of a single term* $\mathscr{F}'$ *in degree* 0*, there are canonical functorial isomorphisms*

(6.9.6.2).

$$\begin{aligned}\mathfrak{Tor}_n^{S'}(f'_1,\ldots,f'_m,1_{S'};\mathscr{P}'^{(1)}_\bullet,\ldots,\mathscr{P}'^{(m)}_\bullet,\mathscr{F}') &\xrightarrow{\sim} \\ \mathfrak{Tor}_n^S(f_1,\ldots,f_m;\mathscr{P}_\bullet^{(1)},\ldots,\mathscr{P}_\bullet^{(m)}) &\otimes_{\mathscr{O}_S} \mathscr{F}'.\end{aligned} \tag{6.9.6.2}$$

More particularly, for $\mathscr{F}' = \mathscr{O}_{S'}$*,*

(6.9.6.3).

$$\begin{aligned}\mathfrak{Tor}_n^{S'}(f'_1,\ldots,f'_m;\mathscr{P}'^{(1)}_\bullet,\ldots,\mathscr{P}'^{(m)}_\bullet) &\xrightarrow{\sim} \\ \mathfrak{Tor}_n^S(f_1,\ldots,f_m;\mathscr{P}_\bullet^{(1)},\ldots,\mathscr{P}_\bullet^{(m)}) &\otimes_{\mathscr{O}_S} \mathscr{O}_{S'}.\end{aligned} \tag{6.9.6.3}$$

Proof. The flatness of the terms of $\mathscr{P}_\bullet^{(i)}$ implies that

$$\mathscr{T}\!or_{q_i}^S(\mathscr{P}_\bullet^{(i)},\mathscr{O}_{S'}) = 0 \qquad (q_i \neq 0)$$

by (6.5.8). Thus the sequence (f') degenerates. Hypothesis 2° also makes it biregular ((6.9.3)), so the edge homomorphism

$$\begin{aligned}\mathfrak{Tor}_n^S(f_1,\ldots,f_m,1_{S'};\mathscr{P}_\bullet^{(1)},\ldots,\mathscr{P}_\bullet^{(m)},\mathscr{Q}'_\bullet) &\longrightarrow {}^{(f')}\mathscr{E}^2_{n0} \\ = \mathfrak{Tor}_n^{S'}(f'_1,\ldots,f'_m,1_{S'};\mathscr{P}'^{(1)}_\bullet,\ldots,\mathscr{P}'^{(m)}_\bullet,\mathscr{Q}'_\bullet)\end{aligned} \tag{6.9.6.4}$$

is bijective by (**0**, 11.1.6).

The flatness of the modules

$$\mathfrak{Tor}_n^S(f_1,\ldots,f_m;\mathscr{P}_\bullet^{(1)},\ldots,\mathscr{P}_\bullet^{(m)})$$

implies, by (6.5.8), that ${}^{(e)}\mathscr{E}^2_{pq} = 0$ for $p \neq 0$. Hence (e) also degenerates, and its biregularity makes the edge homomorphism

$$\begin{aligned}\mathfrak{Tor}_n^S(f_1,\ldots,f_m,1_{S'};\mathscr{P}_\bullet^{(1)},\ldots,\mathscr{P}_\bullet^{(m)},\mathscr{Q}'_\bullet) &\longrightarrow {}^{(e)}\mathscr{E}^2_{0n} \\ = \bigoplus_{n'+n''=n} \mathfrak{Tor}_{n'}^S(f_1,\ldots,f_m;\mathscr{P}_\bullet^{(1)},\ldots,\mathscr{P}_\bullet^{(m)}) &\otimes_{\mathscr{O}_S} \mathscr{H}_{n''}(\mathscr{Q}'_\bullet)\end{aligned} \tag{6.9.6.5}$$

bijective. Combining these two edge isomorphisms gives (6.9.6.1). Formula (6.9.6.2) follows because $\mathscr{H}_q(\mathscr{Q}'_\bullet) = 0$ for $q \neq 0$ and $\mathscr{H}_0(\mathscr{Q}'_\bullet) = \mathscr{F}'$. Finally, taking $\mathscr{F}' = \mathscr{O}_{S'}$ in (6.9.6.2) and using (6.7.12) gives (6.9.6.3). □

Corollary (6.9.7). — *Under the hypotheses of* (6.9.1)*, suppose that S and S′ are affine, and choose an integer* d_i *for each* $1 \leqslant i \leqslant m$*. There is an integer N, depending only on S, the* $X^{(i)}$*, and the* d_i*, with the following property. For every integer* n_0*, the canonical isomorphisms* (6.9.6.3) *hold for* $n \leqslant n_0$ *and for every system of complexes* $\mathscr{P}_\bullet^{(i)}$ *satisfying:*

1° $\mathscr{P}_k^{(i)} = 0$ *for* $k < d_i$*;*

2° $\mathscr{P}_k^{(i)}$ *is S-flat for* $k < n_0 + N$;

3°

$$\mathfrak{Tor}_q^S(f_1,\ldots,f_m;\mathscr{P}_\bullet^{(1)},\ldots,\mathscr{P}_\bullet^{(m)})$$

is S-flat for $q < n_0 + N$.

Proof. Suppose that $\mathscr{P}_k^{(i)}$ is S-flat for $k < r$. Then

$$\mathscr{T}\!or_{q_i}^S(\mathscr{P}_k^{(i)},\mathscr{O}_{S'}) = 0 \qquad (k < r,\ q_i \neq 0).$$

Consider

(6.9.7.1)
$$\mathfrak{Tor}_p^{S'}\Big(f_1',\ldots,f_m';\mathscr{T}\!or_{q_1}^S(\mathscr{P}_\bullet^{(1)},\mathscr{O}_{S'}),\ldots,\mathscr{T}\!or_{q_m}^S(\mathscr{P}_\bullet^{(m)},\mathscr{O}_{S'})\Big).$$

Compute (6.9.7.1) by the method of (6.6.2), using the inverse images of fixed finite affine open coverings $\mathfrak{U}^{(i)}$ of $X^{(i)}$. These coverings are independent of S' and of the complexes $\mathscr{P}_\bullet^{(i)}$. The terms $L_{jk}^{(i)}$ vanish for $j < -N_i$, where N_i depends only on $\mathfrak{U}^{(i)}$. **III | 170**

If some q_i is nonzero, the simple complex whose homology in degree p is (6.9.7.1) has zero terms in every degree

$$< r + \sum_{j\neq i} d_j - \sum_{\ell=1}^m N_\ell.$$

Consequently (6.9.7.1) vanishes for $p < r - N$, where

$$N = \max_i \left(\sum_{\ell=1}^m N_\ell - \sum_{j\neq i} d_j\right).$$

It follows that ${}^{(f')}\mathscr{E}_{pq}^2 = 0$ for $q \neq 0$ and $p < r - N$. Since also ${}^{(f')}\mathscr{E}_{pq}^2 = 0$ for $q < 0$, the edge homomorphism (6.9.6.4) is bijective for $n < r - N$ (M, XV, 5.6), when $\mathscr{Q}_\bullet' = \mathscr{O}_{S'}$.

Secondly, if

$$\mathfrak{Tor}_q^S(f_1,\ldots,f_m;\mathscr{P}_\bullet^{(1)},\ldots,\mathscr{P}_\bullet^{(m)})$$

is S-flat for $q < r$, then ${}^{(e)}\mathscr{E}_{pq}^2 = 0$ for $p \neq 0$ and $q < r$; moreover, ${}^{(e)}\mathscr{E}_{pq}^2 = 0$ for $p < 0$. Thus the edge homomorphism (6.9.6.5) is bijective for $n < r$, which completes the proof. □

The case of (6.9.3) most important in applications is $m = 1$, with $\mathscr{Q}_\bullet'$ reduced to a single term $\mathscr{F}'$ in degree 0. We state it again for later reference.[3]

Proposition (6.9.8). — *Let S be a prescheme, let $g : S' \to S$ be a morphism, and let $f : X \to Y$ be a separated and quasi-compact S-morphism of S-preschemes. Let $\mathscr{P}_\bullet$ be a bounded-below complex of quasi-coherent $\mathscr{O}_X$-modules, and let $\mathscr{F}'$ be a quasi-coherent $\mathscr{O}_{S'}$-module. There are two biregular spectral functors in $\mathscr{P}_\bullet$ and $\mathscr{F}'$, with values in the category of quasi-coherent $\mathscr{O}_{Y_{(S')}}$-modules, having the same abutment*

$$\mathfrak{Tor}_\bullet^S(f,1_{S'};\mathscr{P}_\bullet,\mathscr{F}'),$$

and whose E_2-terms are

(6.9.8.1).

(6.9.8.1)
$$'\mathscr{E}_{pq}^2 = \mathscr{T}\!or_p^S(\mathscr{H}^{-q}(f,\mathscr{P}_\bullet),\mathscr{F}')$$

and

(6.9.8.2).

(6.9.8.2)
$$''\mathscr{E}_{pq}^2 = \mathscr{H}^{-p}(f',\mathscr{T}\!or_q^S(\mathscr{P}_\bullet,\mathscr{F}')),$$

where

$$f' = f_{(S')} : X_{(S')} \longrightarrow Y_{(S')}.$$

[3] The case treated in (6.9.8), and in particular the spectral sequences (6.9.8.3) and (6.9.8.4), was pointed out to us by J.-P. Serre in 1957.

Proof. These spectral sequences may also be obtained, without starting from (6.9.3), from the sequences (a) and (b') of (6.7.3), by taking

$$X^{(1)} = X, \quad Y^{(1)} = Y, \quad X^{(2)} = Y^{(2)} = S', \quad f_1 = f, \quad f_2 = 1_{S'}.$$

When $S = S' = Y$ and Y is affine, one obtains two spectral sequences with E_2-terms

(6.9.8.3) $$'\mathcal{E}^2_{pq} = \mathcal{T}or^Y_p(\mathcal{H}^{-q}(f, \mathcal{P}_\bullet), \mathcal{F})$$

and

(6.9.8.4) $$''\mathcal{E}^2_{pq} = \mathcal{H}^{-p}(f, \mathcal{T}or^Y_q(\mathcal{P}_\bullet, \mathcal{F})).$$

By (6.7.6), both abut to the hypercohomology

$$\mathcal{H}^\bullet(f, \mathcal{P}_\bullet \otimes_Y \mathcal{F})$$

of f_* with respect to the complex $\mathcal{P}_\bullet \otimes_Y \mathcal{F}$ of $\mathcal{O}_X$-modules, for every quasi-coherent Y-flat $\mathcal{O}_Y$-module $\mathcal{F}$; if the terms of $\mathcal{P}_\bullet$ are Y-flat, $\mathcal{F}$ may be any quasi-coherent $\mathcal{O}_Y$-module. These spectral sequences are distinct from those of (6.2.1). □

Corollary (6.9.9). — *Under the hypotheses of (6.9.8), suppose that $\mathcal{P}_\bullet$ is bounded below, consists of S-flat modules, and that the $\mathcal{O}_Y$-modules $\mathcal{H}^n(f, \mathcal{P}_\bullet)$ are S-flat.*

Put **III | 171**

$$\mathcal{P}'_\bullet = \mathcal{P}_\bullet \otimes_{\mathcal{O}_S} \mathcal{O}_{S'}.$$

There are canonical functorial isomorphisms

(6.9.9.1).

(6.9.9.1) $$\mathfrak{Tor}^{S'}_n(f', 1_{S'}; \mathcal{P}'_\bullet, \mathcal{F}') \xrightarrow{\sim} \mathcal{H}^{-n}(f, \mathcal{P}_\bullet) \otimes_{\mathcal{O}_S} \mathcal{F}'.$$

In particular, for $\mathcal{F}' = \mathcal{O}_{S'}$,

(6.9.9.2).

(6.9.9.2) $$\mathcal{H}^n(f', \mathcal{P}'_\bullet) \xrightarrow{\sim} \mathcal{H}^n(f, \mathcal{P}_\bullet) \otimes_{\mathcal{O}_S} \mathcal{O}_{S'}.$$

This is the case $m = 1$ of (6.9.6).

Corollary (6.9.10). — *Let S be a prescheme, let $f : X \to Y$ be a separated and quasi-compact S-morphism of S-preschemes, and let $\mathcal{P}_\bullet$ be a bounded-below complex of quasi-coherent $\mathcal{O}_X$-modules that are flat over S. Suppose further that the $\mathcal{O}_Y$-modules $\mathcal{H}^n(f, \mathcal{P}_\bullet)$ are flat over S.*

For $s \in S$, write X_s and Y_s for the fibres

$$X \times_S \operatorname{Spec}(k(s)), \qquad Y \times_S \operatorname{Spec}(k(s)),$$

write

$$f_s : X_s \longrightarrow Y_s$$

for the morphism $f \times_S 1$, and put

$$\mathcal{P}^s_\bullet = \mathcal{P}_\bullet \otimes_{\mathcal{O}_S} k(s).$$

Then there are canonical functorial isomorphisms

(6.9.10.1).

(6.9.10.1) $$\mathcal{H}^n(f_s, \mathcal{P}^s_\bullet) \xrightarrow{\sim} \mathcal{H}^n(f, \mathcal{P}_\bullet) \otimes_{\mathcal{O}_S} k(s).$$

Thus, under suitable flatness hypotheses, the formation of the derived functors $R^n f_*(\mathcal{F})$ "commutes with passage to the fibres." We shall obtain this result again by another method in §7.

6.10. Local structure of certain cohomological functors III | 171

Proposition (6.10.1). — *Let $S = \operatorname{Spec}(A)$ be an affine scheme, and let $Y^{(i)}$ $(1 \leqslant i \leqslant n)$ be a finite family of affine S-schemes, flat over S. For each i, let*

$$f_i : X^{(i)} \longrightarrow Y^{(i)}$$

be a separated and quasi-compact S-morphism, and let $\mathscr{P}_\bullet^{(i)}$ be a bounded-below complex of quasi-coherent $\mathscr{O}_{X^{(i)}}$-modules. Let Y be the product over S of the schemes $Y^{(i)}$.

There exists a complex $\mathscr{R}_\bullet$ of quasi-coherent $\mathscr{O}_Y$-modules, flat over S, with the following property. For every affine S-scheme S' and every bounded-below complex $\mathscr{Q}'_\bullet$ of quasi-coherent $\mathscr{O}_{S'}$-modules, there is an isomorphism

(6.10.1.1).

$$\mathbf{Tor}_\bullet^S(f_1,\ldots,f_n,1_{S'};\mathscr{P}_\bullet^{(1)},\ldots,\mathscr{P}_\bullet^{(n)},\mathscr{Q}'_\bullet) \xrightarrow{\sim} \mathscr{H}_\bullet(\mathscr{R}_\bullet \otimes_{\mathscr{O}_S} \mathscr{Q}'_\bullet), \tag{6.10.1.1}$$

which is an isomorphism of ∂-functors in $\mathscr{Q}'_\bullet$.

Moreover, for every S-morphism $u : S'' \to S'$ of affine S-schemes, the diagram

(6.10.1.2).

$$\begin{array}{ccc}
\begin{array}{c}\mathbf{Tor}_\bullet^S(f_1,\ldots,f_n,1_{S'};\\ \mathscr{P}_\bullet^{(1)},\ldots,\mathscr{P}_\bullet^{(n)},\mathscr{Q}'_\bullet)\end{array} & \xrightarrow{\sim} & \mathscr{H}_\bullet(\mathscr{R}_\bullet \otimes_{\mathscr{O}_S} \mathscr{Q}'_\bullet) \\
\downarrow & & \downarrow \\
\begin{array}{c}\mathbf{Tor}_\bullet^S(f_1,\ldots,f_n,1_{S''};\\ \mathscr{P}_\bullet^{(1)},\ldots,\mathscr{P}_\bullet^{(n)},u^*(\mathscr{Q}'_\bullet))\end{array} & \xrightarrow{\sim} & \mathscr{H}_\bullet(\mathscr{R}_\bullet \otimes_{\mathscr{O}_S} u^*(\mathscr{Q}'_\bullet))
\end{array} \tag{6.10.1.2}$$

commutes. Here the vertical arrows are the canonical $(1_Y \times_S u)$-morphisms defined in (6.7.10).

Proof. Compute the hypertor by the method of (6.6.2), taking account of Remark (6.6.6). With the III | 172
notation used there, each $L_{\bullet\bullet}^{(i)}$ is a bicomplex of A_i-modules, where A_i denotes the ring of $Y^{(i)}$. It therefore has a projective Cartan–Eilenberg resolution $M_{\bullet\bullet\bullet}^{(i)}$ (in the sense of (**0**, 11.7.1)) by projective A_i-modules. By (6.6.6), the left side of (6.10.1.1) is canonically isomorphic to

$$\mathrm{H}_\bullet(M_{\bullet\bullet\bullet} \otimes_A Q'_\bullet),$$

where

$$M_{\bullet\bullet\bullet} = M_{\bullet\bullet\bullet}^{(1)} \otimes_A M_{\bullet\bullet\bullet}^{(2)} \otimes_A \cdots \otimes_A M_{\bullet\bullet\bullet}^{(n)}$$

and $\mathscr{Q}'_\bullet = \widetilde{Q'_\bullet}$.

By hypothesis, the rings A_i are flat A-modules. Hence the $M_{\bullet\bullet\bullet}^{(i)}$ are tricomplexes of flat A-modules (**0**, 6.2.1), and so is $M_{\bullet\bullet\bullet}$. Furthermore, if B is the ring of Y, that is, the tensor product over A of the A_i, then $M_{\bullet\bullet\bullet}$ is a tricomplex of B-modules. Consequently the complex

$$\mathscr{R}_\bullet = \widetilde{M_{\bullet\bullet\bullet}},$$

where $M_{\bullet\bullet\bullet}$ is regarded as a simple complex, has the required properties, as follows immediately from (6.7.10). □

Corollary (6.10.2). — *In Proposition (6.10.1), the complex $\mathscr{R}_\bullet$ may be taken to be bounded below. If the $\mathscr{P}_\bullet^{(i)}$ are bounded above and the $Y^{(i)}$ have finite cohomological dimension, then $\mathscr{R}_\bullet$ may be taken to be bounded above.*

Proof. The first assertion follows because all three degrees of each $M_{\bullet\bullet\bullet}^{(i)}$ are bounded below. On the other hand, if the rings A_i have finite cohomological dimension, the third degree of each $M_{\bullet\bullet\bullet}^{(i)}$ assumes only finitely many values, and by its construction the same is true of its first degree (6.6.2). Its second degree is bounded above precisely when the degree of $\mathscr{P}_\bullet^{(i)}$ is bounded above (6.6.2), which proves the second assertion. □

Remarks (6.10.3). — (i) With the notation of (6.10.1),

$$\mathscr{H}_\bullet(\mathscr{R}_\bullet \otimes_S \mathscr{Q}'_\bullet) \simeq \mathbf{Tor}_\bullet(\mathscr{R}_\bullet, \mathscr{Q}'_\bullet),$$

because $\mathscr{R}_\bullet$ consists of $\mathscr{O}_Y$-modules that are flat over S (6.5.9). It is therefore (6.5.4) the abutment of a regular spectral sequence whose E^2-term is

(6.10.3.1).

$$E^2_{pq} = \bigoplus_{q'+q''=q} \mathscr{T}or^S_p(\mathscr{H}_{q'}(\mathscr{R}_\bullet), \mathscr{H}_{q''}(\mathscr{Q}'_\bullet)). \tag{6.10.3.1}$$

This is precisely the spectral sequence (e) for base change in (6.9.3).

(ii) Let $\mathscr{R}'_\bullet$ be another complex of quasi-coherent $\mathscr{O}_Y$-modules, flat over S, and let

$$g : \mathscr{R}'_\bullet \longrightarrow \mathscr{R}_\bullet$$

be a homomorphism of complexes for which

$$\mathscr{H}_\bullet(g) : \mathscr{H}_\bullet(\mathscr{R}'_\bullet) \longrightarrow \mathscr{H}_\bullet(\mathscr{R}_\bullet)$$

is an isomorphism. By (6.3.3) and (6.5.9), g induces an isomorphism of ∂-functors in $\mathscr{Q}'_\bullet$,

$$\mathscr{H}_\bullet(\mathscr{R}'_\bullet \otimes_S \mathscr{Q}'_\bullet) \xrightarrow{\sim} \mathscr{H}_\bullet(\mathscr{R}_\bullet \otimes_S \mathscr{Q}'_\bullet),$$

such that the diagram

$$\begin{tikzcd}
\mathscr{H}_\bullet(\mathscr{R}'_\bullet \otimes_S \mathscr{Q}'_\bullet) \arrow[r, "\sim"] \arrow[d] & \mathscr{H}_\bullet(\mathscr{R}_\bullet \otimes_S \mathscr{Q}'_\bullet) \arrow[d] \\
\mathscr{H}_\bullet(\mathscr{R}'_\bullet \otimes_S u^*(\mathscr{Q}'_\bullet)) \arrow[r, "\sim"] & \mathscr{H}_\bullet(\mathscr{R}_\bullet \otimes_S u^*(\mathscr{Q}'_\bullet))
\end{tikzcd}$$

commutes. Thus the properties in (6.10.1) do not determine the complex $\mathscr{R}_\bullet$ completely.

(iii) In the proof of (6.10.1), the $M^{(i)}_{\bullet\bullet\bullet}$ may be chosen to consist of free A_i-modules, as follows immediately by dualising the proof of (**0**, 11.5.2.1). The **III | 173**

$$M^{(i)}_{\bullet\bullet\bullet} \otimes_{A_i} B$$

then consist of free B-modules; since $M_{\bullet\bullet\bullet}$ is their tensor product over B, one may in addition require in (6.10.1) that $\mathscr{R}_\bullet$ be associated with a complex of free B-modules.

Moreover, by (M, XVII, 1.2), the tricomplex $M_{\bullet\bullet\bullet}$ depends functorially on each of the bicomplexes $L^{(i)}_{\bullet\bullet}$, hence on each $\mathscr{P}^{(i)}_\bullet$ once a finite covering of each $X^{(i)}$ has been fixed. Here "morphisms" of tricomplexes are understood as homomorphism classes modulo homotopy. Replacing a covering of $X^{(i)}$ by a finer one gives, for $L^{(i)}_{\bullet\bullet}$, homomorphisms defined up to homotopy by (6.6.8); thus, with the preceding convention on morphisms, $M_{\bullet\bullet\bullet}$ is a functor in each $\mathscr{P}^{(i)}_\bullet$.

We shall make this functorial dependence more precise, in particular the behaviour of $\mathscr{R}_\bullet$ with respect to short exact sequences

$$0 \longrightarrow \mathscr{P}'^{(i)}_\bullet \longrightarrow \mathscr{P}^{(i)}_\bullet \longrightarrow \mathscr{P}''^{(i)}_\bullet \longrightarrow 0,$$

in the chapter devoted to a general algebra of cohomological functors mentioned in (6.1.3).

(6.10.4). *Scholium* ***(6.10.4).*** **—** The fact that $\mathscr{R}_\bullet$ consists of $\mathscr{O}_Y$-modules flat over S immediately implies that

$$\mathscr{H}_\bullet(\mathscr{R}_\bullet \otimes_S \mathscr{Q}'_\bullet)$$

is a homological functor in $\mathscr{Q}'_\bullet$; compare the argument of (7.7.1). As noted in (6.1.1), this property motivates the introduction of hypertor.

Put

$$X = X^{(1)} \times_S X^{(2)} \times_S \cdots \times_S X^{(n)}, \qquad f = f_1 \times_S f_2 \times_S \cdots \times_S f_n,$$
$$\mathscr{P}_\bullet = \mathscr{P}^{(1)}_\bullet \otimes_S \mathscr{P}^{(2)}_\bullet \otimes_S \cdots \otimes_S \mathscr{P}^{(n)}_\bullet,$$

and

$$X' = X \times_S S', \qquad Y' = Y \times_S S', \qquad f' = f \times_S 1_{S'}.$$

Problems of base change lead one to study the hypercohomology

$$\mathscr{H}^*_{f'}(\mathscr{P}_\bullet \otimes_S \mathscr{N}')$$

as a functor of the quasi-coherent $\mathscr{O}_{S'}$-module $\mathscr{N}'$, or equivalently the hypercohomology

$$\mathscr{H}^*_{f}(\mathscr{P}_\bullet \otimes_S \mathscr{N})$$

as a functor of the quasi-coherent $\mathscr{O}_S$-module $\mathscr{N}$. When the $\mathscr{P}^{(i)}_\bullet$, and therefore $\mathscr{P}_\bullet$, are flat over S, the preceding results and (6.7.6) show that this is indeed a cohomological functor in $\mathscr{N}$. This is no longer true when the flatness of the $\mathscr{P}^{(i)}_\bullet$ is not assumed, and the study of

$$\mathscr{H}^*_{f}(\mathscr{P}_\bullet \otimes_S \mathscr{N})$$

can then no longer be approached by the usual methods of homological algebra.

We shall chiefly use the case $n = 1$, $Y = S$, in which $\mathscr{P}_\bullet$ consists of $\mathscr{O}_X$-modules flat over Y. In this case we have the following theorem.

Theorem (6.10.5). — *Let $Y = \operatorname{Spec}(A)$ be an affine Noetherian scheme, let $f : X \to Y$ be a proper morphism, and let $\mathscr{P}_\bullet$ be a bounded-below complex of coherent $\mathscr{O}_X$-modules flat over Y. There exists a bounded-below complex $\mathscr{L}_\bullet$ of $\mathscr{O}_Y$-modules, whose terms have the form*

$$\mathscr{L}_i = \mathscr{O}_Y^{m_i},$$

and an isomorphism

(6.10.5.1).

$$\mathscr{H}^*(f, \mathscr{P}_\bullet \otimes_Y \mathscr{Q}_\bullet) \xrightarrow{\sim} \mathscr{H}_\bullet(\mathscr{L}_\bullet \otimes_Y \mathscr{Q}_\bullet) \tag{6.10.5.1}$$

of ∂-functors in the bounded-below complex $\mathscr{Q}_\bullet$ of quasi-coherent $\mathscr{O}_Y$-modules.

Moreover, for every morphism $u : Y' \to Y$, put

$$X' = X_{(Y')}, \qquad f' = f_{(Y')}, \qquad \mathscr{P}'_\bullet = \mathscr{P}_\bullet \otimes_Y \mathscr{O}_{Y'}, \qquad \mathscr{L}'_\bullet = u^*(\mathscr{L}_\bullet).$$

Then $\mathscr{L}'_\bullet$ is a complex of locally free $\mathscr{O}_{Y'}$-modules of finite type, and there is an isomorphism **III | 174**

(6.10.5.2).

$$\mathscr{H}^*(f', \mathscr{P}'_\bullet \otimes_{Y'} \mathscr{Q}'_\bullet) \xrightarrow{\sim} \mathscr{H}_\bullet(\mathscr{L}'_\bullet \otimes_{Y'} \mathscr{Q}'_\bullet) \tag{6.10.5.2}$$

of ∂-functors in the bounded-below complex $\mathscr{Q}'_\bullet$ of quasi-coherent $\mathscr{O}_{Y'}$-modules. These isomorphisms may be chosen so that the diagram

(6.10.5.3).

$$\begin{array}{ccc} \mathscr{H}^*(f, \mathscr{P}_\bullet \otimes_Y \mathscr{Q}_\bullet) & \xrightarrow{\sim} & \mathscr{H}_\bullet(\mathscr{L}_\bullet \otimes_Y \mathscr{Q}_\bullet) \\ \downarrow & & \downarrow \\ \mathscr{H}^*(f', \mathscr{P}'_\bullet \otimes_{Y'} u^*(\mathscr{Q}_\bullet)) & \xrightarrow{\sim} & \mathscr{H}_\bullet(\mathscr{L}'_\bullet \otimes_{Y'} u^*(\mathscr{Q}_\bullet)) \end{array} \tag{6.10.5.3}$$

commutes.

Proof. Proposition (6.10.1) first gives a bounded-below complex $\mathscr{R}_\bullet$ (6.10.2) of quasi-coherent $\mathscr{O}_Y$-modules that are free over Y (6.10.3) (iii), together with an isomorphism

(6.10.5.4).

$$\mathscr{H}^*(f, \mathscr{P}_\bullet \otimes_Y \mathscr{Q}_\bullet) \xrightarrow{\sim} \mathscr{H}_\bullet(\mathscr{R}_\bullet \otimes_Y \mathscr{Q}_\bullet) \tag{6.10.5.4}$$

of ∂-functors in $\mathscr{Q}_\bullet$. The terms of $\mathscr{R}_\bullet$ need not, *a priori*, be $\mathscr{O}_Y$-modules of finite type.

Apply (6.10.5.4) when $\mathscr{Q}_\bullet$ is the complex concentrated in one degree with sole term $\mathscr{O}_Y$. It follows that $\mathscr{H}_\bullet(\mathscr{R}_\bullet)$ is isomorphic to $\mathscr{H}^*(f, \mathscr{P}_\bullet)$ and hence consists of coherent $\mathscr{O}_Y$-modules (6.2.5). By (**0**, 11.9.2), there is therefore a bounded-below complex $\mathscr{L}_\bullet$ whose terms are associated with free A-modules of finite type, and a homomorphism

$$\mathscr{L}_\bullet \longrightarrow \mathscr{R}_\bullet$$

which induces an isomorphism

$$\mathscr{H}_\bullet(\mathscr{L}_\bullet) \xrightarrow{\sim} \mathscr{H}_\bullet(\mathscr{R}_\bullet).$$

Formula (6.10.5.1) now follows from (6.10.3) (ii).

The remaining assertions of (6.10.5) follow from (6.10.1) and (6.10.3) (ii) when Y' is affine. In the general case, cover Y' by affine open subsets (V_α). If (6.10.5.2) is formed over each V_α, its restrictions from V_α and V_β to an affine open subset

$$W \subset V_\alpha \cap V_\beta$$

both coincide with the isomorphism formed over W. This follows from the commutativity of (6.10.1.2), applied to the canonical inclusions $W \to V_\alpha$ and $W \to V_\beta$, and proves that the local isomorphisms glue. ◻

Remark (6.10.6). — In the following chapters we shall apply (6.10.5) chiefly when $\mathscr{P}_\bullet$ is concentrated in one degree and its sole term is a coherent $\mathscr{O}_X$-module $\mathscr{F}$ flat over Y. Since

$$\mathscr{H}_n(\mathscr{L}_\bullet) = \mathscr{H}^{-n}(f, \mathscr{F}) = R^{-n}f_*(\mathscr{F}) \qquad (6.2.1),$$

the $\mathscr{H}_n(\mathscr{L}_\bullet)$ vanish for $n > 0$. We shall see later (7.7.12) (i) that $\mathscr{L}_\bullet$ may then be chosen to have terms only in degrees $\leqslant 0$, and hence only finitely many terms, provided the requirement that its terms be associated with free A-modules of finite type is replaced by the requirement that they be locally free of finite type.

The complex $\mathscr{L}_\bullet$ corresponding to such an $\mathscr{O}_X$-module $\mathscr{F}$ seems to have no special property apart from the preceding restriction on its degrees. One may therefore ask conversely whether, given a bounded-below complex $\mathscr{L}_\bullet$ whose terms are $\mathscr{O}_Y$-modules associated with projective A-modules of finite type and whose terms of degree > 0 vanish, there exist a Y-scheme X, projective and flat over Y, and a locally free $\mathscr{O}_X$-module $\mathscr{F}$ such that there is an isomorphism III | 175

$$\mathscr{H}^*(f, \mathscr{F} \otimes_Y \mathscr{Q}_\bullet) \xrightarrow{\sim} \mathscr{H}_\bullet(\mathscr{L}_\bullet \otimes_Y \mathscr{Q}_\bullet)$$

functorial in $\mathscr{Q}_\bullet$. Such a result would reduce the cohomological theory of coherent modules flat over Y on proper Y-schemes entirely to the theory, "up to homotopy," of complexes of projective A-modules of finite type over a Noetherian ring A.

§7. Base change for homological functors of sheaves of modules

7.1. Functors of A-modules

(7.1.1). Given a ring A, not necessarily commutative, denote by $\mathtt{Ab}_A$ the category of left A-modules, and simply by $\mathtt{Ab}$ the category of $\mathbf{Z}$-modules, which is the same as the category of commutative groups. Let III | 175

$$T : \mathtt{Ab}_A \longrightarrow \mathtt{Ab}$$

be an additive covariant functor, and let M be an (A, A)-bimodule. Then $T(M)$ is naturally endowed with the structure of a right A-module.

Indeed, for every $a \in A$, let $h_{a,M}$, or simply h_a, denote the endomorphism

$$x \longmapsto xa$$

of the left A-module M. By hypothesis, $T(h_a)$ is an endomorphism of the $\mathbf{Z}$-module $T(M)$. Since T is covariant and additive, for $a, b \in A$ one has

$$T(h_{ab}) = T(h_b \circ h_a) = T(h_b) \circ T(h_a)$$

and

$$T(h_{a+b}) = T(h_a + h_b) = T(h_a) + T(h_b).$$

Thus

$$(y, a) \longmapsto T(h_a)(y)$$

defines a right A-module scalar multiplication on $T(M)$. In particular, $T(A_s)$ is a right A-module.

(7.1.2). Suppose that A is commutative. It follows from (7.1.1) that, for every A-module M, the group $T(M)$ is naturally an A-module. Moreover, if $u : M \to N$ is an A-linear homomorphism, then, for every $a \in A$,

$$u \circ h_{a,M} = h_{a,N} \circ u,$$

and hence

$$T(u) \circ T(h_{a,M}) = T(h_{a,N}) \circ T(u).$$

Consequently $T(u) : T(M) \to T(N)$ is A-linear, and T may be regarded as an additive covariant functor from $\mathtt{Ab}_A$ to itself.

More precisely, this defines a canonical equivalence between the category of additive covariant functors

$$\mathtt{Ab}_A \longrightarrow \mathtt{Ab}$$

and the category of A-linear covariant functors

$$T : \mathtt{Ab}_A \longrightarrow \mathtt{Ab}_A,$$

that is, functors satisfying

$$T(h_{a,M}) = h_{a,T(M)} \qquad (a \in A).$$

The forgetful functor

$$I : \mathtt{Ab}_A \longrightarrow \mathtt{Ab},$$

which sends an A-module to its underlying $\mathbf{Z}$-module, is exact and faithful. The two functors corresponding under the preceding equivalence therefore have the same exactness properties.

(7.1.3). Continue to suppose that A is commutative. Let B be an A-algebra, not necessarily commutative, and let **III | 176**

$$\rho : A \longrightarrow B$$

be the ring homomorphism defining the algebra structure. It defines an additive covariant functor

$$\rho_* : M \longmapsto M_{[\rho]}$$

from the category $\mathtt{Ab}_B$ of left B-modules to $\mathtt{Ab}_A$. By composition one obtains the additive covariant functor

$$T_{(B)} : \mathtt{Ab}_B \xrightarrow{\rho_*} \mathtt{Ab}_A \xrightarrow{T} \mathtt{Ab}.$$

For typographical reasons this functor will also be denoted $T^{(B)}$ or $T \otimes_A B$, and is said to be obtained from T by extension of scalars from A to B.

If B is commutative, $T_{(B)}$ may of course be regarded as a functor from $\mathtt{Ab}_B$ to itself (7.1.2). If B is commutative and C is a B-algebra, then

$$T_{(C)} = (T_{(B)})_{(C)}.$$

Extension of scalars is plainly functorial and additive in T. Furthermore, if T commutes with direct limits or direct sums, then so does $T_{(B)}$; and if T is left exact, right exact, or exact, respectively, the same is true of $T_{(B)}$. Indeed, ρ_* is exact and commutes with direct limits and direct sums.

(7.1.4). Suppose again that A is commutative, and let

$$T : \mathtt{Ab}_A \longrightarrow \mathtt{Ab}_A$$

be an additive covariant A-linear functor that commutes with direct limits. For every multiplicative subset S of A and every A-module M, there is a canonical functorial isomorphism of A-modules

(7.1.4.1).

$$T(S^{-1}M) \xrightarrow{\sim} S^{-1}T(M). \tag{7.1.4.1}$$

First suppose that S is the set of powers f^n, $n \geqslant 0$, of an element $f \in A$. Then

$$M_f = \varinjlim M_n,$$

where (M_n, ϕ_{nm}) is the direct system of A-modules with $M_n = M$ and

$$\phi_{nm} : z \longmapsto f^{n-m}z \qquad (\mathbf{0}, 1.6.1).$$

Formula (7.1.4.1) follows in this case from the hypothesis on T. For arbitrary S, the module $S^{-1}M$ is the direct limit of the M_f, for $f \in S$ (**0**, 1.4.5), and the same argument applies.

The functoriality of (7.1.4.1) also shows that it is an isomorphism of $S^{-1}A$-modules. Thus, up to canonical isomorphism, one may write

(7.1.4.2).

$$T_{(S^{-1}A)}(S^{-1}M) = S^{-1}T(M) = T(S^{-1}M). \tag{7.1.4.2}$$

If $S = A - \mathfrak{p}$ is the complement of a prime ideal $\mathfrak{p}$ of A, write $T_{\mathfrak{p}}$ instead of $T_{(A_{\mathfrak{p}})}$.

Proposition (7.1.5). — *Under the hypotheses of (7.1.4), suppose that $T_{\mathfrak{m}}$ is left exact, respectively right exact or exact, for every maximal ideal $\mathfrak{m}$ of A. Then T is left exact, respectively right exact or exact.*

Proof. If two submodules N and P of an A-module M satisfy

$$N_{\mathfrak{m}} = P_{\mathfrak{m}}$$

for every maximal ideal $\mathfrak{m}$ of A, then $N = P$ (Bourbaki, *Algèbre commutative*, chap. II, § 3, no. 3, th. 1). Applying this fact to the kernels and images in question proves the assertion. □

7.2. Characterisation of the tensor product functor

(7.2.1). Let A be a ring, not necessarily commutative; let M (respectively N) be a left (respectively right) A-module, and let P be a **Z**-module. Recall that giving a **Z**-homomorphism

$$v : N \otimes_A M \longrightarrow P$$

is equivalent to giving a **Z**-bilinear map

$$u : N \times M \longrightarrow P$$

such that

$$u(ta, x) = u(t, ax) \qquad (a \in A,\ t \in N,\ x \in M),$$

the two maps being related by

$$v(t \otimes x) = u(t, x).$$

On the other hand, giving u is equivalent to giving a **Z**-homomorphism

$$x \longmapsto f_x$$

from M to $\mathrm{Hom}_{\mathbf{Z}}(N, P)$ such that

$$f_{ax}(t) = f_x(ta) \qquad (a \in A,\ t \in N,\ x \in M),$$

the two maps being related by $u(t, x) = f_x(t)$.

(7.2.2). Let

$$T : \mathtt{Ab}_A \longrightarrow \mathtt{Ab}$$

be an additive covariant functor. For every left A-module M we shall define a canonical homomorphism of **Z**-modules, functorial in M,

(7.2.2.1).

$$t_M : T(A_s) \otimes_A M \longrightarrow T(M). \tag{7.2.2.1}$$

By (7.2.1), it is enough to define a **Z**-homomorphism

$$x \longmapsto t'_M(x)$$

from M to $\mathrm{Hom}_{\mathbf{Z}}(T(A_s), T(M))$ such that

$$t'_M(ax)(y) = t'_M(x)(ya) \qquad (a \in A,\ x \in M,\ y \in T(A_s)).$$

For this purpose, note that $\mathrm{Hom}_{\mathbf{Z}}(T(A_s), T(M))$ is canonically a left A-module, by virtue of the right A-module structure on $T(A_s)$, with scalar multiplication defined by

$$(a \cdot w)(y) = w(ya)$$

for $a \in A$, $w \in \mathrm{Hom}_{\mathbf{Z}}(T(A_s), T(M))$, and $y \in T(A_s)$. Define t'_M to be the composite of the two canonical homomorphisms

$$M \xrightarrow{\sim} \mathrm{Hom}_A(A_s, M) \xrightarrow{T} \mathrm{Hom}_{\mathbf{Z}}(T(A_s), T(M)).$$

The second arrow sends w to $T(w)$; the first is the canonical isomorphism of A-modules sending x to the map $\theta_x : A_s \to M$ defined by **III** | 177

$$\theta_x(\xi) = \xi x \qquad (\xi \in A,\ x \in M).$$

Since $\theta_{ax} = \theta_x \circ h_a$, one has

$$T(\theta_{ax}) = T(\theta_x) \circ T(h_a),$$

and consequently, for $y \in T(A_s)$,

$$T(\theta_{ax})(y) = T(\theta_x)\big(T(h_a)(y)\big) = T(\theta_x)(ya)$$

by the definition of the right A-module structure on $T(A_s)$. This proves the existence of t_M.

It is immediate that this homomorphism is functorial in M: for every homomorphism $w : M \to M'$ of left A-modules, the diagram

(7.2.2.2).

(7.2.2.2)
$$\begin{tikzcd} T(A_s) \otimes_A M \arrow[r, "t_M"] \arrow[d, "1\otimes w"'] & T(M) \arrow[d, "T(w)"] \\ T(A_s) \otimes_A M' \arrow[r, "t_{M'}"] & T(M') \end{tikzcd}$$

is commutative. The functoriality of (7.2.2.1) also shows that, when A is commutative, t_M is a homomorphism of A-modules (7.1.2).

(7.2.3). When A is commutative, one can more generally define a canonical homomorphism of A-modules

(7.2.3.1).

(7.2.3.1)
$$T(N) \otimes_A M \longrightarrow T(N \otimes_A M).$$

for every A-module N: in the construction of (7.2.2), replace θ_x by the homomorphism of A-modules

$$N \longrightarrow N \otimes_A M, \qquad y \longmapsto y \otimes x.$$

It is immediate that this homomorphism is functorial in both M and N.

In particular, if B is an A-algebra, not necessarily commutative, there is a homomorphism, functorial in M,

(7.2.3.2).

(7.2.3.2)
$$(T(M))_{(B)} = T(M) \otimes_A B \longrightarrow T(M \otimes_A B) = T_{(B)}(M_{(B)}).$$

By the functoriality of (7.2.3.1) in M, this is a homomorphism of B-modules.

There is moreover a commutative diagram

(7.2.3.3).

(7.2.3.3)
$$\begin{tikzcd} T(A) \otimes_A M \arrow[r, "t_M"] \arrow[d] & T(M) \arrow[d] \\ T_{(B)}(B_s) \otimes_B M_{(B)} \arrow[r, "t_{M_{(B)}}"] & T_{(B)}(M_{(B)}) \end{tikzcd}$$

The right vertical arrow is the composite

$$T(M) \longrightarrow T(M) \otimes_A B \longrightarrow T(M \otimes_A B) = T_{(B)}(M_{(B)})$$

of the canonical homomorphism and (7.2.3.2). The left vertical arrow in (7.2.3.3) is the homomorphism

$$T(A) \otimes_A M \longrightarrow T_{(B)}(B_s) \otimes_B (B \otimes_A M) = T_{(B)}(B_s) \otimes_A M,$$

where
$$T(A) \longrightarrow T_{(B)}(B_s) = T(B)$$
is $T(\rho)$, with ρ regarded as the homomorphism of A-modules $A \to B_{[\rho]}$.

Lemma (7.2.4). — *If T is an additive covariant functor from $\mathtt{Ab}_A$ to $\mathtt{Ab}$ that commutes with direct sums, then the canonical homomorphism t_L of (7.2.2.1) is an isomorphism for every free A-module L.*

Proof. Write **III | 178**
$$L = \bigoplus_{\alpha \in I} L_\alpha,$$
where every L_α is isomorphic to A_s. The definition of t_M in (7.2.2) shows that
$$t_L = \bigoplus_{\alpha \in I} t_{L_\alpha},$$
because
$$T : \mathrm{Hom}_A(A_s, L) \longrightarrow \mathrm{Hom}_{\mathbf{Z}}(T(A_s), T(L))$$
is the direct sum of the $\mathbf{Z}$-linear maps
$$T_\alpha : \mathrm{Hom}_A(A_s, L_\alpha) \longrightarrow \mathrm{Hom}_{\mathbf{Z}}(T(A_s), T(L_\alpha)),$$
by the hypothesis on T. It therefore remains only to prove the lemma for $L = A_s$; but then t_L is precisely the canonical isomorphism
$$T(A_s) \otimes_A A_s \xrightarrow{\sim} T(A_s),$$
valid for every right A-module. □

Proposition (7.2.5). — *Let T be an additive covariant functor from $\mathtt{Ab}_A$ to $\mathtt{Ab}$ that commutes with direct sums. The following conditions are equivalent:*

- *(a) T is right exact;*
- *(b) the canonical homomorphism t_M of (7.2.2.1) is an isomorphism for every left A-module M;*
- *(b′) T is semi-exact and t_M is surjective for every left A-module M;*
- *(c) T is isomorphic to a functor of M of the form $N \otimes_A M$, where N is a right A-module.*

Proof. It is clear that (b) implies (c), and that (c) implies (a). Let us prove that (a) implies (b). Put
$$T'(M) = T(A_s) \otimes_A M$$
for every left A-module M. There is an exact sequence
$$L' \longrightarrow L \longrightarrow M \longrightarrow 0,$$
where L and L' are free left A-modules. Since T and T' are right exact, there is a commutative diagram
$$\begin{tikzcd} T'(L') \arrow[r] \arrow[d, "t_{L'}"] & T'(L) \arrow[r] \arrow[d, "t_L"] & T'(M) \arrow[r] \arrow[d, "t_M"] & 0 \\ T(L') \arrow[r] & T(L) \arrow[r] & T(M) \arrow[r] & 0 \end{tikzcd}$$
whose rows are exact. By (7.2.4), t_L and $t_{L'}$ are isomorphisms; hence so is t_M, by the five lemma.

It is clear that (b) implies (b′). To show that (b′) implies (a), it is enough to prove the following lemma. □

Lemma (7.2.5.1). — *Let $\mathtt{K}$ and $\mathtt{K}'$ be abelian categories, let F and G be additive covariant functors from $\mathtt{K}$ to $\mathtt{K}'$, and let*
$$f : F \longrightarrow G$$
be a natural transformation (T, I, 1.2) *such that*
$$f_E : F(E) \longrightarrow G(E)$$
is an epimorphism for every object E of $\mathtt{K}$. If F is right exact and G is semi-exact, then G is right exact.

Proof. It is enough to show that, for every epimorphism $v : E' \to E$ in $\mathtt{K}$,

$$G(v) : G(E') \longrightarrow G(E)$$

is an epimorphism. There is a commutative diagram

$$\begin{tikzcd} F(E') \arrow[r, "F(v)"] \arrow[d, "f_{E'}"'] & F(E) \arrow[d, "f_E"] \\ G(E') \arrow[r, "G(v)"] & G(E) \end{tikzcd}$$

Here $F(v)$, $f_{E'}$, and f_E are epimorphisms; consequently $G(v)$ is also an epimorphism. □

Remark (7.2.6). — For every right A-module N, put III | 179

$$T_N(M) = N \otimes_A M$$

for every left A-module M. Thus T_N is an additive covariant functor from $\mathtt{Ab}_A$ to $\mathtt{Ab}$, is right exact, and commutes with direct sums. If $T_N(A_s)$ is canonically identified with N, the corresponding homomorphism (7.2.2.1) is immediately seen to be the identity. It follows that the right A-module N in (7.2.5)(c) is determined up to unique isomorphism and is canonically isomorphic to $T(A_s)$.

Equivalently, the assignments

$$T \longmapsto T(A_s) \qquad \text{and} \qquad N \longmapsto T_N$$

define an equivalence *(T, I, 1.2)* between the category of right A-modules and the category of additive covariant functors

$$\mathtt{Ab}_A \longrightarrow \mathtt{Ab}$$

that are right exact and commute with direct sums.

Proposition (7.2.7). — *Let A be a left Artinian ring whose quotient by its radical $\mathfrak{m}$ is a field k, and let T be an additive covariant functor from $\mathtt{Ab}_A$ to $\mathtt{Ab}$ that commutes with direct sums. The conditions of (7.2.5) are then also equivalent to the following condition:*

(d) T is semi-exact and the homomorphism

$$T(\varepsilon) : T(A_s) \longrightarrow T(k)$$

induced by the canonical homomorphism $\varepsilon : A_s \to k$ is surjective.

Proof. It is clear that condition (b′) of (7.2.5) implies (d). We prove that (d) implies (b′). There is an integer n such that $\mathfrak{m}^n = 0$. For every A-module M, put III | 180

$$M_h = \mathfrak{m}^h M.$$

We shall prove, by descending induction on h, that t_{M_h} is surjective. The assertion is clear for $h = n$. If $h < n$, there is an exact sequence

$$0 \longrightarrow M_{h+1} \longrightarrow M_h \longrightarrow M_h/M_{h+1} \longrightarrow 0,$$

and the induction hypothesis says that $t_{M_{h+1}}$ is surjective. Moreover, M_h/M_{h+1} is annihilated by $\mathfrak{m}$, and is therefore an $(A/\mathfrak{m})$-module; in other words, it is a direct sum of A-modules isomorphic to k. Since T commutes with direct sums, it is therefore enough, in order to prove that $t_{M_h/M_{h+1}}$ is surjective, to prove that t_k is surjective. The diagram

$$\begin{tikzcd} T(A_s) \otimes_A A_s \arrow[r, "t_{A_s}"] \arrow[d, "1\otimes\varepsilon"'] & T(A_s) \arrow[d, "T(\varepsilon)"] \\ T(A_s) \otimes_A k \arrow[r, "t_k"] & T(k) \end{tikzcd}$$

is commutative. Condition (d), together with (7.2.4), therefore shows that t_k is surjective.

To finish the proof, it remains to show that if

$$0 \longrightarrow M' \longrightarrow M \xrightarrow{v} M'' \longrightarrow 0$$

is an exact sequence of A-modules for which $t_{M'}$ and $t_{M''}$ are surjective, then t_M is surjective. There is a commutative diagram

$$\begin{tikzcd} T'(M') \arrow[r] \arrow[d, "t_{M'}"] & T'(M) \arrow[r] \arrow[d, "t_M"] & T'(M'') \arrow[r] \arrow[d, "t_{M''}"] & 0 \arrow[d] \\ T(M') \arrow[r] & T(M) \arrow[r, "T(v)"] & T(M'') \arrow[r] & \mathrm{Coker}(T(v)) \end{tikzcd}$$

whose two rows are exact, by the hypothesis that T is semi-exact. The induction hypothesis says that $t_{M'}$ and $t_{M''}$ are epimorphisms, while the last vertical arrow is a monomorphism. The five lemma *(M, I, 1.1)* therefore shows that t_M is an epimorphism. □

7.3. Exactness criteria for homological functors of modules

Proposition (7.3.1). — *Let A be a ring, not necessarily commutative, and let $T_\bullet$ be a covariant homological functor* (T, II, 2.1) *from $\mathtt{Ab}_A$ to $\mathtt{Ab}$ that commutes with direct sums. Let p be an integer such that T_p and T_{p-1} are defined. The following conditions are equivalent:*

(a) T_p is right exact;
(b) T_{p-1} is left exact;

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(c) for every left A-module M, the canonical functorial homomorphism (7.2.2.1)

(7.3.1.1) $$T_p(A_s) \otimes_A M \longrightarrow T_p(M)$$

is an isomorphism;

(d) for every left A-module M, the homomorphism (7.3.1.1) *is an epimorphism;*
(e) T_p is isomorphic to a functor of the form

$$M \longmapsto N \otimes_A M,$$

where N is a right A-module.

If, in addition, the conditions of (7.2.7) *on A and $\mathfrak{m}$ are satisfied, the preceding conditions are also equivalent to*

(f) the canonical homomorphism

$$T_p(\varepsilon) : T_p(A_s) \longrightarrow T_p(k)$$

is an epimorphism.

Proof. By definition of a homological functor, T_i is semi-exact for every i for which it is defined; moreover, for every exact sequence

$$0 \longrightarrow M' \xrightarrow{u} M \xrightarrow{v} M'' \longrightarrow 0,$$

one has

$$\mathrm{Ker}\big(T_{i-1}(u)\big) = \mathrm{Coker}\big(T_i(v)\big).$$

It is therefore clear that (a) and (b) are equivalent, and the other assertions follow immediately from (7.2.5) and (7.2.7). □

Corollary (7.3.2). — *Let A be a commutative ring. With the notation of* (7.3.1)*, suppose that T_p is right exact. If $f \in A$ annihilates no nonzero element of an A-module M, then it annihilates no nonzero element of $T_{p-1}(M)$. In particular, if A is an integral domain, the A-module $T_{p-1}(A)$ is torsion-free.*

Proof. If h_f denotes the homothety $x \mapsto fx$ of M, the hypothesis says that h_f is injective. Condition (b) of (7.3.1) then shows that $T_{p-1}(h_f)$ is injective as well. □

Proposition (7.3.3). — *Let A be a ring, and let $T_\bullet$ be a covariant homological functor from $\mathtt{Ab}_A$ to $\mathtt{Ab}$ that commutes with direct sums. Let p be an integer such that T_{p-1}, T_p, and T_{p+1} are defined. The following conditions are equivalent:*

(a) T_p is exact;
(b) T_{p+1} and T_p are right exact;
(c) T_p and T_{p-1} are left exact;

(d) *T_{p+1} is right exact and T_{p-1} is left exact;*

(e) *for every A-module M, the canonical homomorphisms*

$$T_i(A_s) \otimes_A M \longrightarrow T_i(M) \tag{7.3.3.1}$$

are isomorphisms for $i = p$ and $i = p+1$;

(e′) *for every A-module M, the homomorphisms (7.3.3.1) are epimorphisms for $i = p$ and $i = p+1$;*

(f) *for every A-module M, the homomorphism (7.3.3.1) is an isomorphism for $i = p$, and $T_p(A_s)$ is flat as a right A-module;*

(f′) *for every A-module M, the homomorphism (7.3.3.1) is an epimorphism for $i = p$, and $T_p(A_s)$ is flat as a right A-module.*

Proof. The equivalence of (a), (b), (c), and (d) follows from the equivalence of conditions (a) and (b) of (7.3.1). The equivalence of (b), (e), and (e′) follows from the equivalence of (a), (c), and (d) in (7.3.1). Finally, to say that $T_p(A_s)$ is flat means that the functor

$$M \longmapsto T_p(A_s) \otimes_A M$$

is left exact; the equivalence of (a), (f), and (f′) again follows from the equivalence of (a), (c), and (d) in (7.3.1). □

Corollary (7.3.4). — *Suppose that A is commutative, that T_p is exact, and that $T_p(A)$ is an A-module of finite presentation. Then the function*

$$x \longmapsto \operatorname{rank}_{k(x)}\big(T_p(k(x))\big)$$

is locally constant on $X = \operatorname{Spec}(A)$, and is therefore constant if $\operatorname{Spec}(A)$ is connected.

Proof. By condition (f) of (7.3.3), $T_p(A)$ is a flat A-module. It is consequently projective of finite type, so $\widetilde{T_p(A)}$ is a locally free $\mathscr{O}_X$-module (Bourbaki, *Algèbre commutative*, chap. II, § 5, no. 2, th. 1). Moreover,

$$T_p(k(x)) = T_p(A) \otimes_A k(x)$$

by condition (e) of (7.3.3), and the rank at x of the A-module $T_p(A)$ is locally constant *(loc. cit.)*. This proves the corollary. □

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Proposition (7.3.5). — *Suppose that A is a left Artinian ring whose quotient by its radical $\mathfrak{m}$ is a field k. The conditions of (7.3.3) are also equivalent to each of the following:*

(g) *the canonical homomorphism*

$$T_i(\varepsilon) : T_i(A_s) \longrightarrow T_i(k)$$

is an epimorphism for $i = p$ and $i = p+1$;

(h) *$T_p(\varepsilon)$ is an epimorphism and $T_p(A_s)$ is flat as a right A-module, or, equivalently, is a free right A-module (Bourbaki,* Algèbre commutative, *chap. II, § 3, no. 2, cor. 2 of prop. 5).*

Suppose in addition that A is commutative and that the A-module $T_p(k)$ has finite length d. The preceding conditions are then also equivalent to each of the following:

(i) *for every A-module M of finite length,*

$$\operatorname{length}\big(T_p(M)\big) = d \operatorname{length}(M); \tag{7.3.5.1}$$

(j)

$$\operatorname{length}\big(T_p(A)\big) = d \operatorname{length}(A). \tag{7.3.5.2}$$

The equivalence of (g) and (h) with the conditions of (7.3.3) follows at once from (7.2.7). For the remaining assertions we shall use the following lemma.

Lemma (7.3.5.3). — *Let K and K' be abelian categories, and let*

$$F : \mathsf{K} \longrightarrow \mathsf{K}'$$

be an additive covariant functor. Suppose that F is semi-exact and that $F(S)$ has finite length in $\mathtt{K}'$ for every simple object S of $\mathtt{K}$. Then $F(E)$ has finite length in $\mathtt{K}'$ for every object E of finite length in $\mathtt{K}$. For every exact sequence

$$0 \longrightarrow E' \xrightarrow{u} E \xrightarrow{v} E'' \longrightarrow 0$$

of objects of finite length in $\mathtt{K}$, one has

$$\text{length}\, F(E) \leqslant \text{length}\, F(E') + \text{length}\, F(E''), \tag{7.3.5.4}$$

and equality in (7.3.5.4) holds if and only if

$$0 \longrightarrow F(E') \longrightarrow F(E) \longrightarrow F(E'') \longrightarrow 0$$

is exact.

Proof. The sequence

$$F(E') \xrightarrow{F(u)} F(E) \xrightarrow{F(v)} F(E'')$$

is exact by hypothesis. If $F(E')$ and $F(E'')$ have finite length, so do $\mathrm{Im}(F(u))$ and $\mathrm{Im}(F(v))$. Since $\mathrm{Ker}(F(v)) = \mathrm{Im}(F(u))$, the object $F(E)$ has finite length, and

$$\begin{aligned} \text{length}\, F(E) &= \text{length}\, \mathrm{Im}(F(u)) + \text{length}\, \mathrm{Im}(F(v)) \\ &\leqslant \text{length}\, F(E') + \text{length}\, F(E''). \end{aligned} \tag{7.3.5.5}$$

Induction on the length of E proves the first assertion. Moreover, equality in (7.3.5.5) can hold only when **III** | 183

$$\text{length}\, \mathrm{Im}(F(u)) = \text{length}\, F(E')$$

and

$$\text{length}\, \mathrm{Im}(F(v)) = \text{length}\, F(E'').$$

The first equality is equivalent to $\text{length}\, \mathrm{Ker}(F(u)) = 0$, hence to $\mathrm{Ker}(F(u)) = 0$; the second is equivalent to $\text{length}\, \mathrm{Coker}(F(v)) = 0$, hence to $\mathrm{Coker}(F(v)) = 0$. This proves the last assertion. □

Now let M be an A-module of finite length, with A commutative. The quotients in a Jordan–Hölder series of M are necessarily isomorphic to the A-module k. Induction on the length of M, using (7.3.5.4), therefore gives

$$\text{length}\, T_p(M) \leqslant d\ \text{length}(M). \tag{7.3.5.6}$$

It also follows from (7.3.5.3) that if T_p is exact, then equality (7.3.5.1) holds. Thus condition (a) of (7.3.3) implies (i), and (i) plainly implies (j). It remains to prove the following lemma.

Lemma (7.3.5.7). — *The equality*

$$\text{length}\, T_p(A) = d\ \text{length}\, A$$

implies that $T_p(\varepsilon)$ is an epimorphism and that $T_p(A)$ is a flat A-module.

Proof. Starting from the exact sequence

$$0 \longrightarrow \mathfrak{m} \longrightarrow A \longrightarrow k \longrightarrow 0,$$

formulas (7.3.5.4) and (7.3.5.6) give

$$\begin{aligned} \text{length}\, T_p(A) &\leqslant \text{length}\, T_p(\mathfrak{m}) + \text{length}\, T_p(k) \\ &\leqslant d\big(\text{length}\, \mathfrak{m} + \text{length}\, k\big) = d\ \text{length}\, A. \end{aligned}$$

By (7.3.5.3), equality can hold only if

$$0 \longrightarrow T_p(\mathfrak{m}) \longrightarrow T_p(A) \longrightarrow T_p(k) \longrightarrow 0 \tag{7.3.5.8}$$

is exact. By (7.2.7) and (7.2.5), T_p is isomorphic to a functor

$$M \longmapsto N \otimes_A M.$$

The exactness of (7.3.5.8), together with the exact sequence of Tor, shows that

$$\mathrm{Tor}_1^A(N, k) = 0.$$

It follows that $N = T_p(A)$ is a flat A-module *(0, 10.1.3)*. □

Lemma (7.3.6). — *Let A be a ring, and let $T_\bullet$ be a covariant homological functor from $\mathtt{Ab}_A$ to $\mathtt{Ab}$ that commutes with direct sums. Suppose that T_p and T_{p+1} are defined and that T_p is left exact. Then T_{p+1} is exact if and only if $T_{p+1}(A_s)$ is flat as a right A-module.*

Proof. By (7.3.1), the canonical homomorphism

$$T_{p+1}(A_s) \otimes_A M \longrightarrow T_{p+1}(M)$$

is an isomorphism of functors. The assertion now follows from the definition of a flat A-module. □

Proposition (7.3.7). — *Let A be a ring, and let $T_\bullet$ be a covariant homological functor from $\mathtt{Ab}_A$ to $\mathtt{Ab}$ that commutes with direct sums. Suppose there is an i_0 such that T_i is exact for $i \leqslant i_0$. Then, for every integer $p > i_0$, the following conditions are equivalent:*

(a) T_q is exact for $q \leqslant p$;
(b) $T_q(A_s)$ is flat as a right A-module for $q \leqslant p$;
(c) for every A-module M, the canonical homomorphism

$$T_q(A_s) \otimes_A M \longrightarrow T_q(M)$$

is surjective for $q \leqslant p + 1$.

Proof. The equivalence of (a) and (b) follows by induction on q from (7.3.6), since T_{i_0} is exact by hypothesis. The equivalence of (a) and (c) follows from the equivalence of conditions (a) and (e′) in (7.3.3). □

(7.3.8). Let A be a commutative ring, let B be an A-algebra, not necessarily commutative, and let $T_\bullet$ be a covariant homological functor from $\mathtt{Ab}_A$ to $\mathtt{Ab}$. It follows from the definitions in (7.1.3) that the functor from $\mathtt{Ab}_B$ to $\mathtt{Ab}$ obtained by extension of scalars from A to B, denoted

$$T_\bullet^{(B)} = \big(T_i^{(B)}\big),$$

is again a homological functor.

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Corollary (7.3.9). — *Suppose that $T_\bullet$ satisfies the general conditions of (7.3.7) and commutes with direct limits. Suppose further that A is an integral domain and that every $T_n(A)$ is an A-module of finite presentation. Then, for every integer N, there is an $f \in A - \{0\}$ such that*

$$T_p^{(A_f)} : \mathtt{Ab}_{A_f} \longrightarrow \mathtt{Ab}$$

is exact for $p \leqslant N$.

Proof. By hypothesis, T_i is exact for $i \leqslant i_0$, so $T_i(A)$ is flat for these values of i. By condition (b) of (7.3.7), it is enough to choose f such that

$$T_i^{(A_f)}(A_f) = T_i(A_f)$$

is a free A_f-module for $i_0 < i \leqslant N$. Now

$$T_i(A_f) = \big(T_i(A)\big)_f,$$

since T_i commutes with direct limits (7.1.4). If x is the generic point of $\operatorname{Spec}(A)$, then $\big(T_i(A)\big)_x$ is a finite-dimensional vector space over the fraction field of A. Since each $T_i(A)$ is of finite presentation, an f with the desired property exists (Bourbaki, *Algèbre commutative*, chap. II, § 5, no. 1, cor. of prop. 2).

If only finitely many indices i satisfy $T_i \neq 0$, one may choose $f \in A - \{0\}$ such that every $T_p^{(A_f)}$ is exact. □

Corollary (7.3.10). — *Suppose that $T_\bullet$ satisfies the general conditions of (7.3.7) and commutes with direct limits, that A is commutative and Noetherian, and that every $T_n(A)$ is an A-module of finite type. Then, for every integer N, there is a dense open subset U of* $\operatorname{Spec}(A)$ *such that, for every integer $p \leqslant N$, the function*

$$x \longmapsto \operatorname{rank}_{k(x)}\big(T_p(k(x))\big)$$

is constant on U.

Proof. Let $\mathfrak{p}$ be a minimal prime ideal of A. The ring $B = A/\mathfrak{p}$ is an integral domain, and $\operatorname{Spec}(B)$ is identified with an irreducible component of the topological space $\operatorname{Spec}(A)$. We prove by induction on $p \leqslant N$ that there is an $f_p \in B - \{0\}$ such that, if $B' = B_{f_p}$, then $T_i^{(B')}$ is exact and $T_i(B')$ is a B'-module of finite type for $i \leqslant p$.

The assertion holds for $p \leqslant i_0$ by hypothesis, with $f_p = 1$, so that $B' = B = A/\mathfrak{p}$. Indeed, T_p is then exact, and $T_p(B)$ is isomorphic to

$$T_p(A)/T_p(\mathfrak{p});$$

it is therefore an A-module, and *a fortiori* a B-module, of finite type.

For the induction step, let f_p be the canonical image in B of an element $g_p \in A$, and put $A' = A_{g_p}$. Then

$$B' = A'/\mathfrak{p}',$$

where $\mathfrak{p}'$ is the minimal prime ideal $\mathfrak{p}A_{g_p}$ of A'. Since

$$T_i(A_{g_p}) = \big(T_i(A)\big)_{g_p},$$

the $T_i(A')$ are A'-modules of finite type. Thus $T_\bullet^{(A')}$ satisfies the same hypotheses as $T_\bullet$, with i_0 replaced by p.

We may therefore reduce to the case $A' = A$ in which T_p is exact. The exact sequence

$$0 \longrightarrow \mathfrak{p} \longrightarrow A \longrightarrow A/\mathfrak{p} \longrightarrow 0$$

then gives an exact sequence

$$T_{p+1}(A) \longrightarrow T_{p+1}(A/\mathfrak{p}) \xrightarrow{\partial} T_p(\mathfrak{p}) \longrightarrow T_p(A).$$

Since T_p is exact, the last arrow is injective. Consequently $T_{p+1}(A/\mathfrak{p})$ is a quotient of $T_{p+1}(A)$, and is therefore of finite type.

The proof of (7.3.9) used the finite-type hypothesis on $T_i(A)$ only for $i \leqslant N$. We may therefore apply that corollary to the integral domain B, the functor $T_\bullet^{(B)}$, and $N = p+1$, which completes the induction. There is consequently an $f_N \in B - \{0\}$ such that $\operatorname{Spec}(B_{f_N})$ is an open subset V, dense in $\operatorname{Spec}(B)$, that meets no other irreducible component of $\operatorname{Spec}(A)$.

If the assertion has been proved for B_{f_N}, there is a dense open subset W of V on which the functions in the statement are constant, since **III** | 185

$$A_x = (B_{f_N})_x \qquad (x \in W).$$

Repeating the same argument for every irreducible component of $\operatorname{Spec}(A)$ proves the corollary. We may therefore reduce to the case where A is an integral domain. The argument of (7.3.9) then gives an $f \in A - \{0\}$ such that the $T_p(A_f)$ are free A_f-modules of finite type for $p \leqslant N$. The conclusion follows from (7.3.4). □

Proposition (7.3.11). — *Let A be a commutative local ring with residue field k, and let $T_\bullet$ be a covariant homological functor from $\mathtt{Ab}_A$ to $\mathtt{Ab}$ that commutes with direct sums. Suppose there is an i_0 such that T_i is exact for $i \leqslant i_0$, and that every $T_n(A)$ is an A-module of finite presentation. The equivalent conditions (a), (b), and (c) of (7.3.7) imply the following two conditions, and are equivalent to them when A is also reduced:*

(d) *for every $x \in \operatorname{Spec}(A)$,*

$$\operatorname{rank}_{k(x)} T_q(k(x)) = \operatorname{rank}_k T_q(k) \qquad (q \leqslant p);$$

(d′) *for every generic point x_j of an irreducible component of $\operatorname{Spec}(A)$,*

$$\operatorname{rank}_{k(x_j)} T_q(k(x_j)) = \operatorname{rank}_k T_q(k) \qquad (q \leqslant p).$$

Proof. Since $T_q(A)$ is an A-module of finite presentation, condition (b) of (7.3.7) is equivalent to saying that $T_q(A)$ is free for $q \leqslant p$ (Bourbaki, *Algèbre commutative*, chap. II, § 3, no. 2, cor. 2 of prop. 5). Condition (c) implies

$$T_q(k(x)) = T_q(A) \otimes_A k(x) \qquad (q \leqslant p).$$

Thus the equivalent conditions of (7.3.7) imply (d), and it is immediate that (d) implies (d′).

It remains to show that (d′) implies (a) when A is reduced. We argue by induction on $q \leqslant p$, since T_q is exact for $q \leqslant i_0$. Suppose that T_k is exact for $k \leqslant q < p$. We shall show that $T_{q+1}(A)$ is a free A-module. By the induction hypothesis, the canonical map

$$T_{q+1}(A) \otimes_A M \longrightarrow T_{q+1}(M)$$

is an isomorphism for every A-module M, by condition (c) of (7.3.7) and (7.3.3). Applying this to $M = k(x_j)$ and $M = k$, condition (d′) gives

$$\operatorname{rank}_{k(x_j)}\left(T_{q+1}(A) \otimes_A k(x_j)\right) = \operatorname{rank}_k T_{q+1}(k)$$

for every generic-point index j. It follows that $T_{q+1}(A)$ is free (Bourbaki, *Algèbre commutative*, chap. II, § 3, no. 2, prop. 7), which completes the proof. □

The preceding results will be considerably strengthened for the homological functors of the special type to be studied in (7.4): the resulting exactness criteria involve only a single one of the T_p.

7.4. Exactness criteria for the functors $H_\bullet(P_\bullet \otimes_A M)$

(7.4.1). Let A be a ring, not necessarily commutative, and let $P_\bullet$ be a complex of flat right A-modules. Since the functor

$$M \longmapsto P_k \otimes_A M$$

is then exact in $\mathtt{Ab}_A$ for every k, the ∂-functor

(7.4.1.1) $$T_\bullet(M) = H_\bullet(P_\bullet \otimes_A M)$$

is a *homological functor* from $\mathtt{Ab}_A$ to $\mathtt{Ab}$; it is plainly A-linear when A is commutative (7.1.2), and it commutes with inductive limits. III | 186

If A is commutative, then, for every A-algebra B, the homological functor $T_\bullet^{(B)}$ (7.3.8) is, by definition, given by

(7.4.1.2) $$T_i^{(B)}(N) = H_i(P_\bullet \otimes_A N_{[\rho]}),$$

where $\rho : A \to B$ is the homomorphism defining the A-algebra structure on B. Since one may also write

$$P_\bullet \otimes_A N_{[\rho]} = P_\bullet \otimes_A (B \otimes_B N)_{[\rho]} = (P_\bullet \otimes_A B) \otimes_B N,$$

it follows that

(7.4.1.3) $$T_i^{(B)}(N) = H_i(P'_\bullet \otimes_B N)$$

for every B-module N, where $P'_\bullet$ is the complex $P_\bullet \otimes_A B$ of flat B-modules (**0**, 6.2.1).

Proposition (7.4.2). — *Under the general hypotheses of (7.4.1), and for a given integer $p \in \mathbf{Z}$, the following conditions are equivalent:*

(a) T_p is left exact, or, equivalently, T_{p+1} is right exact;

(b)

$$Z'_p(P_\bullet) = \operatorname{Coker}(P_{p+1} \longrightarrow P_p)$$

is a flat right A-module;

(c) there exists a complex $P'_\bullet$ of flat right A-modules for which the differential

$$d'_{p+1} : P'_{p+1} \longrightarrow P'_p$$

is zero, together with an isomorphism of homological functors

$$H_\bullet(P_\bullet \otimes_A M) \simeq H_\bullet(P'_\bullet \otimes_A M).$$

Proof. By definition there is a sequence, exact and functorial in M,

$$0 \longrightarrow T_p(M) \longrightarrow Z'_p(P_\bullet) \otimes_A M \longrightarrow P_{p-1} \otimes_A M.$$

Indeed, by right exactness of the tensor product,

$$Z'_p(P_\bullet \otimes_A M) = \operatorname{Coker}(P_{p+1} \otimes_A M \longrightarrow P_p \otimes_A M) = Z'_p(P_\bullet) \otimes_A M.$$

For every homomorphism $f : M \to N$ one therefore has the commutative diagram

(7.4.2.1)
$$\begin{tikzcd} 0 \arrow[r] & T_p(M) \arrow[r] \arrow[d, "u"] & Z'_p(P_\bullet) \otimes_A M \arrow[r] \arrow[d, "v"] & P_{p-1} \otimes_A M \arrow[d, "w"] \\ 0 \arrow[r] & T_p(N) \arrow[r] & Z'_p(P_\bullet) \otimes_A N \arrow[r] & P_{p-1} \otimes_A N, \end{tikzcd}$$

whose rows are exact. If f is a monomorphism, then so is w, because P_{p-1} is flat. If T_p is left exact, u is likewise a monomorphism, and it follows that v is a monomorphism. Thus $Z'_p(P_\bullet)$ is flat. Conversely, if $Z'_p(P_\bullet)$ is flat, then v is a monomorphism for every monomorphism $f : M \to N$, and diagram (7.4.2.1) shows that u is a monomorphism. Hence T_p, which is already half-exact, is left exact. This proves the equivalence of (a) and (b).

It is immediate that (c) implies (a). Indeed, if $d'_{p+1} : P'_{p+1} \to P'_p$ is zero and

$$0 \longrightarrow M' \longrightarrow M \longrightarrow M'' \longrightarrow 0$$

is an exact sequence of A-modules, then the boundary map in the exact sequence

$$H_{p+1}(P'_\bullet \otimes_A M'') \xrightarrow{\partial} H_p(P'_\bullet \otimes_A M') \longrightarrow H_p(P'_\bullet \otimes_A M)$$

is zero by definition *(M, IV, 1)*, so T_p is left exact. **III** | 187

Conversely, let us prove that (b) implies (c). Put

$$Z_{p+1}(P_\bullet) = \mathrm{Ker}(P_{p+1} \longrightarrow P_p).$$

There is an exact sequence

$$0 \longrightarrow Z_{p+1}(P_\bullet) \longrightarrow P_{p+1} \longrightarrow Z'_p(P_\bullet) \longrightarrow 0$$

in which P_{p+1} and $Z'_p(P_\bullet)$ are flat. Hence $Z_{p+1}(P_\bullet)$ is flat (**0**, 6.1.2). Define

$$P'_i = P_i \quad (i \neq p, p+1), \qquad P'_p = Z'_p(P_\bullet), \qquad P'_{p+1} = Z_{p+1}(P_\bullet).$$

For $i \neq p, p+1$, let $d'_i : P'_i \to P'_{i-1}$ be the differential of $P_\bullet$; let d'_p be the homomorphism

$$Z'_p(P_\bullet) \longrightarrow P_{p-1}$$

induced by d_p after passage to the quotient, and set $d'_{p+1} = 0$. From the flatness just established one has

$$\begin{aligned} Z'_i(P_\bullet \otimes_A M) &= Z'_i(P_\bullet) \otimes_A M, \\ Z_i(P_\bullet \otimes_A M) &= Z_i(P_\bullet) \otimes_A M, \\ B_i(P_\bullet \otimes_A M) &= B_i(P_\bullet) \otimes_A M, \end{aligned}$$

where

$$B_i(P_\bullet) = \mathrm{Im}(P_{i+1} \longrightarrow P_i).$$

One immediately obtains functorial isomorphisms

$$H_i(P_\bullet \otimes_A M) \simeq H_i(P'_\bullet \otimes_A M)$$

for every i. That these form an isomorphism of ∂-functors follows directly from the definition of ∂ *(M, IV, 1)*.

We also note that the conditions of (7.4.2) imply that $B_p(P_\bullet)$ is flat, since there is an exact sequence

$$0 \longrightarrow B_p(P_\bullet) \longrightarrow P_p \longrightarrow Z'_p(P_\bullet) \longrightarrow 0$$

in which P_p and $Z'_p(P_\bullet)$ are flat (**0**, 6.1.2). □

Corollary (7.4.3). — *Suppose that A is a regular Noetherian ring of dimension* 1*, in other words, a product of Dedekind rings* (**0**, 17.1.3) *and* (**0**, 17.3.7)*; for example, A may be a principal ideal ring. Then T_p is left exact if and only if $T_p(A)$ is a flat A-module. The functor T_p is exact if and only if both $T_p(A)$ and $T_{p-1}(A)$ are flat A-modules.*

Proof. Recall that, for a module M over a Dedekind ring, being flat is equivalent to being torsion-free (**0**, 6.3.3) and (**0**, 6.3.4). Under the hypotheses of (7.4.3), every submodule of a flat A-module is therefore flat.

The second assertion follows from the first, since saying that T_p is exact amounts to saying that T_p and T_{p-1} are left exact. To prove the first assertion, observe that there is an exact sequence

$$0 \longrightarrow H_p(P_\bullet) \longrightarrow Z'_p(P_\bullet) \longrightarrow B_{p-1}(P_\bullet) \longrightarrow 0,$$

where $B_{p-1}(P_\bullet)$ is flat, being a submodule of the flat A-module P_{p-1}. Hence $H_p(P_\bullet)$ is flat if and only if $Z'_p(P_\bullet)$ is flat (**0**, 6.1.2). □

The most important applications of (7.4.2) are the following.

Proposition (7.4.4). — *Let A be a Noetherian ring and $P_\bullet$ a complex of flat A-modules. Suppose either that the P_i are of finite type, or that the $H_i(P_\bullet)$ are A-modules of finite type and there exists an i_0 such that $H_i(P_\bullet) = 0$ for $i < i_0$. Let $T_\bullet$ be the homological functor defined by* (7.4.1.1)*. Then the set U of points $y \in \operatorname{Spec}(A)$ such that $(T_p)_y$* (7.1.4) *is right exact, respectively left exact or exact, is open in* $\operatorname{Spec}(A)$.

Proof. Under the second hypothesis on $P_\bullet$, one may replace $P_\bullet$ by a complex $P'_\bullet$ of finite free A-modules such that the functor **III | 188**

$$H_\bullet(P'_\bullet \otimes_A M)$$

is isomorphic to $T_\bullet(M)$ as a ∂-functor (**0**, 11.9.3). Thus one may always reduce to the first hypothesis. In that case the modules $Z'_i(P_\bullet)$ are of finite type. Moreover, it is enough to prove the assertions concerning left exactness (cf. (7.4.2), (a)).

Let $x \in U$. Since $M \mapsto M_x$ is exact,

$$\big(Z'_p(P_\bullet)\big)_x = Z'_p((P_\bullet)_x).$$

Taking (7.4.1.3) into account, the hypothesis and (7.4.2), (b), show that $\big(Z'_p(P_\bullet)\big)_x$ is a flat A_x-module. It is therefore free, since it is of finite type and A_x is a Noetherian local ring (Bourbaki, *Algèbre commutative*, chap. II, § 3, no. 2, cor. 2 of prop. 5). It follows that there is an $f \in A$ for which $\big(Z'_p(P_\bullet)\big)_f$ is free over A_f (Bourbaki, *Algèbre commutative*, chap. II, § 5, no. 1, cor. of prop. 2). A fortiori, $\big(Z'_p(P_\bullet)\big)_y$ is free over A_y for every $y \in D(f)$, and (7.4.2), (b), completes the proof. □

Corollary (7.4.5). — *Under the hypotheses of* (7.4.4)*, suppose in addition that A is an integral domain. Then the set U of $x \in \operatorname{Spec}(A)$ such that $(T_p)_x$ is exact is open and nonempty.*

Proof. It is enough to show that $(T_p)_x$ is exact at the generic point x of $\operatorname{Spec}(A)$. This is immediate, since A_x is a field and every additive functor on $\mathtt{Ab}_{A_x}$ is exact. □

Proposition (7.4.6). — *Under the general hypotheses of* (7.4.4)*, conditions (a), (b), and (c) of* (7.4.2) *are also equivalent to:*

(d) there exists an A-module Q and a functorial isomorphism

$$T_p(M) \simeq \operatorname{Hom}_A(Q, M). \tag{7.4.6.1}$$

Moreover, this property determines the A-module Q up to a unique isomorphism, and Q is of finite type.

Proof. The uniqueness of Q is a special case of the uniqueness of a representing object for a representable functor (**0**, 8.1.5). It is clear that the right-hand side of (7.4.6.1) is left exact.

Conversely, suppose that T_p is left exact. As in (7.4.4), one may first reduce to the case where the P_i are flat and of finite type, and hence, since A is Noetherian, projective of finite type (Bourbaki, *Algèbre commutative*, chap. II, § 5, no. 2, cor. of th. 1). The dual $\check{P}_i$ of P_i is then also a projective A-module of finite type, P_i is canonically isomorphic to the dual of $\check{P}_i$, and the canonical homomorphism

$$P_i \otimes_A M \longrightarrow \operatorname{Hom}_A(\check{P}_i, M)$$

is bijective (Bourbaki, *Algèbre*, chap. II, 3rd ed., § 4, no. 2, prop. 2).

By (7.4.2), (c), one may furthermore suppose that $d_{p+1} : P_{p+1} \to P_p$ is zero. There is then an exact sequence

$$0 \longrightarrow T_p(M) \xrightarrow{u} P_p \otimes_A M \xrightarrow{v} P_{p-1} \otimes_A M,$$

where $v = d_p \otimes 1$. Put $Q' = \mathrm{Ker}(d_p)$, so that

$$0 \longrightarrow Q' \longrightarrow P_p \xrightarrow{d_p} P_{p-1}$$

is exact. Transposition gives the exact sequence

$$\check{P}_{p-1} \xrightarrow{{}^t d_p} \check{P}_p \longrightarrow \check{Q}' \longrightarrow 0.$$

We claim that

$$Q = \check{Q}' = \mathrm{Coker}({}^t d_p)$$

has the required property. Indeed, there is an exact sequence

$$0 \longrightarrow \mathrm{Hom}_A(Q, M) \longrightarrow \mathrm{Hom}_A(\check{P}_p, M) \xrightarrow{v'} \mathrm{Hom}_A(\check{P}_{p-1}, M),$$

where $v' = \mathrm{Hom}_A({}^t d_p, 1)$. Under the canonical identifications **III** | 189

$$P_i \otimes_A M \simeq \mathrm{Hom}_A(\check{P}_i, M),$$

the homomorphism v' identifies with $v = d_p \otimes 1$. Thus one obtains the required functorial isomorphism

$$T_p(M) \simeq \mathrm{Hom}_A(Q, M).$$

Finally, Q is a quotient of $\check{P}_p$, and is therefore of finite type. □

Proposition (7.4.7). — *Suppose that the general hypotheses of (7.4.4) hold. For every A-module M of finite type:*

(i) the $T_i(M)$ are A-modules of finite type;

(ii) for every ideal $\mathfrak{m}$ of A, the canonical homomorphism

(7.4.7.1) $$\widehat{T_i(M)} \longrightarrow \varprojlim_n T_i\big(M \otimes_A (A/\mathfrak{m}^{n+1})\big)$$

is bijective, where the left-hand side denotes the separated completion of $T_i(M)$ for the $\mathfrak{m}$-adic topology.

Proof. As in (7.4.4), one reduces first to the case where the P_i are of finite type. Since A is Noetherian, the submodules of the $P_i \otimes_A M$ are of finite type, which immediately proves (i). Assertion (ii) follows from the more general lemma below. □

Lemma (7.4.7.2). — *Let A be a Noetherian ring and $u : E \to F$ a homomorphism of finite-type A-modules. For every finite-type A-module M, put*

$$K(M) = \mathrm{Ker}(u \otimes 1_M), \qquad C(M) = \mathrm{Coker}(u \otimes 1_M).$$

Then, for every ideal $\mathfrak{m}$ of A, the canonical homomorphisms

(7.4.7.3) $$\widehat{K(M)} \longrightarrow \varprojlim_n K(M_n), \qquad \widehat{C(M)} \longrightarrow \varprojlim_n C(M_n)$$

are bijective, where

$$M_n = M \otimes_A (A/\mathfrak{m}^{n+1}) = M/\mathfrak{m}^{n+1}M.$$

Proof. The modules $E \otimes_A M$ and $F \otimes_A M$ are of finite type, and the functor $M \mapsto \widehat{M}$ is exact on the category of finite-type A-modules (**0**, 7.3.3). Consequently, $\widehat{K(M)}$ and $\widehat{C(M)}$ are, respectively, the kernel and the cokernel of

$$\widehat{u \otimes 1} : \widehat{E \otimes_A M} \longrightarrow \widehat{F \otimes_A M}.$$

Left exactness of $\varprojlim$ therefore gives

$$\widehat{K(M)} = \varprojlim_n K(M_n).$$

On the other hand, right exactness of the tensor product gives

$$C(M_n) = C(M) \otimes_A (A/\mathfrak{m}^{n+1}),$$

and hence

$$\widehat{C(M)} = \varprojlim_n C(M_n)$$

by definition. □

Remark (7.4.8). — Taking (6.10.5) and (6.10.6) into account, one sees that, with an additional flatness hypothesis, (7.4.7) recovers the fact that (4.1.7.1) is an isomorphism, which is the essential content of the "first comparison theorem" for proper morphisms. Moreover, the statement applies not only to a coherent $\mathscr{O}_X$-module but to a *complex* of such modules. It would be interesting to obtain a statement that contains both (7.4.7) and (4.1.7.1) as special cases.

Notice also that, when the P_i are of finite type, the proof of (7.4.7) does not use their flatness. It would be worth examining whether the conclusion of (7.4.7) continues to hold when the P_i are assumed neither flat nor of finite type, but the $H_i(P_\bullet)$ are assumed to be of finite type for every i and zero for $i < i_0$. Can one then replace $P_\bullet$ by a complex $P'_\bullet$ of finite-type A-modules in such a way that the functors

$$H_\bullet(P_\bullet \otimes_A M) \quad \text{and} \quad H_\bullet(P'_\bullet \otimes_A M),$$

which are no longer homological, are still isomorphic?

7.5. The case of Noetherian local rings

(7.5.1). Let A be a Noetherian local ring and $\mathfrak{m}$ its maximal ideal. For every A-module M, denote by **III** | 190
$\widehat{M}$ its separated completion for the $\mathfrak{m}$-adic topology, which is isomorphic to

$$\varprojlim_n \big(M \otimes_A (A/\mathfrak{m}^{n+1})\big) = \varprojlim_n \big(M/\mathfrak{m}^{n+1}M\big).$$

Let T be a covariant additive functor from $\mathtt{Ab}_A$ to $\mathtt{Ab}$. The canonical homomorphisms (7.2.3.1)

$$T(M) \otimes_A (A/\mathfrak{m}^{n+1}) \longrightarrow T\big(M \otimes_A (A/\mathfrak{m}^{n+1})\big)$$

plainly form a projective system of A-homomorphisms. On passing to the limit they therefore give an $\widehat{A}$-homomorphism, functorial in M,

$$\widehat{T(M)} \longrightarrow \varprojlim_n T(M_n), \tag{7.5.1.1}$$

where

$$M_n = M \otimes_A (A/\mathfrak{m}^{n+1}), \qquad A_n = A/\mathfrak{m}^{n+1}.$$

Proposition (7.5.2). — *Let A be a Noetherian local ring with maximal ideal $\mathfrak{m}$ and residue field $k = A/\mathfrak{m}$. Let T be a covariant additive functor from $\mathtt{Ab}_A$ to $\mathtt{Ab}$ that is half-exact and commutes with inductive limits. Suppose, moreover, that for every finite-type A-module M, the module $T(M)$ is of finite type and the canonical homomorphism* (7.5.1.1) *is an isomorphism. Then the following conditions are equivalent:*

- *(a) T is right exact;*
- *(b) for every n, the functor $N \mapsto T(N)$ is right exact in the category of finite-type A_n-modules; equivalently, T is right exact in the category of finite-length A-modules;*
- *(c) the canonical homomorphism*

$$T(\varepsilon) : T(A) \longrightarrow T(k)$$

  is surjective;
- *(d) for every sufficiently large n, the canonical homomorphism*

$$T(A_n) \longrightarrow T(k)$$

  is surjective.

Proof. It is clear that (a) implies (b). Let us prove the converse. Suppose that $u : M \to N$ is an epimorphism of A-modules; we must show that $T(u)$ is an epimorphism. Since T commutes with inductive limits and filtered inductive limits are exact in the category of modules, it is enough to consider the case where M and N are of finite type.

The modules $T(M)$ and $T(N)$ are then of finite type. Since A is Noetherian and local, it is enough to prove that

$$\widehat{T(u)} : \widehat{T(M)} \longrightarrow \widehat{T(N)}$$

is surjective (**0**, 7.3.5) and (**0**, 6.4.1). By hypothesis, $\widehat{T(M)}$ and $\widehat{T(N)}$ are, respectively,

$$\varprojlim_n T(M_n) \quad \text{and} \quad \varprojlim_n T(N_n).$$

Thus $\widehat{T(u)}$ is the inverse limit of the projective system of homomorphisms

$$T(u \otimes 1_{A_n}) : T(M_n) \longrightarrow T(N_n).$$

Condition (b) says precisely that these homomorphisms are surjective. Moreover, $T(M_n)$ is a finite-type A_n-module, and A_n is Artinian. It follows that $\widehat{T(u)}$ is surjective (**0**, 13.1.2) and (**0**, 13.2.2).

Clearly (a) implies (c), and (c) implies (d). Finally, (7.2.7) shows that (b) and (d) are equivalent, because T is half-exact in $\mathtt{Ab}_{A_n}$. This completes the proof. □

Corollary (7.5.3). — *Under the general hypotheses of (7.5.2), if $T(k) = 0$, then $T(M) = 0$ for every A-module M.*

Proof. Since k is the only simple A-module, (7.3.5.4) shows that $T(E) = 0$ for every finite-length **III** | 191
A-module E. If M is of finite type, then

$$\widehat{T(M)} \simeq \varprojlim_n T(M_n).$$

Each M_n has finite length, so $\widehat{T(M)} = 0$. By hypothesis $T(M)$ is of finite type, and it embeds in $\widehat{T(M)}$ (**0**, 7.3.5); hence $T(M) = 0$. Finally, for an arbitrary A-module M, the module $T(M)$ is the inductive limit of the $T(N_\alpha)$, where the N_α run through the finite-type submodules of M. The proof is complete. □

Proposition (7.5.4). — *Let A be a Noetherian local ring with maximal ideal $\mathfrak{m}$ and residue field $k = A/\mathfrak{m}$. Let $T_\bullet$ be a homological functor from $\mathtt{Ab}_A$ to $\mathtt{Ab}$ that commutes with inductive limits. Suppose, moreover, that for every i and every finite-type A-module M, the module $T_i(M)$ is of finite type and the canonical homomorphism*

$$\widehat{T_i(M)} \longrightarrow \varprojlim_n T_i(M_n)$$

is bijective. For a given integer p, the following conditions are equivalent:

(a) T_p is exact;
(b) T_p is right exact and $T_p(A)$ is a free A-module;
(c) the canonical homomorphisms

$$T_{p+1}(A) \longrightarrow T_{p+1}(k), \qquad T_p(A) \longrightarrow T_p(k)$$

are surjective;
(d) for every n, the canonical homomorphisms

$$T_{p+1}(A_n) \longrightarrow T_{p+1}(k), \qquad T_p(A_n) \longrightarrow T_p(k)$$

are surjective;
(e) for every n, the functor $N \mapsto T_p(N)$ is exact in the category of finite-type A_n-modules.

Proof. By (7.3.3), condition (a) is equivalent to saying that T_{p+1} and T_p are right exact. Since $T_\bullet$ is a homological functor in $\mathtt{Ab}_{A_n}$, the same argument as in (7.3.1) shows that (e) is equivalent to saying that T_p and T_{p+1} are right exact on the category of finite-type A_n-modules. It follows from (7.5.2) that (a) and (e) are equivalent; the same proposition also gives the equivalence of (a), (c), and (d).

Finally, every finite-type flat module over the local ring A is free (Bourbaki, *Algèbre commutative*, chap. II, § 3, no. 2, cor. 2 of prop. 5). The equivalence of (a) and (b) therefore follows from (7.3.1) and (7.3.3). □

Corollary (7.5.5). — *Suppose that the general hypotheses of (7.5.4) hold.*

(i) If $T_p(k) = 0$, then $T_p = 0$, the functor T_{p+1} is right exact, and T_{p-1} is left exact.
(ii) If $T_{p-1}(k) = T_{p+1}(k) = 0$, then T_p is exact, the canonical homomorphism

$$T_p(A) \otimes_A M \longrightarrow T_p(M)$$

is bijective, and $T_p(A)$ is a free A-module.

Proof. Assertion (i) follows at once from (7.5.3), since T_p is half-exact; the last assertion follows from the definition of a homological functor. Taking (7.3.3) into account, (i) immediately gives the first two assertions of (ii), and the freeness of $T_p(A)$ follows from (7.5.4). □

Corollary (7.5.6). — *Suppose that the general hypotheses of (7.5.4) hold, and suppose in addition that A is a discrete valuation ring.*

(i) The functor T_p is right exact if and only if $T_{p-1}(A)$ is a free A-module.
(ii) The functor T_p is exact if and only if $T_p(A)$ and $T_{p-1}(A)$ are free A-modules.

Proof. It is clear that (ii) implies (i) (cf. (7.3.3)). To prove (i), let f be a generator of the maximal ideal of A, that is, a uniformizer of A. A finite-type A-module M is free, equivalently flat, if and only if the homothety

$$h_f : M \longrightarrow M, \qquad x \longmapsto fx,$$

is injective, since in the present situation this amounts to saying that M is torsion-free (**0**, 6.3.4). The exact sequence

$$0 \longrightarrow A \xrightarrow{h_f} A \longrightarrow k \longrightarrow 0$$

gives the exact homology sequence

$$T_p(A) \longrightarrow T_p(k) \longrightarrow T_{p-1}(A) \xrightarrow{h_f} T_{p-1}(A).$$

Thus $T_{p-1}(A)$ is free if and only if $T_p(A) \to T_p(k)$ is surjective, and the conclusion follows from (7.5.2). □

Remark (7.5.7). — By (7.4.7), the general hypotheses of (7.4.4) imply that the homological functor $T_\bullet$ defined by (7.4.1.1) satisfies the general hypotheses of (7.5.4). In that case, (7.5.6) is therefore contained in (7.4.3).

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7.6. Descent of exactness properties. Semicontinuity theorem and Grauert's exactness criterion

Proposition (7.6.1). — *Under the hypotheses of (7.4.1), let B be a commutative A-algebra. If T_p is right exact (respectively left exact, exact), then the same is true of $T_p^{(B)}$; the converse is true when B is a faithfully flat A-module.*

Proof. The first assertion is a special case of a trivial assertion of (7.1.3). Conversely, suppose first that B is a flat A-module. Then, for every A-module M,

$$H_\bullet\big(P_\bullet \otimes_A (M \otimes_A B)\big) = H_\bullet(P_\bullet \otimes_A M) \otimes_A B,$$

which may also be written, for every p,

$$T_p(M) \otimes_A B \simeq T_p^{(B)}(M_{(B)}) \tag{7.6.1.1}$$

up to canonical isomorphism. Suppose that $T_p^{(B)}$ is right exact (respectively left exact, exact). Since $M \mapsto M_{(B)}$ is an exact functor, the left-hand side of (7.6.1.1) is a right exact (respectively left exact, exact) functor of M. If B is now faithfully flat over A, it follows that T_p has the same exactness property (**0**, 6.4.1). □

Proposition (7.6.2). — *Under the hypotheses of (7.4.1), suppose in addition that A is a reduced Noetherian ring and that the P_i are finite-type A-modules. For T_p to be right exact (respectively left exact, exact), it is necessary and sufficient that $T_p^{(B)}$ have the same property for every A-algebra B that is a discrete valuation ring.*

Proof. By (7.3.1) and (7.3.3), we may restrict attention to right exactness, and only sufficiency has to be proved. By (7.4.2), it is enough to show that $Z'_{p-1}(P_\bullet)$ is a flat A-module. Since P_{p-1} is of finite type, $Z'_{p-1}(P_\bullet)$ is also of finite type; the criterion of (**0**, 10.2.8) therefore shows that it is enough for

$$Z'_{p-1}(P_\bullet) \otimes_A B$$

to be a flat B-module for every A-algebra B that is a discrete valuation ring. Since $P_\bullet$ is a complex of flat A-modules, we have

$$Z'_{p-1}(P_\bullet) \otimes_A B = Z'_{p-1}(P_\bullet \otimes_A B).$$

The complex $P_\bullet \otimes_A B$ consists of flat B-modules (**0**, 6.2.1), and for every B-module N we have **III** | 193

$$H_\bullet(P_\bullet \otimes_A N) = H_\bullet\big((P_\bullet \otimes_A B) \otimes_B N\big),$$

and hence

$$T_p^{(B)}(N) = H_p\big((P_\bullet \otimes_A B) \otimes_B N\big).$$

Applying (7.4.2) to $T_p^{(B)}$, we see that the hypothesis that $T_p^{(B)}$ is right exact is equivalent to $Z'_{p-1}(P_\bullet \otimes_A B)$ being a flat B-module.

The preceding criterion leads us to examine the case of discrete valuation rings more closely. □

Proposition (7.6.3). — *Under the hypotheses of (7.4.1), suppose that A is a regular Noetherian ring of dimension 1 (in other words, that A is Noetherian and, for every $x \in \operatorname{Spec}(A)$, the ring A_x is a field or a discrete valuation ring). Then, for every integer p and every A-module M, there is a canonical exact sequence, functorial in M,*

$$0 \longrightarrow T_p(A) \otimes_A M \xrightarrow{t_M} T_p(M) \longrightarrow \operatorname{Tor}_1^A\big(T_{p-1}(A), M\big) \longrightarrow 0. \tag{7.6.3.1}$$

Proof. In what follows, to simplify the notation we omit the complex $P_\bullet$ from the usual homological notations $H_p(P_\bullet)$, $B_p(P_\bullet)$, $Z_p(P_\bullet)$, and $Z'_p(P_\bullet)$. We have the three exact sequences

$$0 \longrightarrow H_p \longrightarrow Z'_p \longrightarrow B_{p-1} \longrightarrow 0,$$
$$0 \longrightarrow B_{p-1} \longrightarrow Z_{p-1} \longrightarrow H_{p-1} \longrightarrow 0,$$
$$0 \longrightarrow Z_{p-1} \longrightarrow P_{p-1} \longrightarrow B_{p-2} \longrightarrow 0.$$

Since P_{p-1} and P_{p-2} are flat, so are their respective submodules B_{p-1}, Z_{p-1}, and B_{p-2}: for every $x \in \operatorname{Spec}(A)$, flat A_x-modules and torsion-free A_x-modules are the same. Tensoring with M therefore gives the exact sequences

$$0 = \operatorname{Tor}_1^A(B_{p-1}, M) \longrightarrow H_p \otimes_A M \longrightarrow Z'_p \otimes_A M \xrightarrow{u} B_{p-1} \otimes_A M \longrightarrow 0, \tag{7.6.3.2}$$

$$0 = \operatorname{Tor}_1^A(Z_{p-1}, M) \longrightarrow \operatorname{Tor}_1^A(H_{p-1}, M) \longrightarrow B_{p-1} \otimes_A M \xrightarrow{v} Z_{p-1} \otimes_A M, \tag{7.6.3.3}$$

and

$$0 = \operatorname{Tor}_1^A(B_{p-2}, M) \longrightarrow Z_{p-1} \otimes_A M \xrightarrow{w} P_{p-1} \otimes_A M. \tag{7.6.3.4}$$

By definition,

$$T_p(M) = \operatorname{Ker}(d_p \otimes 1) / \operatorname{Im}(d_{p+1} \otimes 1).$$

It is therefore the kernel of the homomorphism

$$(P_p \otimes_A M) / \operatorname{Im}(d_{p+1} \otimes 1) \longrightarrow P_{p-1} \otimes_A M$$

induced by $d_p \otimes 1$. By the definition $Z'_p = P_p / B_p$, this homomorphism may also be written $Z'_p \otimes_A M \to P_{p-1} \otimes_A M$; it is the composite

$$Z'_p \otimes_A M \xrightarrow{u} B_{p-1} \otimes_A M \xrightarrow{v} Z_{p-1} \otimes_A M \xrightarrow{w} P_{p-1} \otimes_A M.$$

Since w is injective by (7.6.3.4), there is an exact sequence

$$0 \longrightarrow \operatorname{Ker} u \longrightarrow T_p(M) \longrightarrow \operatorname{Ker} v \longrightarrow 0.$$

In view of (7.6.3.2) and (7.6.3.3), and since $H_p = T_p(A)$ by definition, this is precisely (7.6.3.1). □

Remarks (7.6.4). — (i) The group $H_\bullet(P_\bullet \otimes_A M)$ is the homology of the bicomplex $P_\bullet \otimes_A M$, where M is regarded as a complex concentrated in degree 0. It is consequently (6.3.6)(6.3.2) the abutment of the regular spectral sequence whose E_2-term is

$$E^2_{pq} = \operatorname{Tor}_p^A\big(H_q(P_\bullet), M\big) = \operatorname{Tor}_p^A\big(T_q(A), M\big).$$

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The hypothesis on A implies that $\mathrm{Tor}_p^A(E, F) = 0$ for $p \geqslant 2$ and arbitrary A-modules E, F (**0**, 17.2.2). It is known (M, XV) that this implies the exactness of

$$0 \longrightarrow E^2_{0,q} \longrightarrow H_q(P_\bullet \otimes_A M) \longrightarrow E^2_{1,q-1} \longrightarrow 0,$$

which is precisely (7.6.3.1).

(ii) Taking (7.3.1) into account, the exact sequence (7.6.3.1) recovers (7.4.3) as a special case.

Corollary (7.6.5). — *Under the hypotheses of (7.4.1), suppose that A is a discrete valuation ring with field of fractions K and residue field k, and that the $T_i(A)$ are finite-type A-modules. Then*

(7.6.5.1) $$\mathrm{rank}_k\, T_p(k) \geqslant \mathrm{rank}_k\big(T_p(A) \otimes_A k\big) \geqslant \mathrm{rank}_A\, T_p(A) = \mathrm{rank}_K\, T_p(K).$$

Moreover, equality of the two extreme terms of this inequality holds if and only if T_p is exact, or, equivalently, if and only if $T_p(A)$ and $T_{p-1}(A)$ are free A-modules.

Proof. Set $M = k$ in the exact sequence (7.6.3.1). Since the terms are k-vector spaces, this gives

$$\mathrm{rank}_k\, T_p(k) = \mathrm{rank}_k\big(T_p(A) \otimes_A k\big) + \mathrm{rank}_k\, \mathrm{Tor}_1^A\big(T_{p-1}(A), k\big).$$

On the other hand, since $T_p(A)$ is a finite-type module over the local integral domain A, we have (Bourbaki, *Algèbre commutative*, chap. II, § 3, no. 2, cor. 1 of prop. 4)

(7.6.5.2) $$\mathrm{rank}_k\big(T_p(A) \otimes_A k\big) \geqslant \mathrm{rank}_A\, T_p(A) = \mathrm{rank}_K\big(T_p(A) \otimes_A K\big),$$

and the two sides of (7.6.5.2) are equal if and only if $T_p(A)$ is a free A-module (*loc. cit.*, prop. 7). Since K is a flat A-module, we also have, by definition,

$$T_p(A) \otimes_A K = H_p(P_\bullet) \otimes_A K = H_p(P_\bullet \otimes_A K) = T_p(K).$$

This proves (7.6.5.1). It also shows that equality can hold only if, first, $T_p(A)$ is free, and, second,

$$\mathrm{Tor}_1^A\big(T_{p-1}(A), k\big) = 0,$$

the latter condition being equivalent to $T_{p-1}(A)$ being free (**0**, 10.1.3). Finally, for finite-type modules over A, flatness and freeness are equivalent (Bourbaki, *Algèbre commutative*, chap. II, § 3, no. 2, cor. 2 of prop. 5), and the result follows from (7.4.3). □

(7.6.6). The hypotheses remaining those of (7.4.1), for every $x \in \mathrm{Spec}(A)$ set

(7.6.6.1) $$d_p(x) = d_p^T(x) = \mathrm{rank}_{k(x)}\, T_p(k(x)).$$

Lemma (7.6.7). — *Let $\varphi : A \to A'$ be a ring homomorphism, and let*

$$f = {}^a\varphi : \mathrm{Spec}(A') \longrightarrow \mathrm{Spec}(A)$$

be the corresponding map (**I**, 1.2.1). *If $T'_\bullet = T_\bullet^{(A')}$ (7.1.3), then*

(7.6.7.1) $$d_p^{T'} = d_p^T \circ f.$$

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Proof. For $x' \in \mathrm{Spec}(A')$, put $x = f(x')$. Then

$$\begin{aligned} H_\bullet(P_\bullet \otimes_A k(x')) &= H_\bullet\big((P_\bullet \otimes_A k(x)) \otimes_{k(x)} k(x')\big) \\ &= H_\bullet(P_\bullet \otimes_A k(x)) \otimes_{k(x)} k(x'), \end{aligned}$$

because $k(x')$ is flat over $k(x)$. This proves (7.6.7.1). □

Lemma (7.6.8). — *If A is Noetherian and $P_\bullet$ consists of finite-type A-modules, the function $x \mapsto d_p^T(x)$ on* $\mathrm{Spec}(A)$ *is constructible.*

Proof. For every irreducible closed subset Y of $X = \mathrm{Spec}(A)$, we must show that there is a nonempty open subset U of Y on which d_p is constant (**0**, 9.2.2). Write $Y = \mathrm{Spec}(A/\mathfrak{a})$, where $\mathfrak{a}$ is an ideal of A such that $A/\mathfrak{a}$ is reduced. By (7.6.7), we may reduce to the case $Y = X$ and A a Noetherian integral domain; the assertion then follows from (7.4.5). □

Theorem (7.6.9). — *Let A be a Noetherian ring, let $P_\bullet$ be a complex of finite-type flat A-modules, and let*

$$T_\bullet(M) = H_\bullet(P_\bullet \otimes_A M)$$

be the homological functor defined by $P_\bullet$. For $x \in \operatorname{Spec}(A)$, set

$$d_p(x) = \operatorname{rank}_{k(x)} T_p(k(x)).$$

Then:

(i) The function d_p is constructible and upper semicontinuous on $\operatorname{Spec}(A)$.
(ii) If T_p is exact, d_p is continuous (and hence locally constant) on $\operatorname{Spec}(A)$. The converse is true when A is reduced.

Proof. (i) Constructibility was proved in (7.6.8). To prove upper semicontinuity, it is therefore enough (**0**, 9.2.4) to show that if $x' \neq x$ is a generalization of x in $\operatorname{Spec}(A)$, then $d_p(x') \leqslant d_p(x)$. There is a discrete valuation ring B and a morphism

$$f : \operatorname{Spec}(B) \longrightarrow \operatorname{Spec}(A)$$

such that, if a is the closed point of $\operatorname{Spec}(B)$ and b its generic point, then $f(a) = x$ and $f(b) = x'$ (**II**, 7.1.9). By (7.6.7.1), we are reduced to proving $d_p(a) \geqslant d_p(b)$ on $\operatorname{Spec}(B)$, which is exactly (7.6.5.1).[4]

(ii) The first assertion was already proved in (7.3.4). For the converse, use the valuative criterion (7.6.2). In view of (7.6.7.1), we are reduced to the case in which A is a discrete valuation ring. Since $\operatorname{Spec}(A)$ then has only two points, the hypothesis that d_p is constant implies that T_p is exact, by (7.6.5). □

Corollary (7.6.10). — *Let A be a Noetherian ring, let $\mathfrak{p}_i$ $(1 \leqslant i \leqslant r)$ be its minimal prime ideals, and let k_i be the residue field of $A_{\mathfrak{p}_i}$.*

(i) For every $x \in \operatorname{Spec}(A)$, there is an index i such that

$$d_p(x) \geqslant \operatorname{rank}_{k_i} T_p(k_i). \tag{7.6.10.1}$$

In particular, if A is an integral domain and K is its field of fractions, then

$$d_p(x) \geqslant \operatorname{rank}_K T_p(K) \tag{7.6.10.2}$$

for every $x \in \operatorname{Spec}(A)$.

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(ii) Suppose in addition that A is local and reduced, and let k be its residue field. Then T_p is exact if and only if

$$\operatorname{rank}_k T_p(k) = \operatorname{rank}_{k_i} T_p(k_i) \qquad (1 \leqslant i \leqslant r). \tag{7.6.10.3}$$

Proof. Assertion (i) is immediate: every neighbourhood of x contains one of the $\mathfrak{p}_i$, and it is enough to apply the definition of semicontinuity. If A is local, the only neighbourhood of the maximal ideal $\mathfrak{m}$ in $\operatorname{Spec}(A)$ is all of $\operatorname{Spec}(A)$; hence

$$d_p(x) \leqslant \operatorname{rank}_k T_p(k)$$

for every $x \in \operatorname{Spec}(A)$. Together with (i), this shows that (7.6.10.3) implies that d_p is constant on $\operatorname{Spec}(A)$, and hence that T_p is exact by (7.6.9, (ii)). The converse follows at once from the same result. □

Remarks (7.6.11). — One may ask whether (7.6.9, (i)) can be strengthened to the inequality

$$\operatorname{rank}_{k(x)} T_p(k(x)) \geqslant \operatorname{rank}_{k(x)} \big(T_p(A) \otimes_A k(x)\big) \tag{7.6.11.1}$$

for every $x \in \operatorname{Spec}(A)$. This inequality does hold when A is a discrete valuation ring and x is its maximal ideal (7.6.5). Restrict now to the case in which A is a Noetherian local ring with maximal ideal $\mathfrak{m}$ and residue field k. The following conditions are equivalent:

[4] The principle of proving (i) by reduction to the case of a discrete valuation ring was communicated to us orally by Hironaka.

a) For every complex $P_\bullet$ of finite-type flat A-modules,

$$\operatorname{rank}_k T_p(k) \geqslant \operatorname{rank}_k\bigl(T_p(A) \otimes_A k\bigr) \qquad \text{for every integer } p. \tag{7.6.11.2}$$

b) For every finite-type A-module M,

$$\operatorname{rank}_k(M \otimes_A k) \geqslant \operatorname{rank}_k(\check{M} \otimes_A k). \tag{7.6.11.3}$$

c) For every finite-type A-module N,

$$\operatorname{rank}_k \operatorname{Tor}_1^A(N,k) \geqslant \operatorname{rank}_k \operatorname{Tor}_2^A(N,k). \tag{7.6.11.4}$$

Equivalently, by dimension shifting (M, V, 7.2), one may require, for every $i \geqslant 1$,

$$\operatorname{rank}_k \operatorname{Tor}_i^A(N,k) \geqslant \operatorname{rank}_k \operatorname{Tor}_{i+1}^A(N,k). \tag{7.6.11.5}$$

Here are brief indications of the proof. To see that a) implies b), consider an exact sequence

$$L_1 \xrightarrow{d} L_0 \longrightarrow M \longrightarrow 0,$$

where L_0 and L_1 are finite free, and apply a) to the complex

$$P_1 \xrightarrow{{}^t d} P_0, \qquad P_0 = \check{L}_1, \quad P_1 = \check{L}_0,$$

all its other terms being zero. Then

$$T_1(A) = \check{M}$$

and

$$T_1(k) = \operatorname{Hom}_A(M,k) = \operatorname{Hom}_k(M/\mathfrak{m}M,k).$$

Thus $T_1(k)$ is the dual of the vector space $M \otimes_A k$ and has the same rank. To prove that b) implies c), we first establish the following lemma.

Lemma (7.6.11.6). — *Given a complex of flat A-modules*

$$\cdots \longrightarrow 0 \longrightarrow P_1 \xrightarrow{d} P_0 \longrightarrow 0 \longrightarrow \cdots,$$

there is an exact sequence

$$0 \longrightarrow \operatorname{Tor}_2^A(Z_0',k) \longrightarrow T_1(A) \otimes_A k \longrightarrow T_1(k) \longrightarrow \operatorname{Tor}_1^A(Z_0',k) \longrightarrow 0. \tag{7.6.11.7}$$

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Proof. From the exact sequence

$$0 \longrightarrow Z_1 \longrightarrow P_1 \longrightarrow B_0 \longrightarrow 0$$

we obtain

$$0 \longrightarrow \operatorname{Tor}_1^A(B_0,k) \longrightarrow Z_1 \otimes_A k \longrightarrow P_1 \otimes_A k \xrightarrow{u} B_0 \otimes_A k \longrightarrow 0.$$

The exact sequence

$$0 \longrightarrow B_0 \longrightarrow Z_0 \longrightarrow Z_0' \longrightarrow 0$$

and the flatness of $Z_0 = P_0$ give

$$\operatorname{Tor}_1^A(B_0,k) = \operatorname{Tor}_2^A(Z_0',k).$$

By definition, $Z_1 = T_1(A)$. Moreover, $T_1(k) = \operatorname{Ker}(d \otimes 1)$, and $d \otimes 1$ factors as

$$P_1 \otimes_A k \xrightarrow{u} B_0 \otimes_A k \xrightarrow{v} Z_0 \otimes_A k = P_0 \otimes_A k.$$

Thus $T_1(k) = u^{-1}(R)$, where $R = \operatorname{Ker} v$. Since u is surjective, $R = u(T_1(k))$; finally, $R = \operatorname{Tor}_1^A(Z_0',k)$ by the definition of v. This proves (7.6.11.7). ☐

To deduce c) from b), take an exact sequence

$$L_1 \xrightarrow{d} L_0 \longrightarrow N \longrightarrow 0$$

with L_0, L_1 finite free, and consider the functor T associated with the complex formed by L_1 and L_0. Since L_0 and L_1 are free, they identify with their biduals. If

$$M = \operatorname{Coker}({}^t d),$$

then
$$T_1(A) = \operatorname{Ker} d = \check{M}.$$
Furthermore,
$$M \otimes_A k = \operatorname{Coker}({}^t d \otimes 1_k)$$
has the same rank over k as $\operatorname{Ker}(d \otimes 1_k)$. Hypothesis b) therefore gives
$$\operatorname{rank}_k\big(T_1(A) \otimes_A k\big) \leqslant \operatorname{rank}_k T_1(k).$$
Since $Z'_0 = N$, inequality (7.6.11.4) follows from the exact sequence (7.6.11.7).

Finally, to prove that c) implies a), apply (7.6.11.6) to the two-term complex
$$P_p \longrightarrow P_{p-1}.$$
Hypothesis c), applied to Z'_{p-1}, gives
$$\operatorname{rank}_k R \geqslant \operatorname{rank}_k S,$$
where
$$R = \operatorname{Ker}(P_p \otimes_A k \longrightarrow P_{p-1} \otimes_A k), \qquad S = Z_p \otimes_A k.$$
Factor $d_{p+1} : P_{p+1} \to P_p$ as
$$P_{p+1} \xrightarrow{v} Z_p \xrightarrow{j} P_p.$$
Then
$$\operatorname{Im}(d_{p+1} \otimes 1) = (j \otimes 1)\big(\operatorname{Im}(v \otimes 1)\big).$$
Since
$$T_p(A) \otimes_A k = (Z_p/B_p) \otimes_A k = (Z_p \otimes_A k)/\operatorname{Im}(v \otimes 1)$$
and
$$T_p(k) = R/\operatorname{Im}(d_{p+1} \otimes 1),$$
inequality (7.6.11.2) follows.

Now suppose that the local ring A is regular of dimension n. It is known [Ser55] that the A-module $\operatorname{Tor}_i^A(k,k)$ is isomorphic to the exterior power
$$\bigwedge^i(\mathfrak{m}/\mathfrak{m}^2).$$
Thus (7.6.11.4) fails for $N = k$ as soon as $n \geqslant 4$. On the other hand, suppose that A is a local integral domain such that every finite-type reflexive A-module is free, as happens when A is a regular ring of dimension 2. Then (7.6.11.3) holds. Indeed, the dual $\check{M}$ of a finite-type A-module M is reflexive, hence free, and consequently
$$\operatorname{rank}_k(\check{M} \otimes_A k) = \operatorname{rank}_K(\check{M}) = \operatorname{rank}_K(M),$$
where K is the field of fractions of A. Every basis of $M \otimes_A k$ over k consists of the images of a system of generators of M (Bourbaki, *Algèbre commutative*, chap. II, § 3, no. 2, cor. 2 of prop. 4). Hence
$$\operatorname{rank}_K(M) \leqslant \operatorname{rank}_k(M \otimes_A k),$$
which proves the assertion.

7.7. Application to proper morphisms: I. The exchange property The next three paragraphs are, for the most part, translations into the language of morphisms of preschemes of the results of the preceding paragraphs.

(7.7.1). Let $f : X \to Y$ be a quasi-compact separated morphism of preschemes, and let $\mathscr{P}_\bullet$ be a complex of quasi-coherent $\mathscr{O}_X$-Modules whose differential has degree -1. Suppose in addition that the $\mathscr{O}_X$-Modules $\mathscr{P}_i$ are Y-flat (**0**, 6.7.1).

We shall consider the ∂-functor **III | 198**

$$\mathscr{M} \longmapsto \mathscr{T}_\bullet(\mathscr{M}),$$

also denoted $\mathscr{T}_\bullet(\mathscr{P}_\bullet, \mathscr{M})$, from the category of quasi-coherent $\mathscr{O}_Y$-Modules to itself (by (6.2.3)), defined by

$$\mathscr{T}_n(\mathscr{P}_\bullet, \mathscr{M}) = \mathscr{T}_n(\mathscr{M}) = \mathscr{H}^{-n}(f, \mathscr{P}^\bullet \otimes_{\mathscr{O}_Y} \mathscr{M}) \qquad (n \in \mathbf{Z}), \tag{7.7.1.1}$$

where $\mathscr{P}^\bullet$ is the complex whose term of degree j is $\mathscr{P}_{-j}$, so that its differential has degree $+1$. The functor $\mathscr{T}_\bullet$ thus defined is a *homological functor* in $\mathscr{M}$ (6.2.6).

(7.7.2). Let $g : Y' \to Y$ be a morphism, and set

$$X' = X_{(Y')} = X \times_Y Y', \qquad f' = f_{(Y')} : X' \longrightarrow Y'.$$

Thus f' is quasi-compact and separated. Put also

$$\mathscr{P}'_\bullet = \mathscr{P}_\bullet \otimes_{\mathscr{O}_Y} \mathscr{O}_{Y'}.$$

This is a complex of quasi-coherent $\mathscr{O}_{X'}$-Modules that are Y'-flat, by (**I**, 9.1.12) and (**0**, 6.2.1). With the same conventions on degrees, set

$$\mathscr{T}_\bullet^{Y'}(\mathscr{M}') = \mathscr{H}^\bullet(f', \mathscr{P}'^\bullet \otimes_{\mathscr{O}_{Y'}} \mathscr{M}') = \mathscr{H}^\bullet(f', \mathscr{P}^\bullet \otimes_{\mathscr{O}_Y} \mathscr{M}'). \tag{7.7.2.1}$$

This is a homological functor of the quasi-coherent $\mathscr{O}_{Y'}$-Module $\mathscr{M}'$. When Y' is an affine scheme with ring A', we write $\mathscr{T}_\bullet^{A'}$ in place of $\mathscr{T}_\bullet^{Y'}$. For every A'-module M',

$$\mathscr{T}_\bullet^{A'}(\widetilde{M'}) = \big(\Gamma(Y', \mathscr{T}_\bullet^{Y'}(\widetilde{M'}))\big)^\sim.$$

We set

$$T_\bullet^{A'}(M') = \Gamma(Y', \mathscr{T}_\bullet^{Y'}(\widetilde{M'}));$$

this is a homological functor from A'-modules to A'-modules.

If $Y = \operatorname{Spec}(A)$ is also affine, the functor $T_\bullet^{A'}$ on A'-modules is obtained from the homological functor $T_\bullet^{A}$ on A-modules by extension of scalars from A to A' (7.1.3). Indeed, let $g : Y' \to Y$ correspond to the ring homomorphism $A \to A'$, and let $g' : X' \to X$ be the corresponding morphism, which is affine (**II**, 1.6.2). If $\mathfrak{U}$ is an affine open covering of X, then $\mathfrak{U}' = g'^{-1}(\mathfrak{U})$ is an affine open covering of X'. By (6.2.2), it is enough to check

$$C^\bullet(\mathfrak{U}, \mathscr{P}_\bullet \otimes_{\mathscr{O}_Y} g_*(\mathscr{M}')) = C^\bullet(\mathfrak{U}', \mathscr{P}_\bullet \otimes_{\mathscr{O}_Y} \mathscr{M}').$$

Ultimately, if U is any affine open subset of X and $U' = g'^{-1}(U)$, this reduces to

$$\Gamma(U, \mathscr{P}_\bullet \otimes_{\mathscr{O}_Y} g_*(\mathscr{M}')) = \Gamma(U', \mathscr{P}_\bullet \otimes_{\mathscr{O}_Y} \mathscr{M}'),$$

which is immediate from (**I**, 1.3) and (3.2).

In particular, if U is an open subset of Y, then for every quasi-coherent $\mathscr{O}_Y$-Module $\mathscr{M}$,

$$\mathscr{T}_\bullet^{U}(\mathscr{M}|U) = \mathscr{T}_\bullet(\mathscr{M})|U. \tag{7.7.2.2}$$

(7.7.3). For every quasi-coherent $\mathscr{O}_Y$-Module $\mathscr{M}$ there is a canonical homomorphism, functorial in $\mathscr{M}$,

$$\mathscr{T}_p(\mathscr{O}_Y) \otimes_{\mathscr{O}_Y} \mathscr{M} \longrightarrow \mathscr{T}_p(\mathscr{M}). \tag{7.7.3.1}$$

Indeed, when Y is affine this homomorphism was defined in (7.2.2). The definition extends immediately to the general case: if U and V are affine open subsets of Y with $V \subset U$, the diagram

$$\begin{array}{ccc}
(\mathscr{T}_p(\mathscr{O}_Y) \otimes_{\mathscr{O}_Y} \mathscr{M})|U & & \mathscr{T}_p^U(\mathscr{M}|U) \\
= \mathscr{T}_p^U(\mathscr{O}_Y|U) \otimes_{\mathscr{O}_Y|U} (\mathscr{M}|U) & \longrightarrow & = \mathscr{T}_p(\mathscr{M})|U \\
\downarrow & & \downarrow \\
(\mathscr{T}_p(\mathscr{O}_Y) \otimes_{\mathscr{O}_Y} \mathscr{M})|V & & \mathscr{T}_p^V(\mathscr{M}|V) \\
= \mathscr{T}_p^V(\mathscr{O}_Y|V) \otimes_{\mathscr{O}_Y|V} (\mathscr{M}|V) & \longrightarrow & = \mathscr{T}_p(\mathscr{M})|V
\end{array}$$

commutes by (7.2.3.3).

For every morphism $g : Y' \to Y$ there is a canonical homomorphism **III** | 199

(7.7.3.2) $$\mathscr{T}_p(\mathscr{O}_Y) \otimes_{\mathscr{O}_Y} \mathscr{O}_{Y'} \longrightarrow \mathscr{T}_p^{Y'}(\mathscr{O}_{Y'}),$$

which is precisely the special case of (6.7.11.2), for the abutments, in which $S = Y$, $v_1 = f$, $v_2 = 1_Y$, and $\mathscr{P}_\bullet^{(2)}$ is concentrated in degree 0 with sole term $\mathscr{M}$.

When $Y = \operatorname{Spec}(A)$ and $Y' = \operatorname{Spec}(A')$ are affine, (7.7.3.2) is the homomorphism of sheaves corresponding to the canonical homomorphism of A'-modules defined in (7.2.2),

$$T_p^A(A) \otimes_A A' \longrightarrow T_p^{A'}(A') = T_p^A(A').$$

This follows immediately from (6.7.11), since in the situation at hand one may take $\mathfrak{U}'^{(i)} = u_i^{-1}(\mathfrak{U}^{(i)})$ there.

(7.7.4). Suppose that f is proper, that $Y = \operatorname{Spec}(A)$ is an affine Noetherian scheme, and that $\mathscr{P}_\bullet$ is a bounded-below complex of coherent, Y-flat $\mathscr{O}_X$-Modules. We saw in (6.10.5) that, up to isomorphism, one may write

$$\mathscr{T}_p(\mathscr{M}) = \mathscr{H}_p(\mathscr{L}_\bullet \otimes_{\mathscr{O}_Y} \mathscr{M}), \qquad \mathscr{L}_\bullet = \widetilde{L_\bullet},$$

where $L_\bullet$ is a bounded-below complex of finite-type free A-modules. Thus $\mathscr{T}_\bullet$ is of the kind studied in detail in (7.4) and (7.6). We shall now translate the results of that study.

Theorem (7.7.5). — *Let Y be a locally Noetherian prescheme, (U_α) a covering of Y by affine open subsets, $f : X \to Y$ a proper morphism, and $\mathscr{P}_\bullet$ a bounded-below complex of coherent, Y-flat $\mathscr{O}_X$-Modules. The homological functor $\mathscr{T}_\bullet(\mathscr{M})$ defined by* (7.7.1.1) *then has the following properties.*

I) (The semicontinuity property).[5] The function

(7.7.5.1) $$y \longmapsto d_p(y) = \operatorname{rank}_{k(y)} T_p^{k(y)}(k(y))$$

is upper semicontinuous.

II) (The exchange property). For a fixed integer p, the following conditions are equivalent: **III** | 200

- *a)* *$\mathscr{T}_p$ is right exact.*
- *a′)* *$\mathscr{T}_p(\mathscr{M})$ is isomorphic to a functor of the form $\mathscr{N} \otimes_{\mathscr{O}_Y} \mathscr{M}$, where necessarily*

$$\mathscr{N} \simeq \mathscr{T}_p(\mathscr{O}_Y) = \mathscr{H}^{-p}(f, \mathscr{P}^\bullet).$$

- *a″)* *The canonical functorial homomorphism* (7.7.3.1) *is an isomorphism.*
- *b)* *$\mathscr{T}_{p-1}$ is left exact.*
- *b′)* *There is an $\mathscr{O}_Y$-Module $\mathscr{Q}$, necessarily coherent and determined up to unique isomorphism, together with an isomorphism of functors*

(7.7.5.2) $$\mathscr{T}_{p-1}(\mathscr{M}) \xrightarrow{\sim} \mathscr{H}om_{\mathscr{O}_Y}(\mathscr{Q}, \mathscr{M}).$$

[5] A special case of this theorem already occurs in note [3] of Chow–Igusa. The semicontinuity property was discovered, for analytic spaces and under rather special hypotheses, by Kodaira–Spencer, *On the variations of almost-complex structures*, in *Algebraic Geometry and Topology: A Symposium in Honor of S. Lefschetz*, Princeton Series no. 12, pp. 139–150, Princeton, 1957; the general version was proved by Grauert [5].

c) *If A_α denotes the ring of the affine open subset U_α, then for every α the functor $T_p^{A_\alpha}$ on A_α-modules is right exact.*

d) *For every morphism $g : Y' \to Y$, the canonical homomorphism*

(7.7.5.3) $$\mathscr{T}_p(\mathscr{O}_Y) \otimes_{\mathscr{O}_Y} \mathscr{O}_{Y'} \longrightarrow \mathscr{T}_p^{Y'}(\mathscr{O}_{Y'})$$

is an isomorphism.

Proof. The semicontinuity property is local on Y, and therefore follows from (7.7.4) and (7.6.9). It is clear that a$''$) implies a$'$), and that a$'$) implies a). When Y is affine, the equivalence of a), a$''$), b), and b$'$) was proved in (7.3.1) and (7.4.6), in view of (7.7.4).

To pass to the general case, first observe that a) is equivalent to c); this also proves that a) is local on Y. The same argument proves the locality of a$''$) and b). Condition c) plainly implies a), so it remains to prove the converse. It is enough to show that, for every affine open subset U of Y and every exact sequence

$$0 \longrightarrow \mathscr{F}' \longrightarrow \mathscr{F} \longrightarrow \mathscr{F}'' \longrightarrow 0$$

of quasi-coherent $(\mathscr{O}_Y|U)$-Modules, there is an exact sequence

$$0 \longrightarrow \mathscr{G}' \longrightarrow \mathscr{G} \longrightarrow \mathscr{G}'' \longrightarrow 0$$

of quasi-coherent $\mathscr{O}_Y$-Modules such that

$$\mathscr{F}' = \mathscr{G}'|U, \qquad \mathscr{F} = \mathscr{G}|U, \qquad \mathscr{F}'' = \mathscr{G}''|U.$$

This follows immediately from the hypothesis that Y is locally Noetherian and from (**I**, 9.4.2): extend $\mathscr{F}$ to a quasi-coherent $\mathscr{O}_Y$-Module $\mathscr{G}$, extend $\mathscr{F}'$ to a quasi-coherent submodule $\mathscr{G}'$ of $\mathscr{G}$, and take $\mathscr{G}'' = \mathscr{G}/\mathscr{G}'$.

To prove the equivalence of b) and b$'$) in the general case, note first that, when Y is affine, $\mathscr{Q}$ is determined up to unique isomorphism. If U is an affine open subset of the affine scheme Y, it follows that there is a functorial isomorphism

$$\mathscr{T}_{p-1}^U(\mathscr{M}|U) \simeq \mathscr{H}om_{\mathscr{O}_Y|U}(\mathscr{Q}|U, \mathscr{M}|U).$$

For arbitrary Y and each affine open subset U, there is a coherent $(\mathscr{O}_Y|U)$-Module $\mathscr{Q}_U$ and a functorial isomorphism

$$\mathscr{T}_{p-1}^U(\mathscr{M}|U) \simeq \mathscr{H}om_{\mathscr{O}_Y|U}(\mathscr{Q}_U, \mathscr{M}|U).$$

The preceding observation shows that, whenever $V \subset U$ is affine, $\mathscr{Q}_U|V$ is canonically identified with $\mathscr{Q}_V$. Hence there exists a unique $\mathscr{O}_Y$-Module $\mathscr{Q}$ satisfying (7.7.5.2).

It remains to prove the equivalence of a) and d). Condition d) is local on Y, and we have just seen that the same is true of a); moreover, d) is also local on Y'. When $Y = \operatorname{Spec}(A)$ and $Y' = \operatorname{Spec}(A')$, the functor $T_\bullet^{A'}$ is obtained from $T_\bullet^A$ by extension of scalars to A'. It is then clear that a$'$) implies that (7.7.5.3) is an isomorphism.

Conversely, continue to suppose that $Y = \operatorname{Spec}(A)$ is affine and take for A' the A-algebra $A \oplus M$, **III** | 201
where M is an arbitrary A-module and multiplication is given by

$$(a_1, m_1)(a_2, m_2) = (a_1 a_2, a_1 m_2 + a_2 m_1).$$

Then

$$T_p^{A'}(A') = T_p(A \oplus M) = T_p(A) \oplus T_p(M).$$

If (7.7.5.3) is bijective, the same is therefore true of the canonical homomorphism

$$T_p(A) \otimes_A M \longrightarrow T_p(M).$$

Thus d) implies a$''$), and the proof is complete. □

Theorem (7.7.6). — *Let Y be a locally Noetherian prescheme, let $f : X \to Y$ be proper, and let $\mathscr{F}$ be a coherent, Y-flat $\mathscr{O}_X$-Module. There is a coherent $\mathscr{O}_Y$-Module $\mathscr{Q}$, determined up to unique isomorphism, and an isomorphism of functors in the quasi-coherent $\mathscr{O}_Y$-Module $\mathscr{M}$,*

(7.7.6.1) $$f_*(\mathscr{F} \otimes_{\mathscr{O}_Y} \mathscr{M}) \xrightarrow{\sim} \mathscr{H}om_{\mathscr{O}_Y}(\mathscr{Q}, \mathscr{M}).$$

Consequently there is an isomorphism of functors

(7.7.6.2) $$\Gamma(X, \mathscr{F} \otimes_{\mathscr{O}_Y} \mathscr{M}) \xrightarrow{\sim} \operatorname{Hom}_{\mathscr{O}_Y}(\mathscr{Q}, \mathscr{M}).$$

Proof. The functor $\mathscr{M} \mapsto \mathscr{F} \otimes_{\mathscr{O}_Y} \mathscr{M}$ is exact (**0**, 6.7.4), while f_* is left exact. Hence

$$\mathscr{M} \longmapsto f_*(\mathscr{F} \otimes_{\mathscr{O}_Y} \mathscr{M})$$

is left exact. Apply the equivalence of b) and b′) in (7.7.5) with $p = 1$. □

Corollary (7.7.7). — *Let Y be a locally Noetherian prescheme, let $f : X \to Y$ be proper, let $\mathscr{F}$ and $\mathscr{F}'$ be coherent, Y-flat $\mathscr{O}_X$-Modules, and let $u : \mathscr{F} \to \mathscr{F}'$ be a homomorphism. Consider the two functors of the quasi-coherent $\mathscr{O}_Y$-Module $\mathscr{M}$,*

$$\begin{aligned}\mathscr{T}(\mathscr{M}) &= \operatorname{Ker}\left(f_*(\mathscr{F} \otimes_{\mathscr{O}_Y} \mathscr{M}) \longrightarrow f_*(\mathscr{F}' \otimes_{\mathscr{O}_Y} \mathscr{M})\right), \\ T(\mathscr{M}) &= \Gamma(Y, \mathscr{T}(\mathscr{M})) \\ &= \operatorname{Ker}\left(\Gamma(X, \mathscr{F} \otimes_{\mathscr{O}_Y} \mathscr{M}) \longrightarrow \Gamma(X, \mathscr{F}' \otimes_{\mathscr{O}_Y} \mathscr{M})\right).\end{aligned}$$

There is a coherent $\mathscr{O}_Y$-Module $\mathscr{R}$, determined up to unique isomorphism, and isomorphisms of functors

(7.7.7.1) $$\mathscr{T}(\mathscr{M}) \xrightarrow{\sim} \mathscr{H}om_{\mathscr{O}_Y}(\mathscr{R}, \mathscr{M}),$$

(7.7.7.2) $$T(\mathscr{M}) \xrightarrow{\sim} \operatorname{Hom}_{\mathscr{O}_Y}(\mathscr{R}, \mathscr{M}).$$

Proof. It is enough to prove (7.7.7.2). This proves (7.7.7.1) when Y is affine, and the passage to the general case proceeds as in the proof of the equivalence of b) and b′) in (7.7.5), using the uniqueness, up to unique isomorphism, of a representing object of a representable functor (**0**, 8.1.8).

By (7.7.6), there are coherent $\mathscr{O}_Y$-Modules $\mathscr{Q}$ and $\mathscr{Q}'$ and functorial isomorphisms

$$\Gamma(X, \mathscr{F} \otimes_{\mathscr{O}_Y} \mathscr{M}) \xrightarrow{\sim} \operatorname{Hom}_{\mathscr{O}_Y}(\mathscr{Q}, \mathscr{M}),$$

$$\Gamma(X, \mathscr{F}' \otimes_{\mathscr{O}_Y} \mathscr{M}) \xrightarrow{\sim} \operatorname{Hom}_{\mathscr{O}_Y}(\mathscr{Q}', \mathscr{M}).$$

The homomorphism $u : \mathscr{F} \to \mathscr{F}'$ canonically defines a morphism of functors between the two left-hand sides.

It corresponds to a unique homomorphism $v : \mathscr{Q}' \to \mathscr{Q}$ of $\mathscr{O}_Y$-Modules for which **III** | 202

$$\begin{tikzcd} \Gamma(X, \mathscr{F} \otimes_{\mathscr{O}_Y} \mathscr{M}) \arrow[r] \arrow[d, "\sim"'] & \Gamma(X, \mathscr{F}' \otimes_{\mathscr{O}_Y} \mathscr{M}) \arrow[d, "\sim"] \\ \operatorname{Hom}_{\mathscr{O}_Y}(\mathscr{Q}, \mathscr{M}) \arrow[r] & \operatorname{Hom}_{\mathscr{O}_Y}(\mathscr{Q}', \mathscr{M}) \end{tikzcd}$$

commutes (**0**, 8.1.4). The contravariant functor

$$\mathscr{N} \longmapsto \operatorname{Hom}_{\mathscr{O}_Y}(\mathscr{N}, \mathscr{M})$$

is left exact in the category of $\mathscr{O}_Y$-Modules. Taking $\mathscr{R} = \operatorname{Coker}(v)$ therefore gives (7.7.7.2). □

Corollary (7.7.8). — *Under the hypotheses of (7.7.6) on X, Y, and f, let $\mathscr{F}$ and $\mathscr{G}$ be coherent $\mathscr{O}_X$-Modules satisfying:*

(i) $\mathscr{F}$ is Y-flat;
(ii) $\mathscr{G}$ is isomorphic to the cokernel of a homomorphism $\mathscr{E}_1 \to \mathscr{E}_0$ between finite-type locally free $\mathscr{O}_X$-Modules.

Consider the two functors of the quasi-coherent $\mathscr{O}_Y$-Module $\mathscr{M}$,

$$\begin{aligned}\mathscr{T}(\mathscr{M}) &= f_*\mathscr{H}om_{\mathscr{O}_X}(\mathscr{G}, \mathscr{F} \otimes_{\mathscr{O}_Y} \mathscr{M}), \\ T(\mathscr{M}) &= \Gamma(Y, \mathscr{T}(\mathscr{M})) \\ &= \operatorname{Hom}_{\mathscr{O}_X}(\mathscr{G}, \mathscr{F} \otimes_{\mathscr{O}_Y} \mathscr{M}).\end{aligned}$$

There is a coherent $\mathscr{O}_Y$-Module $\mathscr{N}$, determined up to unique isomorphism, and isomorphisms of functors

(7.7.8.1) $$\mathscr{T}(\mathscr{M}) \xrightarrow{\sim} \mathscr{H}om_{\mathscr{O}_Y}(\mathscr{N}, \mathscr{M}),$$

(7.7.8.2) $$T(\mathscr{M}) \xrightarrow{\sim} \operatorname{Hom}_{\mathscr{O}_Y}(\mathscr{N}, \mathscr{M}).$$

Proof. The functorial isomorphism (**0**, 5.4.2.1) gives, functorially in $\mathscr{M}$,

$$\begin{aligned}\mathscr{H}om_{\mathscr{O}_X}(\mathscr{E}_i, \mathscr{F} \otimes_{\mathscr{O}_Y} \mathscr{M}) &\simeq \mathscr{E}_i^\vee \otimes_{\mathscr{O}_X} (\mathscr{F} \otimes_{\mathscr{O}_Y} \mathscr{M}) \\ &\simeq (\mathscr{E}_i^\vee \otimes_{\mathscr{O}_X} \mathscr{F}) \otimes_{\mathscr{O}_Y} \mathscr{M} \\ &\simeq \mathscr{H}om_{\mathscr{O}_X}(\mathscr{E}_i, \mathscr{F}) \otimes_{\mathscr{O}_Y} \mathscr{M}\end{aligned}$$

for $i = 0, 1$. Put

$$\mathscr{F}_i = \mathscr{H}om_{\mathscr{O}_X}(\mathscr{E}_i, \mathscr{F}) \qquad (i = 0, 1).$$

These are coherent (**0**, 5.3.5) and Y-flat (**0**, 5.4.2). If $v : \mathscr{E}_1 \to \mathscr{E}_0$ is the given homomorphism, put

$$u = \mathscr{H}om(v, 1_{\mathscr{F}}) : \mathscr{F}_0 \longrightarrow \mathscr{F}_1.$$

Since

$$\mathscr{H} \longmapsto \mathscr{H}om_{\mathscr{O}_X}(\mathscr{H}, \mathscr{F} \otimes_{\mathscr{O}_Y} \mathscr{M})$$

is left exact, there are functorial isomorphisms

$$\begin{aligned}\mathscr{H}om_{\mathscr{O}_X}(\mathscr{G}, \mathscr{F} \otimes_{\mathscr{O}_Y} \mathscr{M}) &\simeq \operatorname{Ker}\left(\mathscr{H}om_{\mathscr{O}_X}(\mathscr{E}_0, \mathscr{F} \otimes_{\mathscr{O}_Y} \mathscr{M}) \longrightarrow \mathscr{H}om_{\mathscr{O}_X}(\mathscr{E}_1, \mathscr{F} \otimes_{\mathscr{O}_Y} \mathscr{M})\right) \\ &\simeq \operatorname{Ker}\left(\mathscr{F}_0 \otimes_{\mathscr{O}_Y} \mathscr{M} \longrightarrow \mathscr{F}_1 \otimes_{\mathscr{O}_Y} \mathscr{M}\right).\end{aligned}$$

As f_* is left exact, it follows that

$$f_* \mathscr{H}om_{\mathscr{O}_X}(\mathscr{G}, \mathscr{F} \otimes_{\mathscr{O}_Y} \mathscr{M}) \simeq \operatorname{Ker}\left(f_*(\mathscr{F}_0 \otimes_{\mathscr{O}_Y} \mathscr{M}) \longrightarrow f_*(\mathscr{F}_1 \otimes_{\mathscr{O}_Y} \mathscr{M})\right).$$

The assertion now follows from (7.7.7). □

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Remarks (7.7.9). — (i) In (7.7.6), (7.7.7), and (7.7.8), the formation of the $\mathscr{O}_Y$-Modules $\mathscr{Q}, \mathscr{R}$, and $\mathscr{N}$ commutes with base change. For example, retaining the notation of (7.7.2), in the situation of (7.7.6) one has, for every quasi-coherent $\mathscr{O}_{Y'}$-Module $\mathscr{M}'$,

$$f'_*(\mathscr{F} \otimes_{\mathscr{O}_Y} \mathscr{M}') \simeq \mathscr{H}om_{\mathscr{O}_{Y'}}(g^*(\mathscr{Q}), \mathscr{M}').$$

Indeed, by the observation in (7.7.2), it is enough to use

$$\operatorname{Hom}_{\mathscr{O}_Y}(\mathscr{Q}, g_*(\mathscr{M}')) = \operatorname{Hom}_{\mathscr{O}_{Y'}}(g^*(\mathscr{Q}), \mathscr{M}'),$$

which is (**0**, 4.4.3.1).

Likewise, if in (7.7.7) one replaces $Y, f, \mathscr{M}, \mathscr{F}, \mathscr{F}'$ by

$$Y', f', \mathscr{M}', \mathscr{F} \otimes_{\mathscr{O}_X} \mathscr{O}_{X'}, \mathscr{F}' \otimes_{\mathscr{O}_X} \mathscr{O}_{X'},$$

then one must replace $\mathscr{R}$ by $g^*(\mathscr{R})$. Finally, in (7.7.8), if $X, Y, f, \mathscr{M}, \mathscr{F}$ are replaced by

$$X', Y', f', \mathscr{M}', \mathscr{F} \otimes_{\mathscr{O}_X} \mathscr{O}_{X'},$$

and $\mathscr{G}$ by

$$\mathscr{G}' = \mathscr{G} \otimes_{\mathscr{O}_X} \mathscr{O}_{X'},$$

then $\mathscr{N}$ must be replaced by $g^*(\mathscr{N})$. This follows because there remains an exact sequence

$$\mathscr{E}'_1 \longrightarrow \mathscr{E}'_0 \longrightarrow \mathscr{G}' \longrightarrow 0, \qquad \mathscr{E}'_i = \mathscr{E}_i \otimes_{\mathscr{O}_X} \mathscr{O}_{X'},$$

and because, for $i = 0, 1$,

$$(\mathscr{E}'_i)^\vee \otimes_{\mathscr{O}_{X'}} (\mathscr{F} \otimes_{\mathscr{O}_Y} \mathscr{M}') = \mathscr{E}_i^\vee \otimes_{\mathscr{O}_X} (\mathscr{F} \otimes_{\mathscr{O}_Y} \mathscr{M}').$$

(ii) Condition (ii) in (7.7.8) is satisfied by every coherent $\mathscr{O}_X$-Module $\mathscr{G}$ whenever there exists a Y-ample invertible $\mathscr{O}_X$-Module; this holds, for example, when Y is affine and $f : X \to Y$ is projective. Indeed, by (**II**, 5.5.1) and (**II**, 2.7.10), there is a finite-type locally free $\mathscr{O}_X$-Module $\mathscr{E}_0$ with $\mathscr{G}$ isomorphic to a quotient of $\mathscr{E}_0$. Since $\mathscr{E}_0$ and $\mathscr{G}$ are coherent, so is the kernel $\mathscr{G}_1$ of $\mathscr{E}_0 \to \mathscr{G}$. Applying the same result to $\mathscr{G}_1$ gives an exact sequence

$$\mathscr{E}_1 \longrightarrow \mathscr{E}_0 \longrightarrow \mathscr{G} \longrightarrow 0$$

with $\mathscr{E}_0$ and $\mathscr{E}_1$ finite-type locally free.

(iii) We shall prove in Chapter V that the restrictive hypothesis (ii) of (7.7.8) is redundant.

Proposition (7.7.10). — (Local criteria for the exchange property). *Under the general hypotheses of (7.7.5), let $y \in Y$ and let p be an integer. The following conditions are equivalent:*

a) The functor $T_p^{\mathscr{O}_y}$ is right exact.
b) The canonical homomorphism

$$T_p^{\mathscr{O}_y}(\mathscr{O}_y) \longrightarrow T_p^{\mathscr{O}_y}(k(y))$$

is surjective.
c) For every integer n, the canonical homomorphism

$$T_p^{\mathscr{O}_y}(\mathscr{O}_y/\mathfrak{m}_y^{n+1}) \longrightarrow T_p^{\mathscr{O}_y}(k(y))$$

is surjective.

Moreover, the set of points $y \in Y$ satisfying these conditions is the largest open subset U of Y for which $\mathscr{T}_p^U$ is right exact.

Proof. In view of (7.7.4), the equivalence of a), b), and c) follows from (7.4.7) and (7.5.2). That the locus U on which $T_p^{\mathscr{O}_y}$ is right exact is open follows from (7.4.4). Conversely, if $\mathscr{T}_p^V$ is right exact, then $T_p^{\mathscr{O}_y}$ is right exact for every $y \in V$, by condition c) of (7.7.5) and (7.6.1). □

Corollary (7.7.11). — *If $\mathscr{T}_p$ is right exact (respectively left exact), then for every morphism $g : Y' \to Y$ the functor $\mathscr{T}_p^{Y'}$ is right exact (respectively left exact). The converse holds when g is faithfully flat.*

Proof. The first assertion follows immediately from (7.6.1) and from the fact that the question is local III | 204
on Y and Y', by conditions c) and b) of (7.7.5). For the converse, it is enough, by (7.7.10), to show that $T_p^{\mathscr{O}_y}$ is right exact (respectively left exact) for every $y \in Y$. By hypothesis there is a point $y' \in Y'$ with $g(y') = y$, and $\mathscr{O}_{y'}$ is faithfully flat over $\mathscr{O}_y$. The conclusion follows from the hypothesis and (7.6.1). □

Remarks (7.7.12). — (i) Under the hypotheses of (7.7.4), suppose in addition that $\mathscr{P}_\bullet$ is a finite complex. Since one may choose a finite affine open covering $\mathfrak{U}$ of X, it follows from (7.7.1) that the bicomplex

$$C^\bullet(\mathfrak{U}, \mathscr{P}^\bullet \otimes_{\mathscr{O}_Y} \mathscr{M})$$

is finite as well. More precisely, there is a finite set E of pairs, independent of $\mathscr{M}$, such that

$$C^h(\mathfrak{U}, \mathscr{P}^k \otimes_{\mathscr{O}_Y} \mathscr{M}) = 0 \qquad \text{for } (h,k) \notin E.$$

Consequently there is an integer i_1 such that

$$\mathscr{T}_i(\mathscr{M}) = 0 \qquad (i \geqslant i_1)$$

for every quasi-coherent $\mathscr{O}_Y$-Module $\mathscr{M}$. Thus $\mathscr{T}_i$ is trivially exact for these values of i, and (7.4.1) shows that $Z_i(L_\bullet)$ is a flat finite-type A-module, hence a finite-type projective A-module since A is Noetherian.

Define a complex $L'_\bullet$ of A-modules by

$$L'_i = L_i \quad (i < i_1), \qquad L'_{i_1} = Z_{i_1}(L_\bullet), \qquad L'_i = 0 \quad (i > i_1),$$

and put $\mathscr{L}'_\bullet = \widetilde{L'_\bullet}$. Plainly

$$\mathscr{H}_i(\mathscr{L}'_\bullet \otimes_{\mathscr{O}_Y} \mathscr{M}) = \mathscr{H}_i(\mathscr{L}_\bullet \otimes_{\mathscr{O}_Y} \mathscr{M})$$

for $i < i_1 - 1$ and for $i \geqslant i_1$, the two sides being zero in the latter range. Since

$$\operatorname{Im}(Z_{i_1} \otimes_A M) = \operatorname{Im}(L_{i_1} \otimes_A M)$$

by definition, the same equality holds for $i = i_1 - 1$. Thus in (7.7.4) one may also take $\mathscr{L}_\bullet$ to be finite, provided only that its terms are locally free $\mathscr{O}_Y$-Modules associated to finite-type projective A-modules.

This applies in particular when $\mathscr{P}_\bullet$ has only one nonzero term $\mathscr{F}$, in degree 0. In that case

$$\mathscr{T}_n(\mathscr{M}) = R^{-n} f_*(\mathscr{F} \otimes_{\mathscr{O}_Y} \mathscr{M}),$$

and one may take $\mathscr{L}_i = 0$ for $i > 0$. In this case it is preferable to use cohomological notation, writing $\mathscr{T}^{-p}$ in place of $\mathscr{T}_p$.

(ii) If the terms $\mathscr{P}_i$ in (7.7.5) are no longer assumed Y-flat, the conclusions remain valid provided one defines

$$\mathscr{T}_p(\mathscr{M}) = \mathbf{Tor}_p^Y(f, 1_Y; \mathscr{P}_\bullet, \mathscr{M}). \tag{7.7.12.1}$$

Indeed,

$$\mathbf{Tor}_n^Y(f, 1_Y; \mathscr{P}_\bullet, \mathscr{O}_Y)$$

is then a coherent $\mathscr{O}_Y$-Module by (6.7.9). The proof of (6.10.5) applies without change, in view of (6.10.1), and also shows that, when $Y = \operatorname{Spec}(A)$ is affine,

$$\mathscr{T}_p(\mathscr{M}) = \mathscr{H}_p(\mathscr{L}_\bullet \otimes_{\mathscr{O}_Y} \mathscr{M}), \qquad \mathscr{L}_\bullet = \widetilde{L_\bullet},$$

where $L_\bullet$ is a complex of finite-type free A-modules. This proves the assertion.

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7.8. Application to proper morphisms: II. Cohomological flatness criteria

Definition (7.8.1). — Let X and Y be two preschemes, let $f : X \to Y$ be a quasi-compact separated morphism, let $\mathscr{P}_\bullet$ be a complex of quasi-coherent, Y-flat $\mathscr{O}_X$-Modules, and let $\mathscr{T}_\bullet$ be the homological functor on quasi-coherent $\mathscr{O}_Y$-Modules defined by (7.7.1.1). Let y be a point of Y.

We say that $\mathscr{P}_\bullet$ is *homologically flat over Y at y, in dimension p* (or *cohomologically flat over Y at y, in* III | 205 *dimension $-p$*) if there is an open neighbourhood U of y in Y such that

$$\mathscr{T}_p^U = \mathscr{T}_U^{-p}$$

is exact.

We say that $\mathscr{P}_\bullet$ is *homologically flat in dimension p over Y* (or *cohomologically flat in dimension $-p$ over Y*) if it is homologically flat over Y at every point $y \in Y$, in dimension p.

When $\mathscr{P}_\bullet$ is homologically flat over Y (respectively over Y at y) in *every* dimension p, we say simply that $\mathscr{P}_\bullet$ is *homologically flat over Y* (respectively over Y at y), or *cohomologically flat over Y* (respectively over Y at y).

(7.8.2). By definition, the notion of homological flatness over Y is *local on Y*.

If Y is locally Noetherian, or if Y is a scheme, then for $\mathscr{P}_\bullet$ to be homologically flat over Y in dimension p, it is necessary and sufficient that the functor $\mathscr{T}_p$ be exact. The proof for locally Noetherian Y was given in the course of the proof of (7.7.5); when Y is a scheme the argument is the same, based on (**I**, 9.4.2) applied to an affine open subset of a quasi-compact scheme.

Proposition (7.8.3). — *With the notation and hypotheses of (7.8.1), the following conditions are equivalent:*

a) $\mathscr{P}_\bullet$ is homologically flat over Y at y, in dimension p.
b) There is an open neighbourhood U of y in Y such that $\mathscr{T}_p^U$ and $\mathscr{T}_{p+1}^U$ are right exact.
c) There is an open neighbourhood U of y in Y such that $\mathscr{T}_p^U$ and $\mathscr{T}_{p-1}^U$ are left exact.
d) There is an open neighbourhood U of y in Y such that $\mathscr{T}_{p+1}^U$ is right exact and $\mathscr{T}_{p-1}^U$ is left exact.

Proof. In view of the interpretation of $\mathscr{T}_p$ when Y is affine, this is merely a translation of part of (7.3.3). □

Proposition (7.8.4). — *Let Y be a locally Noetherian prescheme, let $f : X \to Y$ be a proper morphism, let $\mathscr{P}_\bullet$ be a bounded-below complex of coherent, Y-flat $\mathscr{O}_X$-Modules, and let $\mathscr{T}_\bullet$ be the functor defined by* (7.7.1.1)*. For every $y \in Y$, the following conditions are equivalent:*

a) $\mathscr{P}_\bullet$ is homologically flat over Y at y, in dimension p.
b) The functor $\mathscr{T}_p^{\mathscr{O}_y}$ is exact.
c) There is an integer n_0 such that, for $n \geqslant n_0$,

$$\operatorname{length} \mathscr{T}_p^{\mathscr{O}_y}(\mathscr{O}_y/\mathfrak{m}_y^{n+1}) = \operatorname{length} \mathscr{T}_p^{\mathscr{O}_y}(k(y)) \cdot \operatorname{length}(\mathscr{O}_y/\mathfrak{m}_y^{n+1}). \tag{7.8.4.1}$$

Here all lengths are lengths of $\mathscr{O}_y$-modules.

d) There is an open neighbourhood U of y such that

$$\mathscr{H}^{-p}(f,\mathscr{P}^\bullet)|U$$

is isomorphic to an $(\mathscr{O}_Y|U)$-Module of the form $(\mathscr{O}_Y|U)^m$ and, for every quasi-coherent $(\mathscr{O}_Y|U)$-Module $\mathscr{M}$, the canonical homomorphism

$$\mathscr{H}^{-p}(f,\mathscr{P}^\bullet)|U\otimes_{\mathscr{O}_Y|U}\mathscr{M}\longrightarrow\mathscr{H}^{-p}(f,(\mathscr{P}^\bullet|U)\otimes_{\mathscr{O}_Y|U}\mathscr{M}) \tag{7.8.4.2}$$

is bijective.

When these conditions hold, one also has:

e) There is a neighbourhood of y on which the function $z\mapsto d_p(z)$, defined in (7.7.5.1), is constant.

*Moreover, if Y is reduced at y (**0**, 4.1.4), condition e) is equivalent to the other conditions.*

Proof. Indeed, condition b) is equivalent to saying that $\mathscr{T}_p^{\mathscr{O}_y}$ and $\mathscr{T}_{p+1}^{\mathscr{O}_y}$ are right exact (7.3.3). The equivalence of a) and b) then follows from (7.7.10) and (7.8.3). **III** | 206

Since $\mathscr{O}_y/\mathfrak{m}_y^{n+1}$ is Artinian, and

$$\mathscr{T}_p^{\mathscr{O}_y}(\mathscr{O}_y/\mathfrak{m}_y^{n+1})\quad\text{and}\quad\mathscr{T}_p^{\mathscr{O}_y}(k(y))$$

are finite-type $(\mathscr{O}_y/\mathfrak{m}_y^{n+1})$-modules (7.7.4), hence have finite length, the equivalence of b) and c) follows again from (7.7.10) and (7.3.5.7).

That a) implies e), and that a) and e) are equivalent when $\mathscr{O}_y$ is reduced, follows from (7.6.9). Finally, a) implies that (7.8.4.2) is bijective, by the definition (7.8.1) and (7.7.5). It also implies that

$$\left(\mathscr{H}^{-p}(f,\mathscr{P}^\bullet)\right)_y$$

is a flat $\mathscr{O}_y$-module by condition (f) of (7.3.3), hence a free $\mathscr{O}_y$-module of finite type (**0**, 10.1.3), since it is of finite type by (7.7.4). As $\mathscr{H}^{-p}(f,\mathscr{P}^\bullet)$ is a coherent $\mathscr{O}_Y$-Module (7.7.4), it is locally free in a neighbourhood of y (**0**, 5.2.7). Conversely, d) plainly implies a), by the definition of the functor $\mathscr{T}_p^U$ (7.7.2.2). □

Proposition (7.8.5). — *Under the hypotheses of (7.8.4), the following conditions are equivalent:*

a) $\mathscr{P}_\bullet$ is homologically flat over Y in every dimension $i\leqslant p$.
b) For $i\leqslant p+1$, the functors $\mathscr{T}_i$ are right exact.
c) For $i\geqslant -p$, the $\mathscr{O}_Y$-Modules $\mathscr{H}^i(f,\mathscr{P}^\bullet)$ are locally free.

Proof. The equivalence of a) and b) is immediate from (7.8.3), and a) implies c) by (7.8.4).

Conversely, suppose that c) holds. Recall also that $\mathscr{L}_i=0$ for $i\leqslant i_0$ (7.7.4), and hence $\mathscr{T}_i=0$ for $i\leqslant i_0$. Every point $y\in Y$ therefore has an affine neighbourhood $U=\operatorname{Spec}(A)$ such that $T_i^A(A)$ is a free A-module for $i\leqslant p$. It follows from (7.3.7) that $\mathscr{T}_i^U=T_i^A$ is exact for $i\leqslant p$. □

We shall chiefly apply the criteria for cohomological flatness to the case where the complex $\mathscr{P}_\bullet$ is concentrated in a *single* coherent $\mathscr{O}_X$-Module $\mathscr{F}$, flat over Y, taken as $\mathscr{P}_0$. Recall that in this case

$$\mathscr{T}_p(\mathscr{M})=R^{-p}f_*(\mathscr{F}\otimes_{\mathscr{O}_Y}\mathscr{M}).$$

Proposition (7.8.6). — *Let Y be a locally Noetherian prescheme, let $f:X\to Y$ be a proper flat morphism, and let y be a point of Y. Denote the fibre $f^{-1}(y)=X\otimes_Y k(y)$ by X_y. Suppose that*

$$\Gamma(X_y,\mathscr{O}_{X_y})=R$$

is a separable $k(y)$-algebra (Bourbaki, Algèbre*, chap. VIII, § 7, no. 5), that is, a direct sum of finitely many finite-degree separable extensions of $k(y)$. Then $\mathscr{O}_X$ is cohomologically flat over Y at y, in dimension* 0.

Proof. By (7.8.4), we may restrict to the case where Y is the spectrum of the local ring $A=\mathscr{O}_y$. The flatness of f implies $\mathscr{T}_{-1}=0$, so that T_0^A is already left exact and it remains only to prove that it is right exact. By condition c) of (7.7.10), we may even reduce to the case where $A=\mathscr{O}_y$ is Artinian.

Let k' be a finite extension of $k(y)$ which splits R, so that

$$R\otimes_{k(y)}k'$$

is a direct sum of finitely many fields isomorphic to k'. There is a local homomorphism from A to a local ring A' such that A' is a finite free A-algebra and its residue field is isomorphic to k' (**0**, 10.3.2).

By (7.6.1), it is therefore enough to prove that $T_0^{A'}$ is right exact; in other words, we may suppose that R is a direct sum of m fields isomorphic to $k(y)$.

We shall use the following elementary lemma.

Lemma (7.8.6.1). — *Let Z be a locally ringed space. For Z to be connected, it is necessary and sufficient that the ring $\Gamma(Z, \mathscr{O}_Z)$ not be a product of two nonzero rings.* **III | 207**

Proof. It is clear that if Z is the union of two disjoint nonempty open subsets Z_1 and Z_2, then $\Gamma(Z, \mathscr{O}_Z)$ is isomorphic to the product of the two nonzero rings $\Gamma(Z_1, \mathscr{O}_Z)$ and $\Gamma(Z_2, \mathscr{O}_Z)$.

Conversely, to say that $\Gamma(Z, \mathscr{O}_Z)$ is such a product is equivalent to saying that it contains an idempotent s distinct from 0 and 1. For every $z \in Z$, the germ s_z is then an idempotent in the local ring $\mathscr{O}_z$, and hence equals 0 or 1. The set of z for which $s_z = 0$ is open. On the other hand, $s_z = 1$ implies, by definition, that $s(z) \neq 0$, and the set of z for which $s_z = 1$ is open as well (**0**, 5.5.2). The conclusion follows. □

It follows from the lemma that X_y has exactly m connected components X'_i and that

$$\Gamma(X'_i, \mathscr{O}_{X'_i}) = k(y)$$

for every i. Since A is local and Artinian, its spectrum consists of one point, and consequently X and X_y have the same underlying space. Thus X has m connected components X_i such that

$$X'_i = X_i \otimes_Y k(y).$$

We are finally reduced to the case $R = k(y)$. By condition b) of (7.7.10), it remains to prove that the canonical homomorphism

$$\Gamma(X, \mathscr{O}_X) \longrightarrow \Gamma(X_y, \mathscr{O}_{X_y})$$

is surjective. This is immediate, since the composite

$$\Gamma(Y, \mathscr{O}_Y) = A \longrightarrow \Gamma(X, \mathscr{O}_X) \longrightarrow \Gamma(X_y, \mathscr{O}_{X_y}) = k(y)$$

is already surjective. □

Corollary (7.8.7). — *Under the hypotheses of (7.8.6), there is an open neighbourhood U of y such that:*

(i) $f_*(\mathscr{O}_X)|U$ is isomorphic to an $(\mathscr{O}_Y|U)$-Module of the form $(\mathscr{O}_Y|U)^m$;
(ii) for every $z \in U$, the canonical homomorphism

$$\big(f_*(\mathscr{O}_X)\big)_z \otimes_{\mathscr{O}_z} k(z) \longrightarrow \Gamma(X_z, \mathscr{O}_{X_z})$$

is bijective.

Proof. Assertion (i) follows from (7.8.6) and (7.8.4). Assertion (ii) follows from the exactness of $\mathscr{T}_0^U$, for a suitable choice of U, and from (7.7.5.3). □

Corollary (7.8.8). — *Suppose that the hypotheses of (7.8.6) hold and, in addition, that*

$$\Gamma(X_y, \mathscr{O}_{X_y}) = k(y).$$

Then there is an open neighbourhood U of y such that the canonical homomorphism

$$\mathscr{O}_Y|U \longrightarrow f_*(\mathscr{O}_X)|U$$

is bijective.

Proof. Indeed, it follows from assertion (ii) of (7.8.7) that the integer m occurring in assertion (i) of that corollary is necessarily equal to 1. □

Corollary (7.8.9). — *Under the hypotheses of (7.8.6), there are an open neighbourhood U of y, a coherent $\mathscr{O}_U$-Module $\mathscr{Q}$, determined up to unique isomorphism, and an isomorphism of functors in the quasi-coherent $\mathscr{O}_U$-Module $\mathscr{M}$,*

(7.8.9.1) $$R^1 f_*(f^*(\mathscr{M})) \simeq \mathscr{H}om_{\mathscr{O}_U}(\mathscr{Q}, \mathscr{M}).$$

Proof. The hypothesis implies that $\mathscr{T}_0^U$ is exact for a suitable U. It is therefore enough to apply the equivalence of conditions a) and b') in (7.7.5), with $p = 0$ and with $\mathscr{P}_\bullet$ the complex concentrated in degree 0 with term $\mathscr{O}_X$. □

Remarks (7.8.10). — (i) Under the hypotheses of (7.8.6), consider the Stein factorisation of f (4.3.3),

$$X \xrightarrow{f'} Y' \xrightarrow{g} Y.$$

Here **III | 208**

$$Y' = \operatorname{Spec}\big(f_*(\mathscr{O}_X)\big).$$

The finite morphism g is such that

$$g_*(\mathscr{O}_{Y'}) = f_*(\mathscr{O}_X)$$

is locally free near y, and its fibre over y is the spectrum of a separable algebra over $k(y)$ (**II**, 1.5.1). We shall deduce in Chapter IV that there is an open neighbourhood U of y in Y such that, for the restriction $g^{-1}(U) \to U$ of g, *every* fibre $g^{-1}(z)$, $z \in U$, is the spectrum of a separable algebra over $k(z)$. This is what we shall call an *étale covering* of U. It will then follow from assertion (ii) of (7.8.7) that the hypothesis on the point y in (7.8.6) also holds at every point of a neighbourhood of y.

(ii) We shall see in Chapter V that, even if X is projective over Y (and even if it is also "simple" over Y, a property to be defined in Chapter IV), the $\mathscr{O}_U$-Module $\mathscr{Q}$ of (7.8.9) need not be locally free. In other words, under these hypotheses $\mathscr{O}_X$ need not be cohomologically flat in *dimension* 1 over Y at y. In Chapter V we shall interpret $\mathscr{Q}$ as the sheaf of 1-differentials of the Picard scheme of X relative to Y, along the unit section.

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7.9. Application to proper morphisms: III. Invariance of the Euler–Poincaré characteristic and the Hilbert polynomial

(7.9.1). Let A be a ring and let M be a projective A-module of finite type. Recall (Bourbaki, *Alg. comm.*, chap. II, §5, no. 2) that this is equivalent to saying that the $\mathscr{O}_X$-Module $\widetilde{M}$ associated with M on $X = \operatorname{Spec}(A)$ is locally free of finite type. For every $\mathfrak{p} \in \operatorname{Spec}(A)$, the *rank of M at* $\mathfrak{p}$, denoted $\operatorname{rank}_{\mathfrak{p}}(M)$, is the rank of the free $A_{\mathfrak{p}}$-module $M_{\mathfrak{p}}$ (or, equivalently, the rank at $\mathfrak{p}$ of the locally free $\mathscr{O}_X$-Module $\widetilde{M}$). Thus

(7.9.1.1) $$\operatorname{rank}_{\mathfrak{p}} M = \operatorname{rank}_{\mathfrak{p}}(M_{\mathfrak{p}}) = \operatorname{rank}_{k(\mathfrak{p})}\big(M \otimes_A k(\mathfrak{p})\big).$$

Proposition (7.9.2). — *Let $P_\bullet$ be a finite complex of projective A-modules of finite type, and, for every A-module M, set*

$$T_\bullet(M) = H_\bullet(P_\bullet \otimes_A M).$$

Then, for every $\mathfrak{p} \in \operatorname{Spec}(A)$,

(7.9.2.1) $$\sum_i (-1)^i \operatorname{rank}_{k(\mathfrak{p})} T_i\big(k(\mathfrak{p})\big) = \sum_i (-1)^i \operatorname{rank}_{\mathfrak{p}}(P_i).$$

Proof. Indeed, $T_i(k(\mathfrak{p})) = H_i(P_\bullet \otimes_A k(\mathfrak{p}))$ by definition. In view of (7.9.1.1), formula (7.9.2.1) is simply the invariance of the Euler–Poincaré characteristic of a finite complex of finite-dimensional vector spaces upon passage to homology (**0**, 11.10.2). ☐

Corollary (7.9.3). — *The function*

$$\mathfrak{p} \longmapsto \sum_i (-1)^i \operatorname{rank}_{k(\mathfrak{p})} T_i\big(k(\mathfrak{p})\big)$$

is locally constant on $\operatorname{Spec}(A)$.

Theorem (7.9.4). — *Let Y be a locally Noetherian prescheme, let $f : X \to Y$ be a proper morphism, and let $\mathscr{P}_\bullet$ be a finite complex of coherent, Y-flat $\mathscr{O}_X$-Modules. If*

$$\mathscr{T}_\bullet(\mathscr{M}) = \mathscr{H}^{-\bullet}\big(f, \mathscr{P}^\bullet \otimes_{\mathscr{O}_Y} \mathscr{M}\big) \qquad \text{(cf. (7.7.1.1))},$$

then the function

(7.9.4.1) $$y \longmapsto \sum_i (-1)^i \operatorname{rank}_{k(y)} \mathscr{T}_i\big(k(y)\big)$$

is locally constant on Y.

Proof. We may restrict ourselves to the case where $Y = \operatorname{Spec}(A)$ is affine and A is Noetherian. Since the complex $\mathscr{P}_\bullet$ is finite, (7.7.12)(i) shows that

$$\mathscr{T}_p(\mathscr{M}) = \mathscr{H}_p(\mathscr{L}_\bullet \otimes_{\mathscr{O}_Y} \mathscr{M}),$$

where $\mathscr{L}_\bullet = \widetilde{L}_\bullet$ and $L_\bullet$ is a finite complex of projective A-modules of finite type. The theorem now follows from (7.9.3). □

(7.9.5). Under the hypotheses of (7.9.4), the function (7.9.4.1) is *constant* when Y is *connected*. When Y is connected and nonempty, its unique (integral) value is denoted

$$\mathrm{EP}(f,\mathscr{P}_\bullet), \qquad \mathrm{EP}(Y,\mathscr{P}_\bullet), \qquad \text{or simply} \qquad \mathrm{EP}(\mathscr{P}_\bullet)$$

when no confusion is possible; this integer is called the *Euler–Poincaré characteristic* of $\mathscr{P}_\bullet$ relative to f (or to Y). In general, we likewise write

$$\mathrm{EP}(f,\mathscr{P}_\bullet;y), \qquad \mathrm{EP}(Y,\mathscr{P}_\bullet;y), \qquad \mathrm{EP}(\mathscr{P}_\bullet;y)$$

for the right-hand side of (7.9.4.1).

(7.9.6). Under the hypotheses of (7.9.4) concerning X, Y, and f, let

$$0 \longrightarrow \mathscr{P}'_\bullet \xrightarrow{u} \mathscr{P}_\bullet \xrightarrow{v} \mathscr{P}''_\bullet \longrightarrow 0$$

be an exact sequence of finite complexes of coherent, Y-flat $\mathscr{O}_X$-Modules, where the homomorphisms u and v have even degrees $2d$ and $2d'$, respectively. Since $\mathscr{T}_\bullet$ is a homological functor (7.7.1), there is an exact homology sequence

$$\cdots \longrightarrow \mathscr{T}_i(\mathscr{P}'_\bullet,k(y)) \longrightarrow \mathscr{T}_{i+2d}(\mathscr{P}_\bullet,k(y)) \longrightarrow \mathscr{T}_{i+2d+2d'}(\mathscr{P}''_\bullet,k(y)) \longrightarrow \mathscr{T}_{i-1}(\mathscr{P}'_\bullet,k(y)) \longrightarrow \cdots,$$

which moreover has only finitely many terms. On writing that the Euler–Poincaré characteristic of this complex is zero (**0**, 11.10.1), we obtain at once

$$\mathrm{EP}(\mathscr{P}_\bullet;y) = \mathrm{EP}(\mathscr{P}'_\bullet;y) + \mathrm{EP}(\mathscr{P}''_\bullet;y) \tag{7.9.6.1}$$

for every $y \in Y$.

Suppose, for example, that $\mathscr{P}_\bullet = (\mathscr{P}_i)$ and that $\mathscr{P}_i = 0$ for $i < 0$. We then have the following exact sequence of complexes:

$$\begin{tikzcd}
\cdots \arrow[r] & 0 \arrow[r] \arrow[d] & 0 \arrow[r] \arrow[d] & 0 \arrow[r] \arrow[d] & 0 \arrow[r] & \cdots \\
\cdots \arrow[r] & 0 \arrow[r] \arrow[d] & 0 \arrow[r] \arrow[d] & \mathscr{P}_1 \arrow[r] \arrow[d] & \mathscr{P}_2 \arrow[r] & \cdots \\
\cdots \arrow[r] & 0 \arrow[r] \arrow[d] & \mathscr{P}_0 \arrow[r] \arrow[d] & \mathscr{P}_1 \arrow[r] \arrow[d] & \mathscr{P}_2 \arrow[r] & \cdots \\
\cdots \arrow[r] & 0 \arrow[r] \arrow[d] & \mathscr{P}_0 \arrow[r] \arrow[d] & 0 \arrow[r] \arrow[d] & 0 \arrow[r] & \cdots \\
\cdots \arrow[r] & 0 \arrow[r] & 0 \arrow[r] & 0 \arrow[r] & 0 \arrow[r] & \cdots
\end{tikzcd}$$

the nonzero vertical arrows being identity automorphisms. Applying (7.9.6.1) to this exact sequence and arguing by induction on the length of $\mathscr{P}_\bullet$ gives

$$\mathrm{EP}(\mathscr{P}_\bullet;y) = \sum_i (-1)^i \,\mathrm{EP}(\mathscr{P}_i;y). \tag{7.9.6.2}$$

Here, for each coherent $\mathscr{O}_X$-Module $\mathscr{F}$ that is flat over Y, the notation $\mathrm{EP}(\mathscr{F};y)$ (or $\mathrm{EP}(f,\mathscr{F};y)$, or **III** | 210
$\mathrm{EP}(Y,\mathscr{F};y)$) means $\mathrm{EP}(\mathscr{Q}_\bullet;y)$ (with the corresponding relative notation) for the complex $\mathscr{Q}_\bullet$ whose only nonzero term is $\mathscr{F}$ in degree 0. Thus the study of Euler–Poincaré characteristics of complexes reduces to that of complexes with a single term.

Proposition (7.9.7). — *Under the hypotheses of (7.9.4), let Y' be a locally Noetherian prescheme, let $g : Y' \to Y$ be a morphism, put*

$$X' = X \times_Y Y', \qquad f' = f_{(Y')} : X' \longrightarrow Y',$$

and let

$$\mathscr{P}'_\bullet = \mathscr{P}_\bullet \otimes_{\mathscr{O}_Y} \mathscr{O}_{Y'}$$

be the resulting finite complex of $\mathscr{O}_{X'}$-Modules. Then $\mathscr{P}'_\bullet$ consists of coherent, Y'-flat $\mathscr{O}_{X'}$-Modules, and, for every $y' \in Y'$,

$$\mathrm{EP}(\mathscr{P}'_\bullet; y') = \mathrm{EP}(\mathscr{P}_\bullet; g(y')). \tag{7.9.7.1}$$

Proof. The $\mathscr{O}_{X'}$-Modules $\mathscr{P}'_i$, being inverse images of the $\mathscr{P}_i$ under the projection $X' \to X$, are coherent. They are Y'-flat by (**0**, 6.2.1) and (**I**, 4.14.5), the question being local on X, Y, and Y'. Finally, f' is proper by (**II**, 5.4.2), so the left-hand side of (7.9.7.1) is defined. Formula (7.9.7.1) now follows from (6.10.4.2), (7.7.2), and (7.6.7), after reducing, as one may always do, to the case where Y and Y' are affine. □

Proposition (7.9.8). — *Suppose that the hypotheses of (7.9.4) hold and, in addition, that there is an integer i_0 such that*

$$\mathscr{T}_i(k(y)) = 0 \qquad \text{for } i \neq i_0 \text{ and every } y \in Y.$$

Then

$$\mathscr{T}_{i_0}(\mathscr{O}_Y) = \mathscr{H}^{-i_0}(f, \mathscr{P}^\bullet)$$

is a locally free $\mathscr{O}_Y$-Module, and its rank at $y \in Y$ is

$$(-1)^{i_0}\,\mathrm{EP}(f, \mathscr{P}_\bullet; y).$$

Proof. First observe that the hypotheses of (7.4.4) are satisfied by the functors $\mathscr{T}_i^{\mathscr{O}_y}$, so that (7.4.7) applies to them. The hypothesis and (7.5.3) imply that $\mathscr{T}_i^{\mathscr{O}_y}$ is zero for $i \neq i_0$. Consequently $\mathscr{T}_{i_0}$ is exact by (7.3.3), and (7.8.4) shows that $\mathscr{H}^{-i_0}(f, \mathscr{P}^\bullet)$ is locally free. Its rank at $y \in Y$ is

$$\mathrm{rank}_{k(y)}\, \mathscr{T}_{i_0}(k(y)) = (-1)^{i_0}\,\mathrm{EP}(f, \mathscr{P}_\bullet; y),$$

by definition, since $\mathscr{T}_i(k(y)) = 0$ for $i \neq i_0$.[6] □

Corollary (7.9.9). — *Let Y be a locally Noetherian prescheme, let $f : X \to Y$ be a proper morphism, and let $\mathscr{F}$ be a coherent, Y-flat $\mathscr{O}_X$-Module. Suppose that there is an integer i_0 such that*

$$H^i\big(f^{-1}(y), \mathscr{F} \otimes_{\mathscr{O}_Y} k(y)\big) = 0$$

for every $i \neq i_0$ and every $y \in Y$. Then $R^{i_0} f_*(\mathscr{F})$ is a locally free $\mathscr{O}_Y$-Module, and its rank at y is $(-1)^{i_0}\,\mathrm{EP}(f, \mathscr{F}; y)$.

In particular:

Corollary (7.9.10). — *Under the preliminary hypotheses of (7.9.9) concerning X, Y, and $\mathscr{F}$, suppose that $R^i f_*(\mathscr{F}) = 0$ for every $i > 0$. Then $f_*(\mathscr{F})$ is a locally free $\mathscr{O}_Y$-Module, and its rank at y is $\mathrm{EP}(f, \mathscr{F}; y)$.*

Proof. In view of (7.9.9), it is enough to prove the following lemma. □

Lemma (7.9.10.1). — *Under the hypotheses of (7.9.10),*

$$H^i\big(f^{-1}(y), \mathscr{F} \otimes_{\mathscr{O}_Y} k(y)\big) = 0$$

for every $i > 0$ and every $y \in Y$.

Proof. We may restrict ourselves to the case where $Y = \mathrm{Spec}(A)$ is affine. With the notation of (7.9.4), and with $\mathscr{P}_\bullet$ reduced to its only nonzero term $\mathscr{F}$ in degree 0, the hypothesis gives $\mathscr{T}_p(\mathscr{O}_Y) = 0$ for $p < 0$. It follows from (7.3.7) that $\mathscr{T}_p$ is exact for $p < 0$, and the lemma then follows from the equivalence of conditions a) and d) in (7.7.5). □

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[6] The printed French proof omits the factor $(-1)^{i_0}$ in this displayed equality; it is required both by the definition and by the statement of the proposition.

Proposition (7.9.11). — *Under the hypotheses of (7.9.4), let $\mathscr{L}$ be an invertible $\mathscr{O}_X$-Module that is very ample over Y, and, for every $n \in \mathbf{Z}$, put*

$$\mathscr{P}_\bullet(n) = \mathscr{P}_\bullet \otimes_{\mathscr{O}_X} \mathscr{L}^{\otimes n}.$$

Then, for every $y \in Y$, the function

$$n \longmapsto \mathrm{EP}(f, \mathscr{P}_\bullet(n); y) \tag{7.9.11.1}$$

is a polynomial with rational coefficients, and is the same for all points in any one connected component of Y.

Proof. It is clear that $\mathscr{P}_\bullet(n)$ is a complex of Y-flat $\mathscr{O}_X$-Modules. By (7.9.6.2), we may reduce to the case where $\mathscr{P}_\bullet$ has a single nonzero term $\mathscr{F} \neq 0$ in degree 0. Since the questions are local on Y, we may moreover suppose that Y is affine and that f is projective (**II**, 5.5.3). Put

$$X_y = f^{-1}(y), \qquad \mathscr{L}_y = \mathscr{L} \otimes_{\mathscr{O}_Y} k(y);$$

then $\mathscr{L}_y$ is a very ample $\mathscr{O}_{X_y}$-Module (**II**, 4.4.10). By (7.7.2), for the functor $\mathscr{T}_\bullet$ associated with $\mathscr{P}_\bullet(n)$ we have

$$\mathscr{T}_i(k(y)) = H^{-i}\big(X_y, \mathscr{F}_y \otimes \mathscr{L}_y^{\otimes n}\big), \qquad \mathscr{F}_y = \mathscr{F} \otimes_{\mathscr{O}_Y} k(y).$$

Thus $\mathrm{EP}(f, \mathscr{F}(n); y)$ is precisely the Euler–Poincaré characteristic $\chi_{k(y)}(\mathscr{F}_y(n))$ defined in (2.5.1). It follows from (2.5.3) that (7.9.11.1) is a polynomial. For each fixed n, its value is constant on each connected component of Y by (7.9.4); this completes the proof. □

(7.9.12). The polynomial (7.9.11.1), with rational coefficients, will be denoted

$$\mathrm{PH}(f, \mathscr{P}_\bullet; y) \qquad \text{or} \qquad \mathrm{PH}(\mathscr{P}_\bullet; y),$$

and will be called the *Hilbert polynomial at y* relative to $\mathscr{P}_\bullet$, f, and $\mathscr{L}$ (or simply the *Hilbert polynomial at y* of $\mathscr{P}_\bullet$, or of f, when no confusion is possible). When Y is connected and nonempty, the reference to y is omitted from the notation and terminology. This invariant will play an essential role in Chapter V, in the theory of "moduli" of coherent quotient sheaves of a given coherent sheaf.

With the notation of (7.9.6) and (7.9.11), we have

$$\mathrm{PH}(\mathscr{P}_\bullet; y) = \mathrm{PH}(\mathscr{P}'_\bullet; y) + \mathrm{PH}(\mathscr{P}''_\bullet; y), \tag{7.9.12.1}$$

and in particular

$$\mathrm{PH}(\mathscr{P}_\bullet; y) = \sum_i (-1)^i \, \mathrm{PH}(\mathscr{P}_i; y). \tag{7.9.12.2}$$

These formulas follow immediately from (7.9.6.1) and (7.9.6.2). Likewise, with the notation and hypotheses of (7.9.7),

$$\mathrm{PH}(\mathscr{P}'_\bullet; y') = \mathrm{PH}(\mathscr{P}_\bullet; g(y')). \tag{7.9.12.3}$$

Formula (7.9.12.2) reduces the study of the Hilbert polynomials of a complex to that of the Hilbert polynomial of a single Y-flat $\mathscr{O}_X$-Module. Such polynomials have the following remarkable interpretation, independent of homological considerations.

Corollary (7.9.13). — *Let Y be a Noetherian prescheme, let $f : X \to Y$ be a proper morphism, let $\mathscr{L}$ be an invertible $\mathscr{O}_X$-Module that is very ample over Y, and let $\mathscr{F}$ be a coherent, Y-flat $\mathscr{O}_X$-Module. There is an integer n_0 such that, for $n \geqslant n_0$, $f_*(\mathscr{F}(n))$ is a locally free $\mathscr{O}_Y$-Module whose rank at $y \in Y$ is*

$$\mathrm{PH}(f, \mathscr{F}; y)(n).$$

Proof. Since f is projective (**II**, 5.5.3), there is an integer n_0 such that, for $n \geqslant n_0$, **III** | 212

$$R^i f_*(\mathscr{F}(n)) = 0 \qquad \text{for every } i > 0$$

by (2.2.1). The conclusion follows from (7.9.10). □

The following flatness criterion will be important in Chapter V, in the theory of "moduli" of coherent sheaves.

Proposition (7.9.14). — *Let Y be a locally Noetherian prescheme, let $f : X \to Y$ be a projective morphism, and let $\mathscr{L}$ be an invertible $\mathscr{O}_X$-Module that is ample for f. For every $\mathscr{O}_X$-Module $\mathscr{F}$ and every $n \in \mathbf{Z}$, put*

$$\mathscr{F}(n) = \mathscr{F} \otimes \mathscr{L}^{\otimes n}.$$

A coherent $\mathscr{O}_X$-Module $\mathscr{F}$ is Y-flat if and only if there is an integer n_0 such that, for every $n \geqslant n_0$, $f_*(\mathscr{F}(n))$ is a locally free $\mathscr{O}_Y$-Module.

Proof. The necessity of the condition is proved as in (7.9.13); the result of (2.2.1) applies to an ample sheaf because f is projective.

Conversely, we may reduce to the case where $Y = \operatorname{Spec}(A)$ is affine. By the hypothesis and (2.2.2)(i), the A-modules $\Gamma(X, \mathscr{F}(n))$ are projective and of finite type (Bourbaki, *Alg. comm.*, chap. II, §5, no. 2, th. 1). Let

$$S = \bigoplus_{n \geqslant 0} \Gamma(X, \mathscr{L}^{\otimes n})$$

be the associated graded ring. We know that X is canonically identified with $\operatorname{Proj}(S)$ by (**II**, 4.5.2)(b) and (5.4.4). Let

$$M = \bigoplus_{n \geqslant n_0} \Gamma(X, \mathscr{F}(n)).$$

After replacing $\mathscr{L}$, if necessary, by a power $\mathscr{L}^{\otimes d}$, we may suppose that S is generated by finitely many elements of degree 1 (2.3.5.1); it then follows from (**II**, 2.7.5) and (2.7.2) that $\mathscr{F}$ is identified with $\mathscr{P}roj_0(M)$.

For every homogeneous element $g \in S$ of positive degree, we therefore have

$$\Gamma(X_g, \mathscr{F}) = M_{(g)}.$$

Now M, being a direct sum of projective A-modules, is a flat A-module. Hence M_g is flat by (**0**, 6.3.2), and so is $M_{(g)}$, which is a direct factor of M_g by (**0**, 6.1.2). We conclude from (**I**, 4.14.5) that $\mathscr{F}$ is Y-flat at every point of X_g. Since the open subsets X_g cover X, the proposition is proved. □

(To be continued.)

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